

Logistics and Supply Chain Management

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Preface

The terms logistics and supply chain management are sometimes used interchangeably. Some say there is no difference between the two terms, that supply chain management is the “new” logistics.

The objective behind logistics is to make sure the customer receives the desired product at the right time and place with the right quality and price. This process can be divided into two subcategories: inbound logistics and outbound logistics.

Logistics is concerned with getting the products and services where they are needed and when they are desired. It is difficult to accomplish any marketing or manufacturing without logistical support. It involves the integration of information, transportation, inventory, warehousing, material handling, and packaging. The operating responsibility of logistics is the geographical repositioning of raw materials, work in process, and finished inventories where required at the lowest cost possible. The formal definition of the word ‘logistics’ as per the perception of Council of Logistics Management is the process of planning, implementing and controlling the efficient, effective flow and storage of goods, services and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements.

"Logistics typically refers to activities that occur within the boundaries of a single organization and Supply Chain refers to networks of companies that work together and coordinate their actions to deliver a product to market. Also, traditional logistics focuses its attention on activities such as procurement, distribution, maintenance, and inventory management. Supply Chain Management (SCM) acknowledges all of traditional logistics and also includes activities such as marketing, new product development, finance, and customer service"

The Logistics Information System allows you not only to evaluate actual data, but also to create planning data. The information systems provide an easy-to-use planning functions that are also supported by a forecasting function. The planning functionality of the information systems and the component Sales and Operations Planning (SOP) were combined and enhanced to make one central planning and forecasting tool. You can use the Logistics Data Warehouse in Customizing to design the Logistics Information System to meet your own requirements. This tool allows you to customize the setup of the data basis for your information system, to define the rules for updating the data and to generate the standard analyses for evaluating the data. The documentation for this tool can be found in the Implementation Guide for the Logistics Information System under “Data basis” and “Updating”. The Early Warning System is integrated in all of the information systems and is based on the key figures of the information system. The Early Warning System supports the decision-making process by allowing you to target and monitor weak areas in logistics. It searches for exceptional situations and helps in the early detection and correction of undesirable situations.

The term supply management describes the methods and processes of modern corporate or institutional buying. This may be for the purchasing of supplies for internal use referred to as indirect goods and services, purchasing raw materials for the consumption during the manufacturing process, or for the purchasing of goods for inventory to be resold as products in the distribution and retail process.

Supply management deals primarily with the oversight and management of materials and services inputs, management of the suppliers who provide those inputs, and support of the process of acquiring those inputs. The performance of supply management departments and supply management professionals is commonly measured in terms of amount of money saved for the organization. However, managing risk is one of the other critical aspects of supply management; especially the risk of non-availability at the required time of quality goods and services critical for an organization’s survival and growth.

The book is designed to contribute to the existing body of literature available on supply chain management.

– *Author*

Introduction

The scope of logistics management is wider than just optimizing transportation costs. Decisions about logistics have their effect on inventories, on packaging, on materials handling and on purchasing and marketing (sales) decisions and vice versa.

For instance, what is the point in producing small (but) economic batch quantities when the distribution facility dictates that a number of these batches should be accumulated before they can be shipped to the customers? Instead of carrying inventory inside the plant, one will carry it outside in the warehouse.

Moreover inventory control decisions have to take into account the inventories 'in transit' or in the pipeline. The latter depends much on the transport facilities or logistics decisions regarding transport. Decisions regarding the number of warehouse, their location, the number of customers serviced by each, etc. will have effects on inventory levels to be held at each warehouse. Also, depending upon the transport activity envisaged the number of warehouses and the inventory levels can be lower or higher. With more volume of transportation and a quicker mode of transportation, one may be able to manage with smaller investment in warehousing for the same level of service to the customers.

Decisions about the mode of packaging depend on the mode of transport, and the converse also is true. Quantities to purchase, quantities to sell, time of purchase and time of sales are well linked with the decision regarding the transport and bulk storage of materials. In short, the organization's goal is to provide good service to the customers in terms of quantity, quality, timely delivery and therefore various aspects such as inventory control, service levels, package design, warehousing (capacity, number location), the mode frequency and routes of transport will all have to be considered. Thus, Logistics is an integral part of the materials management system.

CONCEPTS OBJECTIVES AND ELEMENTS OF LOGISTICS

Logistics is new unique, it never stops! Logistics is happening around the globe 24 hours days Seven days a week during fifty-two weeks a year. Few areas of business involve the complexity or span the geography typical of logistics. Logistics is concerned with getting products and services where they are needed whenever they are desired. Most consumers take a high level of logistical competency for granted. When they go to store, they expect products to be available and fresh. It is rather difficult to visualize any marketing or manufacturing without logistical support Modern logistics is also a paradox. Logistics has been performed since the beginning of civilization: it's hardly new. However implementing best practice of logistics has become one of the most exciting and challenging operational areas of business and public sector management According to Council of logistics management: "Logistics is the process of planning, implementing and controlling the efficient, effective flow and storage of goods, services and related information from point of origin to point of consumption for the purpose of conforming the customer requirement".

Logistical management includes the design and administration of systems to controls the flow of material, work- in – process, and finished inventory to support business unit strategy. Logistics is the designing and managing of a system in order to control the flow of material throughout a corporation. This is a very important part of an international company because of geographical barriers. Logistics of an international company includes movement of raw materials, coordinating flows into and out of different countries, choices of transportation, and cost of the transportation, packaging the product for shipment, storing the product, and managing the entire process.

THE OBJECTIVES OF LOGISTICS

OPERATING OBJECTIVES

In terms of logistical system design and administration, each firm must simultaneously achieve at least six different operational objectives. These operational objectives, which are the primary determinants of logistical performance, include rapid response, minimum variance, minimum inventory, movement consolidation, quality, and life-cycle support.

RAPID RESPONSE

Rapid response is concerned with a firm's ability to satisfy customer service requirements in a timely manner. Information technology has increased the capability to postpone logistical operations to the latest possible time and then accomplish rapid delivery of required inventory. The result is elimination of excessive inventories

traditionally stocked in anticipation of customer requirements. Rapid response capability shifts operational emphasis from an anticipatory posture based on forecasting and inventory stocking to responding to customer requirements on a shipment-to-shipment basis. Because inventory is typically not moved in a time-based system until customer requirements are known and performance is committed, little tolerance exists for operational deficiencies

MINIMUM VARIANCE

Variance is any unexpected event that disrupts system performance. Variance may result from any aspect of logistical operations. Delays in expected time of customer order receipt, an unexpected disruption in manufacturing, goods arriving damaged at a customer's location, or delivery to an incorrect location-all result in a time disruption in operations that must be resolved. Potential reduction of variance' relates to both internal and external operations.

Operating areas of a logistical system are subject to potential variance. The traditional solution to accommodating variance was to establish safety stock inventory or use high-cost premium transportation. Such practices, given their expense and associated risk, have been replaced by using information technology to achieve positive logistics Control. To the extent that variances are minimized, logistical productivity improves as a result of economical operations. Thus, a basic objective of overall logistical performance is to minimize variance.

MINIMUM INVENTORY

The objective of minimum variance involves assess commitment and relative turn velocity. Total commitment is the financial value of inventory deployed throughout the logistical system. Turn velocity involves the rate of inventory usage over time. High turn rates, coupled with inventory availability, means that assets devoted to inventory are being effectively utilized. The objective is to reduce inventory deployment to the lowest level consistent with customer service goals to achieve the lowest overall total logistics cost. Concepts like zero inventories have become increasingly as managers seek to reduce inventory deployment.

The reality of reengineering a system is that operational defects do not become apparent until inventories are reduced to their lowest possible level. While the goal of eliminating all inventories is attractive, it is important to remember that inventory can and does facilitate some important benefits in a logistical system. Inventories can provide improved return on investment when they result in economies of scale in manufacturing or procurement. The objective is to reduce and manage inventory to the lowest possible level while simultaneously achieving desired operating objectives. To achieve the objective of minimum inventory, the logistical system design must control commitment and turn velocity for the entire firm, not merely for each business location.

MOVEMENT CONSOLIDATION

One of the most significant logistical costs is transportation. Transportation cost is directly related to the type of product, size of shipment, and distance. Many Logistical systems that feature premium service depend on high-speed, small-shipment transportation. Premium transportation is typically high-cost. To reduce transportation cost, it is desirable to achieve movement consolidation. As a general rule, the larger the overall shipment and the longer the distance it is transported, the lower the transportation cost per unit. This requires innovative programmes to group small shipments for consolidated movement. Such programmes must be facilitated by working arrangements that transcend the overall supply chain.

QUALITY IMPROVEMENT

A fifth logistical objective is to seek continuous quality improvement. Total quality management (TQM) has become a major commitment throughout all facets of industry. Overall commitment to TQM is one of the major forces contributing to the logistical renaissance. If a product becomes defective or if service promises are not kept, little, if any, value is added by the logistics. Logistical costs, once expended, cannot be reversed. In fact, when quality fails, the logistical performance typically needs to be reversed and then repeated. Logistics itself must perform to demanding quality standards. The management challenge of achieving zero defect logistical performance is magnified by the fact that logistical operations typically must be performed across a vast geographical area at all times of the day and night. The quality challenge is magnified by the fact that most logistical work is performed out of a supervisor's vision. Reworking a customer's order as a result of incorrect shipment or in-transit damage is far more costly than performing it right the first time. Logistics is a prime part of developing and maintaining continuous TQM improvement.

LIFE-CYCLE SUPPORT

The final logistical design objective is life-cycle support. Few items are sold without some guarantee that the product will perform as advertised over a specified period. In some situations. The normal value-added inventory flow towards customers must be reversed. Product recall is a critical competency resulting from increasingly rigid quality standards, product expiration dating and responsibility for hazardous consequences. Return logistics requirements also result from the increasing number of laws prohibiting disposal and encouraging recycling of beverage containers and packaging materials. The most significant aspect of reverse logistical operations is the need for maximum control when a potential health liability exists (*i.e.*, a contaminated product). In this sense, a recall programme is similar to a strategy of maximum customer service that must be executed regardless of cost. Johnson and Johnson's classical response to the Tylenol crisis is an example of turning adversity into advantage. The operational

requirements of reverse logistics range from lowest total cost, such as returning bottles for recycling, to maximum performance solutions for critical recalls. The important point is that sound logistical strategy cannot be formulated without careful review of reverse logistical requirements.

Some products, such as copying equipment, derive their primary profit from selling supplies and providing aftermarket. Service. The importance of service support logistics varies directly with the product and buyer. For firms marketing consumer durables or industrial equipment, the commitment to life-cycle support constitutes a versatile and demanding operational requirement as well as one of the largest costs of logistical operations. The life-cycle support capabilities of a logistical system must be carefully designed. As noted earlier, reverse logistical competency, as a result of worldwide attention to environmental concerns, requires the capacity to recycle ingredients and packaging materials. Life-cycle support, in modern terms, means cradle-to-cradle logistical support.

CONCEPT OF LOGISTICS

The concept of logistics is fairly new in the business world. The theoretical development was not used until 1966. Since then, many business practices have evolved and logistics currently costs between 10 and 25 percent of the total cost of an international purchase. There are two main phases that are important in the movement of materials: material management and physical distribution; Materials management is the timely movement of raw materials, parts, and supplies. The physical distribution is the movement of the firm's finished products to the customers. Both phases involve every stage of the process including storage. The ultimate goal of logistics is: "To coordinate all efforts of the company to maintain a cost effective flow of goods."

Word, 'Logistics' is derived from French word 'loger', which means art of war pertaining to movement and supply of armies.

A military concept, fighting a war requires:

- i. Setting of an objective
- ii. Meticulous planning to achieve the objective
- iii. Troops properly deployed
- iv. Supply line consisting weaponry, food, medical assistance, etc. maintained
- v. Plan should be such that there is minimum loss to men and material

Like fighting a war in the battlefield, the marketing managers also need a suitable logistics plan that is capable of satisfying the company objective of meeting profitably the demand of targeted customers. Inbound logistics + Material Management + Physical Distribution = Logistics

Discussion of each and every term in this above summation

- i. Inbound logistics covers the movement of materials received from suppliers
- ii. Material management describes the movements of material and components within a firm

- iii. Physical distribution refers to movement of goods outward from the end of the assembly line to the customer.
- iv. Supply-chain management is somewhat larger than logistics and it links logistics more directly within the user's total communication network and with the firm engineering staff. It includes manufacturer and suppliers but also transporters, warehouses, retailers and customers themselves.

Importance of logistics:

- i. Transportation cost rose rapidly due to the rise in fuel prices
- ii. Production efficiency was reaching a peak
- iii. Fundamental change in inventory philosophy
- iv. Product line proliferated
- v. Computer technology
- vi. Increased use of computers
- vii. Increased public concern of products Growth of several new, large retail chains or mass merchandise with large demands and very sophisticated logistics services, bypass traditional channel and distribution
- viii. Reduction in economic regulation
- ix. Growing power of retailers
- x. Globalization

The interrelation of different logistics element and their costs should be based on total cost rather than individual costs.

WHY TYLENOL REMAINS NUMBER ONE

Johnson and Johnson's McNeil Consumer Products Division was hit with a major crisis in September 1982. Their top-selling product line, Tylenol, was linked to seven deaths in the Chicago area. At the time of the incident, Tylenol enjoyed 35 percent of the \$1 billion analgesic market, but by the end of September, this market share had dropped 80 percent. Currently, Tylenol is again the top-selling brand with approximately 30 percent of the now \$2.7 billion analgesic market. How Was Johnson and Johnson (J&J) able to regain market share and a leading image after such a damaging tragedy? Its recovery was successful because of reverse logistics capability coupled with a marketing strategy that focused on protecting the consumer and going above and beyond what was necessary to instill trust and an image of security. This recovery plan is a positive prototype for other corporations to follow, which, in effect, may increase the potential for voluntary product recalls across a variety of industries.

When the List news reports hit about cyanide-tainted Extra-Strength Tylenol capsules, J&J was unsure whether the tampering occurred in its manufacturing operations or at the retail level. As such, its first efforts were directed at pinning down the problem. As soon as the lot numbers were identified from the first few deaths, J&J stopped production in the plant responsible. At the same time, it halted all Tylenol commercials nationwide and began recalls that eventually involved 31 million bottles of product, which had a

retail value of \$100 million. Another strategy that J&J took was to work openly and closely with the media. J&J has traditionally maintained a distance from the press, but in this case it felt that openness and honesty would help reduce consumer panic and provide a vehicle for disseminating critical information. A crisis team was put together that included J&J as well as McNeil executives and top managers. This team was quite sure that the tampering had occurred at the retail level since the incident was isolated to Chicago's West Side and other samples from the same lot were normal. Regardless, they began the recall with the remaining 93,000 bottles from this lot.

The expenses of this first phase of the recall included \$1 million just for phone calls and telegrams to doctors, hospitals, and distributors. The sixth poisoning ensured that the tampering was at the retail level since the bottle came from a lot manufactured at its second plant. Since the cause was now isolated, J&J could concentrate on containment. The first step was to advocate a total recall.

While this step was in some ways unnecessary, J&J felt it was a key step to ensure consumer confidence. At first, the FBI and FDA advised against a total recall because of the potential psychological response of the person who tampered with the product and the response of consumers in general. However, after copycat strychnine poisoning in California, all parties agreed that complete removal was the best solution.

This total recall entailed the following

- i. Advertisements stating that McNeil would exchange tablets for capsules,
- ii. Thousands of letters to the trade to explain the incident and recall procedures,
- iii. Media statements,
- iv. A sales force of over 2,000 employees to contact doctors and pharmacists to regain trust and restore their recommendations that had traditionally served as the main promotional avenue for Tylenol products,
- v. An extensive reverse logistics system that included buying products back from retailers and consumers and shipping returns to disposal centres, and
- vi. Creating a tamperproof package. It is estimated that recall costs were at least \$100 million, most of which involved the reverse logistics operations.

By January 1983, the new tamperproof bottles of Tylenol were on the retail shelf. Consumer confidence was obviously regained as a result of the extensive voluntary recall programme, effective public relations, and sales programmes and repack operations. This confidence was shown by the fact that at the end of the year, Tylenol had regained almost 30 percent of the market although market share has remained at about 30 percent, sales dollars have more than doubled. Since the total industry sales were about \$1 billion in the early 1980s but are now \$2.7 billion.

OVERVIEW OF LOGISTICS FUNCTION

Logistical History of India: India was a maritime power since about 300 BC, trading with several countries of the world bringing prosperity home. Traders of Surat brought

riches to the country by extensive maritime trade. Like many of our excellent practices, logistical efficiency also faded away over a period of time.

Some important logistical feats in history:

1. Berlin Airlift – 1945: A study in logistics. When the city of Berlin was blockaded by Soviets and all supply lines were cut off, Americans planned and executed a major logistics operation to feed the city from air.
2. Indians in the Gulf countries – 1991:
In 1991, when gulf war broke out, Indian Government evacuated thousands of Indians from the gulf countries and brought them home in a massive exercise employing Indian airlines planes.
3. Operation Overlord-1945: Allies' invasion of Europe and subsequent victory in II World War.
4. American war of Independence

Keeping 12,000 soldiers armed and fed from England was a big task; British lost the American war of independence due to bad logistics.

INTEGRATES LOGISTICAL ACTIVITIES

In conventional management environment, various activities of logistics work in isolation under different management functions. Each pocket trying to sub optimize its objectives at the cost of overall organizational objectives. Purchasing trying to purchase at minimum price at the cost of what is needed by operations. Operations produce large quantities at minimum production cost ignoring demand leading to doom inventory. Logistics function of management brings all such functions under one umbrella pulling down inter departmental barriers.

Competitive edge: In the fiercely competitive environment logistics provides the edge. Due to technological revolution most of the products are moving into commodity markets. In a commodity market where price is controlled by competition, where there is no product differentiation in terms of quality parameters like performance and reliability, where brands are almost irrelevant, competitive edge is that of availability of product and service in terms of time, place and quantity.

Logistics wins or loses wars

- British lost American war of independence due to poor logistics
- Rommel was beaten in the desert by superior logistics of Allies

Supports critical functions like operations and marketing

Strong logistics support enables a company to move towards JUST IN TIME production system for survival in a highly competitive market

INTERFACE WITH MARKETING

These days marketing a product is increasingly on the strength of availability and flexibility as we discussed earlier. Stronger emphasis is on the last of four Ps of marketing

[product, price, promotion and place]. Logistics provides the interface between production function and marketing function. Marketing is trying to sell the product in the market place. Logistics makes the product accessible to marketing by acting as interface between the function that produces it and the function that makes the consumer buy it.

This interface is gaining importance due to following changes that are sweeping the market making many companies adopt JUST IN TIME production system.

- a. Change in the customer: demanding, knowledgeable, conscious of rights, lacking in brand loyalty, changes preferences very fast, expects very high degree of service
- b. Many products are moving towards commodities market: product differentiation in terms of quality of performance is vanishing and brands are losing their magic.

As a result of above we find that availability is an important determinant of purchasing decision.

Logistical costs: For individual businesses logistics expenditures are 5% to 35% of sales depending on type of business, geographical areas of operation, weight/value ratios of products and materials. This is an expensive operation. Improvement in the efficiency of logistics function yields savings as well as customer satisfaction

RAPID RESPONSE

F-flexibility objective of an organization: Some companies measure this as response time to customer's order. On an average how much time do we need to fulfill one particular type of customer's order in a year? This is a measure of Rapid response

Logistics should ensure that the supplier is able to respond to the change in the demand very fast. Entire production should change from traditional push system to pull system to facilitate rapid response. Instead of stocking the goods and supplying on demand, orders are executed on shipment-to-shipment basis. Information Technology plays an important role here as an enabler. IT helps management in producing and delivering goods when the consumer needs them. This results into reduction of inventory and exposes all operational deficiencies. Now the management resolves these deficiencies and slashes down costs. [Concept of SMED and KANBAN as practiced by JIT companies in Japan or elsewhere]

MINIMUM VARIANCE

D-delivery objective of an organization, this can be measured as 'On Time Delivery' or OTD. If 100 deliveries are made in a month/quarter/year how many reached as per the commitment made to the customer? This percentage is OTD.

Any event that disrupts a system is variance. Logistics operations are disrupted by events like delays due to obstacles in information flow, traffic snarls, acts of god, wrong

dispatches, damage in transit. Traditional approach is to keep safety stocks and transport the goods by high cost mode. The cost of this approach is huge. Logistics is expected to minimize these events, thereby minimize and improve on OTD

MINIMUM INVENTORY

This is component of cost objective of a company. Inventory is associated with a huge baggage of costs. It is termed as a necessary evil. Objective of minimum inventory is measured as Inventory Turns or Inventory Turnover Ratio. Americans call this measure as turn velocity.

Logistics management increases these turns without sacrificing customer satisfaction. Higher turns ensure effective utilization of assets devoted to stock. [Concept of single piece flow as practiced by JIT companies in Japan or elsewhere]. Logistical management should keep the overall well being of a company in view and fix a minimum inventory level without trying to minimize the inventory level as an isolated objective

MOVEMENT CONSOLIDATION

Transportation is the biggest contributor to logistics cost. Transportation cost depends on product type, size, weight, distance to be transported, etc. for transporting small shipments just in time [reduction in inventory costs] expensive transport modes are used which again tend to hike the costs.

Movement consolidation is planning several such small shipments together [of different types of shipments] by integrating interests of several players in the supply chain. Generally, large shipment size and long distances reduce transportation cost per unit. Movement consolidation shall result into reduction in transportation costs.

QUALITY

If the quality of product fails logistics will have to ship the product out of customers premises and repeat the logistics operation again. This adds to costs and customer dissatisfaction. Hence logistics should contribute to TQM initiative of management. In fact, commitment to TQM has made the managements world over wake up to the significance of logistics function. Logistics can play a significant role in total quality improvement by improving the quality of logistics performance continuously and continually.

LOGISTICS MANAGEMENT SOFTWARE

Software is used for logistics automation which helps the supply chain industry in automating the work flow as well as management of the system. There are very few generalized software available in the new market in the said topology. This is because there is no rule to generalize the system as well as work flow even though the practice

is more or less the same. Most of the commercial companies do use one or the other of the custom solutions. But there are various software solutions that are being used within the departments of logistics. There are a few departments in Logistics, namely: Conventional Department, Container Department, Warehouse, Marine Engineering, Heavy Haulage, etc.

Software used in these departments:

- Conventional department : CVT software/CTMS software
- Container Trucking: CTMS software
- Warehouse : WMS/WCS.

Improving Effectiveness of Logistics Management:

1. Logistical Network
2. Information
3. Transportation
4. Sound Inventory Management
5. Warehousing, Materials Handling and Packaging.

LOGISTICS OFFICER

A Logistics Officer, known in the past as a Supply Officer, is a member of an armed force responsible for overseeing the supplying of an army, both at home and abroad. Logistics Officers can be stationary on military bases or deployed as an active part of a field army, air wing, or naval force. The responsibilities of Logistics Officers vary, depending on where they are deployed and what tasks they are assigned. In addition, different countries, have different roles for Logistics Officers. The main role of these officers remains the same regardless of where they are stationed; to ensure that the fighting force is supplied with enough food, water, fuel and ammunition to complete the task at hand.

CANADA

In Canada, Logistic officers work in five main disciplines, Supply chain, Management, Transportation, Human resource management, Finance and food services. Their main objective is to ensure that all Canadian military operations and training exercises are properly supported.

UNITED STATES

In the United States, the work of the logistics officer includes providing Strategic, Operational, or Tactical level logistical support and managing the overall logistic processes in support of mission objectives; leveraging existing networks and expanding others; anticipating mission needs and operational requirements and offering alternatives and advice; and evaluating, redesigning and implementing logistics processes for mission support.

UNITED KINGDOM

In the United Kingdom, a Logistics Officer in the Royal Air Force is colloquially known as a ‘Stacker’ (which is an abbreviation of either ‘Blanket Stacker’ or ‘Duvet Stacker’). This is a reference to the traditionally-held perception of the role of the Logistics Officer from the point of view of an Engineering Officer.

AUSTRALIA

In Australia each of the three independent Services of the Australian Defence Force (ADF) have different definitions for a ‘Logistics Officer’. Logistics Officers lead and coordinate planning and execution of the operations support components of the military effects which include the administrative (1) and logistics (4) functions, as well as major enabling support in the capability support dimension of force generation.

Royal Australian Navy (RAN) logisticians are called a Supply Officer, the commissioned element of the Supply Branch. RAN “Loggies” conduct the non-technical component of 1/4 support effects. Australian Army logisticians are commissioned officers in the Australian Regular Army (ARA) or Army Reserve (ARES) normally drawn from the Ordnance (RAAOC), Transport (RACT), Catering (RAACC), Engineer (RAE), Electrical and Mechanical Engineer (RAEME), or Medical (RAAMC) corps. Army defines engineering as a subset of logistics.

Royal Australian Air Force (RAAF) logisticians are called Logistics Officers (LOG), the commissioned element of the Logistics Employment Group (LEG). As in the RAF they are generally focused on the maintenance of the supply chain (in both Joint and Air environments), equipment governance, catering, movements and transport, the non-technical logistics disciplines. Unlike the Army RAAF “Loggies” work in parallel to RAAF engineers (Aeronautical [AERO], both Ground and Air Electrical [ELECTR], Armament [ARM] and Airfield [AFENG]) who comprise the specialist technical engineering and maintenance components of the RAAF integrated logistics workforce.

LOG and ENG personnel (except AFENG) are managed by Logistics Branch-Air Force within Air Force Headquarters (AFHQ). AFENG are managed by the RAAF’s Headquarters Combat Support Group. Additionally unlike their Navy and Army counterparts RAAF “Loggies” don’t manage administrative, personnel or finance functions which are provided by Administrative Officers (ADMIN) who perform the 1 shop functions of the 1/4 support environment leaving “Loggies” to perform the 4 shop function.

BELGIUM

In Belgium, the army has its own logistics branch. This branch is responsible for supporting the army in the wide logistics spectrum: supply, maintenance and contracting.

MANAGEMENT INFORMATION SYSTEM

A management information system (MIS) is a system or process that provides information needed to manage organizations effectively. Management information systems are regarded to be a subset of the overall internal controls procedures in a business, which cover the application of people, documents, technologies, and procedures used by management accountants to solve business problems such as costing a product, service or a business-wide strategy. Management information systems are distinct from regular information systems in that they are used to analyze other information systems applied in operational activities in the organization. Academically, the term is commonly used to refer to the group of information management methods tied to the automation or support of human decision making, *e.g.*, Decision Support Systems, Expert systems, and Executive information systems.

OVERVIEW

At the start, in businesses and other organizations, internal reporting was made manually and only periodically, as a by-product of the accounting system and with some additional statistic(s), and gave limited and delayed information on management performance. Previously, data had to be separated individually by the people as per the requirement and necessity of the organization. Later, data was distinguished from information, and instead of the collection of mass of data, important, and to the point data that is needed by the organization was stored.

Early on, business computers were mostly used for relatively simple operations such as tracking sales or payroll data, often without much detail. Over time these applications became more complex and began to store increasing amounts of information while also interlinking with previously separate information systems. As more and more data was stored and linked man began to analyze this information into further detail, creating entire management reports from the raw, stored data. The term “MIS” arose to describe these kinds of applications, which were developed to provide managers with information about sales, inventories, and other data that would help in managing the enterprise. Today, the term is used broadly in a number of contexts and includes (but is not limited to): decision support systems, resource and people management applications, ERP, SCM, CRM, project management and database retrieval application.

An ‘MIS’ is a planned system of the collecting, processing, storing and disseminating data in the form of information needed to carry out the functions of management. In a way it is a documented report of the activities that were planned and executed. According to Philip Kotler “A marketing information system consists of people, equipment, and procedures to gather, sort, analyze, evaluate, and distribute needed, timely, and accurate information to marketing decision makers.”

The terms MIS and information system are often confused. Information systems include systems that are not intended for decision making. The area of study called MIS

is sometimes referred to, in a restrictive sense, as information technology management. That area of study should not be confused with computer science. IT service management is a practitioner-focused discipline. MIS has also some differences with Enterprise Resource Planning (ERP) as ERP incorporates elements that are not necessarily focused on decision support.

Any successful MIS must support a businesses Five Year Plan or its equivalent. It must provide for reports based up performance analysis in areas critical to that plan, with feedback loops that allow for titivation of every aspect of the business, including recruitment and training regimens. In effect, MIS must not only indicate how things are going, but why they are not going as well as planned where that is the case. These reports would include performance relative to cost centres and projects that drive profit or loss, and do so in such a way that indentifies individual accountability, and in virtual real-time.

Professor Allen S. Lee states that “...research in the information systems field examines more than the technological system, or just the social system, or even the two side by side; in addition, it investigates the phenomena that emerge when the two interact.”

DATA MINING

Data mining is the process of extracting patterns from data. Data mining is becoming an increasingly important tool to transform this data into information. It is commonly used in a wide range of profiling practices, such as marketing, surveillance, fraud detection and scientific discovery.

Data mining can be used to uncover patterns in data but is often carried out only on samples of data. The mining process will be ineffective if the samples are not a good representation of the larger body of data. Data mining cannot discover patterns that may be present in the larger body of data if those patterns are not present in the sample being “mined”. Inability to find patterns may become a cause for some disputes between customers and service providers. Therefore data mining is not foolproof but may be useful if sufficiently representative data samples are collected. The discovery of a particular pattern in a particular set of data does not necessarily mean that a pattern is found elsewhere in the larger data from which that sample was drawn. An important part of the process is the verification and validation of patterns on other samples of data.

The related terms data dredging, data fishing and data snooping refer to the use of data mining techniques to sample sizes that are (or may be) too small for statistical inferences to be made about the validity of any patterns discovered. Data dredging may, however, be used to develop new hypotheses, which must then be validated with sufficiently large sample sets.

BACKGROUND

Humans have been “manually” extracting patterns from data for centuries, but the increasing volume of data in modern times has called for more automated approaches. Early methods of identifying patterns in data include Bayes’ theorem (1700s) and regression analysis (1800s). The proliferation, ubiquity and increasing power of computer technology has increased data collection and storage. As data sets have grown in size and complexity, direct hands-on data analysis has increasingly been augmented with indirect, automatic data processing. This has been aided by other discoveries in computer science, such as neural networks, clustering, genetic algorithms (1950s), decision trees (1960s) and support vector machines (1980s). Data mining is the process of applying these methods to data with the intention of uncovering hidden patterns. It has been used for many years by businesses, scientists and governments to sift through volumes of data such as airline passenger trip records, census data and supermarket scanner data to produce market research reports. (Note, however, that reporting is not always considered to be data mining.)

A primary reason for using data mining is to assist in the analysis of collections of observations of behaviour. Such data are vulnerable to collinearity because of unknown interrelations. An unavoidable fact of data mining is that the (sub-)set(s) of data being analysed may not be representative of the whole domain, and therefore may not contain examples of certain critical relationships and behaviours that exist across other parts of the domain. To address this sort of issue, the analysis may be augmented using experiment-based and other approaches, such as Choice Modelling for human-generated data. In these situations, inherent correlations can be either controlled for, or removed altogether, during the construction of the experimental design.

There have been some efforts to define standards for data mining, for example the 1999 European Cross Industry Standard Process for Data Mining (CRISP-DM 1.0) and the 2004 Java Data Mining standard (JDM 1.0). These are evolving standards; later versions of these standards are under development. Independent of these standardization efforts, freely available open-source software systems like the R Project, Weka, KNIME, Rapid Miner and others have become an informal standard for defining data-mining processes. The first three of these systems are able to import and export models in PMML (Predictive Model Markup Language) which provides a standard way to represent data mining models so that these can be shared between different statistical applications. PMML is an XML-based language developed by the Data Mining Group (DMG), an independent group composed of many data mining companies. PMML version 4.0 was released in June 2009.

RESEARCH AND EVOLUTION

In addition to industry driven demand for standards and interoperability, professional and academic activity have also made considerable contributions to the evolution and

rigour of the methods and models; an article published in a 2008 issue of the International Journal of Information Technology and Decision Making summarises the results of a literature survey which traces and analyses this evolution.

PROCESS

Pre-processing: Before data mining algorithms can be used, a target data set must be assembled. As data mining can only uncover patterns already present in the data, the target dataset must be large enough to contain these patterns while remaining concise enough to be mined in an acceptable time-frame. A common source for data is a datamart or data warehouse. Pre-process is essential to analyse the multivariate datasets before clustering or data mining.

The target set is then cleaned. Cleaning removes the observations with noise and missing data. The clean data are reduced into feature vectors, one vector per observation. A feature vector is a summarised version of the raw data observation. For example, a black and white image of a face which is 100px by 100px would contain 10,000 bits of raw data. This might be turned into a feature vector by locating the eyes and mouth in the image. Doing so would reduce the data for each vector from 10,000 bits to three codes for the locations, dramatically reducing the size of the dataset to be mined, and hence reducing the processing effort. The feature(s) selected will depend on what the objective(s) is/are; obviously, selecting the “right” feature(s) is fundamental to successful data mining. The feature vectors are divided into two sets, the “training set” and the “test set”. The training set is used to “train” the data mining algorithm(s), while the test set is used to verify the accuracy of any patterns found.

RESULTS VALIDATION

The final step of knowledge discovery from data is to verify the patterns produced by the data mining algorithms occur in the wider data set. Not all patterns found by the data mining algorithms are necessarily valid. It is common for the data mining algorithms to find patterns in the training set which are not present in the general data set, this is called over fitting. To overcome this, the evaluation uses a test set of data which the data mining algorithm was not trained on. The learnt patterns are applied to this test set and the resulting output is compared to the desired output. For example, a data mining algorithm trying to distinguish spam from legitimate e-mails would be trained on a training set of sample e-mails. Once trained, the learnt patterns would be applied to the test set of e-mails which it had not been trained on, the accuracy of these patterns can then be measured from how many e-mails they correctly classify. A number of statistical methods may be used to evaluate the algorithm such as ROC curves.

If the learnt patterns do not meet the desired standards, then it is necessary to re-evaluate and change the preprocessing and data mining. If the learnt patterns do meet the desired standards then the final step is to interpret the learnt patterns and turn them into knowledge.

NOTABLE USES

GAMES

Since the early 1960s, with the availability of oracles for certain combinatorial games, also called table bases (*e.g.*, for 3x3-chess) with any beginning configuration, small-board dots-and-boxes, small-board-hex, and certain endgames in chess, dots-and-boxes, and hex; a new area for data mining has been opened up. This is the extraction of human-usable strategies from these oracles. Current pattern recognition approaches do not seem to fully have the required high level of abstraction in order to be applied successfully. Instead, extensive experimentation with the table bases, combined with an intensive study of table base-answers to well designed problems and with knowledge of prior art, *i.e.*, pre-table base knowledge, is used to yield insightful patterns. Berlekamp in dots-and-boxes, etc. and John Nunn in chess endgames are notable examples of researchers doing this work, though they were not and are not involved in table base generation.

BUSINESS

Data mining in customer relationship management applications can contribute significantly to the bottom line. Rather than randomly contacting a prospect or customer through a call centre or sending mail, a company can concentrate its efforts on prospects that are predicted to have a high likelihood of responding to an offer. More sophisticated methods may be used to optimise resources across campaigns so that one may predict which channel and which offer an individual is most likely to respond to — across all potential offers. Additionally, sophisticated applications could be used to automate the mailing. Once the results from data mining (potential prospect/customer and channel/offer) are determined, this “sophisticated application” can either automatically send an e-mail or regular mail. Finally, in cases where many people will take an action without an offer, uplift modelling can be used to determine which people will have the greatest increase in responding if given an offer. Data clustering can also be used to automatically discover the segments or groups within a customer data set.

Businesses employing data mining may see a return on investment, but also they recognise that the number of predictive models can quickly become very large. Rather than one model to predict which customers will churn, a business could build a separate model for each region and customer type. Then instead of sending an offer to all people that are likely to churn, it may only want to send offers to customers that will likely take to offer. And finally, it may also want to determine which customers are going to be profitable over a window of time and only send the offers to those that are likely to be profitable. In order to maintain this quantity of models, they need to manage model versions and move to automated data mining.

Data mining can also be helpful to human-resources departments in identifying the characteristics of their most successful employees. Information obtained, such as universities attended by highly successful employees, can help HR focus recruiting efforts accordingly. Additionally, Strategic Enterprise Management applications help a company translate corporate-level goals, such as profit and margin share targets, into operational decisions, such as production plans and workforce levels.

Another example of data mining, often called the market basket analysis, relates to its use in retail sales. If a clothing store records the purchases of customers, a data-mining system could identify those customers who favour silk shirts over cotton ones. Although some explanations of relationships may be difficult, taking advantage of it is easier. The example deals with association rules within transaction-based data. Not all data are transaction based and logical or inexact rules may also be present within a database. In a manufacturing application, an inexact rule may state that 73% of products which have a specific defect or problem will develop a secondary problem within the next six months.

Market basket analysis has also been used to identify the purchase patterns of the Alpha consumer. Alpha Consumers are people that play a key roles in connecting with the concept behind a product, then adopting that product, and finally validating it for the rest of society. Analysing the data collected on these type of users has allowed companies to predict future buying trends and forecast supply demands.

Data Mining is a highly effective tool in the catalogue marketing industry. Catalogers have a rich history of customer transactions on millions of customers dating back several years. Data mining tools can identify patterns among customers and help identify the most likely customers to respond to upcoming mailing campaigns.

Related to an integrated-circuit production line, an example of data mining is described in the paper “Mining IC Test Data to Optimize VLSI Testing.” In this paper the application of data mining and decision analysis to the problem of die-level functional test is described. Experiments mentioned in this paper demonstrate the ability of applying a system of mining historical die-test data to create a probabilistic model of patterns of die failure which are then utilised to decide in real time which die to test next and when to stop testing. This system has been shown, based on experiments with historical test data, to have the potential to improve profits on mature IC products.

SCIENCE AND ENGINEERING

In recent years, data mining has been widely used in area of science and engineering, such as bio-informatics, genetics, medicine, education and electrical power engineering.

In the area of study on human genetics, the important goal is to understand the mapping relationship between the inter-individual variation in human DNA sequences and variability in disease susceptibility. In lay terms, it is to find out how the changes in an individual's DNA sequence affect the risk of developing common diseases such as cancer. This is very important to help improve the diagnosis, prevention and treatment

of the diseases. The data mining technique that is used to perform this task is known as multifactor dimensionality reduction. In the area of electrical power engineering, data mining techniques have been widely used for condition monitoring of high voltage electrical equipment. The purpose of condition monitoring is to obtain valuable information on the insulation's health status of the equipment. Data clustering such as self-organizing map (SOM) has been applied on the vibration monitoring and analysis of transformer on-load tap-changers (OLTCs). Using vibration monitoring, it can be observed that each tap change operation generates a signal that contains information about the condition of the tap changer contacts and the drive mechanisms. Obviously, different tap positions will generate different signals. However, there was considerable variability amongst normal condition signals for the exact same tap position. SOM has been applied to detect abnormal conditions and to estimate the nature of the abnormalities.

Data mining techniques have also been applied for dissolved gas analysis (DGA) on power transformers. DGA, as a diagnostics for power transformer, has been available for many years. Data mining techniques such as SOM has been applied to analyse data and to determine trends which are not obvious to the standard DGA ratio techniques such as Duval Triangle.

A fourth area of application for data mining in science/engineering is within educational research, where data mining has been used to study the factors leading students to choose to engage in behaviours which reduce their learning and to understand the factors influencing university student retention. A similar example of the social application of data mining is its use in expertise finding systems, whereby descriptors of human expertise are extracted, normalised and classified so as to facilitate the finding of experts, particularly in scientific and technical fields. In this way, data mining can facilitate Institutional memory.

Other examples of applying data mining technique applications are biomedical data facilitated by domain ontologies, mining clinical trial data, traffic analysis using SOM, et cetera. In adverse drug reaction surveillance, the Uppsala Monitoring Centre has, since 1998, used data mining methods to routinely screen for reporting patterns indicative of emerging drug safety issues in the WHO global database of 4.6 million suspected adverse drug reaction incidents. Recently, similar methodology has been developed to mine large collections of electronic health records for temporal patterns associating drug prescriptions to medical diagnoses.

LOGISTICAL FUNCTIONS [COMPONENTS OF LOGISTICS OR ELEMENTS OF LOGISTICS]

Management is appreciating importance of information as an element of logistics of late, now. The role of information is vital in order processing. Quality of information is critical as error in composition of information requirement creates potential disturbance

in the supply chain. Incorrect order processing due to erroneous information will result into product recall and reshipment if the sales opportunity still exists. Faster and quality information flow from customer to processor results into cost effective logistics. Forecasting and order management are two areas of logistical work dependent on information.

Forecasting is an effort to estimate future requirements to position inventory or assets devoted to inventory. As forecasting becomes unreliable in a fast changing environment, control strategies like JIT, Quick Response and Continuous Replenishment came into being. Now it is the task of the logistics function to use information technology to strengthen operation control and forecasting to the best advantage of the organization.

Leading firms typically have information systems capable of monitoring logistical performance on a real time basis giving them the capability to identify potential operational breakdowns and take corrective actions prior to customer service failure. In situations where timely corrective action is not possible, customers can be notified in advance and thereby taking the surprise out of forthcoming service failures

INVENTORY CONTROL

Keeping the stock levels in such a position, so that neither stock out nor stock piling takes place is Inventory control. While formulating inventory policies find out 20% of the products marketed that account for 80% of the profit.

TRANSPORTATION

Transportation is the most visible of all elements of logistics and high contributor to logistics expenditure. Costs of transportation are mainly as follows

- a. Movement costs: money paid for moving material across geographical terrain
- b. Preservation costs: money spent on preserving the material during transit
- c. Cost of idle asset: inventory is unavailable for conversion during transit. This results into costs for organization
- d. Administration costs: money spent on administration

Transportation is accomplished in three ways

- a. One's own fleet – private carriage
- b. Contract with specialists on long term basis – contract carriage
- c. Contract on individual shipment basis – common carriage

Expectations from transportation service are

- a. minimum cost – transportation costs are explained earlier
- b. Speed: speed of transport means the speed with which goods reach the destination.
- c. Consistency: consistency in speed is achieving the same speed over a long period of time. Consistency reflects on the reliability of carrier. Any unexpected variance can play havoc with logistics. Modern information technology has

made continuous tracking of consignments possible. This takes the element of surprise out. IT has helped logistics managers to seek out ways and means to improve speed and consistency. What is becoming important is a combination of speed and consistency.

Requirement of speed depends on type of industry. In some situations speed may not be important. Then transportation service offering high speed increases cost. So logistics managers have to strike a balance between service and cost. Three important aspects of transportation are facility location, transportation cost and consistency. Design of logistics system should consider total costs rather than elemental cost of transportation.

WAREHOUSING

Warehousing is holding material before dispatch after it is produced. Although warehousing is conventionally considered to be a storage facility, it plays a much higher role from logistics viewpoint. It is perceived to be a switching facility rather than a storage facility. Warehouse ownership can be private, public or third party contract. Warehouse provides economic and service benefits to the logistical system. Economic benefits are Movement Consolidation, Break-bulk, Cross-dock, Processing/Postponement and stock piling. Service benefits are spot stocking, assortment, mixing and production support

MATERIAL HANDLING

Material handling covers receiving, moving, storing, dispatching activities. It has an impact on cost [capital as well as running], quality and safety. One of the principles of material handling is minimum movement. Commonly used material handling equipment are forklifts, EOT Cranes, hoists, pulley blocks, trolleys, railroad cars, conveyers, ropes and slings, etc.

PACKAGING

Packaging is done to make handling and transporting cost effective. It protects the product in transit and handling. Packing is expected to facilitate lifting and moving by providing easy access to forks or hooks. Packing is also expected to display universal symbols and other instructions for handling. Eg. Pallets and containers, wooden boxes, wrapping, etc.

Types of packaging: consumer packaging and industrial packaging

Consumer packaging - There is no focus on logistics. Importance is given to marketing appeal and packaging the finished product.

Industrial packaging - importance is given to logistics considerations handling and moving. Individual parts are packed in cartons or bags and grouped together as master cartons. Master cartons are grouped into units for handling. This concept leads to unitization and subsequently to containerization.

SUPPLY CHAINS AND PHYSICAL DISTRIBUTION MANAGEMENT (PDM)

The organization's objective of providing service to the customer would be better served if the organization realizes the essential 'supply chain' aspect of today's business. Physical distribution function has a big role to play in the efficiency of the supply chain as this function is at one interface between the supplying and receiving organization. PDM should result in optimality for the chain as a whole.

Of course the ideal situation is the one where the inventories carried in any part or location on the supply chain should be minimal. That is, they should be minimal while being optimal for the entire chain and near optimal (if optimally is not possible) for the different members on the supply chain. This requirement for optimality or near optimality suggests that the material supplies in the pipeline, *i.e.*, those that are on the trucks or in the goods train or other transportation mode, should be small in quantity. It means the materials should land 'Just-in-time' at the point of use and should not lie in large quantities at any point. The task before PDM is to make deliveries /pick ups in small quantities that are economical to the transport service provider. All this flows from the supply chain concept.

MAKING FREQUENT SMALL DELIVERIES / PICK UPS THAT ARE ECONOMICAL

The distance between the supplier and the receiver (buyer) should be short. This would help in minimizing the time spent in travel and would cut down the travel time related cost. However, most of the times it is not possible to find suppliers who are only a short distance away and who also meet the other criteria such as quality production, capability to produce enough quantity and price economy. This is similar to asking for a cow that yield much milk per season, the season lasting long, and one that has short horns which do not hurt. Since such a breed of a cow may not exist, the cow has to be genetically engineered or developed.

Companies having the Just-in-Time system, therefore develop or prepare their suppliers from the small area surrounding the company most of the time within a radius of 40 kilometres. This involves a yeoman effort on the part of the company selecting training and nurturing the locally available talent.

Logistics is concerned with getting the products and services where they are needed when they are desired. It is difficult to accomplish any marketing or manufacturing without logistical support. It involves the integration of information, transportation, inventory, warehousing, material handling, and packaging.

The operating responsibility of logistics is the geographical repositioning of raw materials, work in process, and finished inventories where required at the lowest cost possible.

The formal definition of the word 'logistics' is: - it is the process of planning, implementing and controlling the efficient, effective flow and storage of goods, services and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements.

Within the firm the challenge is to coordinate individual job expertise into an integrated competency focused on servicing customers. In most situations the desired scope of such coordination transcends the individual enterprise, reaching out to include customers as well as material and service suppliers. In a strategic sense, the senior logistics officer leads a boundary spanning initiative to facilitate effective supply chain relationships. The excitement of contemporary logistics is found in making the combined results of internal and external integration one of the core competencies of an enterprise. Throughout the history of mankind wars have been won and lost through logistical strengths and capabilities or the lack of them. Even though the generals of the past have understood the critical role of logistics it only in the recent past that the big organizations have realized its role in the achievement of competitive advantage.

Arth Shaw in 1915 pointed out that: the relations between the activities of demand creation and physical supply... illustrate the existence of the 2 principles of interdependence and balance. Failure to co-ordinate any one of these activities with its group-fellows and also with those in the other group, or undue emphasis or outlay put upon any one of these activities, it is certain to upset the equilibrium of forces which mean efficient distribution. The physical distribution of the goods is a problem distinct from the creation of demand. There are many ways of defining logistics but the underlying concept might be defined as follows: 'Logistics is the process of strategically managing the procurement, movement and storage of materials, parts and finished inventory through the organization and its marketing channels in such a way that current and future profitability are maximized through the cost-effective fulfilment of orders.'

COMPETITIVE ADVANTAGE

Effective logistics management can provide a major source of competitive advantage. The bases for successes in the marketplace are numerous, but a simple model has been based around the three C's – Customer, Company and Competitor. The source of competitive advantage is found firstly in the ability of the organization to differentiate itself, in the eyes of the customer, from its competition and secondly by operating at a lower cost and hence at greater profit.

Seeking a sustainable competitive advantage has become the concern of every manager who realizes the marketplace and it is no longer acceptable to assume that the goods will sell themselves. An elemental, commercial success is derived either from a cost advantage or a value advantage or, ideally both. The greater the profitability of the company the lesser is the production of cost. Also a value advantage gives the product an advantage over the competitive offerings. Successful companies either have a

productivity advantage or they have a value advantage or maybe a combination of the two. There are two main vectors of strategic direction that need to be examined: -

PRODUCTIVITY ADVANTAGE

In many industries there will be a competitor who will be a low cost producer and will have greater sales volume in that sector. This is partly due to economies of scale, which enable fixed costs to spread over a greater volume but more particularly to the impact of the experience curve. It is possible to identify and predict improvements in the rate of output of workers as they become more skilled in the processes and tasks on which they work. Bruce Henderson extended this concept by demonstrating that all costs, not just production costs, would decline at a given rate as volume increased. This cost decline applies only to value added, *i.e.*, costs other than bought in supplies. Traditionally it has been suggested that the main route to cost reduction was by gaining greater sales volume and there can be no doubt about the close linkage between relative market share and relative costs. However it must also be recognized that logistics management can provide a multitude of ways to increase efficiency and productivity and hence contribute significantly to reduced unit costs.

VALUE ADVANTAGE

It is a cliché that customers don't buy products they buy benefits. These benefits may be intangible *i.e.*, they relate not to specific product features but to such things as image and reputation. Unless the product or service that we offer can be distinguished in some way from its competitors there is a strong likelihood that the marketplace will view it as a 'commodity' and so the sale will tend to go to the cheapest supplier. Value differentiation can be gained in numerous ways.

When a company scrutinizes markets closely it frequently finds that there are distinct value segments. In other words different groups of customers attach different levels of importance to different benefits. The importance of such benefit segmentation lies in the fact that often there are substantial opportunities for creating differentiated appeals for specific segments. Adding value through differentiation is a powerful means of achieving a defensible advantage in the market. Equally powerful as a means of adding value is service. Increasingly it is the case that markets are becoming more service sensitive and this poses a challenge in management of logistics. It is important to seek differentiation through means other than technology. A number of companies have responded to this by focusing upon service as a means of gaining a competitive edge. Service in this context relates to the process of developing relationships with customers through the provision of an augmented offer. This augmentation can take many forms including delivery service, after sales service, financial packages, technical support and so on.

This matrix is a useful way of examining the options available for value and productivity advantage:

SERVICE LEADER	COST & SERVICE LEADER
COMMODITY MARKET	COST LEADER

In commodity market situations where a company's products are indistinguishable from their competitors' offerings the only strategy is to move towards being a cost leader or towards being a service leader. Often the leadership route is not available. This particularly will be the case in a mature market where substantial market share gains are difficult to achieve. Cost leadership strategies have been based upon the economies of scale, gained through greater volume of sales. This is why market share is considered to be so important in many industries. This cost advantage can be used strategically to assume a position of price leader and make it difficult for high cost competitors to survive. This cost advantage can come through effective logistics management. In many industries logistics cost represents such a large part of total costs that it is possible to make major cost reductions through fundamentally reengineering logistics processes. The other way to come out of the commodity quadrant of the matrix is to seek a strategy of differentiation through service excellence. Customers in all industries are seeking greater responsiveness and reliability from suppliers; they are looking for reduced lead times, just-in-time delivery and value added services that help them do a better job of serving their customers.

GAINING COMPETITIVE ADVANTAGE THROUGH LOGISTICS

A firm can gain competitive advantage only when it performs its strategically important activities (designing, producing, marketing delivering and supporting its product) more cheaply or better than its competitors.

Value chain activity disaggregates a firm into its strategically relevant activities in order to understand behaviour of costs and existing and potential sources of differentiation. They are further categorized into two types

- (i) Primary - inbound logistics, operation outbound logistics, marketing and sales, and service
- (ii) Support – infrastructure, human resource management, technology development and procurement

To gain competitive advantage over its rivals, a firm must deliver value to its customers through performing these activities more efficiently than its competitors or by performing

these activities in a unique way that creates greater differentiation. Logistics management has the potential to assist the firm in the achievement of both a cost/productivity advantage and a value advantage. The underlying philosophy behind the logistics concept is that of planning and coordinating the materials flow from source to user as an integrated system rather than, as was so often the case in the past, managing the goods flow as a series of independent activities. Thus under a logistics management regime the goal is to link the marketplace, the distribution network, the manufacturing process and the procurement activity in such a way that customers are service at higher levels and yet at lower cost.

ORIGINS AND DEFINITION

The term “logistics” comes from the French *loger*, which means “to lodge” or “to house”. It originally referred to the billeting of troops in private dwellings or other housing.

Logistics is considered to have originated in the military’s need to supply themselves with arms, ammunition and rations as they moved from their base to a forward position. In ancient Greek, Roman and Byzantine empires, there were military officers with the title ‘Logistikas’ who were responsible for financial and supply distribution matters.

The Oxford English dictionary defines logistics as: “The branch of military science having to do with procuring, maintaining and transporting material, personnel and facilities.” Another dictionary definition is: “The time-related positioning of resources.” As such, logistics is commonly seen as a branch of engineering which creates “people systems” rather than “machine systems.”

MILITARY LOGISTICS

ILS Integrated Logistics Support is a discipline used in military industries to ensure an easy supportable system with a robust customer service (logistic) concept at the lowest cost and in line with (often high) reliability, availability, maintainability and other requirements as defined for the project.

In military logistics, logistics officers manage how and when to move resources to the places they are needed. In military science, maintaining one’s supply lines while disrupting those of the enemy is a crucial—some would say the most crucial—element of military strategy, since an armed force without resources and transportation is defenceless.

The defeat of the British in the American War of Independence, and the defeat of the Axis in the African theatre of World War II, have been largely attributed to logistical failure. The historical leaders Hannibal Barca, Alexander the Great, and the Duke of Wellington are considered to have been logistical geniuses. Another field within logistics is called Medical logistics.

MEDICAL LOGISTICS

Medical logistics is the logistics of pharmaceuticals, medical and surgical supplies, medical devices and equipment, and other products needed to support doctors, nurses, and other health and dental care providers. Because its final customers are responsible for the lives and health of their patients, medical logistics is unique in that it seeks to optimize effectiveness rather than efficiency. Medical logistics functions comprise an important part of the health care system: after staff costs, medical supplies are the single most expensive component of health care. To drive costs out of the health-care sector, medical logistics providers are adopting supply chain management theories.

This organizational chart is as follows and separated into three keys areas.

1. Medical Material
2. Biomedical Engineering (BMET)) or Clinical Engineering
3. Facilities Management.

These areas are managed by a qualified Director of Logistics. The Director of Logistics educational background holds some type of accredited graduate degree (MBA or M.S.)

LOGISTICS MANAGEMENT

Logistics management is that part of the supply chain which plans, implements and controls the efficient, effective forward and reverse flow and storage of goods, services and related information between the point of origin and the point of consumption in order to meet customer and legal requirements. A professional working in the field of logistics management is called a logistician. The Chartered Institute of Logistics and Transport (CILT) was established in the United Kingdom in 1919 and was granted the Royal Charter in 1926. The Chartered Institute is one of the professional bodies or institutions, for the logistics and transport sectors, that offers professional qualification or degree in logistics management.

THIRD-PARTY LOGISTICS

Third-party logistics involves the utilization of external organizations to execute logistics activities that have traditionally been performed within an organization itself. According to this definition, third party logistics includes any form of outsourcing of logistics activities previously performed in-house. If, for example, a company with its own warehousing facilities decides to employ external transportation, this would be an example of third party logistics. Logistics is one of the emerging business area in many countries.

WAREHOUSE MANAGEMENT SYSTEM AND WAREHOUSE CONTROL SYSTEM

Although there is some functionality overlap, the differences between warehouse management systems (WMS) and warehouse control systems (WCS) can be significant.

To put it simply, the WMS plans a weekly activity forecast, based on such factors as statistics, trends, and so forth, whereas a WCS acts like a floor supervisor, working in real time to get the job done by the most effective means. For instance, a WMS can tell the system it's going to need five of SKU A and five of SKU B, hours in advance, but by the time it acts, other considerations may have come into play or there could be a potential logjam on a conveyor. A WCS can prevent that problem by working in real time and adapting to the situation by making a 'last-minute decision' based on current activity and operational status. Working synergistically, WMS and WCS can resolve these issues and maximize efficiency for companies that rely on the effective operation of their warehouse or distribution centre.

BUSINESS LOGISTICS

Logistics as a business concept evolved only in the 1950s. This was mainly due to the increasing complexity of supplying one's business with materials and shipping out products in an increasingly globalized supply chain, calling for experts in the field who are called Supply Chain Logisticians. This can be defined as having the right item in the right quantity at the right time at the right place for the right price in the right condition to the right customer and is the science of process and incorporates all industry sectors. The goal of logistics work is to manage the fruition of project life cycles, supply chains and resultant efficiency.

In business, logistics may have either internal focus (in bound logistics), or external focus (outbound logistics) covering the flow and storage of materials from point of origin to point of consumption.

The main functions of a qualified logistician include inventory management, purchasing, transportation, warehousing, consultation and the organizing and planning of these activities. Logisticians combine a professional knowledge of each of these functions so that there is a coordination of resources in an organization. There are two fundamentally different forms of logistics. One optimizes a steady flow of material through a network of transport links and storage nodes. The other coordinates a sequence of resources to carry out some project.

PRODUCTION LOGISTICS

The term is used for describing logistic processes within an industry. The purpose of production logistics is to ensure that each machine and workstation is being fed with the right product in the right quantity and quality at the right point in time. The issue is not the transportation itself, but to streamline and control the flow through the value adding processes and eliminate non-value adding ones. Production logistics can be applied in existing as well as new plants. Manufacturing in an existing plant is a constantly changing process. Machines are exchanged and new ones added, which gives the opportunity to improve the production logistics system accordingly. Production

logistics provides the means to achieve customer response and capital efficiency. Production logistics is getting more and more important with the decreasing batch sizes. In many industries (*e.g.*, mobile phone) batch size one is the short term aim. This way even a single customer demand can be fulfilled in an efficient way. Track and tracing, which is an essential part of production logistics-due to product safety and product reliability issues-is also gaining importance especially in the automotive and the medical industry.

PROFESSIONAL LOGISTICIAN

A logistician is a professional logistics practitioner. Professional logisticians are often certified by professional associations. Some universities and academic institutions train students as logisticians, by offering undergraduate and postgraduate programmes.

LIQUID LOGISTICS

Liquid Logistics is a special category of logistics that relates to liquid products, and is used extensively in the “Supply Chain for Liquids” discipline. Standard logistics techniques are generally used for discrete or unit products. Liquid products have logistics characteristics that distinguish them from discrete products. Some of the major characteristics of liquid products that impact their logistics handling are:

- Liquids flowing from a higher level to a lower level provide the ability to move the liquids without mechanical propulsion or manual intervention.
- Liquids’ adaptation to the shape of the container they are in provides a great deal of flexibility in the design of storage systems and the use of “dead” space for storage.
- The level of a liquid as it has settled in a tank may be used to automatically and continuously know the quantity of liquid in the tank.
- Liquids provide indications through changes in their characteristics that may be sensed and translated into measures of the quality of the liquid.
- Many security and safety risks are significantly reduced or eliminated using liquid logistics techniques. Tools such as liquid level sensors and flow metres can be useful in reducing security risk by providing directly, near real-time and accurate measurements of product’s movement and balance along the supply-chain flow. The safety risk is reducing as product movement through the process of supply stream is independent and controlled.
- Liquids may in some cases be “processed” well downstream from the original production facility and thus offer the opportunity for improved efficiency throughout the supply stream together with more flexibility as to the nature of the product at the point of final usage.

Each of these points represents a differentiation of liquid logistics from logistics techniques used for discrete items. When properly planned for and handled these points of differentiation may lead to business advantages for companies that produce, process, move, or use liquid products.

LOGISTIC ENGINEERING

Logistics Engineering is the scientific organization of the purchase, transport, storage, distribution, and warehousing of materials and finished goods. Logistics is generally a Cost centre service activity, but provides value via improved efficiency and customer satisfaction. It can quickly lose that value if the customer becomes dissatisfied. The end customer can include another process or work centre inside of the manufacturing facility, a warehouse where items are stocked or the final customer who will use the product.

Another much more popular derivative and a complete usage of the logistic term which has appeared in recent years is the supply chain. The supply chain also looks at an efficient chaining of the supply/purchase and distribution sides of an organisation. While Logistics looks at single echelons with the immediate supply and distribution linked up, supply chain looks at multiple echelons/stages, right from procurement of the raw materials to the final distribution of finished goods up to the customer. It is based on the basic premise that the supply and distribution activities if integrated with the manufacturing/logistic activities, can result in better profitability for the organisation. The local minima of total cost of the manufacturing operation is getting replaced by the global minima of total cost of the whole chain, resulting in better profitability for the chain members and hence lower costs for the products.

“Logistics Engineering” as a discipline is also a very important aspect of systems engineering that includes reliability engineering. It is the science and process whereby reliability, maintainability, and availability are designed into products or systems. It includes the supply and physical distribution considerations above as well as more fundamental engineering considerations. For example, if we want to produce a system that is 95% reliable (or improve a system to achieve 95% reliability), a logistics engineer understands that total system reliability can be no greater than the least reliable subsystem or component. Therefore our logistics engineer must consider the reliability of all sub-components or subsystems and modify system design accordingly. If a subsystem is only 50% reliable, one can concentrate on improving the reliability of that subsystem, design in multiple subsystems in parallel (5 in this case would achieve approximately 97% reliability of that subsystem), purchase and store spare subsystems for rapid change out, establish repair capability that would get a failed subsystem back in operation in the required amount of time, and/or choose any combination of those approaches to achieve the optimal cost vs. reliability solution. Then the engineer moves onto the next subsystem.

Logistics Engineers work with complex mathematical models that consider elements such as Mean Time Between Failures (MTBF), Mean Time To Failure (MTTF), Mean Time to Repair (MTTR), Failure Mode and Effects Analysis (FMEA), statistical distributions, queuing theory, and a host of other considerations. Obviously, logistics engineering is a complex science that considers tradeoffs in component/system design,

repair capability, training, spares inventory, demand history, storage and distribution points, transportation methods, etc., to ensure the “thing” is where it’s needed, when it’s needed, and operating the way it’s needed all at an acceptable cost.

PERFORMANCE METRICS

Different performance measures are used to examine the efficiency of an organisation’s logistics. The most popular and widely used performance measure is the landed cost. The landed cost is the total cost of purchasing, transporting, warehousing and distributing raw materials, semi-finished and finished goods.

Another performance measure equally important is the end customer fill rate. It is the percentage of customer demand which is satisfied immediately off-the-shelf (from on-site inventory). An alternative to fill rate, is system availability.

In recent years, the United States Department of Defence (DoD) has advocated the use of Performance Based Logistics (PBL) contracts to manage costs for support of weapon systems.

LOGISTICS MANAGEMENT

The term Logistics Management is that part of Supply Chain Management that plans, implements, and controls the efficient, effective, forward, and reverse flow and storage of goods, services, and related information between the point of origin and the point of consumption in order to meet customers’ requirements.

DEVELOPMENTS IN SUPPLY CHAIN MANAGEMENT

Six major movements can be observed in the evolution of supply chain management studies: Creation, Integration, and Globalization (Movahedi *et al.*, 2009), Specialization Phases One and Two, and SCM 2.0.

CREATION ERA

The term supply chain management was first coined by a U.S. industry consultant in the early 1980s. However, the concept of a supply chain in management was of great importance long before, in the early 20th century, especially with the creation of the assembly line. The characteristics of this era of supply chain management include the need for large-scale changes, re-engineering, downsizing driven by cost reduction programmes, and widespread attention to the Japanese practice of management.

INTEGRATION ERA

This era of supply chain management studies was highlighted with the development of Electronic Data Interchange (EDI) systems in the 1960s and developed through the

1990s by the introduction of Enterprise Resource Planning (ERP) systems. This era has continued to develop into the 21st century with the expansion of internet-based collaborative systems. This era of supply chain evolution is characterized by both increasing value-adding and cost reductions through integration. In fact a supply chain can be classified as a Stage 1, 2 or 3 network. In stage 1 type supply chain, various systems such as Make, Storage, Distribution, Material control, etc. are not linked and are independent of each other. In a stage 2 supply chain, these are integrated under one plan and is ERP enabled. A stage 3 supply chain is one in which vertical integration with the suppliers in upstream direction and customers in downstream direction is achieved. An example of this kind of supply chain is Tesco.

GLOBALIZATION ERA

The third movement of supply chain management development, the globalization era, can be characterized by the attention given to global systems of supplier relationships and the expansion of supply chains over national boundaries and into other continents. Although the use of global sources in the supply chain of organizations can be traced back several decades (*e.g.*, in the oil industry), it was not until the late 1980s that a considerable number of organizations started to integrate global sources into their core business. This era is characterized by the globalization of supply chain management in organizations with the goal of increasing their competitive advantage, value-adding, and reducing costs through global sourcing.

SPECIALIZATION ERA—PHASE ONE: OUTSOURCED MANUFACTURING AND DISTRIBUTION

In the 1990s, industries began to focus on “core competencies” and adopted a specialization model. Companies abandoned vertical integration, sold off non-core operations, and outsourced those functions to other companies. This changed management requirements by extending the supply chain well beyond company walls and distributing management across specialized supply chain partnerships.

This transition also re-focused the fundamental perspectives of each respective organization. OEMs became brand owners that needed deep visibility into their supply base. They had to control the entire supply chain from above instead of from within. Contract manufacturers had to manage bills of material with different part numbering schemes from multiple OEMs and support customer requests for work -in-process visibility and vendor-managed inventory (VMI).

The specialization model creates manufacturing and distribution networks composed of multiple, individual supply chains specific to products, suppliers, and customers who work together to design, manufacture, distribute, market, sell, and service a product. The set of partners may change according to a given market, region, or channel, resulting in a proliferation of trading partner environments, each with its own unique characteristics and demands.

SPECIALIZATION ERA—PHASE TWO: SUPPLY CHAIN MANAGEMENT AS A SERVICE

Specialization within the supply chain began in the 1980s with the inception of transportation brokerages, warehouse management, and non-asset-based carriers and has matured beyond transportation and logistics into aspects of supply planning, collaboration, execution and performance management.

At any given moment, market forces could demand changes from suppliers, logistics providers, locations and customers, and from any number of these specialized participants as components of supply chain networks. This variability has significant effects on the supply chain infrastructure, from the foundation layers of establishing and managing the electronic communication between the trading partners to more complex requirements including the configuration of the processes and work flows that are essential to the management of the network itself.

Supply chain specialization enables companies to improve their overall competencies in the same way that outsourced manufacturing and distribution has done; it allows them to focus on their core competencies and assemble networks of specific, best-in-class partners to contribute to the overall value chain itself, thereby increasing overall performance and efficiency. The ability to quickly obtain and deploy this domain-specific supply chain expertise without developing and maintaining an entirely unique and complex competency in house is the leading reason why supply chain specialization is gaining popularity. Outsourced technology hosting for supply chain solutions debuted in the late 1990s and has taken root primarily in transportation and collaboration categories. This has progressed from the Application Service Provider (ASP) model from approximately 1998 through 2003 to the On-Demand model from approximately 2003-2006 to the Software as a Service (SaaS) model currently in focus today.

SUPPLY CHAIN MANAGEMENT 2.0 (SCM 2.0)

Building on globalization and specialization, the term SCM 2.0 has been coined to describe both the changes within the supply chain itself as well as the evolution of the processes, methods and tools that manage it in this new “era”.

Web 2.0 is defined as a trend in the use of the World Wide Web that is meant to increase creativity, information sharing, and collaboration among users. At its core, the common attribute that Web 2.0 brings is to help navigate the vast amount of information available on the Web in order to find what is being sought. It is the notion of a usable pathway. SCM 2.0 follows this notion into supply chain operations. It is the pathway to SCM results, a combination of the processes, methodologies, tools and delivery options to guide companies to their results quickly as the complexity and speed of the supply chain increase due to the effects of global competition, rapid price fluctuations, surging oil prices, short product life cycles, expanded specialization, near-/far- and off-shoring, and talent scarcity.

SCM 2.0 leverages proven solutions designed to rapidly deliver results with the agility to quickly manage future change for continuous flexibility, value and success. This is delivered through competency networks composed of best-of-breed supply chain domain expertise to understand which elements, both operationally and organizationally, are the critical few that deliver the results as well as through intimate understanding of how to manage these elements to achieve desired results.

Finally, the solutions are delivered in a variety of options, such as no-touch via business process outsourcing, mid-touch via managed services and software as a service (SaaS), or high touch in the traditional software deployment model.

SUPPLY CHAIN BUSINESS PROCESS INTEGRATION

Successful SCM requires a change from managing individual functions to integrating activities into key supply chain processes. An example scenario: the purchasing department places orders as requirements become known.

The marketing department, responding to customer demand, communicates with several distributors and retailers as it attempts to determine ways to satisfy this demand. Information shared between supply chain partners can only be fully leveraged through process integration.

Supply chain business process integration involves collaborative work between buyers and suppliers, joint product development, common systems and shared information. According to Lambert and Cooper (2000), operating an integrated supply chain requires a continuous information flow.

However, in many companies, management has reached the conclusion that optimizing the product flows cannot be accomplished without implementing a process approach to the business. The key supply chain processes stated by Lambert (2004) are:

- Customer relationship management
- Customer service management
- Demand management
- Order fulfilment
- Manufacturing flow management
- Supplier relationship management
- Product development and commercialization
- Returns management.

Much has been written about demand management. Best-in-Class companies have similar characteristics, which include the following: a) Internal and external collaboration b) Lead time reduction initiatives c) Tighter feedback from customer and market demand d) Customer level forecasting.

One could suggest other key critical supply business processes which combine these processes stated by Lambert such as:

CUSTOMER SERVICE MANAGEMENT PROCESS

Customer Relationship Management concerns the relationship between the organization and its customers. Customer service is the source of customer information. It also provides the customer with real-time information on scheduling and product availability through interfaces with the company's production and distribution operations. Successful organizations use the following steps to build customer relationships:

- Determine mutually satisfying goals for organization and customers
- Establish and maintain customer rapport
- Produce positive feelings in the organization and the customers.

PROCUREMENT PROCESS

Strategic plans are drawn up with suppliers to support the manufacturing flow management process and the development of new products. In firms where operations extend globally, sourcing should be managed on a global basis. The desired outcome is a win-win relationship where both parties benefit, and a reduction in time required for the design cycle and product development. Also, the purchasing function develops rapid communication systems, such as electronic data interchange (EDI) and Internet linkage to convey possible requirements more rapidly. Activities related to obtaining products and materials from outside suppliers involve resource planning, supply sourcing, negotiation, order placement, inbound transportation, storage, handling and quality assurance, many of which include the responsibility to coordinate with suppliers on matters of scheduling, supply continuity, hedging, and research into new sources or programmes.

PRODUCT DEVELOPMENT AND COMMERCIALIZATION

Here, customers and suppliers must be integrated into the product development process in order to reduce time to market. As product life cycles shorten, the appropriate products must be developed and successfully launched with ever shorter time-schedules to remain competitive. According to Lambert and Cooper (2000), managers of the product development and commercialization process must:

1. Coordinate with customer relationship management to identify customer-articulated needs;
2. Select materials and suppliers in conjunction with procurement, and
3. Develop production technology in manufacturing flow to manufacture and integrate into the best supply chain flow for the product/market combination.

MANUFACTURING FLOW MANAGEMENT PROCESS

The manufacturing process produces and supplies products to the distribution channels based on past forecasts. Manufacturing processes must be flexible to respond to market

changes and must accommodate mass customization. Orders are processes operating on a just-in-time (JIT) basis in minimum lot sizes. Also, changes in the manufacturing flow process lead to shorter cycle times, meaning improved responsiveness and efficiency in meeting customer demand. Activities related to planning, scheduling and supporting manufacturing operations, such as work-in-process storage, handling, transportation, and time phasing of components, inventory at manufacturing sites and maximum flexibility in the coordination of geographic and final assemblies postponement of physical distribution operations.

PHYSICAL DISTRIBUTION

This concerns movement of a finished product/service to customers. In physical distribution, the customer is the final destination of a marketing channel, and the availability of the product/service is a vital part of each channel participant's marketing effort.

It is also through the physical distribution process that the time and space of customer service become an integral part of marketing, thus it links a marketing channel with its customers (*e.g.*, links manufacturers, wholesalers, retailers).

OUTSOURCING/PARTNERSHIPS

This is not just outsourcing the procurement of materials and components, but also outsourcing of services that traditionally have been provided in-house. The logic of this trend is that the company will increasingly focus on those activities in the value chain where it has a distinctive advantage, and outsource everything else.

This movement has been particularly evident in logistics where the provision of transport, warehousing and inventory control is increasingly subcontracted to specialists or logistics partners. Also, managing and controlling this network of partners and suppliers requires a blend of both central and local involvement. Hence, strategic decisions need to be taken centrally, with the monitoring and control of supplier performance and day-to-day liaison with logistics partners being best managed at a local level.

PERFORMANCE MEASUREMENT

Experts found a strong relationship from the largest arcs of supplier and customer integration to market share and profitability. Taking advantage of supplier capabilities and emphasizing a long-term supply chain perspective in customer relationships can both be correlated with firm performance. As logistics competency becomes a more critical factor in creating and maintaining competitive advantage, logistics measurement becomes increasingly important because the difference between profitable and unprofitable operations becomes more narrow. A.T. Kearney Consultants (1985) noted

that firms engaging in comprehensive performance measurement realized improvements in overall productivity. According to experts, internal measures are generally collected and analysed by the firm including:

1. Cost
2. Customer Service
3. Productivity measures
4. Asset measurement, and
5. Quality.

External performance measurement is examined through customer perception measures and “best practice” benchmarking, and includes 1) customer perception measurement, and 2) best practice benchmarking.

WAREHOUSING MANAGEMENT

As a case of reducing company cost and expenses, warehousing management is carrying the valuable role against operations. In case of perfect storing and office with all convenient facilities in company level, reducing manpower cost, dispatching authority with on time delivery, loading and unloading facilities with proper area, area for service station, stock management system, etc.

Components of supply chain management are as follows:

1. Standardization;
2. Postponement;
3. Customization.

Logistics Management

INTEGRATED LOGISTICS

Some qualifications are in order regarding the formal organization of human resources devoted to logistics. Managers are acutely interested in organizational structure because it directly reflects responsibility, title, compensation, and power. Many managers have the perception that grouping responsibility for all logistical activity into a single organizational unit will automatically stimulate effective integration. This perception is wrong because it emphasizes structure over managerial practice. Formal organization structure alone is not sufficient to guarantee integrated logistical performance. Some of the most highly integrated logistical operations exist without organizational accountability to a single executive. Other enterprises that have highly formalized logistics reporting arrangements also achieve superior results. Generalization regarding how logistics organizations should ideally be structured is premature at this point of subject development. Logistical organization structures vary significantly depending on specific mission, type of business, and available human resources. The goal in creating logistical sensitivity is to stimulate all managers within an enterprise to think and act in terms of integrated capabilities and economies.

Logistical competency is achieved by coordinating

- i. Network design
- ii. Information
- iii. Transportation

- iv. Inventory
- v. Warehousing
- vi. Material handling
- vii. Packaging

Work related to these functional areas is combined to create the capabilities needed to achieve logistical requirements. Attention is directed to an introductory discussion of each facet of logistical work and how they interact in a typical business.

Two qualifications are important when discussing logistical work from the vantage point of a single enterprise. First, all firms require the support and cooperation of many other businesses to complete their overall logistical process. Such cooperation unites the firms in terms of common goals, policies, and programmes. From the perspective of the total supply chain, efficiency is improved by eliminating duplication and waste. However, cross-organizational coordination requires joint planning and relationship management.

Second, there are service firms that perform logistical work on behalf of their customers such as transportation carriers or warehouse firms. Such specialists are supplemental to or may be substitutes for a customer's employees performing the involved work. When outside specialists are used in a logistical system, they must be willing to accept reasonable managerial direction and control from their customers. Therefore, while the performance of a specific task may be outsourced to specialists, the contracting firm's management remains responsible for successful performance of the required work.

NETWORK DESIGN

Classical economics neglected the importance of facility location and overall network design. When economists originally discussed supply-and-demand relationships, facility location and transportation cost differentials were assumed to be either non-existent or equal among competitors. However, the number, size, and geographical relationship of facilities used to perform logistical operations directly affect customer service capabilities and cost. Network design is a primary responsibility of logistical management since a firm's facility structure is used to provide products and materials to customers.

Typical logistics facilities are manufacturing plants, warehouses, cross-dock operations, and retail stores. Determining how many of each type of facility are needed, their geographic locations, and the work to be performed 'at each is a significant part of network design. In specific situations, facility operation may be outsourced to service specialists.

Regardless of who does the actual work, all facilities must be managed as an integral part of a firm's logistical network. The network design requirement is to determine the number and location of all types of facilities required to perform logistics work. It is also necessary to determine what inventory and how much to stock at each facility and

where to assign customer orders for shipment. The network of facilities forms a structure from which logistical operations are performed. Thus the network incorporates information and transportation capabilities. Specific work tasks related to processing customer orders, maintaining inventory, and material handling are all performed within the network design framework. The design of a network must consider geographical variations. The fact that a great deal of difference exists between geographical markets is easy to illustrate. The fifty largest United States metropolitan markets in terms of population account for over 55 percent of all product sales.' Therefore an enterprise, marketing on a national scale, must establish logistical capabilities to service these prime markets. A similar geographic disparity exists in typical material and component part source locations. When a firm is involved in global logistics, issues related to network design become increasingly more complex. The sidebar discussion of Laura Ashley's network design highlights such complexity.

INFORMATION

The importance of information to logistical performance has historically not been highlighted. This neglect resulted from the lack of suitable technology to generate desired information. Management also lacked full appreciation and in-depth understanding of how fast and accurate communication could improve logistical performance. Both of these historical deficiencies have been eliminated. Current technology is capable of handling the most demanding information requirements. If desired, information can be obtained on a real-time basis. Managers are learning how to use such information technology to devise new and unique logistical solutions.

LOCATION REDESIGN

Laura Ashley, based in the United Kingdom, produces women's and children's fashions, curtain and upholstery fabric, wallpaper, linens, and decorating accessories in trademark floral patterns. While it had always maintained excellence in product design and development, Laura Ashley suffered sinking profits because of its complex, costly, and inefficient logistics system. Laura Ashley found that too many carriers and too many systems were resulting in an overall loss of managerial control. To regain control, Laura Ashley had to reorganize its logistical operations. Implementation of Laura Ashley's new logistics structure began with the transfer of all in-house logistics operations to Business Logistics, a division of Federal Express. Business Logistics' task was to restructure, improve, and manage every aspect of the flow of goods and information within the Laura Ashley supply chain.

Prior to reorganization, Laura Ashley had five major warehouses, eight principal carriers, and ten unconnected management systems. The result was extremely long lead times for customers, huge inventories, and too many stock outs. A customer looking for a fast selling item from a warehouse in Germany would be told that the item was out

of stock and that new supplies would not arrive for several months. At the same time, the item could be overstocked in a warehouse in Wales. On average, 16 percent of all product lines were out of stock at the retail stores.

Laura Ashley realized that it needed to reanalyze the location of its current facilities. The recommendation was to close all warehouses except one in the United Kingdom, which would be converted from serving local customers only to serving global customers. The single location, at Newtown, allows close proximity to the manufacturing sites in the United Kingdom. The Newtown facility is a world “processing centre,” acting as cost would be offset by increased efficiency. In the past, the problem of unpredictable demand resulted in higher inventory to cover up uncertainty and maintain customer service.

Laura Ashley knew that it would have more predictable flow with a single service location as opposed to a number of small locations. Random demand could now be pooled over the entire market area, allowing spikes in one area to level out low demand in another. Transportation costs were offset by turn rate in inventory. In fact, Laura Ashley discovered that the single hub system actually decreased transportation costs by reducing the amount of cross shipping. By shipping direct to the retail store from the United Kingdom warehouse, the lead time from order to shipment was about the same but the product was shipped only once instead of being shipped and handled at many different locations.

Laura Ashley has taken its reorganization beyond just cost reduction. The firm now sees the opportunity to increase service and flexibility, and it plans to re supply shops anywhere in the world within twenty-four to forty-eight hours. Advanced systems and communications will be used to monitor and control world inventory. Federal Express’s global carrier networks will ensure that goods arrive at their destination in time. Laura Ashley also plans to launch a mail-order business that will feature forty-eight hour delivery to the end consumer’s door anywhere in the world. Its current \$10 million mail-order business has been growing strong, but until now the company had to limit growth because it could not keep up with expanding orders. The new superior location network will make this growth possible and profitable

However, the technology is only as good as the quality of information. Deficiencies in the quality of information can create countless operational problems. Typical deficiencies fall into two broad categories. First of all, information received may be incorrect with respect to trends and events. Because a great deal of logistics takes place in anticipation of future requirements, an inaccurate appraisal or forecast can cause inventory shortage or over commitment. Overly optimistic forecasts may result in improper inventory positioning. Second, information related to order processing may be inaccurate with respect to a specific customer’s requirements. The processing of an incorrect order creates all the cost of logistics but typically does not result in a sale. Indeed, logistics costs are often increased by the expense of inventory return and, if the sales opportunity still exists, the cost of once again trying to provide the

desired service. Each error in the composition of information requirements creates potential disturbance for the total supply chain.

The benefit of fast information flow is directly related to the balance of work procedures. It makes little sense for a firm to accumulate orders at a local sales office for a week, mail them to a regional office, process the orders on a batch basis, assign them to a distribution warehouse, and then ship them via air to achieve fast delivery. Electronic data interchange (EDI) from the customer's office with slower surface transportation may have achieved even faster overall delivery at a lower total cost. The key objective is to balance components of the logistical system.

Forecasting and order management are two areas of logistical work that depend on information. The logistics forecast is an effort to estimate future requirements. The forecast is used to guide the positioning of inventory to satisfy anticipated customer requirements. Logistics managers' track record in forecasting is not impressive. Therefore, one of the main reasons managers use information to achieve positive control of logistical operations is their desire to replace forecasting inaccuracy with faster response to customer requirements. Control concepts such as just-in-time (JIT), quick response (QR), and continuous replenishment (CR) represent approaches to positive logistical control made possible by the application of recently developed information technology. One of the main jobs of logistics managers is to plan and implement their firm's strategy regarding the desired combination of forecasting and operational control.

INVENTORY

The inventory requirements of a firm depend on the network structure and the desired level of customer service. Theoretically, a firm could stock every item sold in a facility dedicated to service each customer. Few business operations could afford such a luxurious inventory commitment because the risk and total cost would be prohibitive. The objective is to achieve the desired customer service with the minimum inventory commitment, consistent with lowest total cost. Excessive inventories may compensate for deficiencies in basic design of a logistics network and to some degree inferior management. However, excessive inventory used as a crutch will ultimately result higher than necessary total logistics cost. Logistical strategies are designed to maintain the lowest possible financial assets in inventory. The basic goal of inventory management is to achieve maximum turnover while satisfying customer commitments. Inventory strategies need to be focused on meeting requirements of such core customers. The key to effective segmented logistics rests in the inventory priorities designed to support core customers.

Most enterprises experience a substantial variance in volume and profitability across product lines. If no restrictions are applied, a firm may find that less than 20 percent of all products marketed account for more than 80 percent of total profit. While the so-called 80/20 rule or Pareto principle is common, management can avoid excessive cost by implementing inventory strategies that consider fine-line product classification. A realistic assessment of which low-profit or low-volume products should be carried is

the key to avoiding excessive cost. For obvious reasons, an enterprise wants to offer high availability and consistent delivery on more profitable products. High-level support of less profitable items, however, may be necessary to provide full-line service to core customers. The trap to avoid is high service performance on less profitable items purchased by fringe or non-core customers. Therefore, product line profitability must be considered when developing a selective inventory policy. Many enterprises find it desirable to hold slow moving or low-profit items at a central distribution warehouse. The actual delivery performance can be matched to customer importance when orders are received. Core customers may be serviced by fast, reliable air service, while other orders to fringe customers are delivered by less expensive ground transportation.

Selection of the product assortment to be stocked at a specific facility has a direct impact on transportation performance. Most transportation rates are based on the volume and size of specific shipments. Thus, it may be sound strategy to stock sufficient products at a warehouse to be able to arrange consolidated shipments to a customer or geographic area. The corresponding savings in transportation may more than offset the increased cost of holding the inventory. Commitments to deliver products rapidly to fulfill customer requirements are important drivers of logistics. Such time-based arrangements seek to reduce overall inventories by developing the capability to respond rapidly to exact requirements of manufacturing or retail customers. If products and materials can be delivered quickly, it may not be necessary to maintain inventories at manufacturing plants. Likewise, if retail stores can be replenished rapidly, less safety stock must be maintained forward in the supply chain. The alternative to stockpiling and holding safety stock is to receive the exact quantity of inventory at the time required. While such time-based programmes reduce customer inventory to absolute minimums, the savings must be balanced against other costs incurred in the time-sensitive logistical process. For example, time-based programmes tend to reduce shipment sizes, which increase the number, frequency, and cost of shipments. This, in turn, can result in higher transportation costs. For a logistical arrangement to be effective and efficient it must achieve trade-offs that result in the desired customer service at the lowest total cost.

Finally, inventory strategies cannot be created in a competitive vacuum. A firm is typically more desirable to do business with if it can promise and perform rapid and consistent delivery. Therefore, it may be necessary to position inventory in a specific warehouse to provide logistical service even if such commitment increases total cost. Sound inventory policies are essential to gain a customer service advantage or to neutralize a strength that a competitor currently enjoys. Material and component inventories exist in a logistical system for different reasons than finished product inventory. Each type of inventory and the level of commitment must be viewed from a total cost perspective.

TRANSPORTATION

Given a facility network and information capability, transportation is the operational area of logistics that geographically positions inventory. Because of its fundamental

importance and visible cost, transportation has received considerable managerial attention over the years: Almost all enterprises, big and small, have managers responsible for transportation. Transportation requirements can be accomplished in three basic ways. First of all, a private fleet of equipment may be operated. Second, contracts may be arranged with transport specialists. Third, an enterprise may engage the services of a wide variety of carriers that provide different transportation services on an individual shipment basis. These three forms of transport are typically referred to as private, contract and common carriage. From the logistical system viewpoint, three factors are fundamental to transportation performance: cost, speed, and consistency.

The cost of transport is the payment for movement between two geographical locations and expenses related to administration and maintaining in-transit inventory. Logistical systems should be designed to utilize transportation that minimizes total system cost. This means that the least expensive transportation does not always result in the lowest total cost of movement.

Speed of transportation is the time required to complete a specific movement. Speed and cost of transportation are related in two ways. First, transport firms, capable of providing faster service, typically charge higher rates. Second, the faster the transportation service, the shorter the time interval during which inventory is in transit and unavailable, thus, a critical aspect of selecting the most desirable method of transportation is to balance speed and cost of service.

Consistency of transportation refers to variations in time required to perform a specific movement over a number of shipments. Consistency is a reflection of the dependability of transportation. For years, transportation managers have considered consistency the most important characteristic of quality transportation. If a given movement takes two days one time and six days the next, the unexpected variance can create serious logistical operational problems. If transportation lacks consistency, inventory safety stocks will be required to protect against unpredictable service breakdowns. Transportation consistency affects both the seller's and the buyer's overall inventory commitment and related risk. With the advent of new information technology to control and report shipment status, logistics managers have begun to seek faster service while maintaining consistency. The value of time is important and will be discussed repeatedly. It is also important to understand that the quality of transportation performance is critical to time-sensitive operations. Speed and consistency combine to create the quality aspect of transportation.

In the design of a logistical system, a delicate balance must be maintained between transportation cost and quality of service. In some circumstances low-cost, slow transportation will be satisfactory. In other situations, faster service may be essential to achieve operating goals. Finding and managing the desired transportation mix is a primary responsibility of logistics.

There are three aspects of transportation that managers should keep in mind concerning the logistical network. First, facility selection establishes a network structure

that creates the framework of transportation requirements and simultaneously limits alternatives. Second, the total cost of transportation involves more than the freight bill. Third, the entire effort to integrate transport capability into a logistical system may be defeated if delivery service is sporadic and inconsistent.

WAREHOUSING, MATERIAL HANDLING, AND PACKAGING

Four of the functional areas of logistics-network design, information, transportation, and inventory-can be engineered into a variety of different operational arrangements. Each arrangement will have the potential to achieve a level of customer service at an associated total cost. In essence, these four functions combine to create a system solution for integrated logistics. The final function of logistics warehousing, material handling, and packaging-also represent an integral part of an operating solution. However, these functions do not have the independent status. Warehousing, material handling, and packaging are an integral part of other logistics areas. For example, merchandise typically needs to be warehoused at selected times during the logistics process. Transportation vehicles require material handling for efficient loading and unloading. Finally, the individual products are most efficiently handled when packaged together into shipping cartons or other types of containers.

When warehouses are required in a logistical system, a firm can choose between obtaining the services of a specialist or operating its own facility. The decision is broader than simply selecting a facility to store inventory, since many activities essential to the overall logistical process are typically performed while products are warehoused. Examples of such activities are sorting, sequencing, order selection, transportation consolidation and, in some cases, product modification and assembly.

Within the warehouse, material handling is an important activity. Products must be received, moved, sorted, and assembled to meet customer order requirements. The direct labour and capital invested in material-handling equipment are a major part of total logistics cost. When performed in an inferior manner, material handling can result in substantial product damage. It stands to reason that the fewer times a product is handled, the less potential exists for product damage, and the overall efficiency of the warehouse is increased. A variety of mechanized and automated devices exist to assist in material handling. In essence, each warehouse and its material-handling capability represent a mini system within the overall logistical process.

To facilitate handling efficiency, products in the form of cans, bottles, or boxes are typically combined into larger units. The initial unit, the master carton, provides two important features. First of all, it serves to protect the product during the logistical process. Second, the master carton facilitates ease of handling by creating one large package rather than a multitude of small, individual products. For efficient handling and transport, master cartons are typically consolidated into larger units. The most

common units for master carton consolidation are pallets, slip sheets, and various types of containers. When effectively integrated into an enterprise's logistical operations, warehousing, material handling, and packaging facilitate the speed and overall ease of product flow throughout the logistical system. In fact, several firms have engineered devices to move, broad product assortments from manufacturing plants directly to retail stores without intermediate handling.

INTEGRATED LOGISTICS

The conceptualization of integrated logistics is illustrated in the shaded area. Logistics is viewed as the competency that links an enterprise with its customers and suppliers. Information from and about customer's flows through the enterprise in the form of sales activity, forecasts, and orders. The information is refined into specific manufacturing and purchasing plans. As products and materials are procured, a value-added inventory flow is initiated that ultimately results in ownership transfer of finished products to customers. Thus, the process is viewed in terms of two interrelated efforts, inventory flow and information flow. Prior to discussing each flow in greater detail, two observations are in order.

First, viewing internal operations in isolation is useful to elaborate the fundamental importance of integrating all functions and work involved in logistics. While such integration is prerequisite to success, it is not sufficient to guarantee that a firm will achieve its performance goals. To be fully effective in today's competitive environment, firms must expand their integrated behaviour to incorporate customers and suppliers.

Second, the basic process is not restricted to for-profit business, nor is it unique to manufacturing firms. The need to integrate requirements and operations occurs in all businesses as well as within public sector organizations. For example, retailing or wholesaling firms typically link physical distribution and purchasing, since traditional manufacturing is not required. Nevertheless, retailers and wholesalers must complete the logistics value-added process. The same is true of all public sector organizations that manufacture products or provide other services.

INFORMATION FLOW

Information flow identifies specific locations within a logistical system that have requirements. Information also integrates the three operating areas. The primary objective of developing and specifying requirements is to plan and execute integrated logistical operations. Within individual logistics areas, different movement requirements exist with respect to size of order, availability of inventory, and urgency of movement. The primary objective of information sharing is to reconcile these differentials. In the discussion that follows it is important to stress that information requirements parallel the actual work performed in physical distribution, manufacturing support, and procurement. Whereas these areas contain the actual logistics work, information

facilitates coordination of planning and control of day-to-day operations. Without accurate information the effort involved in the logistical system can be wasted. Logistical information involves two major types of flows: coordination flows and operational flows. The objective at this point is to provide an introductory overview of the information requirements necessary to drive an integrated logistics system.

PLANNING AND COORDINATION FLOWS

Coordination is the backbone of overall information system architecture among value chain participants. Coordination results in plans specifying

- i. Strategic objectives
- ii. Capacity constraints
- iii. Logistical requirements
- iv. Inventory deployment
- v. Manufacturing requirements
- vi. Procurement requirements, and
- vii. Forecasting.

The primary drivers of the overall value chain are the strategic objectives that result from marketing and financial goals. Strategic objectives detail the nature and location of customers, which are matched to the required products and services to be performed. The financial aspect of strategic plans detail resources required to support inventory, receivables, facilities, equipment, and capacity. Capacity constraints coordinate internal and external manufacturing requirements. For non-manufacturing participants in the value chain, this form of capacity planning is not required. Given strategic objectives, capacities constraints identify limitations, barriers, or bottlenecks within basic manufacturing capabilities and determine appropriate outsource requirements.

To illustrate, whereas Kellogg owns the brand and distributes Cracklin Oat Bran, a third party on a contract basis performs all manufacturing. The result of capacity constraints is a plan that places strategic objectives in a time-phased schedule that details facility utilization, financial resources, and human requirements.

Logistics requirements specify the work that distribution facilities, equipment, and labour must perform to implement the capacity plan. Using inputs from forecasting, promotional scheduling, customer orders, and inventory status, logistics requirements specify value chain performance.

Inventory deployments are the interfaces between planning/coordination and operations that detail the timing and composition of where inventory will be positioned. A major concern of deployment is to balance timing and consolidation to create efficiency as inventory flows through the value chain. Inventory is unique in that it is an integral part of the planning/coordination and operational flows involved in logistics. From an information perspective, deployment specifies the what, where, and when of the overall logistics processes. From an operational viewpoint, inventory management is performed as a day-to-day event. Because of this duality, inventory deployment and management

the planning and coordination and operational information flows. In production situations, manufacturing plans are derived from logistical requirements and typically result in inventory deployment. The primary deliverable is a statement of time-phased inventory requirements that drives master production scheduling (MPS) and manufacturing requirements planning (MRP). The deliverable from manufacturing requirements is a day-to-day production schedule that can be used to specify material and component requirements.

Procurement requirements schedule material and components for inbound shipment to support manufacturing requirements. In retailing and wholesaling situations, procurement involves maintaining product supplies. In manufacturing situations, purchasing must facilitate inbound materials and component parts from suppliers. Regardless of the situation, purchasing coordinates decisions concerning supplier qualifications, degree of desired speculation, third-party arrangements, and feasibility of long-term contracting.

Forecasting utilizes historical data, current activity levels, and planning assumptions to predict future activity levels. Logistical forecasting is generally concerned with relatively short-term predictions (*i.e.*, less than ninety days). The forecasts predict periodic (usually monthly or weekly) sales levels for each product, forming the basis of logistics requirement and operating plans. The overall purpose of information planning/coordination flow is to integrate specific activities within a firm and to facilitate overall integrated performance. Unless a high level of coordination is achieved, the potential exists for operating inefficiencies and excessive inventory. Planning/coordination is illustrated in the health care business by the sidebar discussing how hospitals use information to improve efficiency and customer service.

Operational Requirements the second aspect of information requirements is concerned with directing operations to receive, process, and ship inventory as required supporting customer and purchasing orders. Operational information requirements deal with

- i. Order management,
- ii. Order processing,
- iii. Distribution operations,
- iv. Inventory management,
- v. Transportation and shipping, and
- vi. Procurement

INVENTORY FLOW

The operational management of logistics is concerned with movement and storage of materials and finished products. Logistical operations start with the initial shipment of a material or component part from a supplier and are finalized when a manufactured or processed product is delivered to a customer. From the initial purchase of a material or component, the logistical process adds value by moving inventory when and where needed. Providing all goes well, a material gains value at each step of its transformation

into finished inventory. In other words, a Customer individual part has greater value after it is incorporated into a machine. Likewise, the machine has greater value once it is delivered to a buyer.

To support manufacturing, work-in-process inventory must be moved to support final assembly. The cost of each component and its movement becomes part of the value-added process. The final or meaningful value that is added occurs only with final ownership transfer of products to customers when and where specified.

For a large manufacturer, logistical operations may consist of thousands of movements, which ultimately culminate in the delivery of products to an industrial user, retailer, wholesaler, dealer, or other customer. For a large retailer, logistical operations may commence with the procurement of products for resale and may terminate with consumer pickup or delivery. For a hospital, logistics starts with procurement and ends with full support of patient surgery and recovery, the significant point is that regardless of the size and type of enterprise, logistics is essential and requires continuous management attention. For better understanding it is useful to divide logistical operations into three areas: physical distribution, manufacturing support, and procurement.

PHYSICAL DISTRIBUTION

The area of physical distribution concerns movement of a finished product to customers. In physical distribution, the customer is the final destination of a marketing channel. The availability of the product is a vital part of each channel participant's marketing effort. Even a manufacturer's agent, which typically does not own inventory, must depend on inventory availability to perform expected marketing responsibilities.

Unless a proper assortment of products is efficiently delivered when and where needed, a great deal of the overall marketing effort can be jeopardized. It is through the physical distribution process that the time and space of customer service become an integral part of marketing. Thus physical distribution links a marketing channel with its customers. To support the wide variety of marketing systems that exist in a highly commercialized nation, many different physical distribution systems are utilized. All physical distribution systems have one common feature: they link manufacturers, wholesalers, and retailers into marketing channels that provide product availability as an integral aspect of the overall marketing process.

MANUFACTURING SUPPORT

The area of manufacturing support concentrates on managing work-in-process inventory as it flows between stages of manufacturing. The primary logistical responsibility in manufacturing is to participate in formulating a master production schedule and to arrange for timely availability of materials, component parts, and work-in-process inventory. Thus, the overall concern of manufacturing support is not how production occurs but rather what, when and where products will be manufactured.

Manufacturing support has one significant difference when compared with physical distribution. Physical distribution attempts to service the desires of customers and therefore must accommodate the uncertainty of consumer and industrial demand. Manufacturing support involves movement requirements that are under the control of the manufacturing enterprise. The uncertainties introduced by random customer ordering and erratic demand accommodated by physical distribution are not present in most manufacturing operations. From the viewpoint of overall planning, the separation of manufacturing support from outbound (physical distribution) and inbound (procurement) activities provides opportunities for specialization and improved efficiency.

PROCUREMENT

Procurement is concerned with purchasing and arranging inbound movement of materials, parts, and/or finished inventory from suppliers to manufacturing or assembly plants, warehouses, or retail stores. Depending on the situation, the acquisition process is commonly identified by different names. In manufacturing, the process of acquisition is typically called purchasing. In government circles, acquisition has traditionally been referred to as procurement. In retailing and wholesaling, buying is the most widely used term. In many circles, the process is referred to as inbound logistics. Although differences do exist concerning acquisition situations, the term procurement is used here to include all types of purchasing. The term material is used to identify inventory moving inbound to an enterprise, regardless of its degree of readiness for resale. The term product is used to identify inventory that is available for consumer purchase. In other words, materials are involved in the process of adding value through manufacturing, whereas products are ready for consumption. The fundamental distinction is that products result from the value added to material during manufacturing” sorting, or assembly.

Procurement is concerned with availability of the desired material assortments where and when needed. Whereas physical distribution is concerned with outbound product shipments, purchasing is concerned with inbound materials, sorting, or assembly. Under most marketing situations involving consumer products, such as a grocery manufacturer that ships to a retail food chain, the manufacturer’s physical distribution is the same process as a retailer’s procurement operations. Although similar or even identical transportation requirements may be involved, the degree of managerial control and risk related to performance failure varies substantially between physical distribution and procurement. Within a typical enterprise, the three areas of logistics overlap. Viewing each as an integral part of the overall value-adding process creates an opportunity to capitalize on the unique attributes of each while facilitating the overall process. The prime concern of an integrated logistical process is to coordinate overall value added inventory movement. The three areas combine to provide integrated management of materials, semi finished components, and products moving between locations, supply sources, and customers of the enterprise. In this sense, logistics is concerned with strategic management of total movement and storage.

BARRIERS TO INTERNAL INTEGRATION

Organizations do not implement internal logistics integration in a vacuum. It is important to recognize obstacles, or barriers, that often serve to inhibit internal process integration. Integration barriers originate in traditional practices related to organization structure; measurement systems, inventory ownership, information technology, and knowledge transfer capability.

ORGANIZATION STRUCTURE

The traditional organization structure for conducting business prevents any cross-functional process from being implemented. Most traditional organizations are structured to divide authority and responsibility according to functional work. In essence, both structure and budget closely follow the work to be performed. The traditional practice is to assemble all persons related to performing specific work into a functional department such as inventory control, warehousing operations, or transportation. Each of these organizations becomes concerned with achieving its own functional excellence. Since the goal of integration is cooperation among functional areas, the formal organizational structure can hinder success. Popular terms to describe traditional functions are the sandbox or silo mentality. In part, this managerial preoccupation with function is caused by the fact that most managers are rewarded for achieving functional excellence. The general belief that prevailed was that functions, excellently executed, would combine to create overall superior performance. Successful integration of a process such as logistics requires managers to look beyond their organizational structure and facilitate cross-functional coordination. This may or may not be best accomplished by creating a new organization structure. However, regardless of whether the organization structure is realigned, significant modification of how an organization deals with cross-functional matters is essential for successful process integration.

MEASUREMENT SYSTEMS

Traditional measurement systems have also made cross-functional coordination difficult. Most measurement systems mirror organization structure. To successfully facilitate integration of logistics functions, a new scorecard must be developed. Managers must be encouraged to view their specific functions as part of a process rather than as stand-alone activities. Managers may, at times, have to assume increased costs within their functional area for the sake of lower costs throughout the process. Unless a measurement system is created that does not penalize managers, logistical integration will be more theory than practice.

INVENTORY OWNERSHIP

It is a fact that inventory can help a specific function achieve its mission. The traditional approach to inventory ownership is to maintain adequate supply to gain

comfort and protect against demand and operational uncertainty. The availability of inventory, for example, can support long manufacturer runs resulting in maximum economy of scale. Forward commitment of inventory to local markets can also serve to facilitate sales. While such practices create benefits, they have a related cost. The critical issue is the cost-benefit relationship and the risks related to incorrectly located or obsolete inventory.

INFORMATION TECHNOLOGY

Information technology is the key resource to achieve integration. However, similar to performance measurement, information system applications tend to be designed along organization lines. Many databases are limited to specific functions and are not easily accessed on a cross-functional basis. The need to share information has resulted in the development of data warehouses that exist for the sole purpose of sharing information between systems. Until schemes are developed to transfer information, the existing applications can serve as a barrier to process integration because critical data cannot be readily shared.

KNOWLEDGE TRANSFER CAPABILITY

Knowledge is power in most business situations. An additional barrier to integration is limitation in the ability to share experience. Failure to transfer information or knowledge containment tends to foster the functional orientation by developing a workforce composed of specialists. The failure to transfer knowledge can also create a barrier to continued integration when an experienced employee retires or for some other reason leaves the firm. In many cases, replacement personnel are not available to “learn” from the experienced worker. The more serious situation is a failure of many firms to develop procedures and systems for transferring cross functional knowledge. Process work often involves many employees and is not limited to any specific functional area. Transfer of this type of knowledge and experience is difficult to standardize.

IMPLICATIONS ON THE BENEFITS OF ONLINE SHOPPING

The benefits of online shopping stem from a weakening or elimination of the richness-reach trade-off, as explained in the preceding analysis. The Internet allows non-store shoppers to obtain both richness and reach without the tyranny of the trade-off. The weakening or elimination of the ‘richness/reach’ trade-off enables shoppers to browse through a large number of stores (a high level of ‘reach’) to gain a high level of product information (a high level of ‘richness’). The hierarchical search process that is inherent in traditional shopping is no longer necessary with online shopping, saving the shopper

time and effort. The constraint of “bounded rationality” in shoppers is consequently relaxed, as shoppers acquire an increased capability of more optimal problem solving in regard to their purchases.

INTERNET-BASED BENEFITS

The Internet permits a separation between the “economics of information” and the “economics of things” referred to previously. Amazon.com offers three million books ‘located’ on some 25 million computer screens (“economics of information”) as compared to the average bookstore, which carries an average of 80,000 titles (“economics of things”). A similar separation is Dell’s Internet site offers over 10 million configurations of the PC by letting the consumer permute hardware configurations, as compared to 20 configurations offered in the average computer superstore.

With the aforementioned separation, online retailers are able to offer a greater selection of products online as compared to in physical stores, while allowing better product navigation for shoppers. The Internet store can provide a larger inventory of products and sizes, and can virtually guarantee the availability of any type and size of merchandise. New advances in Internet technology such as electronic shopping navigators or ‘shopping bots’ assist shoppers in their product search by enabling the separation of the economics of information and the economics of things. These ‘shopping bots’ take a query, visit shops that may have the product sought, bring back the results and present them in a consolidated compact format that allows comparison shopping. Examples of popular ‘shopping bots’ are Mysimon.com and Netmarket.com.

The weakening of the richness-reach trade-off also manifests itself in the ability of shoppers to browse and purchase goods on the Internet anytime, unlike traditional brick-and-mortar storefronts that have fixed opening hours. The Internet operates 24 hours a week, seven days a week, and can be accessed anywhere in the connected world. Shoppers can also purchase goods that are unavailable at their location, as the Internet allows shoppers to make purchases from vendors in other locations around the world.

SHOPPER-BASED BENEFITS

Shopping using the Internet by overcoming the trade-off between ‘richness’ and ‘reach’ saves shoppers time and pecuniary costs of traditional shopping; shoppers can shop from the comfort and convenience of home, and need not travel to physical storefronts.

As compared to traditional shopping, online shopping provides a greater reach of information that benefits shoppers in terms of reduced search costs. This could be a major saving for time-poor consumers, depending on their opportunity cost of time. Shoppers are able to locate many vendors online using search engines and websites designed to navigate shoppers, view detailed product information from a variety of vendor’s websites, compare price and quality among different vendors, and make

purchases online. Shoppers are able to find the lowest prices due to the wider reach of information facilitated by the Internet and by using navigator websites. With online shopping, shoppers no longer have to suffer the costs and incomplete information of traditional hierarchical search, making product searches easier and more effective.

BUSINESS-BASED BENEFITS

Like shoppers, corporations are burdened with the trade-off between richness and reach in conventional channels and markets. The wider the reach to customers that the corporation tries to obtain, the lower the information richness it is able to transmit, and vice-versa. As a result, corporations are forced to use intermediaries to enhance its customer reach and information richness.

With the Internet, corporations are able to practice disintermediation by obtaining both richness and reach without the limitations of the trade-off. Opportunities are created to rationalise the logistical value chain. This means that corporations can save different types of costs. Such costs savings would be transmitted to shoppers depending on the competitive pressures in the marketing channel and the relative bargaining powers between buyers and sellers. Corporations can eliminate supply intermediaries as they use the Internet to transmit information to a wide reach of consumers using online stores. This way, corporations save on the costs of supporting intermediaries such as agency fees and commissions. This could lower prices for shoppers, implying that a win-win situation is created between the corporation and customer as the savings on intermediary costs are divided between them, with the division of the gain depending on the competitive pressures in the marketplace.

When the underlying enabling technologies are moving an order of magnitude faster, the story is very different. The disintermediating business model can beat the old in both richness and reach. This can happen because technology permits more rich information to be delivered to consumers directly. It can also happen because technology permits such a thorough deconstruction of the old value chain that new combinations of free standing players can match the capabilities of the old vertically integrated business model. Another type of business cost saving is the reduced need for brick-and-mortar facilities. Traditional retailing is burdened with higher costs due to the capital cost of the physical premises and the attendant operational costs such as wages of sales staff. The Internet store has lower fixed costs, as it does not have to be physically located close to the customer. Also, since technology replaces most functions carried out by retail salespeople (such as check-out and verification), the costs are lower.

The high fixed capital costs associated with the physical facilities of conventional channels have in the past operated as formidable barriers to entry limiting competitive pressures for lower consumer prices. These physical facilities typically take years to build, and their presence automatically disadvantages potential new entrants. With the weakening or elimination of the trade-off between 'richness' and 'reach', competitive entry is made possible because the high fixed capital facilities no longer provide a

protective barrier. In fact, they could provide a burden that disadvantages incumbent businesses, while advantaging new entrants. The upshot is the creation of competitive pressures for the passing on to consumers of the savings in capital and operational costs in the form of lower prices for online products.

In the last three years, the threat of the Internet has increased the competitive pressure that retailers face. Online vendors are aware that due to the wider informational reach of consumers on the Internet, which leads to low levels of pricing secrecy, they have to focus on low prices to gain the competitive advantage. Prospective shoppers viewed price savings and selection as more important benefits than convenience, which was ranked third.

The Internet weakens or eliminates the trade-off between 'richness' and 'reach', creating benefits for both shoppers and corporations. With Internet shopping, shoppers are able to obtain a higher reach of alternative products and richer product information enabling an improved purchase decision. Shoppers no longer have to search in a hierarchical manner and be constrained by bounded rationality inherent in traditional shopping. Corporations are also able to reap similar benefits like the ability to obtain a wider reach of customers and provide greater information richness. This enables savings on intermediary costs and a reduced need for brick-and-mortar facilities, creating competitive pressures in the retail industry. This implies a competitive advantage for corporations (B2B and B2C) that use the Internet as a medium to reach its customers.

What is the impact of web service technology on supply chain management? Specifically, how can a third-party logistics provider use this technology to either improve services or provide new services?

WHAT IS SUPPLY CHAIN MANAGEMENT?

Supply chain management (SCM) is the combination of art and science that goes into improving the way your company finds the raw components it needs to make a product or service and deliver it to customers. The following are five basic components of SCM.

1. *Plan:* This is the strategic portion of SCM. Companies need a strategy for managing all the resources that go towards meeting customer demand for their product or service. A big piece of SCM planning is developing a set of metrics to monitor the supply chain so that it is efficient, costs less and delivers high quality and value to customers.
2. *Source:* Next, companies must choose suppliers to deliver the goods and services they need to create their product. Therefore, supply chain managers must develop a set of pricing, delivery and payment processes with suppliers and create metrics for monitoring and improving the relationships. And then, SCM managers can put together processes for managing their goods and services inventory, including receiving and verifying shipments, transferring them to the manufacturing facilities and authorizing supplier payments.

3. *Make:* This is the manufacturing step. Supply chain managers schedule the activities necessary for production, testing, packaging and preparation for delivery. This is the most metric-intensive portion of the supply chain—one where companies are able to measure quality levels, production output and worker productivity.
4. *Deliver:* This is the part that many SCM insiders refer to as logistics, where companies coordinate the receipt of orders from customers, develop a network of warehouses, pick carriers to get products to customers and set up an invoicing system to receive payments.
5. *Return:* This can be a problematic part of the supply chain for many companies. Supply chain planners have to create a responsive and flexible network for receiving defective and excess products back from their customers and supporting customers who have problems with delivered products..

Supply chain management software is possibly the most fractured group of software applications on the planet. Each of the five major supply chain steps previously outlined is comprised of dozens of specific tasks, many of which have their own specific software. Some vendors have assembled many of these different chunks of software together under a single roof. For example, most companies need to track demand, supply, manufacturing status, logistics (*i.e.*, where things are in the supply chain), and distribution. They also need to share data with supply chain partners at an ever increasing rate. While products from large ERP vendors like SAP's Advanced Planner and Optimizer (APO) can perform many or all of these tasks, because each industry's supply chain has a unique set of challenges, many companies decide to go with targeted best of breed products instead, even if some integration is an inevitable consequence.

It's worth mentioning that the old adage about systems only being as good as the information that they contain applies doubly to SCM. If the information entered into a demand forecasting application is not accurate, then you will get an inaccurate forecast. Similarly, if employees bypass the supply chain systems and try to manage things manually (using the fax machine or spreadsheets), then even the most expensive systems will provide an incomplete picture of what is happening in a company's supply chain.

What is the goal of installing supply chain management software?

Before the Internet came along, the aspirations of supply chain software devotees were limited to improving their ability to predict demand from customers and make their own supply chains run more smoothly. But the cheap, ubiquitous nature of the Internet, along with its simple, universally accepted communication standards, have thrown things wide open. Now, companies can connect their supply chain with the supply chains of their suppliers and customers together in a single vast network that optimizes costs and opportunities for everyone involved. This was the reason for the B2B explosion; the idea that everyone a company does business with could be connected together into one big happy, cooperative family.

Of course, reality isn't quite that happy and cooperative. But today most companies share at least some data with their supply chain partners. The goal of these projects is greater supply chain visibility. The supply chain in most industries is like a big card game: the players don't want to show their cards because they don't trust anyone else with the information, but if they showed their hands they could all benefit. Suppliers wouldn't have to guess how many raw materials to order, and manufacturers wouldn't have to order more than they need from suppliers to make sure they have enough on hand if demand for their products unexpectedly increases. And retailers would have fewer empty shelves if they shared the information they had about sales of a manufacturer's product in all their stores with the manufacturer. The Internet makes showing your hand to others possible, but centuries of distrust and lack of coordination within industries make it difficult. The payoff of timely and accurate supply chain information is the ability to make or ship only as much of a product as there is a market for. This is the practice known as just-in-time manufacturing, and it allows companies to reduce the amount of inventory that they keep. This can cut costs substantially, since you no longer need to pay to produce and store excess goods. But many companies and their supply chain partners have a long way to go before that level of supply chain flexibility can be achieved.

WEB SERVICES AS A STRATEGIC ELEMENT OF YOUR SUPPLY CHAIN

Web services are seen as a wonderful advancement in technology enabling all manners of integration and information exchange between multiple platforms, operating systems, databases and systems. Recent adoptions of Web service based strategies by software companies, portals and systems integrators supports the life of Web services as a viable toolset. To date, however, the sparkling examples of Web services have been currency converters, stock quote services, etc.

DEFINITION

A Supply Chain is comprised of suppliers, partners, customers and employees. These separate companies work in concert to deliver information and products/services to fulfill demand of the market. Each of these companies, and sometimes the departments or divisions within a company, utilize widely disparate technologies and systems to support their participation in the aggregate model. This has been a significant barrier to deploying true, collaborative business processes that improve the exchange of information and product.

These existing systems (ERP, SCM, CRM, etc.) are data rich, secure and scalable relative to the scope of their design intent. Yet this scope is too narrow for today's opportunities that demand intra and inter-company collaboration. Add to this the

technology utilized to develop these systems is not a service-oriented architecture and as such, introduce tremendous risk, cost and time in modifying them to address the pressing issues of your Supply Chain. Web services allows you to protect your existing system investments and play to their advantages while creating the common infrastructure, framework and application set that can exploit the opportunities of your Supply Chain as you deploy a Collaborative Supply Chain.

What are the opportunities in my Supply Chain?

A Collaborative Supply Chain yields the following results

Decrease inventory levels 10-20%

Reduce inventory carrying costs 5-15%

Increase sales by 10-30%

Increase margins by 5-15%

The financial impact is significant if even one of these improvements is realized. Web services provide the ability to realize these benefits by enabling you to utilize the data from your existing systems to improve the flow of information and product that creates these benefits. Web services for the Supply Chain represent a new way of deploying software. Historically, an entire solution was implemented that required replacing existing systems and process. This is very disruptive to the business and can result in headlines about your projects failure. Web services are an incremental solution that take advantage of your existing systems and don't put your business at risk. Web services projects should be directly aligned with a benefit and managed in smaller, discrete projects with short duration and clearly measurable benchmarks and results. Let's look at each opportunity separately to understand how the benefit can be achieved.

STRUCTURE OF THE DOCUMENTATION

Information systems (Purchasing Information System, Sales Information System, Inventory Controlling, Shop Floor Information System, Plant Maintenance Information System, Quality Management Information System, Retail Information System) The information systems have a uniform, logical structure as well as uniform user guidance. They use a number of central techniques for evaluating data. For this reason, they are described in one document.

SUCH FEATURES APPEARS FOR ALL THE OTHER INFORMATION SYSTEMS

FUNCTION OF THE INFORMATION SYSTEMS

An information system uses informative key figures to aid the rapid detection of existing problem areas and analyze their origin. Business transactions that are often

performed thousands of times in one day form the basis for these key figures. An information system therefore has to access aggregated data in order to gain a quick overview.

The information systems in logistics are flexible tools for collecting, aggregating and analyzing data from the operative applications (Purchasing, Sales, Production, Inventory Controlling, Plant Maintenance, Quality Management/Inspection Processing). They thus enable you to continually control target criteria and to react in time to exceptional situations.

The function of the information systems is to allow you to view the information from your operative applications from any perspective you wish. You are free to define the level of detail in which the information is displayed.

In the Logistics Information System, data can be analysed either as a standard analysis or as a flexible analysis.

Standard analyses are based on statistical data in the Logistics Information System, which is contained in “information structures”. Important key figures are directly updated from the operative application to the information structures.

Flexible analyses allow you to evaluate any SAP data structures and can be used for ad-hoc analyses.

The comparison of planned and actual data plays a vital part in the decision-making process. For this reason, the information systems give you the option of entering planning data, in addition to retrieving and aggregating actual data.

The integrated Early Warning System enables you to recognize exceptional situations at an early stage (for example, a positive trend in delivery times). Undesirable situations can be detected and rectified in time.

The Early Warning System also facilitates the selection of potential weak points (for example, you can select all of the materials with an order value higher than \$5,000).

The information systems in logistics can be used on a variety of levels in the decision-making process as a management, control, and planning instrument.

Let us illustrate this through a hypothetical scenario. Let us say that you are operating a RESTUARANT.

Of course, you would need to have an effective delivery system, to get your MENU to your waiting customers.

How then can you ensure that this SERVICE DELIVERY system in your RESTUARANT is as efficient as it should be?

Here lies the importance of managing logistics. It can help you to provide such a service, ensuring that your business does have proper management of its resources and people so that you can achieve the goal of customer satisfaction. Some of the activities included in business logistics are customer service, documentation flow, inventory management, packaging, plant and warehouse site selection, demand forecasting, interplant movements, order processing, parts and service support, and production scheduling. There are also other factors to consider when undergoing business logistics

as well. Supply of materials, for one, should be considered in the sense that you need to make sure there is sufficient supplies of needed materials so that your restaurant can meet the needed quota to garner profit. Facilities should also be considered in the sense that the different needs of the restaurant would entail different facilities so that they would be carried out properly. These facilities should be custom-made so that the specifications of both the restaurant and the client can be met.

1. You must know the market demand for various items required to satisfy the customers.
2. You must estimate the raw materials required.
3. You must plan to source/ order the raw material from the suppliers.
4. You must have facility to stock the raw materials.
5. You must have the human resources to prepare the dishes.
6. You must have the facility to store /serve the customers.
7. You must have the delivery system to serve the customers.

In the end, implementing proper logistics management can only be beneficial for your restaurant. This is precisely why such incorporation is recommended.

Why is it important for a firm to periodically review its logistics network design? How do a firm's requirements for its logistics network change over time.

THE LIMITATIONS OF TRADITIONAL SHOPPING

Buyers engage in external search for problem solving purposes. External search can maximize satisfaction to consumers by providing the benefits of lower prices, preferred styles, higher quality, reduced risk and greater confidence. However, this external search is limited by the trade-off between 'richness' and 'reach' present in traditional shopping. The trade-off between information richness and reach of alternative shops forces shoppers to search in a hierarchical manner, which limits their search capabilities.

In terms of theoretical expectations, shoppers would begin their search by looking at high reach sources such as the Ads, which offer contacts to many shops but little or no information about its products. The shopper then selects a few shops to visit to gain detailed information about the product before making the purchase. Shoppers have to navigate their way from high reach/low richness information sources (such as a phone book) to high richness/low reach sources (such as the sales assistant; a sales assistant would offer a single customer richly detailed, interactive, and personalized information about his own limited range of wares).

The compromise between the "economics of information" and the "economics of things" makes it difficult for stores to display a large variety of products while giving customers the information they want to know about them.

For instance, a bookstore may be unable to carry extremely large volumes of books due to the cost of inventory (economics of things). However, a smaller inventory would

mean that shoppers have a lesser variety of books to choose from; the number of books in the store also serves as a list for informing customers about the number of books the store carries (economics of information).

Searching in a hierarchical manner involves crawling along the richness/reach trade-off. As shoppers obtain higher informational richness, they lower their reach of alternative shops. This trade-off prevents shoppers from obtaining a high reach of alternative sources and rich product information, hence limiting their search capabilities.

Searching in a hierarchical manner takes time and effort, and risks a sub-optimal purchase. Due to the limited number of shops the shopper eventually visits before making the purchase, as a result of time and pecuniary cost constraints, it is very unlikely that the shopper is able to maximize satisfaction by obtaining the best deal in terms of price and quality. Shoppers act with “bounded rationality”: they make decisions that are sensible given the incomplete information that they possess and the high cost of getting better information. In other words, physical information channels result in asymmetrical and limited information flows that could cause sub-optimal customer choices leading to post purchase dissonance.

Marketing Management in Logistics

MARKETING LOGISTICS

In 1991 the Council of Logistics Management (CLM) a prestigious, professional organization, defined logistics as “the process of planning, implementing and controlling the efficient, effective flow the point of origin to the point of consumption for the purpose of conforming to customer requirements”. “Logistics means the art of managing the flow of raw materials and finished goods from the source to the user” To get goods from where they arise to the right place in the right form, at the right time, at the right cost, “Logistics or physical distribution or distribution logistics is an integral part of Marketing Process”.

Essence of logistics in marketing

- i. Marketing Process is successfully completed when
- ii. Products are produced and priced to satisfy the identified needs of the segment of buyers ? Arrangements are made to supply these goods through selected distribution channels
- iii. An awareness is created among the buyers about the availability of the goods through information facilitation and
- iv. Goods are physically supplied to the buyers at the place and time selected by them.
- v. Besides satisfying the customers need, the marketing process must be profitable to the seller.

So in the Marketing sense, utility is not merely the usefulness of a product to satisfy the customer needs but also moving the product from a manufacturing facility to the user. “Thus, Logistics is a link between the manufacturing and selling process that leads to the creation of place and time utility”

While the production element in the marketing – mix (product, price, place and promotion) leads to creation of ‘form’ utility by taking decisions as product line variety, design, colour brand, service, etc. the distribution element comprising distribution channel fixation and physical movement, creates ‘time’ and ‘place’ utility by ensuring that the produced goods reach the place and time chosen by the buyer.

Logistics is the designing and managing of a system in order to control the flow of material throughout a corporation. This is a very important part of an international company because of geographical barriers.

LOGISTICAL INTEGRATION

The Phrase ‘Supply Chain Management’ is widely used. It is a management of upstream and downstream of relationships with suppliers and customers to deliver superior customer value at less cost to the supply chain as a whole.

So supply chain management focuses on the management of relationships in order to achieve a more profitable outcome for all parties in the chain. It is argued that it should really be ‘Demand Chain Management’ as a chain should be driven by the market, and not by suppliers. So also the word ‘chain’ should be replaced by ‘network’. Since there will normally be multiple suppliers and, indeed, suppliers to suppliers as well as multiple customers and customers’ customers to be included in the total system.

S = Supplier; SS = Suppliers’ Supplier;

C = Customer; CC = Customers’ Customer

TRANSPORTATION

TRANSPORT FUNCTIONALITY

Transportation is one of the most visible elements of logistics operations. Transportation provides 2 major functions: product movement and product storage.

PRODUCT MOVEMENT

Whether the product is in the form of materials, components, assemblies, work-in-process, or finished goods, transportation is necessary to move it to the next stage of the manufacturing process or physically closer to the ultimate consumer. A primary transportation function of product movement is moving up and down the value chain. Since transportation utilizes temporal, financial, and environmental resources, it is

important that items be moved only when it truly enhances the product value. Transportation involves the use of temporal resources because product is inaccessible during the transportation process. Such product, commonly referred to as in-transit inventory, is becoming a significant consideration as a variety of supply chain strategies such as just – in – time and quick response practices reduce manufacturing and distribution centre inventories.

Transportation uses financial resources because internal expenditures are necessary for private fleets or external expenditures are required for commercial or public transportation. Transportation uses environment resources both directly and indirectly.

In direct terms, it is one of the largest consumers of energy (fuel and oil) in the domestic United States economy. In fact, it accounts for close to 67% of all domestic oil use.

Indirectly, transportation creates environmental expense through congestion, air pollution and noise pollution.

The major objective is to move product from an origin location to a prescribed destination while minimizing temporal, financial and environmental resource costs. Loss and damage expenses must also be minimized. At the same time the movement must take place in such a manner that meets customer demands regarding delivery performance and shipment information availability.

PRODUCT STORAGE

A less common transportation function is temporary storage. Vehicles make rather expensive storage facilities. However, if the in-transit product requires storage but will be moved again shortly (*e.g.*, in a few days), the cost of unloading and reloading the product in a warehouse may exceed the per-daily charge of storage in the transportation vehicle.

In circumstances where warehouse space is limited, utilizing transportation vehicles may be a viable option.

One method involves loading on the vehicle and then having it take an indirect route to its destination. This is desirable when the origin or destination warehouse has limited storage capacity.

A second method to achieve temporary product storage is diversion. This occurs when an original shipment destination is changed while the delivery is in transit.

E.g., suppose a product is initially scheduled to be shipped from Chicago to Los Angeles. However, if during the delivery process, it is determined that San Francisco is in greater need of the product or has available storage capacity; the product could be delivered to the alternative destination of San Francisco.

In short, although product storage in transportation vehicles can be costly, it may be justified from a total cost or performance perspective when loading or unloading costs, capacity constraints, or the ability to extend lead times are considered.

PRINCIPLES

There are two fundamental principles guiding transportation management and operations. They are economy of scale and economy of distance.

Economy of scale refers to the characteristic that transportation cost per unit of weight decreases when the size of the shipment increases.

E.g., truckload shipments cost less per pound than less-than-truckload shipments. It is also generally true that larger capacity transportation vehicles such as rail or water are less expensive per unit of weight than smaller capacity vehicles like motor or air. Transportation economies of scale exist because fixed expenses associated with moving a load can be spread over the load's weight. The fixed expenses include administrative costs of taking the order; time to position the vehicle for loading or unloading, invoicing and equipment cost. These costs are fixed because they do not vary with shipment volume.

E.g., suppose the cost to administer a shipment is \$ 10.00. Then the 1-pound shipment has a per unit of weight cost of \$10.00, while the 1,000 pound shipment has a per unit of weight cost of \$0.01. Thus, it can be said that an economy of scale exists for the 1000-pound shipment. Economy of distance refers to the characteristic that transportation cost per unit of distance decreases as distance increases.

E.g., a shipment of 800 miles will cost less than two shipments (of the same combined weight) of 400 miles. Transportation economy of distance is also referred to as a tapering principle since rates or charges taper with distance. The rationale of distance economies is similar to that for economies of scale.

Longer distances allow the fixed expenses to be spread over more miles, resulting in lower overall per mile charge.

These principles are important considerations when evaluating alternative transportation strategies or operating practices. The objective is to maximize the size of the load and the distance that is shipped while still meeting customer service expectations.

TRANSPORT INFRASTRUCTURE

Transportation infrastructure consists of the rights-of-ways, vehicles, and carrier organizations that offer transportation services on a for-hire or internal basis. The nature of the infrastructure also determines a variety of legal and economic characteristics for each mode or multimodal system. A mode identifies the basic transportation method or form.

MODAL CHARACTERISTICS

RAIL NETWORK

Since olden times, railroads have handled the largest number of ton-miles. As a result of the early establishment of a comprehensive rail network connecting almost all

the cities and towns, railways dominated the intercity freight tonnage till World War II and in some cases of Europe, Asia and Africa they even connected the countries. This early superiority enabled railways to transport large shipments very economically.

Railroads have come a long way, as we can see by just the US figures that 54.0% of intercity transport in 1947 was by Railways, 39.2% in 1958, 36.4 % in 1980 and almost 37% in 1992. The period from 1950-1970 saw a tremendous decline in use of railways.

The capability to efficiently transport large tonnage over long distances is the main reason railroads continue to handle significant intercity tonnage and revenue. Railroad operations incur high fixed costs because of expensive equipment, right-of-way (railroads must maintain their own track), switching yards, and terminals. However, rail experiences relatively low operating costs. The replacement of steam by diesel power reduced the railroads' variable cost per ton-mile, and electrification offers potential for more reductions. New labour agreements have reduced workforce requirements, further decreasing variable costs

These days' rail transport only focuses on transporting specific products, which are best, suited to the requirement. Greatest railroad tonnage comes from raw material extractive industries located at considerable distances. Despite problems, Rail fixed-variable costs are still superior for long distances. Railroads basically concentrate on the container traffic and are becoming more responsive of the customer needs, emphasizing bulk industries and heavy manufacturing.

They have expanded their intermodal operations through alliances and motor carrier ownership. Railroads are even concentrating on development of special equipment. There are unit trains which are entire train carrying the same commodity, which are bulk products such as coal or grain. Unit trains are faster, less expensive to operate and quick as it can bypass rail yards and go direct to the product's destination.

There are also various different types, such as articulated cars for extended Rail chassis, double-stack railcars, have 2 levels of containers, thereby doubling the capacity of each car. It also reduces chances of damage because of their design. These technologies have are being applied by railroads to reduce weight, increase carrying capacity, and facilitate interchange.

After a major decline till 1970, Railroads have emerged as a major mode of transport once again since the early 80's because of revitalization of rail network. Evidence suggests that Rail network will be used extensively once again in the 1990's and early 21st century.

MOTOR CARRIERS

Highway transportation has increased rapidly since the end of World War II. This is because Motor carrier industry results from door-to-door operating flexibility and speed of intercity movement. They are even flexible because they can operate on each and every kind of roadways.

In comparison to railroads, motor carriers have relatively small fixed investments in terminal facilities and operate on publicly maintained highways. Although the cost of license fees, user fees, and tolls are considerable, these expenses are directly related to the number of over-the-road units and miles operated. The variable cost per mile for motor carriers is high because a separate power unit and driver are required for each trailer or combination of tandem trailers. Labour requirements are also high because of driver safety restrictions and the need for substantial dock labour. Motor carriers are best suited to handle small shipments moving short distances.

The characteristics of motor carriers favour manufacturing and distributive trades, short distances, and high-value products. Motor carriers have made significant inroads into rail traffic for medium and light manufacturing. This is also because of delivery flexibility, that they have captured a major chunk of the market. In short, the prospect for maintaining a stable market share in highway transport remains bright.

This industry even has a few problems, and one of the primary difficulties relate to increasing cost to replace equipment, maintenance, driver wages, and platform and dock wages. Although accelerating, labour rates influence all modes of transport; motor carriers are more labour-intensive, which causes higher wages to be a major concern. One more threat for hire-motor carrier industry is over-the-road transportation by shipper-owned trucks or by specialized carriers under contract to perform transport services for shippers.

Since 1980, the industry segments have become more definitive since deregulation, and include truckload (TL), less than truckload (LTL), and specialty carriers. TL segment includes loads over 15,000 pounds that generally do not require intermediate stops for consolidation. LTL segment of the industry loads less than 15,000 pounds that generally requires stops at intermediate terminals for consolidation. Because of terminal costs and relatively higher marketing expenses, LTL experiences a higher percentage of fixed costs than TL.

These characteristics have caused extensive industry consolidation, since deregulation has resulted in small number of relatively large carriers worldwide.

Specialty carriers include package haulers such as Federal Express and United Parcel Service. These firms focus on specific requirements of the market or product. It is quite apparent that highway transportation will continue to function as the backbone of logistical operations for the foreseeable future.

WATER TRANSPORT

It is the oldest mode of transportation. First it was the sailing vessels, which was replaced by steamboats in early 1800's and by diesel power in the 1920's.

Domestic water transportation – involves the Great Lakes, canals, and navigable rivers. In every country, fewer system miles exist for inland water than any other transportation mode.

The main advantage of water transportation is the capacity to move extremely large shipments. Water transport employs 2 types of vessels. Deep-water vessels, which are generally designed for Ocean and Great Lakes use, and are restricted to deep-water ports for access. In contrast, diesel-towed barges, which generally operate on rivers and canals, have considerably more flexibility.

Water transport ranks between rail and motor carrier in the fixed cost aspect. Although water carriers must develop and operate their own terminals, the right-of-way is developed and maintained by the government and results in moderate fixed costs as compared to railways and highways.

The main disadvantage of water transport is the limited range of operation and speed. Unless the origin and destination are adjacent, supplement haul by rail or truck is required. The capability to carry very high cargo at an extremely low variable cost places this mode of transport in demand when low freight rates are desired and speed of transit is a secondary consideration.

But, water transport on the other hand isn't all that flexible. Labour restrictions on loading and unloading at docks create operational problems and tend to reduce the potential range of available traffic. Also, a highly competitive situation has developed between railroads and inland water carriers in areas where parallel routes exist.

Great lakes are concentrating towards transportation of bulk products while Deep-water vessels transport a significant high volume of non-bulk items. Containerized cargo facilitates vessel loading and unloading and enhances intermodal capability by increasing the efficiency of cargo transfer between highway, rail and water.

Inland and great lakes will continue to be a viable option in future logistical systems while the slow passage of inland river transport can provide a form of warehousing in transit integrated into overall system design.

PIPELINES

Surprisingly, Pipelines are also one of the major form of transportation medium throughout the world. In 1989, in USA over 53% of all crude and petroleum ton-mile movements were through Pipelines.

In addition to Petroleum, other important product transported by pipeline is the natural gas. Pipelines are owned and operated privately in most of the countries and many gas companies act as both gas distributors and contract transportation providers. Pipelines are also utilized for transport of manufacturing chemicals, pulverized dry bulk materials such as cement, flour via hydraulic suspension, and sewage and water within cities and municipalities.

Pipelines are very unique in nature compared to other types of transport, such as; they operate on 24-hour basis, seven days a week, and are limited by commodity changeover and maintenance. Unlike other modes, there is no empty "Container" or "vehicle" that must be returned.

Pipelines have highest fixed cost and lowest variable cost among transport modes. High fixed costs result from right-of-way, construction and requirements for control stations, and pumping capacity. Since pipelines are not labour-intensive, the variable operating cost is extremely low once the pipeline has been constructed. An obvious disadvantage of these pipelines is that they are not flexible and are limited with respect to commodities that can be transported. Experiments regarding transport of solid products are still going on. Coal slurry pipelines are proving to be economical over long distances but it concerns environmentalists, as it requires massive quantities of water.

AIR TRANSPORT

Air transport is the newest and the least utilized mode of transport. Its major advantage being its speed, which is accompanied by high costs. A coast-to-coast shipment via air requires only a few hours contrast to days taken by other mean of transportation. The high cost of transport can be traded off for high speed, which allows other elements of logistical design, such as warehousing, inventory to be reduced or eliminated. But still air transport remains more of a potential opportunity than a reality because it is very much under utilized.

The high cost of jet aircraft, coupled with erratic nature of freight demand, has limited the assignment of dedicated planes to all-freight operations. However premium carriers provide planes dedicated for freight operations. This premium service started off with documents and has moved onto large parcels, which is an ideal service for firms with a large number of high-value products and time-sensitive service requirements.

The fixed cost of air transport is low as compared to rails, water and pipeline. In fact, air transport ranks second only to highway with respect to low fixed cost. Airways and airports are maintained by public funds and terminals are by local communities. The fixed costs of airfreight are associated with aircraft purchase and the requirement for specialized handling systems and cargo containers. But the air freight variable cost is extremely high as a result of fuel, maintenance, and labour intensity of both in-flight and ground crews.

Since they require wide-open space, airports are generally not integrated with other means of transport. However more “all freight” airports are being developed so as to reduce conflict with passenger operations.

No particular commodity dominates the traffic carried by airfreight operations. These operations are carried more on emergency basis than routine basis. Firms usually utilize scheduled or non-scheduled air cargo movements when the situation justifies high cost. Products with greatest potential for regular air movements are those having high value or extremely perishable. When the marketing period for an item is extremely limited, air transport comes into the picture, as it may be the only practical method for logistical operations.

TRANSPORTATION HIDDEN COSTS

The physical distribution component of a major project, including transportation of raw materials, project materials, machinery and equipment and such infrastructure facilities as roads, vehicles, etc., usually accounts for 20 to 30% of the total capital cost.

The general scarcity of various goods, unpredictable nature of the economy and economic behaviour on the part of the business community and the bureaucracy in India make it all the more necessary for one to plan well ahead. In fact, planning of transportation and infrastructure must be done well ahead of general planning, so that resources spent on other parts of the project do not result in - fruituous expenditure.

In developed countries, these facilities are already available in abundance or are provided for well in advance. In developing countries, action is generally initiated only after the project has been partially put through or when it becomes totally inescapable to do so.

On the other hand, because of lack of these facilities, such problems are faced even during construction.

Eg. Trucks get stuck in muddy roads, work sites remain unapproachable, and serious vehicular accidents are caused near the project areas.

In developing countries, for the supervision of construction of a building, usually no qualified individual is appointed for the co-ordination and planning of transportation infrastructure, which forms a major fraction of the total cost of a project running into crores.

For a major project, the average total cost to the economy of a project costing Rs. 100 crores which is delayed by one year from the date of targeted completion is 39% more than the original budgeted cost on account of the following:

1. The rate of interest on capital may be taken at nearly 12%.
2. The profit on income per year about 12%
3. The cost of depreciation on account of obsolescence or rusting without running of the plant at 5%.
4. The cost of escalation of the project cost at 10%.

The cost of delay in the completion of the project would therefore be roughly about Rs.11 lakhs per day or Rs. 3.25 crores per month. It would be worthwhile to educate senior executives to recognize these facts, for this aspect of the project is usually ignored by project authorities.

Project authorities do not hold themselves responsible for the transportation bottlenecks and resulting delays. These delays are due to non-receipt of equipment, machinery, raw materials, etc. and these delays generally run into months. Due to such delays, the project suffers heavy losses, which occur because of congestion in the ports, traffic jams, railway restrictions, etc. Therefore, it is financially more practical to obtain critical equipment, machinery and raw materials, critical not from the point of view of availability in the market but critical from the standpoint of transportation bottlenecks-

so that the likely delays are avoided. A proper transportation planning of materials, therefore, may well save a project as well as the economy from the ill effects of wasteful expenditure. At the same time it would help speed up production.

TRANSPORTATION COST ELEMENTS

Following are the essential elements of transportation to be taken into account:

1. Transport Mode – The most critical decision is the selection of appropriate mode of transport. This fixes two basic elements of distribution function:
 - a) Transit time or time lapse between production and sale;
 - b) Level of transportation costs.

There is an inverse relationship between transit time and transport cost – the lower the transit time, the higher the transport cost. However, a decision that takes into account only one cost factor cannot be justified. An evaluation of the effect of transit time on other costs must also be considered. Unsold production represents a high cost, and the longer the transit time, the higher the level of unsold production.

2. Inventory Costs – A first class service to clients often requires immediate delivery and, hence a higher level of inventory at the market centre. Economy, on the other hand, calls for minimum inventory. The level of output held in stock is dictated by-
 - a) Transit time: If the time lapse between production and sale is longer, the level of inventory becomes higher.
 - b) Sales pattern: If the pattern of sales is erratic, higher inventory levels are caused.
 - c) Production pattern: If the production pattern is erratic, higher inventory levels have to be maintained to prevent stock –outs.

Assuming that the sales and production patterns are largely fixed, the important variable, which can influence stock, levels in transit time. As transit time is reduced, the level of static stock can be reduced with accompanying stock reduction.

3. Transit Capital – Capital can be released by changing the proportion of the total output in transit. This can be done by adjusting the transit time. As transit time is reduced, the quantity of goods in transit can be decreased with an associated reduction in transit inventory costs. By realizing the capital cost of transit inventory and goods in transit, capital commitments can be reduced, and more capital can be available for other purposes.
4. Obsolescence - When a slow or erratic mode of transport is employed, a higher level of inventory is necessary to ensure continuous, prompt delivery to the customer. However, when designs change rapidly, obsolescence reduces the market value of the products in store. Rapid advances in technology bring about swifter technical obsolescence. Any goods in the pipeline realize a lower

figure when new models are introduced by a company or its competitors. Air distribution can overcome this problem, and the effect of such obsolescence can be minimized.

5. **Packaging** – The nature of packaging of a product is often determined by the mode of its transport. *E.g.*, Because of the dry conditions of carriage, short transit times and minimum handling, air cargo generally requires much less packaging than other forms of long distance transport. Goods dispatched by air may require only a dust cover or even no cover at all. In some cases, savings on the packaging of sophisticated products may more than pay for the actual transport charges. Less packaging may lead to other advantages too. These include lower unpacking costs and lower chargeable weight for freight.
6. **Insurance** – Insurance risks are based on transit time as well as the possibility of damages *en route*. With faster transit times, skilful handling, substantial reduction in damage and greater security in transit, insurance premiums tend to fall substantially.
7. **Breakages** – Cost of breakages is an important factor in any cost benefit analysis. Because breakages may be indemnified by insurance companies, the true cost of damage to cargos can easily be overlooked. In the first place, the vulnerability of various products sent by different modes is reflected in the insurance premium. To high premiums must be added the clerical work involved in establishing claims, making replacements and the loss of customer's goodwill. The replacements themselves will be subject to the same hazards and premiums will require further documentation. Therefore, only that mode of transport must be selected which substantially reduces real damage in transit. This calls for a selection of the routes which are more direct and which avoid transshipment. Handling equipment must also be more sophisticated. Containers can be used by shippers for door-to-door transportation, thereby avoiding all handling of goods by the carrier.
8. **Pilferage** – Many expensive administrative problems associated with breakages also apply to pilferage. This problem is reduced for example, when door-to-door containers are used, a fact which is again reflected in lower insurance rates.
9. **Deterioration** – In many surface cargos, deterioration may be avoided only by complicated and expensive packing to counteract mechanical shock, exposure to weather or unfavourable temperature, etc. Some cannot be stored at all, except at great expense, and others deteriorate slowly. Deterioration can be costly in terms of packing, stock losses and expensive conditioning in store. It can only shut the door on many distant markets. A high speed of transport and the frequency of services can overcome many of these problems.

10. Transport Costs – Transport can be divided into 3 phases:

- (i) Delivery to docks, airport or railway station.
- (ii) Transport from one terminal to another.
- (iii) Delivery from the terminal to the consignee's place.

In the cost-benefit analysis of the turnaround time of a company's delivery and collection vehicles, their man hour costs can be significant. Some customers may find this item to be more costly than the cost of the major journey.

TRANSPORT ELEMENTS

Whether the movement of material and equipment is by rail, sea, air or road, adequate facilities for their free flow to and from the factory must be ensured. The factors which affect progress at the construction stage, and production and dispatches after commission.

Terminal Facilities: Terminal facilities are usually grudgingly provided. One reason for this is that any delay or any in convenience caused to truck operators is not a loss to the project. It is treated as a loss to the carrier. In some cases, this may be true. However, this usual incidence of stoppage or regulation of the production process can be minimized, if not eliminated.

Often extreme stinginess is expressed in planning for these facilities, which include storage space, and loading and unloading arrangements in a suitable area. If the storage space is not adequate or if the traffic is exceptionally heavy, production suffers inevitably. Since transport requirements of each project are different and depend on its location, physical availability of infrastructure, etc., it is not advisable to prescribe one uniform scale of terminal facilities. They must be worked out for an individual project on the basis of its own specific requirements.

Storage, loading and unloading facilities, good quality roads, which are usable throughout the whole year, and suitably, designed yard for railway wagons have to be planned as a part of terminal facilities. It is also essential to pay special attention to the maintenance of loading and unloading equipment, the design, location, length, height and other features of loading and unloading platforms, etc., and the maintenance of circulating area and roads where heavy vehicles ply. The overall savings in transport rates would more than justify the expenditure incurred on the provision of additional facilities. This has not been recognized by the planners of individual projects.

For rail movement, not only sufficient number of loading lines, but also sufficient number of marshalling, examination and holding lines must be planned for. These lines must be suitably connected with one another to ensure smooth shunting operations. The configuration of lines (yard designs) is more important than the number of lines in the yard, for the requirements of prime mover (shunting engines) can also be cut down by a suitable design of yard.

Vehicles: An important feature of movement of finished products of major projects is the type of vehicle used for movement. The vehicle dimensions, capacity. Type and its special characteristics, if any, have to be examined with the reference to the quality

and quantity of goods to be moved. In case of sea transport—the size, speed and the type of ship, in case of road movement—capacity, moving dimensions and speed of the trucks and in case of rail movement—the capacity, type and general availability of wagons must be closely examined.

Planned movement on any section must be taken into account utilisation of the existing sectional capacity, the expected general growth in traffic on the section, and the possible future identifiable streams of new traffic. If movement on a saturated section is inevitable, line capacity of the section must be increased.

Prime Movers: The motive power utilized for the internal handling of vehicles and transportation to destinations is another important component of the total movement system. In the case of rail movement, locomotives required for the shunting and marshalling of wagons within the plant must be of such weight, horsepower and performance characteristics as will match the specific tasks of shunting and reception and dispatch of wagons. In case of road movement, suitable design and layout of conveyors and mechanical loaders can reduce the drudgery of manual labour and make pre-despatch and post-receipt handling operations more efficient.

Routes and Sectional Capacity: Another important aspect of transport planning is the routes for streams of traffic, viz., roadways, railways, waterways and airways. The routes or pathways must have adequate capacities. Generally speaking, because of lack of understanding of the transportation subject, executives take it for granted that capacity of routes is unlimited.

A very important but invisible component of movement activity is sectional capacity, which is dependant on permissible sectional speed and other characteristics of a section. In turn, sectional speed depends on the geometrics of the road (track, sea route, road surface, carriage way, gradients and curves, etc.).

Over a section of railways or roadways between two stations A and B, only a limited number of wagons, trucks or vehicles can be pushed through, depending on the availability of terminal facilities to handle these vehicles, the facilities to enable vehicles to move on the section, and availability of sufficient number of vehicles. Unless sufficient capacity is developed on each of the different routes to move the vehicles, the additional number of vehicles provided would not necessarily lead to higher levels of transport availability. On the contrary, movement may become more sluggish.

Transit Time: The relative locations of a plant and the customers or suppliers determine largely the transit time for raw materials, spare parts and finished products. Transit time generally never receives adequate attention in the planning of major projects. There is a general impression that, if need be, transit time can be drastically cut at any time by air-lifting a consignment. Apart from the fact that the neglect of transportation planning leads to an overall higher cost of transportation, in practice, reduction in transit time actually achieved may not justify the heavy cost of air transport. Rough estimates of transit time from unreliable sources are generally utilized for planning movements of

goods. Although more detailed information sources may be readily available. It is essential therefore, that executives understand clearly the difference between:

- (a) Normal transit time under normal conditions;
- (b) Normal transit time under abnormal conditions;
- (c) Optimal transit time;
- (d) Most optimistic transit time
- (e) Most pessimistic transit time; and
- (f) Desirable transit time.

Because the importance of transit time is not adequately recognized, it is not realistically provided for. Major projects suffer from the heavy delays even before the commencement of construction because of the non-availability of construction equipment and machinery in time. The existing bottlenecks in the fields of transportation are almost always ignored. Construction schedules, inventories, warehousing facilities, order processing or production schedules, etc., are generally planned without the recognition of the inevitable delays that flow from these bottlenecks.

Weigh Bridge: Another usually neglected aspect of industrial transportation activity is the factory weigh bridge. Weigh bridges in factories are generally inaccurate, if not actually out of order. It is seldom appreciated that the losses continuously occurring on this single piece of factory equipment and general inefficiency, which results from its ineffective and inefficient management, can be easily avoided by proper advance planning. Executives ignore the usual traffic jams at factory gates slow down receipts and dispatches, which, in turn, indirectly affect output. The relative advantages of various types of weigh bridges must be properly appreciated by executives, and a weigh bridge which will handle the anticipated volume of traffic expeditiously must be selected.

Distribution Pattern: The pattern of movement of the finished product by road or rail must be planned properly. For example, when the requirements of the number of rail wagons are to be worked out, it is not sufficient to take the average lead or distance for the whole country for calculating fleet requirements. It is also not sufficient to use the figure of the existing average lead of general goods or even that pertaining to a specific commodity.

However, when it comes to actually transport, because of imprecise pre-planning, the manufacturer wants the commodity carrier to transport goods to anywhere and everywhere in the country. This presents a problem. The manufacturer provides information to the common carrier about the quantity of goods to be marketed. But detailed information must be supplied to the carrier so that the carrier can plan the movement in entirety.

NATURE OF PRODUCT

Another aspect, which is often disregarded by project managements as well as common carrier, is the variability arising out of the specialized nature of products to be moved. The generally low level of sophistication in transport planning in the country

had made it difficult for the planners to appreciate the fact that transport capacity is influenced by the nature of goods, their packing and other specialized requirements, such as special handling equipment.etc.

FREIGHT RATE STRUCTURE

Freight rates of any mode of transport are based on the following principles:

1. Freight should be the actual cost of transport operation. The actual cost of operation depends on the following factors:
 - a) Fixed costs - Freight should cover interest on capital, depreciation, registration and insurance expenses of a vehicle, if applicable, general upkeep of the vehicle, administration overheads, and expenditure on other fixed facilities, etc.
 - b) Semi-fixed costs - Freight should cover the salary of the driver, cleaner, conductor and miscellaneous maintenance expenses, which vary partially with the running of the vehicle.
 - c) Vehicle Utilization - A transporter is interested in getting maximum mileage out of his vehicle by moving it at top speed to cover the distance in as short a time as possible.
 - i. If the *consignments loaded or the route covered is not conducive*, the transporter would quote a higher freight rates.
 - ii. Higher freight rates are also quoted when *vehicles are detained at terminals* either for certain formalities, terminal congestion in busy ports or at factory gates, or while waiting for loading or unloading operations. Terminal detentions are invariably accounted for in the freight rates themselves, but they are normally not noticed at all.
 - iii. Freight rates are quoted higher if there is an expectation of obtaining a *return trip with a load* or if considerable *empty movement of vehicles* is involved after unloading.
 - iv. Vehicle Utilization is affected by the nature of goods. *Hazardous goods* that are likely to cause damage to the other consignments or the vehicle itself attract higher freight rates.
 - v. Consignments, which can be loaded *less by weight* in a vehicle, attract higher unit freight rate since they yield poor utilization of the vehicle.
2. Traffic Bearing Capacity: An age-old consideration for the freight rates is the doctrine of “what traffic can bear.” Transportation adds place utility to goods, for it makes them marketable at another place. However, after the addition of the cost of transport, the price of goods should be still attractive to the buyer.
3. Public Use: Freight rates all over the world are governed on human grounds that items of public use should be made available to the common man at the cheapest rate. For example, foodgrains and salt are carried at rock-bottom prices, sometimes even at those, which do not cover the actual cost of operation.

4. Government Policies: Freight rates are often framed on the basis of government objectives, which aim at serving certain points – such as promotion of certain type of trade, development of certain industries, etc. In such cases, freight rates are either depressed to promote the particular traffic or hiked to discourage particular traffic.
5. Reasonable Profit: The transporter must provide for a reasonable profit after covering the cost of operations and capital investment. This margin must give not only return of investment but also compensate him for the entrepreneurial time and effort he puts in, but also provide sufficient funds for future development of his enterprise.

ROAD TRANSPORT

Road transport forms an essential part of any transport activity, whether rail, sea or air. It is essential as a supplementary and complementary mode of transport to complete movement by other modes of transport.

Eg. From one terminal *i.e.*, the railway station the goods have to be carried to the destination like an area by road.

Road transport offers certain advantages like

- i. Door to door service to customers which neither rail nor sea nor air transport can offer.
- ii. On per unit basis, the cost of making a road is 1/6th that of laying a railway line.
- iii. Capital investment in case of railways is much less than railways designed to carry equivalent quantum of traffic.
- iv. Road transport provides employment to six million persons (two million direct and four million indirect)

There are negative points also to be viewed like:

- i. As much as 6200 crores are collected in dues but only 10% of this revenue is used for development of roads, which is a disadvantage as road transport is still not well organized.
- ii. Road transport industry is not of age in India, as there are very few companies, which own the entire fleet of trucks. Single vehicle owners operate most of the road transport.

There is a system of associates, who pool in their vehicles and other resources and share the benefits of vehicle contribution.

The associates operate from different cities. An associate in one city, say, Mumbai, watches the interest of his counterpart in Delhi. Together they operate as a team. They minimize expenses on the maintenance of a branch office at the out-station and yet give the identity of a single branch operator.

PROBLEMS IN ROAD TRANSPORT

Road transport faces a number of problems. This is evident from the following facts:

- (1) There is an occasional storage of diesel fuel in the country.
- (2) Vehicle availability in the country has been problematic. With the recent entry of a number of manufacturers, the situation has improved to some extent.
- (3) The cost of components and accessories, such as tyres and batteries, has escalated tremendously.
- (4) The Octroi and police check posts are too many, resulting in heavy detention to road vehicles.
- (5) The present Motor Vehicle Act regulating the issue of licenses and permits and movement of vehicles is very restrictive. It was decades ago to control and regulate traffic. The current requirements of traffic are for development. There are persistent demands from various transport associations for suitable amendments of the Motor Vehicle act. But they do not seem to receive due attention.
- (6) Conditions on Indian road are very bad and hazardous. They tend to reduce speed of vehicles, which leads to wastage of natural transport capacity.
- (7) Roadside maintenance and service facilities have not developed though found necessary.
- (8) The system of national, zonal and state permits restricts free growth of road transport but the system has to be followed.

TRANSPORTATION NETWORK DESIGN OPTIONS

Classical economists neglected the importance of facility, location and overall network design. Economists, when originally discussed supply – demand relationships, facility, location and transportation cost differentials were assumed to be non-existent or equal among competitors.

- The number, size, geographical relationship of the facilities are used to perform logistics operation directly affect customer service capabilities and cost.
- Network design – primary responsibility of logistics.
- Typical logistics facilities are manufacturing plants, warehouses, gross dock operations and retail stores.
 - Determining
 - (i) How many of each facility are required
 - (ii) Their geographic location
 - (iii) The work to be performed at each is a significant part of network design.
 - In specific situations facility operations may be outsourced to service specialists.
 - No matter who does the actual work, all facilities must be managed as an integral part of a firm's logistical network.

- You have a network for internal customers, external customers as well as suppliers. Suppliers should be made to understand what their commitment is and follow it. You have to co-exist with your suppliers and share information with them.
- Network design must take into account geographical location because a great deal of variation exists between different geographical markets. If you set up your base in a place where your demand is high then after a while the demand shifts and come from another location, then again you are in a dilemma.

In a dynamic, competitive environment

- Product assortment
- Customer supplies
- Manufacturing requirements

All of the above are constantly changing. However re-location of all logistics facilities at one time is inconceivable. Relocation or redesign of specific facilities is possible.

Over a period of time, all facilities should be evaluated, to determine if their location is desirable.

Selection of a superior location network can provide the first step.

Your network of facilities forms a structure from which logistical operations are performed. Your network should incorporate transportation and information. Specific work tasks related to network design are processing customer orders, management of inventory, material handling.

IMPORTANCE OF LOGISTICS AS A STRATEGIC RESOURCE

Logistical Management includes the design and administration of systems to control the flow of material, work-in-progress and finished inventory to support business unit strategy. The concept of logistics, its place in the value-chain process leading to profitability, its contribution as one of the primary functions and its interface with other functions of the firm bring out its importance as a strategic resource. However, to be of a real strategic influence, a good amount of competency has to be achieved and a well-defined logistical mission and objectives has to be committed to, by every one in the firm, especially the top management.

LOGISTICAL COMPETENCY

Logistics involves detailed and complex work. Logistical management starts with how logistical competency fits into a firm's overall strategic. Positioning. It is fundamentally important to view logistics as to how it can be exploited as a core competency. For logistical competency to develop, it is important to develop an integrated framework that defines

and relates key concepts. This integration should be in such a way that competitively superior logistical performance contributes to overall enterprise strategy.

Logistical competency is a relative assessment of a firm's capability to provide competitively superior customer service at the lowest possible total cost. This typically means that logistical performance is dedicated to supporting any or all marketing and manufacturing requirements in a manner that exploits delivery capability. In short, the strategy is to provide superior service at a total cost below industry average.

Alternative logistical capabilities, emphasizing flexibility, time-based performance, operational control, postponement capabilities, and most of all a commitment to perfect service performance typically characterize the service platform of superior logistic achievers.

So we can say that all enterprises must perform logistics to achieve their basic business goals. One of several competencies required to create customer value is logistics. When logistics becomes a cornerstone of basic business strategy, it must be managed as a core competency.

THE LOGISTICAL MISSION

Logistics exists to satisfy customer requirements by facilitating relevant manufacturing and marketing operations. The challenge is to balance service expectations and cost expenditures in a manner that achieves business objectives.

RELEVANCE OF LOGISTICS IN EXPORT MANAGEMENT

International trade is becoming a more important part of the GNP in the industrially advanced countries. Many firms in these countries have production centres world wide for markets all over the world. Lack of local resources, small size of home market and many other reasons has resulted in functional centres being maintained in various countries.

Issues associated with international transportation of finished goods are essentially the same as those that apply to transportation in domestic trade. But, under international operations, goods can be out of exporter's control for longer period of time, more documentation is required, packaging may be more costly and shipping insurance is more costly.

The transportation alternatives include ocean shipping and containerization as well as airfreight. The basic activities involved in the flow of goods, like transportation, warehousing and holding of inventories, should be integrated in a systems approach. The systems approach would recognize the trade-offs, such that sometimes more expensive airfreight may be opted for, instead of less expensive ocean shipping, because of savings in warehouse and inventory costs.

In the field of exports, it should be noted that transport systems in developing countries are generally not as efficient as in the industrially advanced countries. Transportation is often considered to be the most important single determinant of plant location. Firms in international trade also try to reduce amount of unnecessary product packaging, since packing material can account for almost 40 per cent of the weight of the products shipped. A company can reduce inland transportation charges by locating its distribution facilities adjacent to container ports or airports. The burden of documentation can be eased through computerization.

Export management involves marketing in overseas market. Hence the discussions on the interface of logistics with marketing holds good for the relevance of logistics in export management. Yet, in addition, export management has certain unique features, to be understood in the context of relevance of logistics to export management.

BASIC LOGISTICAL SERVICE IS MEASURED IN TERMS OF AVAILABILITY

Availability means having inventory to consistently meet customer material or product requirements.

OPERATIONAL PERFORMANCE

Operational performance deals with the elapsed time from order receipt to delivery. Operational performance involves delivery speed and consistency. A firm's operational performance can be viewed in terms of how flexible it is in accommodating unusual and unexpected customer requests.

SERVICE RELIABILITY

Service reliability involves the quality attributes of logistics. For logistics performance to continuously meet customer expectations, it is essential that management be committed to continuous improvement. Do you know in 1956, in an effort to explain conditions under which high-cost air transport could be justified, Lewis, Colleton and Steele conceptualized the total cost of logistics. Total cost was positioned to include all expenditures necessary to perform logistical requirements. The authors illustrated an electronic parts distribution strategy wherein the high variable cost of direct factory to customer air transport was more than offset by reductions in inventory and field warehouse costs. They concluded that the least total cost logistical way to provide desired customer service was to centralize inventory in one warehouse and make deliveries using air transportation.

The concept of total cost, although basic, had not previously been applied to logistical analysis. Managers typically focused on minimizing functional cost, such as transportation, with the expectation that such effort would achieve the lowest combined cost. The total-

cost concept opened the door to examining how functional costs interrelate. The appropriate level of logistics cost expenditure must be related to desired service performance. The simultaneous attainment of high availability, operational performance, and reliability is expensive. A significant managerial challenge stems from the fact that logistical cost and increased performance have a no proportional relationship. The typical logistical system in an enterprise seeks to develop and implement an overall logistical competency that satisfies key customer expectations at a realistic total-cost expenditure. Overall, logistical management is concerned with operations and coordination. Operations deal with strategic movement and storage. To complete the total operations mission. Attention must be directed to integrating physical distribution, manufacturing support, and procurement into a single logistical process. These three areas, functioning as an integrated and coordinated process, can best provide operational management of materials; semi finished components, and finished products moving between locations, supply sources, and customers of an enterprise. The mission of the logistical system is measured in terms of total cost and performance. Performance measurement is concerned with the availability of inventory, operational capability, and quality of effort. Logistical costs are directly related to desired level of performance. As a general rule, the greater the desired performance, the higher the total logistics cost. The key to effective logistical performance is to develop a balanced effort of service performance and total-cost expenditure. The strategic integration of logistics is fundamental to an enterprise's success. While a firm may not select to differentiate competitively on the basis of logistical competency, it must perform logistical responsibilities as part of the fundamental process of creating customer value. The relative importance that a firm places on logistical competency will determine the degree of emphasis on achieving internal and external integration. Flexibility is key to logistical competency. Logistical flexibility results from integration and from implementing time-based control techniques.

There are four logistics concepts:

- i. The systems concept
- ii. The total cost concept
- iii. The after-tax concept
- iv. The trade-off concept.

The systems concept is based on all functions of a organization working together in order to maximize benefits. This concept sometimes requires certain components of the organization to operate sub optimally in order to achieve maximum goals of the system. The total cost concept is based on the systems concept; however goal achievement is measured in terms of cost.

A variation of the total cost concept is the after-tax concept. This goal of this concept is after-tax profit. This concept is becoming very popular because of the many different national tax policies. The trade-off concept links the system together in a way that is very efficient, but can have trade-offs that might be inefficient. The advantages of such high efficiency must be weighed against the risk involved.

Logistics is a system having number of components, which can be combined in different proportions to achieve a set objective. Long-term objective is profitability; short-term objective is to survive competition by recovering marginal costs.

Elements of logistics system

- i. Transportation
- ii. Warehousing
- iii. Inventory Management
- iv. Packing and Utilization and
- v. Information and Communication

When economists originally discussed supply-and-demand relationships, facility location and transportation cost differentials were assumed either non-existent or equal among competitors.

Given a facility network and information capability, transportation is the operational area of logistics that geographically positions inventory. Because of its fundamental importance and visible cost, transportation has received considerable managerial attention over the years. Almost all enterprises, big and small, have managers responsible for transportation. Finding and managing the desired transportation mix is a primary responsibility of logistics.

Network of three of the functional areas of logistics - information, transportation, and inventory can be engineered into a variety of different operational arrangements. Each arrangement will have the potential to achieve a level of customer service at an associated total cost; In essence, these three functions combine to create a system solution for integrated logistics. The final functions of logistics - warehousing, material handling, and packaging - also represent an integral part of an operating solution. However, these functions do not have the independent status. Warehousing, material handling and packaging are an integral part of other logistics areas. For example, merchandise typically needs to be warehoused at selected times during the logistics process. Transportation vehicles require material handling for efficient loading and unloading. Finally, the individual products are most efficiently handled when packaged together into shipping cartons or other types of containers.

Logistics is viewed as the competency that links an enterprise with its customers and suppliers. Information from and about customer's flows through the enterprise in the form of sales activity, forecasts, and orders. The whole process is viewed in terms of two interrelated efforts, inventory flow and information flow. Information flow is a key element of logistics operations. Paper-based information flow increases both operating cost and decreases customer satisfaction. Electronic information movement and management provide the opportunity to reduce logistics expense through increased coordination and to enhance service by offering better information to customers. Information flow was often overlooked because it was not viewed as being important to customers. The Council of Logistics Management recognized this change in 1988 when it incorporated "material, in-process, finished goods and information" into its

definition of logistics Transportation is a key activity in the logistics value chain as it moves product through the various stages of production and ultimately to the consumer. The primary functions include product movement, product storage and integration of international production and distribution operations. The major transportation principles involve economies of scale and economies of distance.

While effective distribution systems should not be designed to hold inventory for an excessive length of time, there are occasions when inventory storage is justified. While the traditional warehousing role has been to maintain a supply of goods to protect against uncertainty, contemporary warehousing offers many other value-added services. These services can be described in terms of economic and service benefits. Economic benefits include consolidation, break bulk and cross-dock, processing/postponement, and stockpiling. Service benefits include spot stocking, assortment, mixing, product support, and market presence. The handling of products is a key to warehouse productivity. Handling activities include receiving, in storage handling, and shipping. Packaging has a significant impact on the cost and productivity of the logistical system. An integrated logistics approach to packaging operations can yield dramatic savings.

A marketing mix is a compilation of activities designed to attract customers while simultaneously achieving business objectives. The so-called four P's - products/service, promotion, price, and place - constitute a generic marketing mix. The key to formulating an effective mix strategy is to integrate resources committed to these activities into an effort that maximizes customer impact. Logistics ensures that customer requirements involved in timing and location of inventory and other related services are satisfactorily performed. Thus, the output of logistical performance is customer service. Logistical competence is a tangible way to attract customers who place a premium on time and place-related performance.

Thus the discussion on the objectives, logistics interface with marketing and the system elements brings out the depth of the scope of logistics in the efficient functioning of any business entity. The key to excellent logistics is to achieve integration of both internal and external operations. Such integration requires clear identification concerning the role that logistical competency is expected to play in overall enterprise strategy.

TRADE-OFF ANALYSIS

Trade-off analysis is a family of methods by which respondents' utilities for various product features (usually including price) are measured. In some cases, the utilities are measured indirectly. In this case, respondents are asked to consider alternatives and state a likelihood of purchase or preference for each alternative. As the respondent continues to make choices, a pattern begins to emerge which, through complex multiple regression (and other) techniques, can be broken down and analysed as to the individual features that contribute most to the purchase likelihood or preference. The importance or influence contributed by the component parts. *i.e.*, product features, are measured in relative units called "utils" or "utility weights." In other cases, respondents are asked to

tell the interviewer directly how important various product features are to them. For example, they might be asked to rate on a scale of 1 to 100 various product features, where 1 means not at all important to their purchase decision and 100 means extremely important to their purchase decision. Trade-off analyses produce several types of information. First, they tell us what features (and levels of features) are most valued by customers. Second, they allow us to model how likely people will be to purchase various configurations of products, the share of revenue these products will most likely receive and what role price plays in the assessment of acceptability.

There are four main types of trade-off

- i. Conjoint
- ii. Discrete Choice
- iii. Self-explicated
- iv. Hybrid

One additional model, the MACROModel©2, will be discussed which does not fall into any of the above four categories. We will discuss each of these trade-off types after reviewing a few basic concepts.

Experimental Design, A critical issue in most trade-off methods is the selection of product attributes to be combined together to create each product configuration to be tested. If every possible combination of attributes were included in the study, the study would be said to be using a complete or full factorial design. This is desirable but very seldom practical. For example, if we had 6 attributes with 3 levels each, the total number of possible combinations would be 36 or 729. This is much too large to ask one respondent to rate (and 6 attributes with 3 levels each is untypical modest). When a fractional factorial design is used, only a fraction of the total possible number of product combinations needs to be tested, For the above example, a fractional factorial design could be generated (usually with the help of a computer) that would require perhaps as few as 14 product configurations to be rated. It must be kept in mind, however, that whenever a fractional factorial design is used, some information will be lost. It is the job of the researcher creating the experimental design to ensure that the information being sacrificed (usually higher order interaction effects) does not compromise the project's ability to answer the research objectives.

BRIDGING

Occasionally, even with the most efficient fractional factorial design, we still end up with more products than can be practically accommodated. One possible solution to that problem is bridging. Bridging allows the attributes to be divided into two or more sets (with some attributes common to all sets). Each set of attributes is treated like its own trade-off study. A fractional factorial design is created for each set of attributes. Respondents are asked to rate or rank two smaller sets of products rather than one large set. The utilities are calculated for each trade-off exercise independently and bridged together to create one final set of utilities.

COGNITIVE AND NON-COGNITIVE BEHAVIOUR

Critical to the selection of an appropriate trade-off technique is the issue of which type of behaviour, cognitive or non-cognitive, best represents the behaviour being measured. Cognitive behaviour is behaviour that is based on rational, conscious decision-making. Such factors as price, functionality or durability are typically cognitive. Non-cognitive behaviour is behaviour that is based on less tangible or even less conscious factors such as status, aspiration, insecurity, perceived taste, etc. One might argue that the selection of a life insurance policy, a computer or a water heater are all cognitive decisions and that the selection of a beer, a skin cream or a pair of pants are all non-cognitive. One might also argue that all decisions made by humans are non-cognitive.

However, trade-off techniques that employ direct questions (self-explicated and hybrid) all assume that the behaviour being modeled is cognitive, because at least some of the product features are being rated in a way that requires both awareness and honesty from the respondent. That is, the respondent must be aware of the degree to which a product feature affects his or her purchase decision and also be willing to admit to that degree of affect.

Additionally, any data collection methods that rely on verbal or written descriptions of product features all assume that the behaviour being modeled is cognitive, because the process of understanding a verbal or written description is itself a cognitive behaviour.

Non-cognitive trade-off models should be based on an indirect trade-off technique (conjoint or discrete choice) and data collection that relies on experience rather than language to communicate the product choices. For example, if you are modelling the pant selection process, show respondents a variety of pants that they can see and touch. A consumer may respond to the phrase “light blue pants” very differently than he or she would to a particular pair of light blue pants.

THE FOUR MAIN TYPES OF TRADE-OFF

CONJOINT

Conjoint analysis is the original trade-off approach and uses linear models. There is metric conjoint, where respondents monadically rate various product configurations, and non-metric conjoint, where respondents rank a set of product configurations. There are also full-profile conjoint, partial-profile conjoint and pair wise conjoint. Full-profile conjoint uses all product features in every product configuration. Partial profile conjoint uses a smaller subset of available product features in the product configurations. Pair wise conjoint requires the respondent to rate their preference for one product over another in a paired comparison. We will only discuss conjoint methods in general in this paper. Conjoint models are simply regression models which are constructed for each individual respondent. Typically, each respondent rates or ranks 20 to 30 product

configurations. Each product configuration contains different levels of the product attributes being tested. If the product levels are varied appropriately (the role of experimental design), a regression model can be estimated for each individual, using the product ratings as cases. The coefficients from the model are the utilities or utilises. A conjoint approach should be used if a limited number of attributes needs to be tested and utilities need to be estimated for individual respondents, *e.g.*, conjoint-based segmentation.

DISCRETE CHOICE

Discrete choice differs from conjoint in that respondents are shown a set of products from which they pick the one they most want to buy or none if they are not interested in any of the choices shown (rather than rate or rank choices). Respondents are shown several sets of choices sequentially. For each choice set, they are asked to pick one or none. This is in contrast to most forms of conjoint where respondents are not allowed to choose none of the product options (MACRO incorporates no-buy choices into its conjoint models). The discrete choice procedure has the advantage of being more like the actual purchase decision process than does any of the data collection methods used in most Conjoint studies..

Also, in conjoint methods, the mathematical models constructed to simulate market behaviour are based on linear regression models. In discrete choice, the basis is the multinomial logit model, which is non-linear. Another analytical difference is that, in conjoint procedures, the utility weights are estimated for each respondent individually. These weights can often provide the basis for a very powerful customer segmentation. Most commercially available forms of discrete choice do not allow this option, although this may be rapidly changing. Further, because discrete choice models are generally estimated at the aggregate level, there exists the possibility that respondents will have strong but opposite preferences to one another. These preferences will effectively cancel each other out when the model is constructed at the aggregate level, yielding the incorrect conclusion that respondents had no strong preference. This is sometimes referred to as the heterogeneity problem.

There are two basic forms of discrete choice: classic and exploding data. Classic discrete choice involves showing a respondent a series of sets of products (as described above). In exploding data discrete choice, respondents are asked to rank order a set of products based on purchase interest (similar to non-metric conjoint). This rank-ordered data set can be transformed into a format suitable for logic model estimation. Exploding data discrete choice has the advantage of more efficient data collection over classic discrete choice. The exploding data approach creates many times more data points (or cases) than the classic approach with the same interview length. Discrete choice should be used if the primary objective of the study is to estimate market share or price sensitivity, a limited number of attributes need to be tested and the sample population is known to be homogeneous with respect to all product attributes.

SELF-EXPLICATED

Conjoint and discrete choice both determines respondent's utilities indirectly. Self-explicated determines respondents' utilities directly. With self-explicated scales, respondents are asked directly how important all levels of all attributes are to their purchase interest. Despite its conceptual simplicity, self-explicated models have been shown to be comparable to conjoint models. Self-explicated conjoint analysis requires respondents to reveal their utilities directly. Accordingly, standard questionnaire methods can be used to collect the information. The technique involves the following steps:

- i. Respondent are informed about all the attributes and their levels, and the respondents are then asked to identify attribute levels that are totally unacceptable to them
- ii. From among the acceptable levels of the attributes, respondents are asked to indicate which are the most preferred and least preferred levels of each attribute
- iii. Using the respondents' most important attribute as an anchor, elicit importance ratings for the other attributes (on a 0 - 100 scale)
- iv. For each attribute, rate the desirability of the different acceptable levels with the attribute
- v. Utilities for acceptable attribute levels are obtained by multiplying the importance rating and the desirability ratings. The utilities are then entered into a choice simulator programme, and choice information similar to other conjoint programmes can be obtained. Self-explicated approaches are useful when there are a large number of attributes and the decision process being modeled is cognitive.

HYBRID

Hybrid models are models that use a combination of the above techniques. The most famous hybrid model is ACA, Adaptive Conjoint Analysis. Adaptive Conjoint Analysis, in this procedure, a computer programme prompts the interviewer with questions. The procedure is as follows:

Respondents are first walked through a battery of feature-importance ratings and rankings; second, through a series of pair wise trade-offs of different product configurations. The product configurations shown to any one respondent may not include all of the attributes being tested. The configurations to be paired are based on the answers to the importance questions and rankings asked in the beginning of the interview. Items that are considered of little importance show up in the comparisons less often. Items that are considered of greater importance show up in the comparisons more often. For each pair of products being tested, the respondent is to indicate which product they prefer and the degree to which they prefer it.

The software continues prompting with pair wise comparisons of product configurations until enough data has been collected to estimate conjoint utilities for

each level of each feature. Since the procedure is adaptive, only a fraction of the total number of possible product combinations is tested. ACA is an approach that is appropriate for building preference models of cognitive behaviour with large numbers of attributes. It may not be as useful when price sensitivity, non-cognitive purchase decisions or interaction terms are to be modeled.

CAKE METHOD AND LOGIT-CAKE METHOD

Other hybrid models include the Cake Method⁸ and the Logit-Cake Method⁹. Both of these models have been developed by MACRO Consulting and were designed to overcome weaknesses in other models.

CAKE METHOD

The Cake Method⁸ is a unique, proprietary approach to conjoint analysis which offers several advantages over other conjoint methods: A large number of product features (50 or more) can be included in the model First order interactions can be estimated at both the disaggregate and aggregate levels There is complete control over the experimental design, in a full-profile format Since product combinations are specified, via traditional experimental design, before the interview takes place, physical exhibits can be easily incorporated into the interview The approach involves a specific data collection procedure as well as a unique analytic protocol. The basic outline of the approach is to:

- i. Collect self-explicated scales on most of the product attributes tested
- ii. Conduct a full-profile conjoint exercise with a limited number of product attributes, some of which are common to the self-explication exercise
- iii. Estimate conjoint utilities for each respondent
- iv. Bridge self-explicated scales to utility weights

The Cake Method should be used when there are a large number of attributes, utilities need to be estimated for individuals, interaction terms need to be measured and the purchase decision is at least partially cognitive.

LOGIT-CAKE METHOD

The Logit-Cake Method is a unique, proprietary approach to choice-based trade-off analysis which offers several advantages over other conjoint methods:

- i. A large number of product features (50 or more) can be included in the model
- ii. The heterogeneity problem long associated with aggregate logit models is avoided
- iii. The traditional advantages of logit models over conjoint models are maintained
- iv. First order interactions can be estimated
- v. There is complete control over the experimental design, in a full-profile format

Since product combinations are specified, via traditional experimental design, before the interview takes place, physical exhibits can be easily incorporated into the interview,

The approach involves a specific data collection procedure as well as a unique analytic protocol. The basic outline of the approach is to:

- i. Collect self-explicated scales on all product attributes tested
- ii. Conduct a full-profile choice-based exercise with a subset of product attributes
- iii. Segment the sample based on self-explicated scales
- iv. Estimate logit models for each respondent cluster
- v. Bridge self-explicated scales to logit-based utility weights

The Logit-Cake Method© should be used when there are a large number of attributes, market share and price need to be estimated, interaction terms need to be measured and the purchase decision is at least partially cognitive.

MACRO MODEL

It does not fall into any of the four main types of trade-off models. In fact, it is not strictly speaking a trade-off model because it does not estimate utilities for any product attributes. The MACRO Model was developed by MACRO Consulting to address a specific research methods need that frequently occurs in new product development and packaging.

The MACRO Model is a unique approach to new product screening which offers several advantages over other methods:

- i. A large number of concepts or packages (50 or more) can be screened at one time
- ii. Price sensitivity can be calculated for every new product concept screened
- iii. Price/volume can be individually optimized for every product concept tested
- iv. New product concepts can be screened and/or completely rank ordered on consumer appeal, market share, unit volume, gross dollar volume or gross profits.

The approach involves a specific data collection procedure as well as a unique analytic protocol. The basic outline of the approach is to Sort a stack of new product concepts cards (all new product concepts, each at three price points) into two piles: would definitely buy and would not buy. Note: Stack would contain several existing products as reference have them rank order the would buy pile on a continuum from most want to buy to least want to buy.

If the number of items to be sorted is too large for one sorting exercise, the task can be broken down into several smaller exercises, with two or three items common across sorting tasks. After the data are collected for all respondents for the various sorting exercises, a bridging technique can be used to incorporate the data from the separate exercises into one rank ordering of all of the items used in the study. Once the data are combined into one rank order data set for each respondent, the MACRO Model© (a first choice share of preference model) can be constructed. The MACRO Model© should be used when the product is too complex to decompose into attributes, *e.g.*, packaging

graphics, when a large number of highly different products are to be included, *e.g.*, new product screening, when price sensitivity needs to be measured and when products will be screened based on their revenue potential.

FORMS OF LOGISTICS MANAGEMENT

Centralized logistics management provides that managers that also head other divisions of the company head the logistics operations. This type of management helps avoid internal problems by having a central manager that ultimately decides how logistics and operations are coordinated.

DECENTRALIZED LOGISTICS MANAGEMENT

Decentralized logistics management is based on the fact that a company needs to have a division that helps control the local-adaptation needs. Dealing with different cultures requires input from the local branch. The managers that deal with the cultural differences on a daily basis normally know what works and what don't. Outsourcing is the final option for logistics management. When this happens, transportation firms concentrate on logistics, and the company can concentrate on its production. There are many cost savings using this type of programme, however that lack of control can negatively effect many companies. International logistics requires many different options and requirements to be met in order for a company to operate internationally. It's like a big puzzle that must be put together, in order for all the goals to be met. As described above, there are many options to consider, and sometimes what appears to be an option really isn't. It is not difficult to hit a road block, and you must start over with a new plan. Once the logistics plan is in place, you must constantly look for improvements in order to maximize profits and goals.

Supply Chain Management

INTRODUCTION

Supply chain management (SCM) is the management of a network of interconnected businesses involved in the ultimate provision of product and service packages required by end customers (Harland, 1996). Supply chain management spans all movement and storage of raw materials, work-in-process inventory, and finished goods from point of origin to point of consumption (supply chain).

Another definition is provided by the APICS Dictionary when it defines SCM as the “design, planning, execution, control, and monitoring of supply chain activities with the objective of creating net value, building a competitive infrastructure, leveraging worldwide logistics, synchronizing supply with demand and measuring performance globally.”

DEFINITIONS

More common and accepted definitions of supply chain management are:

- Supply chain management is the systemic, strategic coordination of the traditional business functions and the tactics across these business functions within a particular company and across businesses within the supply chain, for the purposes of improving the long-term performance of the individual companies and the supply chain as a whole (Mentzer *et al.*, 2001).

- A customer focused definition is given by Hines “Supply chain strategies require a total systems view of the linkages in the chain that work together efficiently to create customer satisfaction at the end point of delivery to the consumer. As a consequence costs must be lowered throughout the chain by driving out unnecessary costs and focusing attention on adding value. Throughput efficiency must be increased, bottlenecks removed and performance measurement must focus on total systems efficiency and equitable reward distribution to those in the supply chain adding value. The supply chain system must be responsive to customer requirements.”
- Global supply chain forum - supply chain management is the integration of key business processes across the supply chain for the purpose of creating value for customers and stakeholders (Lambert, 2008).
- According to the Council of Supply Chain Management Professionals (CSCMP), supply chain management encompasses the planning and management of all activities involved in sourcing, procurement, conversion, and logistics management. It also includes the crucial components of coordination and collaboration with channel partners, which can be suppliers, intermediaries, third-party service providers, and customers. In essence, supply chain management integrates supply and demand management within and across companies. More recently, the loosely coupled, self-organizing network of businesses that cooperate to provide product and service offerings has been called the *Extended Enterprise*.

A supply chain, as opposed to supply chain management, is a set of organizations directly linked by one or more of the upstream and downstream flows of products, services, finances, and information from a source to a customer. Managing a supply chain is ‘supply chain management’ (Mentzer *et al.*, 2001).

Supply chain management software includes tools or modules used to execute supply chain transactions, manage supplier relationships and control associated business processes.

Supply chain event management (abbreviated as SCEM) is a consideration of all possible events and factors that can disrupt a supply chain. With SCEM possible scenarios can be created and solutions devised.

SUPPLY MANAGEMENT

The term supply management describes the methods and processes of modern corporate or institutional buying. This may be for the purchasing of supplies for internal use referred to as indirect goods and services, purchasing raw materials for the consumption during the manufacturing process, or for the purchasing of goods for inventory to be resold as products in the distribution and retail process.

In many organizations, acquisition or buying of services is called contracting, while that of goods is called purchasing or procurement. The supply management function of an organization is responsible for various aspects of these acquisitions:

- Working with business leaders who have identified a business need or requirement to identify, source, contract, and procure the needed good or service from qualified suppliers
- Managing supplier performance
- Implementing technologies, processes, policies, and procedures to support the purchasing process (Supplier Relationship Management).
- The supplier relationship management process: a process for providing the structure for how relationships with suppliers will be developed and maintained.
- Economic theories of supply and demand.

Supply management is generally regarded as a systematic business process that includes more functions than traditional buying, such as coordinating inbound and internal pre-production logistics and managing inventory.

Supply management deals primarily with the oversight and management of materials and services inputs, management of the suppliers who provide those inputs, and support of the process of acquiring those inputs. The performance of supply management departments and supply management professionals is commonly measured in terms of amount of money saved for the organization. However, managing risk is one of the other critical aspects of supply management; especially the risk of non-availability at the required time of quality goods and services critical for an organization's survival and growth.

HISTORY

Prior to 1900, purchasing was recognized as an independent function by many railroad organizations, but in few other industries.

Prior to World War I, purchasing was regarded as primarily clerical.

During World War I and II: The function increased due to the importance of obtaining raw materials, supplies, and services needed to keep the factories and mines operating.

1950s and 1960s: Purchasing continued to gain stature as the techniques for performing the function became more refined and as the number of trained professionals increased. The emphasis became more managerial.

1970s and 1980s: More emphasis was placed on purchasing strategy as the ability to obtain needed items from suppliers at realistic prices increased.

1983: In September 1983, Harvard Business Review published a ground-breaking article by Peter Kraljic on purchasing strategy that is widely cited today as the beginning of the transformation of the function from "purchasing," something that is viewed as highly tactical to supply management or procurement, something that is viewed as very strategic to the business

1990s: Supply management (procurement) starts to become more integrated into the overall corporate strategy and a broad-based transformation of the business function is ignited, fueled strongly by the development of supply management software solutions which help automate the source-to-settle process.

2000s: The leader of the supply management function within many enterprises is established with a C-Level title - the Chief Procurement Officer (sometimes called the Head of Procurement). Publications, events, and websites that are dedicated solely to the advancement of supply management and Chief Procurement Officers arise. The global recession of 2008-2009 places supply management at the crux of business strategy.

2011: With the advancement of social networks ([facebook.com/supplymanagement](https://www.facebook.com/supplymanagement)), supply management professionals prefer to unite around common interests and values.

GROUPS AND CERTIFICATIONS

The importance of supply management in global business has prompted the formation of professional organizations to address the need for higher levels of supply management skill and expertise. One of the largest of these is the Institute for Supply Management, a United States not-for-profit association that includes more than 40,000 members. It is affiliated with the International Federation of Purchasing and Supply Management, a union of local and national purchasing associations with approximately 200,000 members. For companies seeking to fulfil diversity supplier spend commitments, the National Minority Supplier Development Council with 39 affiliated nation-wide councils, was established in 1972 to assist in promoting supplier diversity, and also provides management training and access to viable minority business enterprises.

Many certification programmes are relevant to the supply management profession. Some are offered through non-profit associations, such as the Certified Purchasing Manager (CPM) and Certified Professional in Supply Management (CPSM) through the Institute for Supply Management. There are also for-profit companies who offer certification programmes, such as Next Level Purchasing, Inc. who offers the Senior Professional in Supply Management (SPSM) Certification.

SUPPLY MANAGEMENT

Supply management is different than supply chain management, though it can be considered a component of supply chain management. Conversely, where the supply management function is established as a C-level strategic effort, supply chain management is but one component of an overall strategic supply management approach. Supply management is a complementary discipline that encompasses the alignment of organizations, processes, and systems for strategic sourcing, contract management, supplier management, spend analysis to continuously improve global supply for best-value performance in support of the strategic objectives of the business.

SUPPLY MANAGEMENT SOFTWARE

Supply Management software comprises all of the different solutions which automate the source-to-settle process and include Spend Analysis, eSourcing, Contracts, Supply Base Management, eProcurement, eCatalogs (for Supplier Enablement), and Accounts Payable or ePayables solutions. Software which helps automate the management of complex services like business travel and temporary labor are also included in this software segment

One report that focuses on a sub-set of the space is the “Magic Quadrant For Strategic Sourcing Application Suites, 2010,” a Gartner research report, summarizes: “Sourcing applications provide a systematic and scalable means for organizations to manage the full sourcing process, including finalizing purchase specifications, selecting suppliers and negotiating prices....Most sourcing solution vendors bundle spend analysis, contract management and supplier performance management tools into their suites.” The Gartner report summarizes, “Best-of-breed providers with suites delivered via software as a service dominate the strategic sourcing application market, while ERP companies with integrated offerings are gaining traction by providing tactical sourcing support.” Gartner estimates the sourcing software market at close to a half-billion dollars in 2007 with an annual growth rate of 5%. According to Gartner, the research firm, leading providers of supply and contract management software include SAP, Ariba, BravoSolution, Ivalua, AECsoft, Rosslyn Analytics and Emptoris.

WAREHOUSE MANAGEMENT SYSTEM

A warehouse management system, or WMS, is a key part of the supply chain and primarily aims to control the movement and storage of materials within a warehouse and process the associated transactions, including shipping, receiving, putaway and picking. The systems also direct and optimize stock putaway based on real-time information about the status of bin utilization.

Warehouse management systems often utilize Auto ID Data Capture (AIDC) technology, such as barcode scanners, mobile computers, wireless LANs and potentially Radio-frequency identification (RFID) to efficiently monitor the flow of products. Once data has been collected, there is either a batch synchronization with, or a real-time wireless transmission to a central database. The database can then provide useful reports about the status of goods in the warehouse.

The objective of a warehouse management system is to provide a set of computerized procedures to handle the receipt of stock and returns into a warehouse facility, model and manage the logical representation of the physical storage facilities (*e.g.*, racking etc), manage the stock within the facility and enable a seamless link to order processing and logistics management in order to pick, pack and ship product out of the facility.

Warehouse management systems can be stand alone systems, or modules of an ERP system or supply chain execution suite.

The primary purpose of a WMS is to control the movement and storage of materials within a warehouse – you might even describe it as the legs at the end-of-the line which automates the store, traffic and shipping management.

In its simplest form, the WMS can data track products during the production process and act as an interpreter and message buffer between existing ERP and WMS systems. Warehouse Management is not just managing within the boundaries of a warehouse today, it is much wider and goes beyond the physical boundaries. Inventory management, inventory planning, cost management, IT applications and communication technology to be used are all related to warehouse management.

The container storage, loading and unloading are also covered by warehouse management today. Warehouse management today is part of SCM and demand management. Even production management is to a great extent dependent on warehouse management. Efficient warehouse management gives a cutting edge to a retail chain distribution company. Warehouse management does not just start with receipt of material but it actually starts with actual initial planning when container design is made for a product. Warehouse design and process design within the warehouse (*e.g.*, Wave Picking) is also part of warehouse management. Warehouse management is part of Logistics and SCM.

Warehouse Management monitors the progress of products through the warehouse. It involves the physical warehouse infrastructure, tracking systems, and communication between product stations. Warehouse management deals with receipt, storage and movement of goods, normally finished goods, to intermediate storage locations or to final customer. In the multi-echelon model for distribution, there are levels of warehouses, starting with the Central Warehouse(s), regional warehouses services by the central warehouses and retail warehouses at the third level services by the regional warehouses and so on. The objective of warehousing management is to help in optimal cost of timely order fulfillment by managing the resources economically. Warehouse management = “Management of storage of products and services rendered on the products within the four walls of a warehouse”

IMPACT OF E-COMMERCE ON SUPPLY CHAIN MANAGEMENT

A production supply chain refers to the flow of physical goods and associated information from the source to the consumer. Key supply chain activities include:

- Production planning
- Purchasing
- Materials management
- Distribution
- Customer service
- Sales forecasting.

These processes are critical to the success of any operation whether they're manufacturers, wholesalers, or service providers.

Electronic commerce and the Internet are fundamentally changing the nature of supply chains, and redefining how consumers learn about, select, purchase, and use products and services. The result has been the emergence of new business-to business supply chains that are consumer-focused rather than product-focused. They also provide customized products and services. E-commerce impacts supply chain management in a variety of keyways. These include:

Cost Efficiency: E-commerce allows transportation companies of all sizes to exchange cargo documents electronically over the Internet. E-commerce enables shippers, freight forwarders and trucking firms to streamline document handling without the monetary and time investment required by the traditional document delivery systems.

By using e-commerce, companies can reduce costs, improve data accuracy, streamline business processes, accelerate business cycles, and enhance customer service. Ocean carriers and their trading partners can exchange bill of lading instructions, freight invoices, container status messages, motor carrier shipment instructions, and other documents with increased accuracy and efficiency by eliminating the need to re-key or reformat documents. The only tools needed to take advantage of this solution are a personal computer and an Internet browser.

Changes in the Distribution System: E-commerce will give businesses more flexibility in managing the increasingly complex movement of products and information between businesses, their suppliers and customers. E-commerce will close the link between customers and distribution centres. Customers can manage the increasingly complex movement of products and information through the supply chain.

Customer Orientation: E-commerce is a vital link in the support of logistics and transportation services for both internal and external customers. E-commerce will help companies deliver better services to their customers, accelerate the growth of the e-commerce initiatives that are critical to their business, and lower their operating costs. Using the Internet for e-commerce will allow customers to access rate information, place delivery orders, track shipments and pay freight bills.

E-commerce makes it easier for customers to do business with companies: Anything that simplifies the process of arranging transportation services will help build companies' business and enhance shareholder value. By making more information available about the commercial side of companies, businesses will make their web site a place where customers will not only get detailed information about the services the company offers, but also where they can actually conduct business with the company.

Ultimately, web sites can provide a universal, self-service system for customers. Shippers can order any service and access the information they need to conduct business with transportation companies exclusively on-line. E-commerce functions are taking companies a substantial step forward by providing customers with a faster and easier way to do business with them.

Shipment Tracking: E-commerce will allow users to establish an account and obtain real-time information about cargo shipments. They may also create and submit bills of lading, place a cargo order, analyze charges, submit a freight claim, and carry out many other functions. In addition, e-commerce allows customers to track shipments down to the individual product and perform other supply chain management and decision support functions. The application uses encryption technology to secure business transactions.

Shipping Notice: E-commerce can help automate the receiving process by electronically transmitting a packing list ahead of the shipment. It also allows companies to record the relevant details of each pallet, parcel, and item being shipped.

Freight Auditing: This will ensure that each freight bill is efficiently reviewed for accuracy. The result is a greatly reduced risk of overpayment, and the elimination of countless hours of paperwork, or the need for a third-party auditing firm. By intercepting duplicate billings and incorrect charges, a significant percent of shipping costs will be recovered. In addition, carrier comparison and assignment allows for instant access to a database containing the latest rates, discounts, and allowances for most major carriers, thus eliminating the need for unwieldy charts and tables.

Shipping Documentation and Labeling: There will be less need for manual intervention because standard bills of lading, shipping labels, and carrier manifests will be automatically produced; this includes even the specialized export documentation required for overseas shipments. Paperwork is significantly reduced and the shipping department will therefore be more efficient.

On-line Shipping Enquiry: This gives instant shipping information access to anyone in the company, from any location. Parcel shipments can be tracked and proof of delivery quickly confirmed. A customer's transportation costs and performance can be analyzed, thus helping the customer negotiate rates and improve service.

PROBLEMS ADDRESSED BY SUPPLY CHAIN MANAGEMENT

Supply chain management must address the following problems:

- **Distribution Network Configuration:** number, location and network missions of suppliers, production facilities, distribution centres, warehouses, cross-docks and customers.
- **Distribution Strategy:** questions of operating control (centralized, decentralized or shared); delivery scheme, *e.g.*, direct shipment, pool point shipping, cross docking, DSD (direct store delivery), closed loop shipping; mode of transportation, *e.g.*, motor carrier, including truckload, LTL, parcel; railroad; intermodal transport, including TOFC (trailer on flatcar) and COFC (container on flatcar); ocean freight; airfreight; replenishment strategy (*e.g.*, pull, push or hybrid); and transportation control (*e.g.*, owner-operated, private carrier, common carrier, contract carrier, or 3PL).

- **Trade-Offs in Logistical Activities:** The above activities must be well coordinated in order to achieve the lowest total logistics cost. Trade-offs may increase the total cost if only one of the activities is optimized. For example, full truckload (FTL) rates are more economical on a cost per pallet basis than less than truckload (LTL) shipments. If, however, a full truckload of a product is ordered to reduce transportation costs, there will be an increase in inventory holding costs which may increase total logistics costs. It is therefore imperative to take a systems approach when planning logistical activities. These trade-offs are key to developing the most efficient and effective Logistics and SCM strategy.
- **Information:** Integration of processes through the supply chain to share valuable information, including demand signals, forecasts, inventory, transportation, potential collaboration, etc.
- **Inventory Management:** Quantity and location of inventory, including raw materials, work-in-progress (WIP) and finished goods.
- **Cash-Flow:** Arranging the payment terms and methodologies for exchanging funds across entities within the supply chain.

Supply chain execution means managing and coordinating the movement of materials, information and funds across the supply chain. The flow is bi-directional.

ACTIVITIES/FUNCTIONS

Supply chain management is a cross-function approach including managing the movement of raw materials into an organization, certain aspects of the internal processing of materials into finished goods, and the movement of finished goods out of the organization and towards the end-consumer. As organizations strive to focus on core competencies and becoming more flexible, they reduce their ownership of raw materials sources and distribution channels.

These functions are increasingly being outsourced to other entities that can perform the activities better or more cost effectively. The effect is to increase the number of organizations involved in satisfying customer demand, while reducing management control of daily logistics operations.

Less control and more supply chain partners led to the creation of supply chain management concepts. The purpose of supply chain management is to improve trust and collaboration among supply chain partners, thus improving inventory visibility and the velocity of inventory movement.

Several models have been proposed for understanding the activities required to manage material movements across organizational and functional boundaries. SCOR is a supply chain management model promoted by the Supply Chain Council.

Another model is the SCM Model proposed by the Global Supply Chain Forum (GSCF). Supply chain activities can be grouped into strategic, tactical, and operational

levels. The CSCMP has adopted The American Productivity and Quality Centre (APQC) Process Classification FrameworkSM a high-level, industry-neutral enterprise process model that allows organizations to see their business processes from a cross-industry viewpoint.

STRATEGIC LEVEL

- Strategic network optimization, including the number, location, and size of warehousing, distribution centres, and facilities.
- Strategic partnerships with suppliers, distributors, and customers, creating communication channels for critical information and operational improvements such as cross docking, direct shipping, and third-party logistics.
- Product life cycle management, so that new and existing products can be optimally integrated into the supply chain and capacity management activities.
- Information technology chain operations.
- Where-to-make and make-buy decisions.
- Aligning overall organizational strategy with supply strategy.
- It is for long term and needs resource commitment.

TACTICAL LEVEL

- Sourcing contracts and other purchasing decisions.
- Production decisions, including contracting, scheduling, and planning process definition.
- Inventory decisions, including quantity, location, and quality of inventory.
- Transportation strategy, including frequency, routes, and contracting.
- Benchmarking of all operations against competitors and implementation of best practices throughout the enterprise.
- Milestone payments.
- Focus on customer demand and Habits.

OPERATIONAL LEVEL

- Daily production and distribution planning, including all nodes in the supply chain.
- Production scheduling for each manufacturing facility in the supply chain (minute by minute).
- Demand planning and forecasting, coordinating the demand forecast of all customers and sharing the forecast with all suppliers.
- Sourcing planning, including current inventory and forecast demand, in collaboration with all suppliers.
- Inbound operations, including transportation from suppliers and receiving inventory.

- Production operations, including the consumption of materials and flow of finished goods.
- Outbound operations, including all fulfillment activities, warehousing and transportation to customers.
- Order promising, accounting for all constraints in the supply chain, including all suppliers, manufacturing facilities, distribution centres, and other customers.
- From production level to supply level accounting all transit damage cases and arrange to settlement at customer level by maintaining company loss through insurance company.

IMPORTANCE OF SUPPLY CHAIN MANAGEMENT

Organizations increasingly find that they must rely on effective supply chains, or networks, to compete in the global market and networked economy. In Peter Drucker's (1998) new management paradigms, this concept of business relationships extends beyond traditional enterprise boundaries and seeks to organize entire business processes throughout a value chain of multiple companies.

During the past decades, globalization, outsourcing and information technology have enabled many organizations, such as Dell and Hewlett Packard, to successfully operate solid collaborative supply networks in which each specialized business partner focuses on only a few key strategic activities (Scott, 1993). This inter-organizational supply network can be acknowledged as a new form of organization. However, with the complicated interactions among the players, the network structure fits neither "market" nor "hierarchy" categories (Powell, 1990). It is not clear what kind of performance impacts different supply network structures could have on firms, and little is known about the coordination conditions and trade-offs that may exist among the players. From a systems perspective, a complex network structure can be decomposed into individual component firms (Zhang and Dilts, 2004). Traditionally, companies in a supply network concentrate on the inputs and outputs of the processes, with little concern for the internal management working of other individual players. Therefore, the choice of an internal management control structure is known to impact local firm performance (Mintzberg, 1979).

In the 21st century, changes in the business environment have contributed to the development of supply chain networks. First, as an outcome of globalization and the proliferation of multinational companies, joint ventures, strategic alliances and business partnerships, significant success factors were identified, complementing the earlier "Just-In-Time", "Lean Manufacturing" and "Agile Manufacturing" practices. Second, technological changes, particularly the dramatic fall in information communication costs, which are a significant component of transaction costs, have led to changes in coordination among the members of the supply chain network (Coase, 1998).

Many researchers have recognized these kinds of supply network structures as a new organization form, using terms such as "Keiretsu", "Extended Enterprise", "Virtual

Corporation”, “Global Production Network”, and “Next Generation Manufacturing System”. In general, such a structure can be defined as “a group of semi-independent organizations, each with their capabilities, which collaborate in ever-changing constellations to serve one or more markets in order to achieve some business goal specific to that collaboration” (Akkermans, 2001).

The security management system for supply chains is described in ISO/IEC 28000 and ISO/IEC 28001 and related standards published jointly by ISO and IEC.

ENVIRONMENTAL SUPPLY CHAIN MANAGEMENT

Environmental concerns are finally finding its way from strategy and board room meetings to operations and to supply chain management (SCM). A growing number of companies have realized the world over that implementing supply chain management is only one of the objectives they need to realize. The other important agenda is to make this SCM environmentally friendly. This has catapulted SCM to a new height that promises to give a major contribution to the companies overall strategy.

The companies are implementing ESCM (Environmental SCM) by looking beyond their own facilities. Ofcourse this is only after they have ensured that their manufacturing facility is friendly to the environment and does not throws or generates harmful waste products into the environment. Companies are also involving their suppliers in environmental initiatives and agendas. This is done by screening suppliers for environmental performance, working collaboratively with them on green design initiatives and providing training and information to build suppliers’ environmental management capacity.

Environmental supply chain management recognizes the crucial role to be played by the purchasing and the function’s involvement in activities that include reduction, recycling, reuse and the substitution of materials. Working with suppliers on environmental issues not only generates significant environmental benefits, but also opportunities for cost containment, improved risk management and enhanced quality and brand image. This will also help companies streamline their supply base and develop more co-operative, long-term relationships with key suppliers, a practice that has fostered greater opportunities to work together on environmental issues.

It is hence important that Indian companies too make ESCM an important part of their agenda. Already Asian Paints, M&M and some other companies have taken great strides in this area. However large scale adoption of environmental aspects within the framework of SCM is essential. The starting point as for any other major strategy is the top management commitment to the environmental issues. This is not only essential but is critical for the overall success of the project. It should be the mission of the company’s top management to sensitize everyone in the company towards environmental issues. It will also be a good idea to have a separate environmental department reporting directly to the top management. This department should look at all the aspects of the

environment barometer and track its progress on a regular basis. Any discrepancy is immediately reported to the CEO. However, it is equally important to make everybody participate in the process with the same gusto. This can happen by empowering the employees to act immediately in response to the environmental problems, rather than simply offering suggestions. The rewards and incentive structure also emphasizes results. This will allow the employees to think and be innovative in implementing environmentally friendly policies and procedures.

It is important to spread the message to the customers and thereby making them participate in the environment journey. This can be achieved by branding the company and its products on environmental lines. This can be done by bringing out the synergy between the company's strategy and its strong environmental image and reputation, and environmentally sensitive products, suppliers, etc. it deals with.

Next it is important to connect with the most vital link of the company – its suppliers. A holistic approach, such as supply chain integration to environmental management links reduction in energy consumption to waste generation and release of pollutants. Setting aggressive and progressive environmental goals is important. This can be done by using tools such as life-cycle management (LCM) and environmental audits (EA) improve environmental and operating performance. Suppliers are asked to keep track of their cost of waste (COW) so that this can be reduced. Supplier negotiations and selection revolve around COW and other environmental issues.

Supplier audits on environmental issues are an absolute must and this process should be a cross-functional initiative. This should involve employees from quality assurance, environmental affairs and purchasing. Similarly, teams should include financial analysts who decide whether suppliers will be the most productive from the perspective of maximizing Environment Value Add (EVA) make-or-buy decisions. Adherence of suppliers to quality and environment standards is a necessary prerequisite for achieving a company's various objectives. Early sourcing and early supplier involvement in basic product developmental work is required and while doing so it will be easy for the companies to strictly adhere to the norms it has prescribed for itself. These sourcing decisions must also take into consideration safety issues, capacity of suppliers, and ability to treat compounds and effluents. Products and processes should be subjected to continual critical analysis at every stage of the value-added process. The early integration of suppliers into all decisions affecting them is critical to environmental effectiveness. The close alignment of supplier capabilities with buying firm's environmental goals is critical to programme success. This alignment can be achieved through an alliance supporting organizational and informational framework and the benchmarking of performance with environmental, quality, and cost parameters.

Purchasing with its proximity to its suppliers needs to play a broader role in the company's environmental agenda. It needs to conduct timely supplier evaluations using criteria such as risk and environmental capability. Because of the emphasis on environmental value-added performance, companies turn to suppliers for use of their

waste treatment facilities. Companies also seek to develop suppliers who can collect, clean and reship process waste back to the company.

Finally, companies must carefully justify all environmental changes through either cost reduction or customer satisfaction issues. The focus of continuous improvement (used so effectively during TQM programme implementation) can be applied quite effectively to improving environmental efficiency and effectiveness. The cross-functional relevance of environmental supply chain management is ensured by its direct impact on the supplier selection and management processes. Change should be viewed as a competitive tool and environmental efficiency viewed as a positive catalyst for change.

SUPPLY CHAIN STRATEGIES

The growth and success of regional economic groupings like the European Union (EU), the North American Free Trade Agreement (NAFTA) and the Association of South East Asian Nations (ASEAN) as the competing economies of the present day world in replacement of nation states underscores the reality of interdependence among national economies. Economics and not politics are in the driver's seat in the inexorable march of political economies of the world towards partnerships transcending national boundaries.

At the functional levels, the corporate world has now discovered the power of teamwork. Restructuring of organisations and businesses from the shop floor to the boardrooms on team basis, dismantling traditional intra organisational boundaries reflects this transition from the individual to the collective and complimentary efforts in enterprises as a route to growth and development in the competitive environment of today.

Under the pressure of these fundamental changes, presence of independent business as competitive units, however large and vertically integrated appears anachronistic in a globalised economy. Business have now come to realise the value of co-operation and partnership and accept the centrality of the supply function in their performance. As a consequence, they will, sooner than later, have to coalesce into supply chains to serve the customers in a world characterised by fierce competition.

The battle of the "giants" will, for sure, yield place to battle of the "joints". This being the trend of developments, need for studying the strategies for successfully forging and positioning such supply chains and then fielding them as integrated wholes, becomes self-evident.

FUNDAMENTAL NEED

Supply chains are nothing if not alliances for mutual benefit of all constituents and like all alliances, their effectiveness is predicated upon mutual trust and visibility. Every downstream link in the chain is the customer of its immediate upstream neighbour and unless they abjure adversarial commercial relationship from link to link, no chain can

endure. The mutually hostile attitudes must give way to a climate of confidence born out of shared information, complementarity of skills, experience and knowledge base applied synergistically in pursuit of customer service to provide a solid foundation for supply chains to grow and succeed.

Trust unfortunately is seldom built through legal provisions so characteristic of the traditional contractual agreements. Trust is underpinned by implicit faith and shared moral as well as ethical codes besides genuine belief in collective good through collective effort. In any case, conflict resolution through legal remedies becomes unworkable in an international environment. Trust is the bedrock of all supply chain management and thus the fundamental need for companies considering this strategy. Those who cannot be team players must not consider supply chain route. No strategy will be of any avail, if trust among constituents does not exist.

FORGING OF SUPPLY CHAINS

That businesses are, in the ultimate analysis, a stylized form of warfare, is clear from the universal acceptance of the Sun Tzu's classic work "The Art of Warfare" as essential reading in management schools the world over. His edict "If you do not seek out helpers and allies, then you will be isolated and weak" might well have been a prescription for modern business to form strategic alliances and develop powerful supply chains for survival.

Lessons from combat have a direct application in the business world and the experience of war must be drawn upon to forge winning alliances and balanced array of forces that are knit together through a unity of purpose and operate as a homogenous whole in pursuit of clear goals. The exercise of forging supply chains should therefore begin with an identification of the core competencies of the individual constituents, which are to form the backbone of the alliance; the principal purpose of the exercise being to so array those competencies that the cumulative result is a pre-eminent market presence. When prudence in procurement, excellence in manufacturing practices utilising frontier technologies, combine with effectiveness in distribution and fulfillment functions handled by people committed to delight the customer, the result cannot but be enhanced market share and growth of all partners.

As a corollary to the identification and arraying of core competencies is the determination of those functions and activities which the constituents perform sub-optimally in comparison with others both within the chain and outside and to carefully weed out or reallocate the same where the application of resources will be most profitable. This is a simple exercise in substitution, enhancing in the process, the overall performance to the point where it peaks. Benchmarking supply chain costs, link by link, element by element and activity by activity from the raw material procurement to the price finally paid by the ultimate customer against the industry best should help not only in re-allocation and refinement of functional responsibilities but also in indicating the standards to be achieved by those entrusted with the relevant functions within the developing, learning and continually fine-tuning supply chain.

At the systems level, potential effectiveness and profitability of a given supply chain can be assessed through a form of systems cost analysis where both the individual cost elements and overall costs are compared with the competitors as well as the industry leaders. The main elements would normally include the material procurement costs, manufacturing set up costs “on materials” by way of inventories carried, materials handling, packaging and unitisation as well as internal movement and external transportation, not to mention the administrative and associated overhead costs. The entire exercise must be aimed at identifying the sources of avoidable costs that need to be reduced if not eliminated altogether. Excessive material and component costs, wastages of all types, duplication of effort, redundancies, defects, re-work, large inventories and inefficient production processes with old and antiquated equipment as well as high marketing and distribution costs, high wage rates coupled with low productivity bear careful examination. Such costs need to be addressed as a supply chain strategy and cannot remain the concern of the individual constitute in whose domain these costs reside. This is where information sharing and co-operative efforts will have to be at their best for the overall health of the supply chain.

Once such costs have been addressed and reduced to the minimum possible levels consistent with the level and quality of service, which the supply chain has chosen to collectively provide, it is then necessary for it to identify its competitive advantages as a chain as apposed to competitive advantages as individual units. Naturally such advantages have to be seen in the light of those of the competing chains. It is in the exploitation of those advantages and safeguarding against identified disadvantages that the secret of success are to be found.

POSITIONING THE SUPPLY CHAINS

Positioning of the supply chain, in essence involves consideration of the following issues:

Geographical Positioning: Strategy and costs are inter-related. No amount of operational or tactical efficiency can make up for the cost ineffective strategies. If the production units and the warehousing facilities are so located in relation to the sources of raw material and the markets for finished products as to involve excessive transit times over tortuous routes and high transportation costs, it is unlikely that the emerging supply chain will ever be able to gain any significant competitive advantage in the market. In choosing supply chain partners, the locations from where they participate, would be a matter for very careful consideration.

Allocation of Production Responsibilities: In multi-location production units, with or without inter unit input/output linkages, the allocation of production responsibilities among units becomes a matter for strategic choice. The product mix of different units has to be so determined as to both maximize use of production capacities and minimise logistic costs involved in attendant inter unit moves as well as distribution and order fulfilments costs. Also relevant in this connection are the decisions relating to product

customisation in relation to market demands and prescribing the plants as well as stages at which such customisation may take place so as to retain a high degree of flexibility in resource utilisation.

Inventory Levels: Inventories have long been recognised as sources of avoidable costs but their optimisation has somehow been left to the operational and tactical levels; believing that the new found techniques applied tactically will discover both the causes and the remedies to generate competitive advantage. This is simplistic view of a very complex problem and indicates a tendency to push down responsibility. Inventory related decisions are as much strategic as geographical positioning and production responsibilities determination and errors at the level can irreversibly add major costs to the entire supply chain operations. With inventory carrying cost exceeding 20% in most cases, it is for the boardrooms to decide whether they wish to operate in high or low inventory environments. Present research suggests that a low inventory environment is more conducive to the health of the supply chain and must be chosen a matter of strategic policy.

Modes of Transportation: There exists a direct relationship between the inventory levels at various locations and the modes of transportation chosen to connect the supply chain end to end. Slower transportation modes involve large stocks at each location in comparison with faster modes, which can replenish more speedily. It becomes a matter of strategic decision making to choose inter location modes that best meet the cost effectiveness criteria consistent with the service and assurance levels desired at each location.

FIELDING SUPPLY CHAINS

In order to best utilise its strengths, every supply chain must at the very outset, carefully formulate and continually review its value proposition to reflect its understanding of the changing customer needs, the competitive environment and its own operating systems and costs in order both to serve and survive as a collective endeavour. It is prudent for the supply chains to segment the customers based on the service levels demanded and tailor their service to meet those expectations thereby optimising total operating costs. Best of structures and organisation need imaginative operational plans to fully develop their potential and apply it in pursuit of organisational goals. Perceptive military commanders know the strengths and weaknesses of their forces in relation of those of the enemy and then proceed to so manoeuvre them into positions of advantage as to strike the enemy when and where he is the weakest.

Supply chains must similarly get the maximum value out of the their areas of excellence while carefully avoiding exposing their weaknesses. “In the wise leader’s plans, considerations of advantage and of disadvantage will be blended together. If our expectations of advantage be tempered in this way, we may succeed in accomplishing the essential parts of our schemes.” If, on the other hand, in the midst of difficulties we are always ready to realise on advantage, we may extricate ourselves from misfortune

“preached Sun Tzu when it came to the tactical handling of troops. In a dynamic market situation, tactics have to be continually adapted to the advantage of the supply chain always using its inherent strength born out of the collective core competencies of the constituents. It is at the cutting edge, where the supply chain faces the ultimate customer that all the performance is to be measured and that will be positive only if the value promises are fulfilled or exceeded. Equally, it is necessary to measure performance not only of the individual links but also of the chain as a whole on systems basis. Customer feedback must be assiduously sought and valued in order to continually improve overall performance.

CUSTOMER CENTRIC MATERIALS MANAGEMENT: A CASE STUDY OF ECIL

ECLL-AN OVERVIEW

Electronics Corporation of India Ltd.(ECIL) is a Public Enterprise under the Department of Atomic Energy established with the purpose of supporting India’s Nuclear Power Programme and help the country achieve self-reliance in professional electronics. Over the years the company evolved itself into a multi-product and multi-disciplinary organisation with focus on Computers, Control Systems and Communications. In the post-liberalisation scenario, the compulsions of global competition on local soil guided its Vision, Mission and Objectives as follows:

VISION

To help the country achieve self-reliance in Strategic Electronics.

MISSION

To strengthen the status as a valued national asset in the area of Strategic Electronics meeting the requirements of Atomic Energy, Defence, Space, Civil Aviation, Security and such other sectors of strategic importance.

OBJECTIVES

- To strengthen the technology base and thereby the capability to combat technology denials
- To promote creativity and innovation and realise higher levels of operational efficiency through actionable learning
- To attain and maintain world-class competitiveness by pursuing global benchmarks

- To lay down plans and programmes for effective succession at senior management level
- To consistently ensure a customer-centric organisational culture
- To achieve steady growth in business performance and generate reasonable internal resources.

The operations are therefore focused towards meeting the requirements of Strategic Electronics in the Nuclear, Defence, Security and such other sectors of National Importance.

THE CRISIS

The post-liberalisation period of 90s was characterised by intense competition from both the MNCs and private sector. The impact of the globalisation process and the sanctions in the wake of Pokhran-11 experiments have brought the company to the brink of sickness in 1998-99. ECIL suffered a loss of Rs. 10 crore in 1997-98 and a substantial loss of Rs. 60 crore in 1998-99. The networth of the company was badly eroded and ECIL had to be reported to BIFR.

THE INITIATIVES IN MATERIALS MANAGEMENT

The company had to initiate a number of innovative measures to tide over the crisis. Material costs constituting around 55-60 % of the total cost, it became imperative for the company to introduce a host of innovative practices in the area of materials management. These initiatives need to be in consonance with the nature of global electronics business characterised by high rate of obsolescence, falling prices, high quality inputs and global sourcing. More over the business environment of ECIL is different even from other PSUs as there is no assured market, customer-driven requirements and threat of denials. It is against this background that the entire supply chain is addressed and briefed below are the salient aspects of this process.

SUPPLIER COMMUNICATION

It was ensured that the requirements of the customers of ECIL are clearly communicated to the suppliers, thus making the latter jointly responsible for ensuring customer satisfaction. This is done through constant touch with the suppliers to indicate the priorities through written and verbal means and by hosting vendor meets Vendor Development and Quality Assurance. The suppliers were continuously provided all the support during product development and engineering, prototype testing, evaluation, qualification and guidance in the implementation of ISO 9000 Quality Management system and other industry standards and practices. Emphasis was on prevention rather than detection and correction. Suppliers are encouraged to imbibe the culture of 'Ownership of Quality' as?

INVENTORY CONTROL

The scheduling of placement of orders and receipt of materials was streamlined to ensure efficient inventory management covering such requirements as timely availability of material, minimisation of waste and surplus due to obsolescence, etc.

NEGOTIATIONS AND PAYMENTS

Suppliers are involved right from the tender stage to offer the most competitive terms to the customers in terms of quality, cost and delivery. Mutually beneficial payment terms are negotiated and strictly adhered to.

IMPLEMENTATION OF IT

All the processes involved in Integrated Materials Management are fully computerised for speedy disposal of material requisitions, resulting in substantial reductions in lead times across various operations.

THE RESULTS

These simple initiatives were implemented effectively resulting in incredible results that culminated in the historic turnaround of the company that brought wide recognition and national awards like SCOPE award for Outstanding Performance and Contribution to Public Sector Management. Specifically the following achievements were realised.

BETWEEN 1998-99 AND 2002-03

- Turnover shot up from Rs. 250 crore to Rs. 1000 crore
- Profitability reversed from a loss of Rs. 60 crore to a profit of 130 crore
- Networth increased from Rs. 7 crore to Rs. 309 crore
- Loans Outstanding and Statutory Outstandings became nil
- Inventory levels and Sundry Debtors demonstrated efficient Working Capital Management
- Substantial cost Reductions were realised through negotiations and improved quality levels across various processes
- Procurement lead-times reduced substantially from about 60 days to 30 days
- Customer Satisfaction Index improved phenomenally from 66% to 95%
- MoU rating of 'Excellent' has been achieved and is maintained and
- Image of the company enhanced remarkably.

FUTURE PLANS

The multi-disciplinary competencies and capabilities of ECIL coupled with the Strategic Sectors it has chosen, is bound to give the company a competitive advantage

to succeed in the national market but explore avenues in the international market. Many of its products have been identified for export promotion and the brilliant performance of the Electronic Voting Machines in the recent general elections established its export worthiness with many countries evincing interest in the product. While all the processes under materials management are fully computerised, plans are underway to instal a fully IT enabled Supply Chain Management including e-procurement. The company is fully aware of the competitive global environment surrounding it in the high technology electronics and is focusing continuously on enhancing its technology base and enriching its skill base to strengthen its status as a valued national asset.

SUPPLY CHAIN DEMYSTIFIED

Supply Chain Management or SCM as it's popularly known seems to have become the all important focal point for business success. It has gained enormous importance and continues to be a buzzword in the Industry as well as in the IT and the student community. Today, people are increasingly realizing that in the intensely competitive world, one has to keep the chain lean and responsive. Is it just something made popular by the consultants and a few people or is it really significant for the industry? And, what really does supply chain mean and why has it been in the limelight? Is it something new that was non-existent earlier or is it just another buzzword which will fade away giving way to something new. In my view, supply chain as a concept is definitely here to stay and it implies a new way of looking at things by questioning the existing boundaries.

WHAT EXACTLY IS SUPPLY CHAIN?

There are various ways to look at supply chain. One can say that it starts from the raw material vendor and ends with the consumer; thus, it includes purchasing, marketing and even consumer buying the product. Therefore all the processes involved in the entire spectrum from demand generation to demand satisfaction can be called as supply chain management. Today, in some advanced companies supply chain extend right from the vendor procuring his raw material to the point of sale where the last sale of the product takes place. This implies that there is transparency and information flow in the entire chain resulting in appropriate action at each point. This action by each entity contributes to the smooth functioning of supply chain. In some companies supply chain could only be internal *i.e.*, across the manufacturing facilities to company owned depots. One can always choose a part of the supply chain that is most relevant and focus resources to achieve increased productivity. The benefits of supply chain could vary from organization to organization. Key benefits could be achieving higher channel partner satisfaction by reducing inventory, reducing inventory levels, reducing stock out situations by balancing inventory, reducing transportation, storage and damage/expiry costs, meeting delivery lead times and achieving greater accuracy in the planning process through automation.

WHY IS SUPPLY CHAIN IN THE LIMELIGHT?

This recent popularity of supply chain is a simple result of advances in technology and competition. In today's world barriers across countries have become almost invisible as far as trade and technology are concerned. Today, the industry is facing immense competition and also has access to the advanced world-class technology. This coupled with the sky-high expectations of the company stakeholders' calls for a new way of doing old things. The emergence of the concept of supply chain is a result of this external pressure. Earlier one would look at supply chain in parts, *i.e.*, marketing, manufacturing, logistics, purchase, distribution, transportation and so on. Each function had a different set of people working with a particular objective in their minds. This objective was not necessarily common across people and led to very little interaction across functions. People were encouraged to be experts in their own functions and this helped in achieving functional excellence. You could hear somebody say that company X excels in manufacturing whereas company Y excels in distribution or purchasing. Being excellent in even one of these functions could give a company an edge over competition. However, a further improvement in any of these functions will no longer yield substantial business benefits.

Therefore, a time has come to integrate all functions with a view to achieve overall excellence. Probably this is why supply chain has suddenly become an all-important arena where all the action is. Simultaneously, advances in technology have fuelled the growth of supply chain solutions. Thus, one can look at overall supply chain by eliminating/simplifying many mundane activities. In fact, if you observe the emergence of supply chain... it has come only after ERP has become a 'must have' from a 'can have' for the corporate world.

IS A SUPPLY CHAIN FOCUS POSSIBLE WITHOUT TECHNOLOGY

The data that was available earlier say only say once in a month is available on-line today. And not only that, all those complex calculations done using paper and pencil are just a matter of seconds. One no longer needs to spend energy on getting and correcting data but correct information is available on-line as per requirements. Also, by just one click one can do calculations that would probably have taken say a day earlier. The results could also be far superior due to the sheer number of permutations that a human mind may not be able to comprehend and also eliminates possibility of human error at various instances. Also, the system facilitates usage of advanced planning or optimization algorithms to achieve a very high quality output.

However, this can be made possible only if one begins to appreciate the huge implications of a supply chain project and invites the change by working towards building a system that fulfils the organizational needs. Even though it may sound cliched, the concept of supply chain comes as a tool or rather a weapon to fight out all the inefficiencies and give real business benefits in measurable numbers. One can rise

above the mundane activities as they can be easily left to the system and give a higher share of mind to more productive activities that include modelling and using the system intelligently to add value to the company.

WHAT DOES A SUPPLY CHAIN JOURNEY IMPLY?

The above holistic perspective is extremely essential when any organization wants to move towards a supply chain approach. Supply chain is after all a way of thinking.... Just doing the same old things in a new way.

However, having said the above, I would also like to add that supply chain is not as rosy or as simple as it appears. Supply chain implies everything right from re-looking at the organization structure to changing people's mindsets across levels. It could also need an advanced technology like an APO or an i2 but this in itself can help an organization only a small way. It also includes empowering people with new skills on aH fronts, technological inputs, analytical inputs, etc. Various training sessions could be required along with change management inputs as this not only calls for learning, mastering and LIKING a new complex system but it also means changing attitudes for which beliefs need to be changed! Over the years, people have formed and arrived at a set way of working which is considered to be the best. Thus, working with a new system which is complex initially and which calls for new learning can face a lot of resistance. A lot of drive and commitment from the top management will be required in the initial phase if the entire organization has to succeed on this supply chain journey. And believe me, it will not be easy at all... it will be really tough as one not only has to cope with the day to day business performance pressures but also the pressures of the new way of working. Thus, till everyone becomes comfortable with this new way of working... the support will be required across levels. Along with this, it could also call for a change in the Key Performance Indicators across the organization.

A lot of questions may be raised about whether the system was required at all and also about the credibility of the system. These questions/doubts need to be addressed in order to ensure that a lot of positive energy is generated. It could also lead to some undercurrents that need to be managed tactfully. If issues are not managed well, this could shake the confidence of the entire organization in the system benefits and derail the supply chain journey.

In order to avoid any jolts midway, a proper homework is absolutely essential on the functional as well as the people front. Once the organization has very thoroughly and logically decided on commencing a supply chain journey and achieved a true consensus within the organization, the likelihood of any stumbling blocks on the way is highly unlikely. The benefits as well as the steps should be clearly shared and agreed among all involved or impacted by the journey. This will give a strong foundation to the project that will go a long way in ensuring success. Needless to say, along with good homework, constant moral boosting and appropriate support at every step will be required.

There are various questions that need to be answered before one starts on a supply chain journey. What is the benefit that the organization is seeking from improvement of the supply chain? Has it been sufficiently debated and is it unanimously agreed among all functions? Is it clearly defined and measurable? Another very important question- is the supply chain strategy clear and is it aligned with corporate strategy? Will an improvement in supply chain really help achieve the targeted sales or profits? Unless one is very clear about all this, a supply chain journey either may not go anywhere as it will not get the desired focus and also it could come to a halt.

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If all the above homework has been done in the true sense, an organization can start planning a supply chain journey with a detailed project plan. It will definitely result in new learning at every stage that the organization will benefit from.

PROSPECTS OF SCM

Change is rapid on the Indian industrial scene. Competition is becoming acute and customers are becoming choosy. The emphasis is on building "Value to the customer". Supply Chain concepts are being practised with increasing frequency by companies, not only in FMCG and Automobile sectors but also in other sectors like steel, cement, etc. There are not many individuals who have direct experience but there are lot of consultancy organisations like Anderson Consulting, Price Waterhouse Coopers, McKinsey, KSA Technopak, etc. who have accumulated a wealth of experience in conceptualising, designing and implementing Supply Chain Systems in different companies in India and around the world. This paper is an attempt to understand the contribution of Supply Chain Management consultants to the Indian industry.

At a first glance of the consultants, client would appear to be faced with a diffuse spectrum of suppliers. Almost every organisation nowadays enlists in some way the services of a Supply Chain Consultant, *e.g.*, management consultants, third party logistics suppliers, supply chain services, software houses, engineering companies, general organisations, university lecturers, etc. Occasionally one gets the impression that 'consultation' is being confused with the term "canvassing". It would be, therefore, more desirable to encourage the use of the term "Supply Chain Consultation"!

The purpose of the following paper, however, is not to lead an academic debate about concept instead to clarify the above mentioned term using personal practical experience as a basis.

WHO EMPLOYS SUPPLY CHAIN CONSULTANTS?

The layperson may be surprised at the following whilst the professional will nod approvingly: it is mostly companies which already have effective supply chain and qualified supply chain experts who enlist the services of supply chain consultants. In other words, companies with modest levels of logistics often turn away any form of external consultation and believe that they will manage better without it. A market survey conducted by a Professor of Technology Management at the Technical College of Darmstadt, Germany confirms the general trend that organisation with progressive attitudes and, more importantly, with good supply chain regularly employs supply chain consultants.

As in every case there are always exceptions. In this instance, it is mostly the larger, more specialised organisations that seldom call upon supply chain advisors. In case of investments, they frequently operate directly with general managers. Consultants function frequently at such times as sparring partners for the company's own supply chain management. Indeed, the diversity of roles of consultant is immense, *e.g.*, they act as rationalizer, investment consultant, technical developer, project coordinator, project manager, scapegoat, generator of ideas, etc. The list is endless. Equally difficult to define are the market limits.

WHO IS A SUPPLY CHAIN CONSULTANT?

Supply Chain Consultants act as competent business partners to both top management and heads of supply chain. They can contribute an external viewpoint to organisational development. Just like other members of this discipline, the supply chain consultant considers, approves or rejects structures, free from hierarchical and historically evolved restrictions. He develops new structures, puts them forward to the company and supports companies in their move to introduce new systems alongside their daily business. What is the difference between this person and a management consultant? Merely the word "supply chain". Yet this in itself is important, because, for the supply chain consultant, technique, investment, and realisation are vital; that is to say areas in which the management consultant is not at home. This is the time for engineers, technological institutes and software developers to perform. Together they cultivate their vast field of technical realisation. And what of the supply chain consultants? They also work on the same task and combine their experience of both organisational development and technique with their special knowledge of the industry and their practical competence.

To put it in a different way: with the service that they offer, supply chain consultants stand between the big management consultants and the organisational consultants, and between institutes affiliated to universities and manufacturers and general managers.

- Their niche lies between organisational strategy and supply chain strategy.
- They are not research oriented, yet they are innovative.
- They are able to realise ideas but are independent of both manufacturers and product.

In short, supply chain consultants are placed beside and behind their customer in the function of temporary supply chain managers. They are trustees of the company's interests, yet they are at the same time more independent than company members are. Their relationship to their client is one of absolute loyalty. One can find supply chain consultants in a wide variety of functions. Some of them are illustrated below.

CONSULTANTS ARE COMPETITORS, SPARING PARTNERS AND SPARE CAPACITY FOR COMPANY MANAGEMENT

Competition stimulates business and the quality of a solution. Large organisations are allowing us more and more opportunities to compete against their own company management. This type of planned competition with their own people is often an important instrument in testing and improving the capabilities of their staff. In case of large investments of strategic decisions, planned competition with the outsider, *i.e.*, the consultant, brings new ideas to the situation, gained through his broad experience of other industries. Finally, consultants act as a pool to assist at times of peak demand. They can be used as a free reserve that does not cause any fixed costs.

THE CONSULTANT SUPPORTS THE POSITIONING OF SUPPLY CHAIN IN THE ORGANISATION

Is supply chain one of the central areas of competence of a company? Would one contract out supply chain management? What does supply chain contribute to the company's profit? Many companies relieve themselves of burden of supply chain management by simply contracting out. With regard to central areas of competence, this is often very sensible. Yet this decision should only be taken if it is workable in the long-term. Supply chain consultants can help, acting here as independent external business partners to answer the question of central competence of supply chain management. For example, they can calculate exactly the contribution of supply chain towards the profit and the cost of supply chain itself, *i.e.*, calculate carefully, make or buy alternatives.

CONSULTANTS INTEGRATE SUPPLY CHAIN AS A BUSINESS MANAGEMENT SYSTEM

Material flow in the warehouse cannot be regarded as an isolated process. Both the systems theorist and also the supply chain practitioner know how much one process depends on the other. For example, the warehouse dimensions depend on the structure of procurement and on stock; the distribution routes depend upon stock location; delivery service and commissioning depend on Last-Order times and agreed delivery dates. Often, the wishes of the sales strategist and the customer coincide, yet these are not always cost effective. The level of service that is regarded by supply chain personnel as

possible is often oriented towards daily business and is dominated by considerations of cost. Supply chain consultants can help sales and logistics personnel to reach a compromise.

Supply Chain Consultants help to Formulate Strategic Goals in Logistics and to Draw up Investment and Rationalisation Programmes.

Those organisations seeking orientation, which want to invest in the future of their supply chain, must clarify first of all their strategic goals. Ordering methods, distribution point, and delivery service requirements must be defined in accord with the aims of the company.

Supply chain consultants are in this instance helpful business partners because they know both the strategic and also the technical side of a distribution system, that is the technical aspect involving knowledge of investment. They can calculate exactly at what cost and with how many employees an imaginary warehouse “X” can be operated at location “Y”.

Hypothetical examples that are unrealistic are virtually impossible with an experienced consultant. As soon as an investment programme is strategically fixed, the next step is to work out the technical details. Whoever desires now well-established standards will work in the main directly together with a manufacturer or general manager. On the other hand, whoever is seeking competitive advantage within a strongly contested market, for example by means of exceptional commitment to schedule and to low rate of failure, whoever finally, is aiming at exceptional growth, then let tailor-made supply chains be recommended to him over and above standards.

SUPPLY CHAIN CONSULTANTS ARE INDEPENDENT AND UNBIASED TOWARDS PRODUCT AND MANUFACTURER

Manufacturers, too frequently offer planning and development resources. It is advisable to use them when optimising subsystems for which the manufacturer is a specialist. An investor, who is certain that, for example, only a new shelf system is required, or the connection of two warehouse areas by conveyors, will be in safe hands with the producer of material flow components. If he wants to know, however, whether he needs a warehouse or rather a service provider or if a new system, conveyor belt or shelf system is at all necessary, then firstly, he should think and plan independently of both the product and manufacturer. One makes a better choice of system.

THE EVOLVING STRUCTURE OF SUPPLY CHAINS

The participants in a supply chain are continuously making decisions that affect how they manage the five supply chain drivers. Each organization tries to maximize its performance in dealing with these drivers through a combination of outsourcing, partnering, and in-house expertise. In the fast-moving markets of our present economy

a company usually will focus on what it considers to be its core competencies in supply chain management and outsource the rest. This was not always the case though. In the slower moving mass markets of the industrial age it was common for successful companies to attempt to own much of their supply chain. That was known as vertical integration. The aim of vertical integration was to gain maximum efficiency through economies of scale.

In the first half of the 1900s Ford Motor Company owned much of what it needed to feed its car factories. It owned and operated iron mines that extracted iron ore, steel mills that turned the ore into steel products, plants that made component car parts, and assembly plants that turned out finished cars. In addition, they owned farms where they grew flax to make into linen car tops and forests that they logged and sawmills where they cut the timber into lumber for making wooden car parts. Ford's famous River Rouge Plant was a monument to vertical integration—iron ore went in at one end and cars came out at the other end. Henry Ford in his 1926 autobiography, *Today and Tomorrow*, boasted that his company could take in iron ore from the mine and put out a car 81 hours later.

This was a profitable way of doing business in the more predictable, one-size-fits-all industrial economy that existed in the early 1900s. Ford and other businesses churned out mass amounts of basic products. But as the markets grew and customers became more particular about the kind of products they wanted, this model began to break down. It could not be responsive enough or produce the variety of products that were being demanded. For instance, when Henry Ford was asked about the number of different colours a customer could request, he said, "they can have any colour they want as long as it's black." In the 1920s Ford's market share was over 50 percent but by the 1940s it had fallen to below 20 percent. Focusing on efficiency at the expense of being responsive to customer desires was no longer a successful business model.

Globalization, highly competitive markets, and the rapid pace of technological change are now driving the development of supply chains where multiple companies work together, each company focusing on the activities that it does best. Mining companies focus on mining, timber companies focus on logging and making lumber, and manufacturing companies focus on different types of manufacturing from making component parts to doing final assembly. This way people in each company can keep up with rapid rates of change and keep learning the new skills needed to compete in their particular business.

Where companies once routinely ran their own warehouses or operated their own fleet of trucks, they now have to consider whether those operations are really a core competency or whether it is more cost effective to outsource those operations to other companies that make logistics the centre of their business. To achieve high levels of operating efficiency and to keep up with continuing changes in technology, companies need to focus on their core competencies. It requires this kind of focus to stay competitive.

Instead of vertical integration, companies now practice “virtual integration.” Companies find other companies who they can work with to perform the activities called for in their supply chains. How a company defines its core competencies and how it positions itself in the supply chains it serves is one of the most important decisions it can make.

PARTICIPANTS IN THE SUPPLY CHAIN

In its simplest form, a supply chain is composed of a company and the suppliers and customers of that company. This is the basic group of participants that creates a simple supply chain. Extended supply chains contain three additional types of participants. First there is the supplier’s supplier or the ultimate supplier at the beginning of an extended supply chain. Then there is the customer’s customer or ultimate customer at the end of an extended supply chain. Finally there is a whole category of companies who are service providers to other companies in the supply chain. These are companies who supply services in logistics, finance, marketing, and information technology.

In any given supply chain there is some combination of companies who perform different functions. There are companies that are producers, distributors or wholesalers, retailers, and companies or individuals who are the customers, the final consumers of a product. Supporting these companies there will be other companies that are service providers that provide a range of needed services.

PRODUCERS

Producers or manufacturers are organizations that make a product. This includes companies that are producers of raw materials and companies that are producers of finished goods. Producers of raw materials are organizations that mine for minerals, drill for oil and gas, and cut timber. It also includes organizations that farm the land, raise animals, or catch seafood. Producers of finished goods use the raw materials and subassemblies made by other producers to create their products.

Producers can create products that are intangible items such as music, entertainment, software, or designs. A product can also be a service such as mowing a lawn, cleaning an office, performing surgery, or teaching a skill. In many instances the producers of tangible, industrial products are moving to areas of the world where labour is less costly. Producers in the developed world of North America, Europe, and parts of Asia are increasingly producers of intangible items and services.

DISTRIBUTORS

Distributors are companies that take inventory in bulk from producers and deliver a bundle of related product lines to customers. Distributors are also known as wholesalers. They typically sell to other businesses and they sell products in larger quantities than an

individual consumer would usually buy. Distributors buffer the producers from fluctuations in product demand by stocking inventory and doing much of the sales work to find and service customers. For the customer, distributors fulfil the “Time and Place” function—they deliver products when and where the customer wants them.

A distributor is typically an organization that takes ownership of significant inventories of products that they buy from producers and sell to consumers. In addition to product promotion and sales, other functions the distributor performs are inventory management, warehouse operations, and product transportation as well as customer support and post-sales service. A distributor can also be an organization that only brokers a product between the producer and the customer and never takes ownership of that product. This kind of distributor performs mainly the functions of product promotion and sales. In both these cases, as the needs of customers evolve and the range of available products changes, the distributor is the agent that continually tracks customer needs and matches them with products available.

RETAILERS

Retailers stock inventory and sell in smaller quantities to the general public. This organization also closely tracks the preferences and demands of the customers that it sells to. It advertises to its customers and often uses some combination of price, product selection, service, and convenience as the primary draw to attract customers for the products it sells. Discount department stores attract customers using price and wide product selection. Upscale speciality stores offer a unique line of products and high levels of service. Fast food restaurants use convenience and low prices as their draw.

CUSTOMERS

Customers or consumers are any organization that purchases and uses a product. A customer organization may purchase a product in order to incorporate it into another product that they in turn sell to other customers. Or a customer may be the final end user of a product who buys the product in order to consume it.

SERVICE PROVIDERS

These are organizations that provide services to producers, distributors, retailers, and customers. Service providers have developed special expertise and skills that focus on a particular activity needed by a supply chain. Because of this, they are able to perform these services more effectively and at a better price than producers, distributors, retailers, or consumers could do on their own.

Some common service providers in any supply chain are providers of transportation services and warehousing services. These are trucking companies and public warehouse companies and they are known as logistics providers. Financial service providers deliver services such as making loans, doing credit analysis, and collecting on past due invoices.

These are banks, credit rating companies, and collection agencies. Some service providers deliver market research and advertising, while others provide product design, engineering services, legal services, and management advice. Still other service providers offer information technology and data collection services.

All these service providers are integrated to a greater or lesser degree into the ongoing operations of the producers, distributors, retailers, and consumers in the supply chain. Supply chains are composed of repeating sets of participants that fall into one or more of these categories. Over time the needs of the supply chain as a whole remain fairly stable. What changes is the mix of participants in the supply chain and the roles that each participant plays. In some supply chains, there are few service providers because the other participants perform these services on their own. In other supply chains very efficient providers of specialized services have evolved and the other participants outsource work to these service providers instead of doing it themselves.

ALIGNING THE SUPPLY CHAIN WITH BUSINESS STRATEGY

A company's supply chain is an integral part of its approach to the markets it serves. The supply chain needs to respond to market requirements and do so in a way that supports the company's business strategy. The business strategy a company employs starts with the needs of the customers that the company serves or will serve. Depending on the needs of its customers, a company's supply chain must deliver the appropriate mix of responsiveness and efficiency. A company whose supply chain allows it to more efficiently meet the needs of its customers will gain market share at the expense of other companies in that market and also will be more profitable.

For example, let's consider two companies and the needs that their supply chains must respond to. The two companies are 7-Eleven and Sam's Club, which is a part of Wal-Mart. The customers who shop at convenience stores like 7-Eleven have a different set of needs and preferences from those who shop at a discount warehouse like Sam's Club.

The 7-Eleven customer is looking for convenience and not the lowest price. That customer is often in a hurry and prefers that the store be close by and have enough variety of products so that they can pick up small amounts of common household or food items that they need immediately. Sam's Club customers are looking for the lowest price. They are not in a hurry and are willing to drive some distance and buy large quantities of limited numbers of items in order to get the lowest price possible.

Clearly the supply chain for 7-Eleven needs to emphasize responsiveness. That group of customers expects convenience and will pay for it. On the other hand, the Sam's Club supply chain needs to focus tightly on efficiency. The Sam's Club customer is very price conscious and the supply chain needs to find every opportunity to reduce costs so that these savings can be passed on to the customers. Both of these companies'

supply chains are well aligned with their business strategies and because of this they are each successful in their markets.

There are three steps to use in aligning your supply chain with your business strategy. The first step is to understand the markets that your company serves. The second step is to define the strengths or core competencies of your company and the role the company can or could play in serving its markets. The last step is to develop the needed supply chain capabilities to support the roles your company has chosen.

UNDERSTAND THE MARKETS YOUR COMPANY SERVES

Begin by asking questions about your customers. What kind of customer does your company serve? What kind of customer does your customer sell to? What kind of supply chain is your company a part of? The answers to these questions will tell you what supply chains your company serves and whether your supply chain needs to emphasize responsiveness or efficiency. Chopra and Meindl have defined the following attributes that help to clarify requirements for the customers you serve. These attributes are:

- *The quantity of the product needed in each lot*—Do your customers want small amounts of products or will they buy large quantities? A customer at a convenience store or a drug store buys in small quantities. A customer of a discount warehouse club, such as Sam's Club, buys in large quantities.
- *The response time that customers are willing to tolerate*—Do your customers buy on short notice and expect quick service or is a longer lead time acceptable? Customers of a fast food restaurant certainly buy on short notice and expect quick service. Customers buying custom machinery would plan the purchase in advance and expect some lead time before the product could be delivered.
- *The variety of products needed*—Are customers looking for a narrow and well-defined bundle of products or are they looking for a wide selection of different kinds of products? Customers of a fashion boutique expect a narrowly defined group of products. Customers of a "big box" discount store like Wal-Mart expect a wide variety of products to be available.
- *The service level required*—Do customers expect all products to be available for immediate delivery or will they accept partial deliveries of products and longer lead times? Customers of a music store expect to get the CD they are looking for immediately or they will go elsewhere. Customers who order a custom-built new machine tool expect to wait a while before delivery.
- *The price of the product*—How much are customers willing to pay? Some customers will pay more for convenience or high levels of service and other customers look to buy based on the lowest price they can get.

- *The desired rate of innovation in the product*—How fast are new products introduced and how long before existing products become obsolete? In products such as electronics and computers, customers expect a high rate of innovation. In other products, such as house paint, customers do not desire such a high rate of innovation.

DEFINE CORE COMPETENCIES OF YOUR COMPANY

The next step is to define the role that your company plays or wants to play in these supply chains. What kind of supply chain participant is your company? Is your company a producer, a distributor, a retailer, or a service provider? What does your company do to enable the supply chains that it is part of? What are the core competencies of your company? How does your company make money? The answers to these questions tell you what roles in a supply chain will be the best fit for your company.

Be aware that your company can serve multiple markets and participate in multiple supply chains. A company like W.W. Grainger serves several different markets. It sells maintenance, repair, and operating (MRO) supplies to large national account customers such as Ford and Boeing and it also sells these supplies to small businesses and building contractors. These two different markets have different requirements as measured by the above customer attributes.

When you are serving multiple market segments, your company will need to look for ways to leverage its core competencies. Parts of these supply chains may be unique to the market segment they serve while other parts can be combined to achieve economies of scale. For example, if manufacturing is a core competency for a company, it can build a range of different products in common production facilities. Then different inventory and transportation options can be used to deliver the products to customers in different market segments.

DEVELOP NEEDED SUPPLY CHAIN CAPABILITIES

Once you know what kind of markets your company serves and the role your company does or will play in the supply chains of these markets, then you can take this last step, which is to develop the supply chain capabilities needed to support the roles your company plays. This development is guided by the decisions made about the five supply chain drivers. Each of these drivers can be developed and managed to emphasize responsiveness or efficiency depending on the business requirements.

1. *Production:* This driver can be made very responsive by building factories that have a lot of excess capacity and that use flexible manufacturing techniques to produce a wide range of items. To be even more responsive, a company could do their production in many smaller plants that are close to

major groups of customers so that delivery times would be shorter. If efficiency is desirable, then a company can build factories with very little excess capacity and have the factories optimized for producing a limited range of items. Further efficiency could be gained by centralizing production in large central plants to get better economies of scale.

2. *Inventory:* Responsiveness here can be had by stocking high levels of inventory for a wide range of products. Additional responsiveness can be gained by stocking products at many locations so as to have the inventory close to customers and available to them immediately. Efficiency in inventory management would call for reducing inventory levels of all items and especially of items that do not sell as frequently. Also, economies of scale and cost savings could be gotten by stocking inventory in only a few central locations.
3. *Location:* A location approach that emphasizes responsiveness would be one where a company opens up many locations to be physically close to its customer base. For example, McDonald's has used location to be very responsive to its customers by opening up lots of stores in its high volume markets. Efficiency can be achieved by operating from only a few locations and centralizing activities in common locations. An example of this is the way Dell serves large geographical markets from only a few central locations that perform a wide range of activities.
4. *Transportation:* Responsiveness can be achieved by a transportation mode that is fast and flexible. Many companies that sell products through catalogues or over the Internet are able to provide high levels of responsiveness by using transportation to deliver their products, often within 24 hours. FedEx and UPS are two companies who can provide very responsive transportation services. Efficiency can be emphasized by transporting products in larger batches and doing it less often. The use of transportation modes such as ship, rail, and pipelines can be very efficient. Transportation can be made more efficient if it is originated out of a central hub facility instead of from many branch locations.
5. *Information:* The power of this driver grows stronger each year as the technology for collecting and sharing information becomes more widespread, easier to use, and less expensive. Information, much like money, is a very useful commodity because it can be applied directly to enhance the performance of the other four supply chain drivers. High levels of responsiveness can be achieved when companies collect and share accurate and timely data generated by the operations of the other four drivers. The supply chains that serve the electronics markets are some of the most responsive in the world. Companies in these supply chains from manufacturers, to distributors, to the big retail stores collect and share data about customer demand, production schedules, and inventory levels.

Where efficiency is more the focus, less information about fewer activities can be collected. Companies may also elect to share less information among themselves so as not to risk having that information used against them. Please note, however, that these information efficiencies are only efficiencies in the short term and they become less efficient over time because the cost of information continues to drop and the cost of the other four drivers usually continues to rise. Over the longer term, those companies and supply chains that learn how to maximize the use of information to get optimal performance from the other drivers will gain the most market share and be the most profitable.

E-Business and Supply Chain

The widespread availability and use of the Internet offers companies opportunities that did not exist before. These opportunities are made possible because it is now so easy and relatively inexpensive for companies to connect to the Internet. Once connected, companies can send data to and receive data from other companies that they do business with regardless of the particular computers or software that individual companies may be using to run their internal operations. Based on this data sharing, opportunities exist to achieve tremendous supply chain efficiencies and significant increases in customer service and responsiveness. These are the results of better supply chain integration.

E-business encompasses the evolving set of principles and practices that companies are employing to gain the benefits inherent in better we go again to build up the scale. We have come to believe that if a customer feels strongly enough that they will pay for a new feature, only then is it worth building that feature. Purely speculative development is a pernicious habit. Frankly, we don't waste a lot of time trying to revolutionize an industry. We work with individual customers to give them what they need to achieve their individual ROIs. The art is combining these solutions into the same code base."

Chris summed up what the last several years have taught them, "We don't want to rebuild systems that are already out there. We are here to connect these different systems and provide the data across the extended supply chain. Tibersoft does this by building private, point-to-point connections between companies using common data protocols. And those connections need to remain just that—private. Many dot com companies got themselves in trouble taking the data generated between their customers and using it to their economic benefit. "Enable, don't participate: that rule we don't break."

In a white paper titled “E-Business and Supply Chain Integration” published by the Stanford Global Supply Chain Management Forum professors Lee and Whang lay out four key dimensions of the impact of e-business on supply chain integration. These four dimensions create a sequence of greater and greater integration and coordination among supply chain participants. This sequence culminates in the creation of whole new ways to conduct business. The four dimensions are:

1. *Information Integration:* Is the ability to share relevant information among companies in a supply chain. This includes data such as: sales history and demand forecasts; inventory status; production schedules; production capacities; sales promotions; and transportation schedules. This data should be available to the people who need it in a real-time, on-line format via the Internet or private network.
2. *Planning Synchronization:* Refers to the joint participation of companies in a supply chain in the demand forecasting and inventory replenishment scheduling. It also includes the collaborative design, development, and bringing to market of new products.
3. *Workflow Coordination:* Is the next step after planning synchronization. It is the streamlining and automation of ongoing business activities across companies in a given supply chain. This includes activities such as purchasing and product design.
4. *New Business Models:* Can emerge as a result of supply chain integration made possible by the Internet. Roles and responsibilities of companies in a supply chain can be redesigned so that each company can truly concentrate on the activities that are its core competencies. Non-core activities can be outsourced to other companies. New capabilities and efficiencies will become possible.

EXPERIENCE OF THE LAST SEVERAL YEARS

Several waves of e-business development have occurred since the late 1990s. The first wave of developments by entrepreneurial start-up companies typically focused on using Internet-based exchanges to improve purchasing efficiencies and drive down the cost of products through the use of online bid/auction techniques. Then industry news, statistics, and reference material were added to many of these exchanges and they were referred to as industry portals. Most of these developments have not lived up to expectations.

The next wave of developments continued to focus on purchasing efficiencies but this time the systems were developed by the purchasing companies themselves instead of by third party entrepreneurs. Often these developments have taken the form of a consortium of big companies in an industry banding together to finance the start up of an Internet-based purchasing platform that will support the whole range of their purchasing functions. Examples of this are Covisint in the automobile industry,

ForestExpress in the forest products industry, and Aero Exchange International in the airline industry. These developments are yielding efficiencies in procurement operations.

The most recent wave of developments are now looking at how to achieve efficiencies in a broad range of supply chain operations such as product design, demand forecasting, inventory management, and customer service. The key to realizing these efficiencies is information sharing between companies in a supply chain. Many current e-business developments are working on methods and standards to share information across multiple companies. Information sharing is the foundation and then cross company coordination is what will deliver the desired efficiencies. Once information integration is in place, the next three dimensions: planning synchronization; workflow coordination; and new business models can evolve much more rapidly. E-business development has only just begun.

A USEFUL MODEL OF MARKETS AND THEIR SUPPLY CHAINS

A supply chain exists to support the market that it serves. To identify the performance that a supply chain should deliver, we need to evaluate the market being served. In support of this analysis we will employ a simple model. The model allows us to categorize a market and identify the requirements and opportunities that each kind of market presents to its supply chains. Reality is, of course, more subtle and more complex than any model can represent but this model can point you in the right direction and guide you through an investigation of the markets your company serves.

Let us start by defining a market using its two most basic components— supply and demand. A market is characterized by its combination of supply and demand. This model defines four basic kinds of markets, or market quadrants. In the first quadrant is a market where both supply and demand for its products are low and unpredictable. Let's call this a *developing* market. In the second quadrant is a market where supply is low and demand is high. This is a *growth* market. The third quadrant contains a market where both supply and demand are high. There is a lot of predictability in this market so call this a *steady* market. In the fourth quadrant, this kind of market supply is higher than demand. This is a *mature* market.

In a developing market, both supply and demand are low and also uncertain. These are usually new markets that are just emerging. These markets are created by new technology becoming available or by social and economic trends that cause a group of customers to perceive some new set of needs. Opportunities in a developing market are in the areas of partnering with other players in the supply chain to gather intelligence about what the market wants. Cost of sales is high in this market and inventories are low.

Growth markets are markets where demand is higher than supply and so supply is often uncertain. If a developing market solidifies and builds up momentum, it can

suddenly take off and for a time there is a surge in demand that suppliers cannot keep up with. Opportunities in a growth market are in providing a high level of customer service as measured by order fill rates and on-time deliveries. Customers in a market like this value a reliable source of supply and will pay premium prices for reliability. Cost of sales should be low since customers are easy to find and inventories can be higher because they are increasing in value.

In a steady market both supply and demand are high and thus relatively predictable. This is an established market where market forces have been at work for a while and have pretty well balanced supply and demand. Opportunities here lie in fine tuning and optimizing internal company operations. Companies should focus on minimizing inventory and cost of sales while maintaining high levels of customer service.

In a mature market, supply has overtaken demand and excess supply capacity exists. Demand is reasonably stable or slowly falling but because of the fierce competition due to oversupply, demand seems uncertain from the point of view of any one supplier in this market. Opportunities in this market are in the area of flexibility as measured by an ability to respond quickly to changes in product demand while maintaining high levels of customer service.

Customers in a market like this value the convenience of “one stop shopping” where they can purchase a wide variety of related products at low prices. Inventories should be minimized and the cost of sales are somewhat higher due to the expense of attracting customers in a crowded market.

MARKET PERFORMANCE CATEGORIES

Markets in each quadrant have their own mix of opportunities for the supply chains that support them. A different mix of performance characteristics is required of companies in the supply chains of each kind of market. In order to thrive, the companies in a supply chain must be able to work together to exploit the opportunities available in their markets. The highest profits go to the companies that can successfully respond to them in more precise terms so that they can be measured objectively. We will use four measurement categories:

CUSTOMER SERVICE

Customer service measures the ability of the supply chain to meet the expectations of its customers. Depending on the type of market being served, the customers in that market will have different expectations for customer service. Customers in some markets both expect and will pay for high levels of product availability and quick delivery of small purchase quantities. Customers in other markets will accept longer waits for products and will purchase in large quantities. Whatever the market being served, the supply chain must meet the customer service expectations of the people in that market.

INTERNAL EFFICIENCY

Internal efficiency refers to the ability of a company or a supply chain to operate in such a way as to generate an appropriate level of profitability. As with customer service, market conditions vary and what is an appropriate level of profit varies from one market to another. In a risky developing market the profit margins need to be higher in order to justify the investment of time and money. In a mature market where there is little uncertainty or risk, profit margins can be somewhat lower. These markets offer the opportunity to do large volumes of business and to make up in gross profit what is given up in gross margin.

DEMAND FLEXIBILITY

This category measures the ability to respond to uncertainty in levels of product demand. It shows how much of an increase over current levels of demand can be handled by a company or a supply chain. It also includes the ability to respond to uncertainty in the range of products that may be demanded. This ability is often needed in mature markets.

PRODUCT DEVELOPMENT

This encompasses a company and a supply chain's ability to continue to evolve along with the markets it serves. It measures the ability to develop and deliver new products in a timely manner. This ability is necessary when serving developing markets.

A FRAMEWORK FOR PERFORMANCE MEASUREMENT

There are other demands that real-world markets place on their supply chains, however, by using these four performance categories we can create a useful framework. This framework describes the mix of performance required from companies and supply chains that serve the four different market quadrants. When a company identifies the markets it serves it can then define the performance mix required by those markets in order to best respond to the opportunities they provide.

Markets in the first quadrant, developing markets, require their supply chains to excel in product development and customer service. Growth markets require very high levels of customer service particularly as measured by order fill rates and on-time delivery. Steady markets require internal efficiency as well as an even broader scope of customer service. Mature markets require all the internal efficiency and customer service called for by steady markets. They also require the highest levels of demand flexibility. The most profitable companies and supply chains are those that deliver the performance called for by their markets. These organizations are the most profitable because they

are the ones most able to respond effectively to the opportunities offered by their markets. Companies should collect and track a handful of performance measures that cover these four areas. This will give them valuable information about how well they are responding to their markets.

The metrics that measure performance in the four areas are applicable to individual companies and also to entire supply chains. It is harder to gather these metrics for entire supply chains because companies are reluctant to share data that may be used against them by their competitors or by their customers or suppliers. There are issues of trust and incentive to work out before these metrics can readily be collected for an entire supply chain. Nonetheless, when these issues are worked out, these metrics will help to guide the behaviour of the entire supply chain and should benefit all the participants in that chain over the long term.

CUSTOMER SERVICE METRICS

In the words of Warren Hausman, a professor at Stanford University, “service relates to the ability to anticipate, capture and fulfil customer demand with personalized products and on-time delivery” (Hausman, Warren H., 2000, “Supply Chain Performance Metrics,” Management Science and Engineering Department, Stanford University). The reason that any company exists is to be of service to its customers. The reason that any supply chain exists is to serve the market it is attached to. These measures indicate how well a company serves its customers and how well a supply chain supports its market.

There are two sets of customer service metrics depending on whether the company or supply chain is in a build to stock (BTS) or build to order (BTO) situation. Popular metrics for a build to stock situation are:

- Complete Order Fill Rate and Order Line Item Fill Rate
- On-Time Delivery Rate
- Value of Total Backorders and Number of Backorders
- Frequency and Duration of Backorders
- Line Item Return Rate

Popular metrics for a build to order situation are:

- Quoted Customer Response Time and On-Time Completion Rate
- On-Time Delivery Rate
- Value of Late Orders and Number of Late Orders
- Frequency and Duration of Late Orders
- Number of Warranty Returns and Repairs

BUILD TO STOCK

A build to stock or BTS situation is one where common commodity products are supplied to a large market or customer base. These are products such as office supplies, cleaning supplies, building supplies, and so on. Customers expect to get these products

right away any time they need them. Supply chains for these products must meet this demand by stocking them in inventory so they are always available.

In a BTS environment a customer wants their complete order to be filled immediately. This may be expensive to provide if customer orders contain a wide range and number of items. It is costly for companies to carry all those items in stock so they may have backup plans to provide expedited delivery of items not in stock or substitution of upgraded items for those not in stock. The order fill rate measures the percentage of total orders where all items on the order are filled immediately from stock. The line item fill rate is the percentage of total line items on all orders that are filled immediately from stock. Used together, these two measures track customer service from two important perspectives.

BUILD TO ORDER

A build to order or BTO situation is one where a customized product is ordered by a customer. This is any situation where a product is built based on a specific customer order and is configured to meet a unique set of requirements defined by the customer. An example of this is the way Boeing builds airplanes for specific customers and their requirements or the way Dell Computer assembles PCs to fit individual customer orders and specifications.

In a BTO environment it is important to track both the quoted customer response time and the on-time completion rate. It is easier for a company to achieve a high on-time completion rate if it quotes longer customer response times.

The question is whether the customer really wants a short response time or will accept a longer response time. The quoted response time needs to be aligned with the company's value proposition and competitive strategy.

INTERNAL EFFICIENCY METRICS

Internal efficiency refers to the ability of a company or a supply chain to use their assets as profitably as possible. Assets include anything of tangible value such as plant, equipment, inventory, and cash. Some popular measures of internal efficiency are:

INVENTORY VALUE

This should be measured both at a point in time and also as an average over time. The major asset involved in a supply chain is the inventory contained throughout the length of the chain. Supply chains and the companies that make them up are always looking for ways to reduce inventory while still delivering high levels of customer service. This means trying to match inventory availability (supply) with sales (demand) and not have excess inventory left over. The only time a company would want to let inventory exceed sales is in a growth market where the value of the inventory will increase. However, markets change and as a rule it is best to avoid excess inventory.

INVENTORY TURNS

This is a way to measure the profitability of inventory by tracking the speed with which it is sold or turned over during the course of a year. This measure is often referred to as T&E or “turn and earn.” It is calculated by the equation:

$\text{Turns} = \text{Annual Cost of Sales} / \text{Annual Average Inventory Value}$ Generally, the higher the turn rate the better, although some lower turning inventory needs to be available in order to meet customer service and demand flexibility.

RETURN ON SALES

Return on sales is a broad measure of how well an operation is being run. It measures how well fixed and variable costs are managed and also the gross profit generated on sales:

RETURN ON SALES = EARNINGS BEFORE INTEREST AND TAX / SALES

Again, as a rule, the higher the return on sales the better. There are times though when a company may deliberately reduce this number in order to gain or defend market share or to incur expenses that are necessary to achieve some other business objective.

CASH-TO-CASH CYCLE TIME

This is the time it takes from when a company pays its suppliers for materials to when it gets paid by its customers. This time can be estimated with the following formula:

$\text{Cash-to-Cash Cycle Time} = \text{Inventory Days of Supply} + \text{Days Sales Outstanding} - \text{Average Payment Period on Purchases}$

The shorter this cycle time the better. A company can often make more improvements in their accounts payable and receivable areas than they can in their inventory levels. Accounts receivable may be large due to late payments caused by billing errors or selling to customers who are bad credit risks. These are things a company can manage as well as inventory.

DEMAND FLEXIBILITY METRICS

Demand flexibility describes a company's ability to be responsive to new demands in the quantity and range of products and to act quickly. A company or supply chain needs capabilities in this area in order to cope with uncertainty in the markets they serve. Some measures of flexibility are:

ACTIVITY CYCLE TIME

The cycle time measures the amount of time it takes to perform a supply chain activity such as order fulfilment, product design, product assembly, or any other activity

that supports the supply chain. This cycle time can be measured within an individual company or across an entire supply chain. Order fulfilment within a single company may be fast but that company may only be filling an order from another company in the supply chain. What is important is the cycle time for order fulfilment to the ultimate end use customer that the entire supply chain is there to serve.

UPSIDE FLEXIBILITY

It is the ability of a company or supply chain to respond quickly to additional order volume for the products they carry. Normal order volume may be 100 units per week for a product. Can an order be accommodated that is 25 percent greater one week or will the extra product demand wind up as a backorder? Upside flexibility can be measured as the percentage increase over the expected demand for a product that can be accommodated.

OUTSIDE FLEXIBILITY

This is the ability to quickly provide the customer with additional products outside the bundle of products normally provided. As markets mature and technologies blend, products that were once considered outside of the range of a company's offerings can become a logical extension of its offerings. There is danger in trying to provide customers with a new and unrelated set of products that has little in common with the existing product bundle. However, there is opportunity to acquire new customers and sell more to existing customers when outside flexibility is managed skilfully.

PRODUCT DEVELOPMENT METRICS

Product development measures a company or a supply chain's ability to design, build, and deliver new products to serve their markets as those markets evolve over time. Technical innovations, social change, and economic developments cause a market to change over time. Measurements in this performance category are often overlooked, but companies do so at their own peril. A supply chain must keep pace with the market it serves or it will be replaced. The ability to keep pace with an evolving market can be measured by metrics such as:

- Percentage of total products sold that were introduced in the last year
- Percentage of total sales from products introduced in the last year
- Cycle time to develop and deliver a new product

OPERATIONS THAT ENABLE SUPPLY CHAIN PERFORMANCE

In order for an organization to meet the performance requirements of the markets it serves it must look to measure and improve its capabilities in the four categories of supply chain operations:

1. Plan
2. Source
3. Make
4. Deliver

The efficiency with which these activities are carried out will ultimately determine how well a company performs as measured by things such as order and line item fill rate, on-time delivery, inventory turns, and cash-to-cash cycle time. Certain activities are directly related to certain performance categories. For instance, inventory management will directly affect a company's order and line item fill rate and its inventory turns. Its procurement activity will directly affect its return on sales and its upside ability. A company needs to collect data about its activities in these four operational areas and monitor results.

The Supply-Chain Council's SCOR model suggests the kind of operational data that should be collected. This data is referred to as "Level 2 Performance Metrics." In the plan operation, useful measures are the cost of planning activities, inventory financing costs, inventory days of supply on hand, and forecast accuracy. In the sourcing operation, it is useful to have data on material acquisition costs, sourcing cycle times, and raw material days of supply. Useful measures in the make operation are the number of product defects/complaints, make cycle times, build order attainment rates, and product quality. Suggested delivery operation measures are fill rates, order management costs, order lead times, and item return rates.

This data should be collected regularly and trends should be watched. When performance targets start to be missed, the next step is to investigate the business operations that support that performance. Again the SCOR model suggests more detailed data that can be collected and analyzed in each of the four supply chain operating areas. This more detailed data is referred to as "Level 3 Diagnostic Metrics."

Diagnostic metrics can be used to analyze the complexity and configuration of the supply chain and also to study specific practices.

In the plan operation, complexity measures are the number and percentage of order changes, number of stock keeping units (SKUs) carried, production volumes, and inventory carrying costs. Configuration measures track things such as product volume by channel, number of channels, and number of supply chain locations. Measures of management practices in the plan operation are such things as planning cycle time, forecast accuracy, and obsolete inventory on hand.

In the source operation, measures of complexity and configuration are number of suppliers, percentage of purchasing spending by distance, and purchased material by geography. Some practice measurements are supplier delivery performance, payment period, and percentage of items purchased by their associated lead time.

The make operation has measures of complexity and configuration such as number of SKUs, upside production flexibility, manufacturing process steps by geographical location, and capacity utilization. Management practice measurements are value added

percentage, build to order percentage, build to stock percentage, percentage of manufacturing order changes due to internal issues, and work in process inventory.

In the fourth supply chain operation, deliver, there are complexity measures that include number of orders by channel, number of line items and shipments by channel, and percentage of line items returned. Configuration measures are delivery locations by geography and number of channels. Practice measures cover things like published delivery lead times, percentage of invoices that contain billing errors, and order entry methods.

COLLECTING AND DISPLAYING PERFORMANCE DATA

Historically, companies based their management decisions on periodic, standard reports that showed what happened during some period in the past. In stable and slow-moving business environments this worked well enough. However, there are not many companies that work in stable and slow-moving environments any more. Working from traditional, periodic, accounting-oriented reports in a fast-paced world is somewhat like trying to drive a car by looking into the rear-view mirror.

The business environments we live in are characterized by shorter product life cycles, mass markets dissolving into smaller niche markets, and new technology and distribution channels constantly opening up new opportunities. The pace of change is both exhilarating and relentless. A company must keep up. To do this, a company needs to build a reporting system that presents data at three levels of detail:

- *Strategic*: To help top management decide *what* to do
- *Tactical*: To help middle management decide *how* to do it
- *Operational*: To help people *actually* do it

THREE LEVELS OF DETAIL

In a supply chain management context, strategic data consists of current actual, as well as plan and historical numbers that show the company's standing in the four performance categories: customer service; internal efficiency; demand flexibility; and product development. In the Supply-Chain Council SCOR model, data of this type is referred to as "Level 1" data. This data is summarized by major business units and for the company as a whole. Strategic data also consists of data from outside the company such as market sizes and growth rates, demographics, and economic indicators such as GNP, inflation rates, and interest rates. There should also be benchmark data from industry trade associations and studies that show the operating standards and financial performance levels that are standard for companies in the markets being served. Tactical data consists of actual, plan, and historical numbers in the four performance categories displayed at the branch office level of detail. This data also includes the performance

metrics labelled “Level 2” in the SCOR model. These metrics monitor the plan, source, make, and deliver operations that every company in a supply chain must perform. Operational data consists of the measures labelled “Level 3” in the SCOR model. These measurements help people who are charged with getting a job done to understand what is happening and to find ways to make improvements where needed to meet the performance targets that have been set. The SCOR model refers to these measurements as diagnostic measures.

We are awash in data. It is important to present it in such a way that it is useful. If people are overwhelmed with data they cannot use it. By organizing data into these three levels, people can quickly access what they need to do their jobs. Upper management uses strategic level data to assess market conditions and set business performance objectives. They can drill down to the tactical level or even the operational levels when necessary. Middle managers use tactical data to do planning and resource allocation to achieve the performance objectives set by upper management. Line managers and their staffs use operational data to solve problems and get things done.

THE DATA WAREHOUSE

To collect this data requires the creation of a data warehouse. This data warehouse is a central repository of data that is drawn from a variety of operating systems and accounting systems in a company. It is important to collect the needed data at its source. Tap into relevant systems within a company and capture needed data automatically as a by-product of daily operations. Avoid having people do manual entry to get data into the data warehouse.

A data warehouse is composed of a database software package and the automated connections to other systems needed to collect the relevant data on a regular and timely schedule. Working in conjunction with the database software is software that allows for the creation of standard predefined reports and graphic displays which people can use to monitor operations. In addition to predefined reports and displays, the software must also allow people to do *ad hoc* queries of the data in the data warehouse to do detailed investigations when necessary.

When designing and building a data warehouse it is best to start quickly with something that is simple and on a smaller scale. This way people can get experience in using data more actively to do their jobs. As they gain experience and can clearly describe the additional features they would like, larger and more complex data warehouses can be built.

Remember, the most important component in any data warehouse system is not the technology, or even the data, but the people who use the system and their ability to use the system effectively and learn from the data and become more efficient at their jobs. Chapter 6 goes into further detail about the design and building of these kinds of systems. In addition to helping people inside of a company to become more efficient in performing their supply chain management jobs, a data warehouse can also be the foundation for

collaboration with other companies in the supply chain. Whatever information is shared between companies in a supply chain should be made available to those other companies electronically. This often takes the form of reports that can be retrieved on demand by other companies who access a company's data warehouse over the Internet using features of the same data reporting software that people inside the company use.

SPOTLIGHTING PROBLEMS AND FINDING OPPORTUNITIES

Depending on the type of markets a company serves, senior management needs to define a handful of key performance targets in the areas of customer service, internal efficiency, demand flexibility, and product development. The task then becomes one of figuring out how to manage operations to achieve the target numbers. The point of collecting performance data is to help monitor and control daily, weekly, and monthly operations.

People in a company need access to a one page display of the key operating or financial measures that they are responsible for achieving. These one page displays are known as "dashboards" because they show a person at a glance the data that is most important to them. The data that is displayed on a senior management dashboard is different from that on an operating manager's dashboard and the data on the dashboard of a staff person in one department is different from a staff person's in another department.

Senior management sets company performance targets and they need access to a dashboard report that shows them the company's current performance against these targets. If things are going well and performance is meeting expectations, then no further attention is called for, but if performance is falling short against one or more of the performance targets, then the senior manager knows right away where more attention is needed.

Middle managers are responsible for managing their operations to achieve one or more of the company's performance targets. Their dashboards need to show them the plan and actual data on company performance targets they are responsible for. They need to see quickly if operations are on target or not and direct their attention accordingly. Once alerted by their dashboard that there is a problem in a particular area, the manager can then drill down into further detail in that area. Staff people in various departments need dashboards that track and illuminate the specific business operations that they are responsible for such as purchasing, credit, inventory management, and so on. These displays should highlight issues needing their attention.

For the most part, people run their business or do their job by keeping track of a handful of key indicators. These indicators tell them where to direct their attention and help them steer through a complex and changing world. When a data warehouse and software reporting tools are in place in a company, people need to experiment with the design of their dashboard displays or reports. As they get better at using their dashboards

to guide their actions, the overall effect will be for the company as a whole to become more efficient and more responsive to its markets. Since very few companies work in stable and slow-moving markets anymore, there is a great need to learn to use data effectively to make decisions and act. Speed is a major competitive advantage. The faster a company can spot problems and fix them or see opportunities and respond to them, the more profitable the company will be. It will also have a much better chance of survival over the long term. Companies that can see their markets change and adjust and follow those markets most efficiently are the ones that will stay in business. Companies that do not notice problems soon enough or that do not see how their markets change are the ones that will get into trouble.

MARKETS MIGRATE FROM ONE QUADRANT TO ANOTHER

Markets migrate from one quadrant to another during the course of their lifecycle. Over time, market forces are always pushing a market towards an equilibrium where supply meets demand. At the same time, other forces also influence a market so it fluctuates back and forth around the equilibrium point. At times demand outstrips supply and at other times there is more supply than there is demand.

Companies in the supply chains that supply a market must be able to adjust their operations over time as their markets migrate from one quadrant to another in order to remain competitive. For instance, in growth markets, supply chains that do the best are the ones that have the highest levels of customer service as measured by order fill rate and on-time delivery. All the companies in the supply chain must focus on delivering this performance in order to succeed.

As a growth market moves on to a steady market, the most profitable companies will be those that are able to maintain high levels of existing customer service and also broaden the scope of their customer services. In addition, profitable companies will be the ones that achieve the best levels of internal efficiency. They can no longer focus only on customer service. As steady markets become mature markets, the supply chains that serve them must again develop their performance in another category. Mature markets require companies to develop the capabilities needed to accommodate high levels of demand flexibility. Then in the midst of mature markets, new developing markets can appear and the ability to create new products and bring them to market becomes critical.

Adaptability itself is now as important to survival and success as the four performance categories. Market evolution is now often measured in years and sometimes in months. Gone are the days when markets changed more slowly over decades. No company has the luxury of being able to focus on optimizing any single mix of performance capabilities over the long term.

A company may become very skilled at internal efficiency and customer service as called for in a steady market. The company needs to remember though that its markets will change. The company will have to add skills in the area of demand flexibility as some of its markets mature. The company may even need to de-emphasize some of its internal efficiency policies in order to emphasize its performance in product development so that it can participate in a promising developing market. The key here is that a company needs to know when to shift its emphasis from one mix of performance categories to another. A ship at sea needs to watch the wind and the waves and respond appropriately when the weather changes. So too must a company watch the supply and demand situation in its markets and respond appropriately when one of its markets enters a new quadrant. If the collection and display of market and company performance data alerts a company to respond sooner to a market change than its competitors, then the company has indeed developed an important tool for its success and survival.

GOOD SUPPLY CHAIN STRATEGY

The configuration components-operations strategy, channel strategy, outsourcing strategy, customer service strategy, and asset network-are the fundamental building blocks of your supply chain strategy. However, to drive forward your strategic business objectives and really gain a competitive edge, these components and the choices you make about each one must be:

- Aligned with your business strategy
- Aligned with your customers' needs
- Aligned with your power position (your influence)
- Adaptive, because competitive advantage is temporary and market conditions change.

These four criteria may sound elementary, but few companies actually follow them. In fact, the practice of developing and managing a supply chain strategy is not widespread. Many of our clients over the years have had only the most rudimentary supply chain strategy process in place, indicating that these concepts are either not well understood or difficult to implement. Let's examine them one by one.

ALIGN WITH YOUR BUSINESS STRATEGY

Your supply chain strategy should directly support and drive forward your business strategy. We believe that an effective business strategy begins with a core strategic vision that lays down the boundary conditions for your business: what you are, what you'll do, and-just as important-what you are not and what you won't do. The core strategic vision clarifies the answers to key business strategy questions such as: What are your overall strategic objectives? What value do you deliver to your customers? How does your company differentiate itself in the marketplace? Unless the answers to these key questions drive your supply chain strategy and configuration, your supply

chain will be operating in a vacuum. Here's one example of many. We worked with an electronics company that had spent millions to improve production and order-fulfilment times. The company's on-time delivery performance was excellent. There was only one problem-delivery performance was no longer the key to profitable growth. Increased competition meant that customers were demanding, and getting, lower prices. What's more, a slowdown in several of the company's primary markets was cutting into revenues and sharply reducing return on assets.

The company's president recognized the need to move to a much lower breakeven point, but supply chain operations were focusing on yesterday's priority-customer delivery excellence. Where does this type of disconnect come from? Simply put, the new business vision of the company's president had not been translated into new objectives for the supply chain. Although the overall business plan clearly showed the volume and price declines and their impact on margins, the implications for supply chain operations had not been considered. It was only when the company began to lose money that a major supply chain reorganization occurred, which led to factory closures, facility consolidation, and outsourcing of manufacturing. Translating your business strategy into an effective plan of action requires communication and discipline.

COMPETING ON COST

Companies that compete on cost offer low prices to attract cost-sensitive buyers or to maintain share in a commodity market. This strategy demands highly efficient, integrated operations, and the supply chain plays a critical role in keeping both product and supply chain costs down. The low-cost supply chain focuses on efficiency-based metrics such as asset utilization, inventory days of supply, product costs, and total supply chain costs. Product standardization and process standardization are critical, as are supplier and production quality and inventory control.

Hewlett-Packard (HP) traditionally pursued an innovation-based strategy-until an upstart competitor changed the industry dynamics. In 1997, HP and other printer manufacturers were taken by surprise by Lexmark's launch of a below-\$100 printer. When Lexmark had doubled its market share by mid-1999, HP embarked on an ambitious programme called "Big Bang" to sharply reduce product costs through new design and supply chain changes. The goal? To compete directly with Lexmark on price. "Big Bang" was a big success. By 2002, HP had won back its market share.

Efficiency and low cost are good things-but not at the expense of service, innovation, or quality if one of those is a key element of your business strategy. Consider the low-cost offshore manufacturing that we discussed earlier. Most apparel manufacturers outsource their production to Southeast Asia, where contract manufacturers insist on fixed production schedules to minimize costs. This low-cost approach also results in low flexibility and can hurt margins at the retail level. If one style lags while another takes off, the retailers are limited in their ability to change volumes and mix. With too many of the wrong garments, the stores end up with marked-down inventory and eroded

margins. Too often the missed revenue and reduced margins associated with a poorly aligned supply chain strategy are not included when assessing the total impact of supply chain strategic choices.

Zara, the retailer owned by Spanish textile giant Inditex, chose a very different model. Zara positioned itself as the designer-boutique alternative for the price-conscious but trendy consumer. To deliver its strategy, it manufactures almost 50 percent of its garments in-house—an industry exception. Although its manufacturing costs are 15 to 20 percent higher than the competition, Zara more than makes up for the cost differential by using its supply chain to ensure that merchandise in the stores matches what customers want.

COMPETING ON INNOVATION

Companies whose primary strategy is innovation focus on developing category killers “must have” products that benefit from significant consumer pull. And because their products are category killers, these companies can command a price premium, the innovator's advantage. Companies such as Sony, Nike, and L'Oréal seem to have a finger on the pulse of the consumer and are fast to market with new products that buyers want. The underlying source of power for such companies is unparalleled marketing and product development. How does the supply chain support a company that competes on innovation? For new products and services, the window of opportunity—before the fast followers start taking market share—can be small. Innovative companies are acutely aware of the benefits of getting into a market early and gaining first-mover advantage, so new product introduction (NPI) is key. By getting new products to market faster, the supply chain can boost revenues and profits. This is why it's important for a company with innovation as a primary strategy to integrate the supply chain with the design chain, which we define as all the parties—both inside and outside the enterprise—that participate in defining and designing a new product or service.

The challenge isn't just time to market, however. Time to volume is critical too. The faster a company can pump up production to meet demand, the greater are the profits, and the less likely imitators are to catch up. Creating strong demand for a new product and then being unable to meet that demand is one of the worst things that can happen to an innovation-driven company. Achieving that time-to-volume advantage is a major competitive weapon.

Design chain/supply chain integration is critical to innovation driven companies, ensuring the fast and sustainable launch of new products. Moving from product development to volume production at the target level of quality requires management of processes, assets, products, and information. Design chain/supply chain integration also ensures that when demand cranks up, the whole supply chain is ready—that suppliers can handle your needs, that order-management systems support the new product information, and that sales channels and service people are trained. Consider again the example of clothing retailer Zara. While most of the industry focuses the supply chain on delivering the lowest purchase price, Zara's supply chain supports its primary

innovation strategy. Designers and planners use point-of-sale information to adjust production plans and designs to focus on best-sellers. This translates into a much shorter time to market, higher revenues, and fewer markdowns. Between 2001 and 2002, when many fashion retailers were struggling to break even, Zara's performance translated into steady double-digit growth and healthy EBIT (earnings before income taxes) margins, growing from 18.1 to 18.5 percent. By combining innovation and cost performance, Zara created an NPI-focused supply chain that delivers spectacular results for the company and its investors.

COMPETING ON QUALITY

Companies that compete on quality are known for the premium nature of their products and services, as well as consistent and reliable performance. Quality products include well-known names such as Lexus automobiles, Maytag appliances, and Tropicana juices. Product development is obviously critical to quality, but so are key supply chain processes such as manufacturing, sourcing, quality assurance, and return. And if a product is perishable or fragile, transportation and storage play an integral role. One key supply chain attribute relative to quality is traceability-the ability to trace a product back to its point of origin-a growing requirement in a number of industries. Concerns about food safety and the booming market for organic and "ethical" products mean that consumers want to be able to trace a product from "farm to plate." In the U.S. tire market, for example, traceability back to the point of manufacture is a legal requirement. Moreover, counterfeiting has emerged in a growing number of sectors, such as luxury goods, entertainment, and pharmaceuticals.

To offset this risk, manufacturers increasingly use special tags, such as RFID (radio frequency identification), that identify merchandise as genuine and closely control product flows to consumers. The example of Barlean's Organic Oils shows how a supply chain strategy can deliver quality as a basis of competition. Barlean's is a \$22 million family-owned company that sells health supplements. The company's flagship product, flaxseed oil, outsells the competition 20 to 1. Freshness makes the difference. Barlean's oil carries a four-month expiration date, compared with other oils that can be up to five months old before they even hit store shelves. Barlean's manufacturing and distribution processes give the company its edge. Conventional manufacturing techniques expose flax seeds to heat, light, air, and overpressing, all of which compromise quality. Barlean's designed a manufacturing process that protects its seeds from the elements. The company doesn't even press the seeds until an order comes in-a "press-to-order" production method. Barlean's also uses Express Mail to ship orders so that they arrive faster and fresher-a more expensive choice, but one that further supports the company's commitment to quality. In the crowded health supplements market, Barlean's uses the supply chain to deliver on its commitment to quality, profitably growing sales every year by 40 percent since 1999.

COMPETING ON SERVICE

Companies that compete on service tailor their offerings to their customers' specific needs and are known for exceptional customer service. These companies customize their products and services to build customer loyalty and lock in repeat sales. To excel at service, all of a company's customer-touching processes and information systems such as order capture, order fulfilment, and invoicing must be fast, consistent, and trouble free. The ability to integrate internal processes and systems with those of key customers is a core skill.

Our research shows that best-in-class deliver companies—those with exceptional order-fulfilment processes overall—are more than 20 percent more profitable than average companies and are growing their top-line sales 25 percent faster. Why does better service lead to such striking financial gains? On a basic level, companies with superior customer service processes avoid the costs related to expediting and the short-term churn that other, less adept companies face. On a more strategic level, companies that excel at customer service develop the ability to segment their customers. They understand the relationship between cost to serve and profitability and can assess the cost of offering customized services. As a result, they avoid offering customized services to customers who don't meet hard business criteria. They also tend to focus on the higher-value segments of an industry and on developing relationships with their priority customers, resulting in lower account turnover and a decrease in customer retention costs—all of which add to the bottom line.

A case in point is Shell Chemical. It offers its manufacturing customers an inventory-management solution called SIMON (supplier inventory management order network) that simplifies their purchasing process and cuts costs from the supply chain at the same time. With Shell's automated replenishment system, customers no longer place orders, run out of inventory, or build "safety cushions" of stock. Instead, they've integrated their information technology (IT) systems with Shell's so that they can share information. Every night, data on consumption, inventory levels, and usage forecasts are recorded at the customer's site and forwarded to Shell. At predetermined inventory trigger points, Shell places a refill order for the customer, schedules order transport, and tracks the shipment until it arrives.

The system is invoiceless. Each month, based on consumption figures that both parties share, the customer sends payment electronically. Shell's solution makes life easier for everyone. By cutting administrative costs for customers and the safety cushions of stock that result from forecast uncertainty, the solution reduces the total amount of inventory in the supply chain and vastly simplifies inventory management.

Shell gains an added bonus. In exchange for using the company's service, and because the solution couldn't work if inventory from different suppliers were mixed in the same tanks, customers agree to use Shell as their exclusive vendor for products managed under SIMON. By offering a value-added service and integrating with its customers' IT systems, Shell is able to forge extremely tight relationships with its customers—a powerful

supply chain-driven, strategic advantage. The best companies understand that they can't be all things to all people. They identify how they will differentiate themselves and drive their supply chain to best-in-class performance for that basis of competition. This sharp focus sets them apart from the competition and helps them gain a competitive edge.

The supply chain supports every business strategy, so make alignment a top priority. Remember, though, that strategy is a balancing act. No cost leader can afford to ignore customer service. Nor can an innovator ignore the price ceiling of a market. Supply chain performance leaders understand the trade-offs among service levels, lead times, assets, and costs and make decisions that best fit their overall strategic mission. By choosing where to focus and achieving best-in-class supply chain performance in those key areas, companies can set themselves apart from the competition.

ALIGN WITH YOUR CUSTOMERS' NEEDS

Barlean's Organic Oils identified freshness as a key differentiator. Shell Chemical realized the advantage of providing a new service concept to its customers. Zara understood the market for high fashion at attractive price points. Each of these companies identified solutions that would create a competitive advantage and aligned their supply chains accordingly. Do you really know what your customers want? Are there opportunities that you're not exploiting simply because you can't envision them? Answering these questions can be a challenge. Often our assumptions about what customers want are wrong. Just as often customers are unaware of or can't fully articulate their needs. Voice of the customer (VoC) is a methodology we use that helps our clients really listen to their customers and understand their environment. VoC can help uncover and translate customer needs into requirements for new products and services that leverage your existing supply chain capabilities or pinpoint changes you can make to exploit new business opportunities.

Aligning with customer needs also means identifying the particular requirements of different market segments. As we shall see, Michelin's passenger-car tire business has two different market segments—automobile makers and the after-market customers. The automobile makers are far more demanding, requiring that tires be delivered directly to the assembly lines right on time because any delay can back up production schedules. Besides being more demanding, the automobile makers deliver only razor-thin margins, compared with healthier margins for the less demanding replacement market. Thus, you'd think that the after-market segment would be a priority, right? Wrong. Tire manufacturers such as Michelin always put the automobile makers first because being on a new car creates a pull for replacement sales and because supply contracts contain severe penalties.

If you're a tire manufacturer, you have to play according to the automobile makers' rules. If reliability and timeliness are key customer requirements, as they are for manufacturers such as Michelin, then you'll make it a priority to continually improve

all the supply chain activities that support those performance goals-or pay the price with high levels of finished-goods inventory. If customers don't get what they need, you'll lose market share. If you don't know what your customers want, then, ask them. Or use VoC to find out. And check back periodically to understand how their needs are changing.

MULTIPLE SEGMENTS, MULTIPLE SUPPLY CHAINS

For some companies, one supply chain with a single set of physical assets, processes, and information systems makes the most sense, even with different products, suppliers, and customers. A company with very flexible operations can "design anywhere, build anywhere," so product assignments are made based on available capacity, product costs, and target markets. Some companies, however, find that they can serve their markets better with separate supply chains. Why? Different products, market segments, or regions may call for very different approaches. Or separate business units may want their own supply chains so that they can have the business control needed to achieve their specific business objectives. There are inherent risks in trying to manage disparate requirements such as increased complexity, increased costs, or unmanageable processes, organizations, and information systems in one total supply chain. When managed correctly, however, these risks can be mitigated while achieving significant benefits through tailoring key supply chain steps to enhance specific business unit (or product line) performance. One positive example of this approach is Seiko Epson Corporation's (Epson's) European Group company. Epson is a global leader in printer and imaging products for both home and business based on its strong technology, brand recognition, and reputation for quality. Behind Epson's overall leadership on imaging products there is a wide range of diversity of technology within its product portfolio. Products include inkjet printers, scanners, laser products, ink and toner cartridges, point-of-sale printers, and liquid-crystal display (LCD) projectors, to name a few.

The back-end supply chain for these various product lines and businesses is often made up of separate global supply chains, each with its own suppliers, factories, and logistics routings. But the front end of the supply chain-encompassing the processes that touch the customer, such as order management and distribution-is common. This derivative approach allows Epson to optimize its back-end supply chain and gain substantial efficiencies in the production processes of its varied products. It also ensures that its customers interact with "one Epson" at the front end and can easily do business with Epson regardless of the mix of products they wish to purchase. According to Ramon Ollé, Epson Europe chairman and chief executive officer (CEO), this combination allows Epson to achieve an overall efficiency in its end-to-end supply chain while increasing customer satisfaction.

Another example is Michelin. A capital-intensive manufacturer, Michelin uses the same factories for customers in both its automobile makers' segment and the after-market segment to increase capacity utilization, along with a single production planning

process. Once the tires leave the factory, however, everything else—from demand planning, order entry, and shipping to invoicing and warehousing—is different. The company uses two separate supply chains to meet the needs of two different markets. For automobile makers, the company delivers tires to mounting centres, which move them directly to the assembly line, because timeliness and reliability are critical. By contrast, tires for the after-market are shipped to distributors and a variety of retailers, such as auto repair shops, car dealerships, and other retail outlets, where on-time delivery is less critical because most hold some level of inventory. Companies with both new product sales and after-sales support businesses often have different supply chains. Besides managing different activities, such as returns, repairs, and refurbishment, the after-sales support supply chain must meet very different customer service expectations.

Although new product sales and after-sales support are usually structured as two distinct business units, both typically use the same production centres to gain economies of scale. Yet customer-facing processes, including demand planning, order entry, and invoicing, are tailored to the two different markets. If your company has multiple business units, identify your greatest point of shared leverage and use that as a starting point. For Michelin's passenger car business, the answer was raw materials and manufacturing assets, so the company structured its supply chains accordingly. If you can standardize within a single supply chain and avoid additional fixed assets and unnecessary variations in processes and information systems, do so. If different processes or multiple locations performing similar activities are the result of acquisitions or historically independent geographic or business units, check to see if these duplications are still needed. In many industries, converging customer requirements and the emergence of regional and global logistics service providers have opened the door to more streamlined and standardized supply chains. Leverage any synergies that you can.

Here's a general rule of thumb: If you can meet the needs of your most profitable customer segments at an acceptable service level and at a reasonable cost with a single supply chain, then do so. If supply chain performance targets such as cost, lead time, and delivery for different products or market segments are truly at odds with each other, or if business control is an issue, then consider partially or completely separate supply chains.

ALIGN WITH YOUR POWER POSITION

A good supply chain strategy is grounded in an understanding of your power and influence relative to customers and suppliers. Why is this so important? Your relative power determines what can be achieved realistically in terms of reconfiguring the supply chain to meet your overall strategic objectives. The reality is that many of the supply chain innovators you read about are in an enviable position: They're big, with enormous market clout. But not every company is a Wal-Mart, able to strong-arm its suppliers to eliminate inefficiencies. When it comes to making fundamental changes in your extended supply chain, you need to understand how much power and influence you really have.

Scale Matters: Big companies can leverage their volume of output to buy inputs more cheaply, boost asset utilization, and reduce the cost of everything from information systems to transportation. Just as important, they can impose their own processes and rules on suppliers and customers. In the automobile industry, any supplier who shuts down the production line by not delivering on time can be subject to a penalty up to the equivalent of the revenue lost while the line is down. It's written into the supply contract. Of course, not every company can do this. It requires power—the power of scale.

Companies with scale can exercise a high degree of control over the supply chain and structure it in a way that supports their own strategic objectives. When a company is bigger than its suppliers or its customers and they need it more than it needs them, the company calls the shots. But scale is relative. Companies often underestimate their own power because they're thinking in broad, global terms instead of narrowing their scope to a country or market segment. Even relatively small companies can find ways to work strategically with select suppliers or customers to gain a competitive edge. The key is to segment, focus, and consolidate. If you're not a priority for your suppliers and aren't getting adequate collaboration and service, rethink your supplier relationships. Consider shifting the power equation by focusing on a few, smaller suppliers and giving them an opportunity to grow their business in exchange for working with you more collaboratively over time to cut costs, boost efficiency, and improve overall performance. Brand also can be a major source of supply chain power, especially in the consumer markets. If your products are highly desired by consumers, you'll have more clout with retailers and other channel partners. Think of Louis Vuitton bags or Rolex watches. Lack of availability only heightens the brand's prestige. Customer service, fast delivery, or squeezing out supply chain costs and inefficiency can seem irrelevant to companies with the hottest brands or products. Buyers of highly desirable products have learned to wait-and to pay a premium.

Make sure that you understand your relative position before trying to exercise your supply chain power. Even consumer packaged-goods brand leaders, those whose product dominates a product category, are modifying their physical logistics to satisfy the demands of key retail accounts. Manufacturers that occupy the less desirable second position on the retailer's shelf typically have to go even further to contribute to the logistics efficiency of their key accounts by, for example, modifying product packaging and providing very high supply flexibility. The hard reality is that second-tier branded-goods suppliers just don't have the supply chain power of category leaders. Changes to their supply chain must accommodate the retailer requirements, even to the detriment of manufacturer supply chain cost and inventory performance.

When you're developing your supply chain strategy, take some time to assess the situation. Is your supply chain brand-led, channelled, or supplier-led? Do you need your channels more than they need you? How about your suppliers? Who has the power? If you're a supplier in an industry that's fragmented on the sell side but concentrated on the buy side, like the automobile industry, your power may be limited. The same is true

if you're one buyer among many in an industry with just a few suppliers- such as an electronics buyer of speciality components. As we've seen from these examples, supply chain control is a possibility. For most companies, though, collaboration is the best bet. Therefore, analyze your position in the supply chain. Explore how you can rethink your interactions to cut costs, eliminate inefficiencies, boost satisfaction, or add value. And since collaboration is tough to do well, pick your targets carefully. Focus on key customers or suppliers, and look for opportunities where collaboration will have a real strategic impact.

BECOME ADAPTIVE

Change is a given. Market conditions shift, business strategies evolve, and new technologies emerge. If you're not paying attention, your supply chain can get out of sync. Your supply chain strategy, just like your business strategy, has to adapt.

Although change is constant, the frequency of significant change will differ by industry. In the PC industry, companies make fundamental changes to their supply chains about every three to five years, driven by the constant pursuit of lower costs and the need for rapid new product introductions. Companies have adopted new operating strategies such as make to order, white-box packaging, and resale from low-cost producers and are opening their own stores, selling directly through the Internet, and exploring other new channels.

In other industries, such as aerospace, fundamental supply chain changes can occur less often-every 10 years or so-and have major consequences. Consider the supply chain strategies of Boeing and Airbus. Airbus's network of partners delivers finished subassemblies direct to the company's assembly line in Toulouse, France, where fewer than 500 workers do final assembly. Airbus's production model requires lower capital spending, spreads development risks, and leverages the know-how of many partners.

Boeing today is in a state of continual change. The company is achieving marked improvements in efficiency by reducing its supply base and is working within the company and with suppliers to implement techniques of lean manufacturing. At the same time, the company is involving more risk-sharing supplier partners in its business and making great progress towards large-scale systems integration. Given the major changes experienced by the airline industry in the late 1990s and early 2000s, including bankruptcies, mergers, and acquisitions, and the impressive growth of low-cost carriers, it is likely that the supply chain strategies of major commercial aircraft manufacturers will evolve further to meet the airline industry's demand for cost-efficient solutions and to deliver profitable growth. Both internal and external factors determine your supply chain's shelf life and can trigger a need for a change in configuration. These factors include:

- *A new technology that transforms the dynamics of your industry.* The Internet, for instance, created a direct link to customers so that companies such as Amazon could sell direct and cut out the middleman.

- *A change in the scope of your business.* If your company is offering new products or services, targeting new markets, or expanding geographically, you may need to expand your manufacturing capacity, add new distribution capabilities, develop new channels, find new suppliers, and rethink your supply chain strategy overall.
- *A change in your basis of competition.* Perhaps a new competitor has emerged with a stronger value proposition, or you need to change the type of service you offer to increase market share, or you want to compete in a new market that requires faster delivery, greater flexibility, or higher quality. Any major change to your company's basis of competition should drive a re-examination of your supply chain strategy and components.
- *The need to assimilate a new acquisition.* Mergers and acquisitions can create a need to reconfigure the supply chain. You'll have to see where it makes sense to eliminate redundancies, where to keep operations separate, and where to integrate.

A company's growth trajectory can have major implications, too. Is your sales volume increasing or decreasing? Is your industry expanding or contracting? An organization and processes designed to manage and support a growing company may no longer be appropriate during retrenchment, when the focus is on cost control. In either case, your company may need to adjust supplier requirements and resize fixed assets. As an example, consider Nokia. Nokia Mobile Phones was facing a crisis in 1995-1996. After ending 1995 with a massive shortfall in profits versus plan, the business was operating at a loss in early 1996, with costs spiralling out of control. Business conditions weren't good either, with a sales slowdown and continuing handset price erosion. Pekka Ala-Pietala, Nokia's president, announced the need to move the company's focus from market share to profitability. Specifically, the company had to review its product portfolio with a critical eye and take actions on a number of supply chain issues, including product quality, forecasting, and manufacturing productivity.

As Nokia discovered, producing 100,000 mobile phones a month is not the same as producing 100,000 phones a day. Explained Pertti Korhonen, vice president of global logistics at the time, "We weren't ready for the transition to a high-volume business and rapidly found ourselves with excess inventory and quality problems with key suppliers."

Nokia turned the situation around by implementing new supply chain disciplines in Europe, Asia, and the United States that included improvements in manufacturing productivity and quality, supplier development, supply chain planning, and integration. The results were impressive. For example, on-hand inventory was reduced from 154 to 68 days in nine months, releasing \$450 million in cash. Taken together, these actions put the company back on track for profitable growth. Supply chain expertise was developed and maintained to build a foundation for future growth. Today, Nokia's supply chain is recognized by the rest of the industry as a true source of competitive advantage.

Cisco is another interesting example. In May 2001, Cisco shocked the markets by announcing a \$2.2 billion provision for inventory write-offs. As the star player of the fast-growing networking equipment sector, Cisco had developed a reputation as one of the most savvy supply chain managers in the world. Outsourcing partners provide Cisco with manufacturing, materials management, warehousing, and transportation services, allowing the company to focus on its core strategy of rapid innovation and rapid growth—without increasing fixed assets.

In 2001, when the whole networking equipment sector experienced an unprecedented collapse in demand, few expected that Cisco would suffer as much as other industry players. But the company had built a complex global supply chain without robust supply chain planning between itself and its partners. This led to duplication of planning based on "second guess" estimates and an inability to see and respond quickly to changes in market demand, which resulted in excess inventory across the supply chain. To minimize the impact of future boom-bust scenarios, Cisco turned to a private Internet-based network called eHub. The result was an innovative, shared environment with top-tier suppliers, distributors, and contract manufacturers. In so doing, Cisco hopes to build one of the most effective supply chain networks in industry.

Given the dynamic nature of industries, businesses, and product life cycles, creating and putting in place a supply chain strategy are not an annual or biannual exercise but an ongoing orchestration of decisions and actions. New opportunities and threats emerge every day. If you can't keep up with these new realities, you risk becoming a footnote in some dusty management text. It's also a fact of life that major supply chain innovations can be copied. Even if you're first out of the gate, your edge will dissipate with time. So stay vigilant, and keep looking for ways to improve and differentiate your supply chain performance.

CHALLENGES OF E-BUSINESS CHANGE

One of the first challenges companies face when attempting to develop online channels is to consider how such a strategy will impact upon their 'bricks and mortar' organization. Developing a 'clicks and mortar' operation may result in a more 'virtual' form of organization in which traditional ways of working are mixed with electronic communications. It is here that a company encounters its first problem, which is usually one of technology. The particular difficulty relates to the attempt to evolve 'legacy' systems (*i.e.*, the technical infrastructure that has accumulated to support the business over time) to an infrastructure that will support e-Business. Few businesses find themselves in the position where they can 'throw away' the old and introduce new, customized computer systems. Legacy systems often perform essential activities upon which daily business processes depend.

While start-up companies can leapfrog these problems, established ones face some difficult challenges. Effective e-Business solutions demand integrated front-and back-

end systems, a process which may demand close co-operation between two groups (or even subcultures) with the organization. As Nigel Waterson of Gemini puts it, 'The front end has quite often been built by guys in ponytails, while the people who understand the back end are often grey haired'. System integration means that when customers interact via the Web, placing orders and purchasing goods, the stock control and financial systems also speak the same language and carry out their part of the transactions. The problem is that many such back-end systems are unlikely to be based on open Internet protocols and may even have been custom built. Nonetheless, such systems may be critical to a company's business, and include such details as bank account data and stock rotation information. As Conway points out,

IT managers are loath to replace them with something new and untested. They may not even fully understand how their legacies work any longer. The people who built the systems may well have left the company, leaving present IT experts reluctant to tinker. Replacing or upgrading such systems also takes time, which may slow up Web developments critical to speedy e-Commerce innovation. Over and above the technological matters identified above, major *organizational* change issues must be recognized and addressed for online strategies to be realized successfully. This is because of the need to redesign business processes and structures, change organizational culture, and engage in education and training. A wide range of stakeholders may be affected, with many personnel needing to 'buy in' to the change. There is only one way to do e-Business: fully committed. Everyone in the company must be dedicated to the effort. You can't have ten people for every thousand working on it. You can't delegate it. You have to encourage everyone to jump into the water and support them in teaching each other to swim. I'm asking for the biggest cultural change in your company's history. These issues will now be considered in turn.

RESISTANCE TO CHANGE

There are a number of reasons why people may be unwilling to accept organizational change, as summarized below:

- Stability and security are threatened.
- Coping strategies and comfort zones are affected.
- The uncertainty of change creates anxiety.
- Imposed change reduces perceived autonomy and control.
- Job content is changed and new skills are demanded.
- Authority structures and reporting lines are altered.
- Work groups and other relationships are disrupted.
- Established routines and practices are abandoned.
- An individual's power and authority is threatened.

According to Markus (1999), workers' reactions to change vary greatly. She suggests that while some workers readily embrace new technologies, some can be hostile, and the reason for this hostility can often be attributed to poor communication and shifts in

organizational power. Effective internal marketing is important here in order to segment employees into 'supporters', 'neutrals' and 'opponents' of change, and then develop appropriately customized communications in order to deal with the differing priorities of each of these groups. Implementation of change involves the disruptive transition from a current state to a future state, as a result individuals or groups can resist change. They claim that there are three types of problems encountered when an organization goes through a significant change:

- *Power*: Change can be viewed as a threat to existing power structures and it creates uncertainties, so the struggle for power escalates as individuals and groups attempt to control their environment by resisting change.
- *Uncertainty*: Anxiety is created as individuals are not sure where they stand and whether at the end of the change process they will still have a role within the organization. As a result, individuals can act irrationally, as they find it difficult to understand and interpret clearly information related to the change.
- *Control*: During the change period, it becomes difficult to maintain control because goals are changing, as are structures and roles, and certain control systems become irrelevant.

These authors also highlight the important point that within large organizations individuals usually do not openly resist change; instead, they 'subtly or passively' resist. The degree of employee involvement and participation can significantly affect the success of change. Markus claims that workers' reaction to change is dependent on

whether they have had a say in selecting the technology or the way it is introduced and used, how the new technology is communicated, how much training and support are provided, and how carefully the roll out is planned and executed.

Involving employees fully means that they become responsible for the success of the change, becoming 'owners' of the change process. IBM Global Services recommends that in order to build a successful e-Business, organizations need to challenge their employees to:

identify the cultural changes that will also be needed and to shape the processes, the linked education, learning and competency development that will be required to deliver real value in the new ways of doing business that they are proposing.

Another critical point is that organizations are bound to encounter complications in asking individuals to make changes that they are incapable of implementing. Adequate investments in training and development programmes are therefore essential, particularly, for example, if new skills in Web design are required. If individuals do not possess the skills and capability necessary to introduced planned changes, the likelihood of success is significantly reduced.

According to Buchanan and Boddy (1992), the more radical change projects are, the more open they are to organizational disruption and failure. Badham *et al.* (1997) point out that there are two aspects to radical change, as illustrated with examples in Figure 3.2. The first concerns the issue of 'breadth' - the degree to which change is central to

the organization's strategy - and demands radical as opposed to incremental modifications throughout the organization. The second relates to the degree to which such modifications mark a significant departure from existing ways of doing things. Both of these certainly apply to e-Business. For example, such changes point to business process redesign, the development of cross-functional team working, and the move towards a customer-focused (instead of management-led) culture. These changes are likely to be politically controversial and threaten the interests of a wide range of stakeholders. Unlike routine change, such initiatives are also likely to be highly complex. As Badham *et al.* note, there may be a high degree of uncertainty as to what to do and how to do it; objectives may be less clear, and resource requirements will be less well known. In addition, it may be less easy to create shared perceptions of goals and build and maintain necessary commitment. For this reason, these authors suggest that more time will need to be spent ensuring effective communication to encourage flexibility, address perceptions, and generate and regenerate involvement. To illustrate the problems that can ensue in such a situation, the authors describe Merrill Lynch's move into online trading:

At the core of the change process was conflict at many levels within Merrill Lynch. There was conflict between the defenders of the brokers and their commissions and proponents of online investing. There was conflict between Merrill brokers who were concerned about losing customers to online brokers and Merrill brokers who were concerned about losing commissions. There was even conflict among Merrill executives between who favoured setting up a separate online unit to compete with the brokers and those who favoured keeping the online unit under the same executive.

LEADERSHIP

Leadership plays a crucial role in effective change management. Hunt (1998) claims that change is a learning process because people have to learn to behave differently, but in order for this process to work and for change to be successful, individuals need to be encouraged, and it is here that leadership becomes very important. Carnall also expresses the need for strong leadership; that is, 'the ability to maintain progress and a facilitative and supportive approach', and suggests that for this to happen, the leader needs to have knowledge of the change area and appropriate skills to encourage learning and change.

Some leaders use 'coercive persuasion' to force change; in other words, 'employees have no alternative but to accept the new reality because they have nowhere else to go'. Although individuals may not like the change, they have no choice, so will accept the change and accommodate the new ideas. He goes on to state that leaders need to be able to 'manipulate people's understandings of what is going on, and who are able to deliver the results quickly'. The role of the leader can be to create conditions under which success can visibly be achieved, even if only in a limited and partial way, and to rationalize and capitalize upon positive events after they have happened. Managers will instead need to be skilled in the art of leadership and corporate politics. The need to enrol and re-enrol support, neutralize dissent and resistance, and secure resources

will demand networking skills and the ability to build consensus and support. They must start by gaining buy-in at the top. As Siegel puts it, 'You can have the world's greatest web strategy, but it won't work unless managers have a stake in the outcome'. To make sure they do, Siegel recommends the formation of a change team, headed by a Chief Net Officer (CNO). Furthermore, he suggests that businesses should 'Strengthen the team with managers who have good relationships with people in other divisions. The CNO will need a lot of favours, so make sure the team is credible in the eyes of the rest of the company'. If leaders recognize the type of change they are faced with, and are familiar with the sort of skills and tactics that may be employed to deal with it, the management of change is more likely to be successful.

RELATIONSHIP-BUILDING AND COMMUNICATIONS

The importance of effective internal marketing to engage staff in the change process was emphasized. Increasingly, such communications are a critical aspect of external relationship building too. According to Symonds (1999a), a crucial misconception is that any business is a 'free-standing entity'. For example, in order to participate in a new development such as e-Procurement or customer relationship management, it is essential that companies involve their suppliers, partners and customers in their processes and allow each party to become familiar with each other's processes. This degree of openness and transparency is new to most organizations, and it requires significant change and high levels of trust between participants.

A genuine e-Business strategy provides electronic links in order to 'foster conversations' with staff, customers and partners. Such 'customer-led' approaches involve listening to customers in a strategic way, deepening relationships and loyalty. The importance of engaging in rich customer conversations is underpinned by a number of recent works on e-Business. The highly influential *Cluetrain Manifesto*, for instance, asserts in the first of its ninety-five theses on the new economy that 'markets are conversations'. Newell's *Loyalty.com* (2000) underlines this point, and highlights the way companies must 'leverage customer information' for the effective management of customer relationships on the Internet. Seybold's *Customers.com* (1998) makes similar points, again focusing on the need for customer-focused strategies that engage customers as parts of a community based around a company's products and personnel. Such strategies suggest something much more radical in terms of change than the mere 'bolt-on' approach of adding an online channel to market, which is often proposed as a straightforward and simple process.

It calls for a re-engineering of processes and structures focused around key customer groups, rather than product or service divisions. It also implies cross-functional, team-based working. As Siegel puts it, The customer-led company has a broad interface across which all employees can get to know their customers. Employees invite customers in to collaborate on new products, support systems, and methodologies. Facilitating those interactions will take new communication skills, new tools, and the ability to

move people in and out of product teams easily. At Dell Computers, for instance, customers are brought into the product planning and manufacturing processes, with all employees encouraged to have contact with customers.

Through effective collaboration across boundaries, ideas can be shared about product designs and value propositions. The result is faster and more customer-focused product and service innovation. To produce the capacity for this, considerable attention must be placed on organizational structures, processes, skills and culture - elements that may need a radical overhaul in established companies.

ROLE OF THE CHANGE AGENT

Buchanan and Boddy emphasize the importance of having an appropriate 'change agent' or 'project champion' in the successful implementation of change. The change agent should be someone who is committed to the success of the project and prepared to 'go the extra mile' in order to motivate, bully or cajole other participants as required. There is a diverse range of skills associated with performing this role; for example:

- Influencing;
- Negotiating;
- Selling;
- Inspiring;
- Commanding respect;
- Political;
- Magic and miracle working.

It is a tall order for any one individual to have all of these skills, and some will be more important than others at different stages of the project. In practice, therefore, more than one change agent may be required.

ADAPTING ORGANIZATIONAL CULTURE

There are many ways in which the terms 'culture' and 'organizational culture' can be understood, but for our purposes we have chosen the following: the system of meanings which are shared by members of a human grouping and which define what is good and bad, right and wrong and what are the appropriate ways for members of that group to think and behave. The pattern of learned basic assumptions that has worked well enough to be considered valid and, therefore, to be taught to new members as the correct way to perceive, think, and feel in relation to the problems of survival and integration.

Schein suggested that employee acceptance of technological change required a change in organizational culture, because practices and values tend to be built around existing technologies that have contributed to the successful development and self image of the organization. One of the strongest elements of culture he identified was the status system attached to these traditions, and the possession by individuals of critical skills. Schein

claimed that innovation was a property of culture, and the potential of information technology as a competitive and strategic weapon would not be fulfilled unless innovative cultures were present or developed, enabling the organization to learn and adapt. This attitude was also noted, whose theory of 'flexible specialization' emerged from their observations that technological progress could be self-blocking, and new products were generally designed to fit existing equipment and procedures. The importance of culture was also emphasized in a study of video disk development by Graham, who found that prevailing cultural values and attitudes in the firm, based upon past experience, were inadequate when it came to dealing with an innovative product in a new market.

Although these studies pre-date the Internet era, they illustrate quite clearly how a change in organizational culture is often necessary for full advantage to be taken of the opportunities presented by new technologies. A vast literature now exists on the subject of how to change culture. As early as 1952, Lewin identified three phases of culture change, namely 'unfreezing', 'change' and 'refreezing'. 'Unfreezing' results from the questioning of norms that have led to a specific failure, thereby sensitizing employees to the need for change, which is then consolidated into new procedures or behaviours during the 'refreezing' phase. This theory seems overly simplistic and linear in its classification of the technological change process, which in practice appears invariably to be a more dynamic and turbulent series of events. Mintzberg (1979) advocated the development of a culture of 'adhocracy' that is organic and decentralized in structure, thereby avoiding the pitfalls that are usually associated with bureaucracy. He claimed that in search of innovation, such organizations minimize planning, control and the division of labour.

In practice, however, while examples do exist of individual companies, such as British Airways, that have 'created' a new corporate culture, these success stories appear to be few and far between. March and Simon, in their revised edition of a seminal text that introduced the concept of 'bounded rationality' (referring to the limited cognition of organizational members), maintained that their theory still holds true today. Despite the vastly increased resources available to the modern organization, these authors believed that potential for change would always be constrained by the conflicting agendas of employees. Green also questioned the ability of organizations to manage culture in a prescriptive way:

If culture could be levered into shape then, by now, someone would have discovered the method. The metaphors of fine-tuning and fit which abound in much of the literature on culture and strategy are altogether inappropriate for something as complex as human social systems.

He noted that while corporate cultures appear to be stable and static when studied at a particular point in time, applying a longitudinal perspective reveals that they are in fact too multidimensional and dynamic to be susceptible to manipulation by management. The key point about culture change, therefore, is that it is rarely as simple a process as is sometimes suggested when recommended as a tool for implementing projects involving new technology.

E-MANAGEMENT E-MAIL

Before proceeding to look at the law of e-commerce, it is useful to consider some particular aspects of e-business management. There is no doubt that, for the managers of any commercial enterprise, the Internet presents the most significant universal challenge that has yet been presented in the history of commerce. It is tempting to think of the Internet as merely an electronic expression of what has gone before: e-mails replace typed letters, websites replace glossy brochures and electronic invoices replace their paper equivalents. But the Internet is not merely a more efficient replacement for existing systems. Its significance lies in its ability to be not merely a communications device, but a market, an information system and a manufacturing tool. It allows managers to do not only what they have done before communicating with suppliers, advertising products and services, collecting customer data and obtaining payment more efficiently, cheaply and comprehensively, but also opens up significant new possibilities for the core operations of every business.

Of course it is the potential to increase productivity and cut costs that is a most attractive feature of the Internet, and this must not be overlooked. Suppose, for example, you are a car manufacturer. Traditionally you have obtained orders for new cars from a network of showrooms that have in turn obtained orders from their customers. Where outsourcing has been required, you have placed orders with your suppliers. Assembly takes place, the car is customised to the customer's requirements and is shipped to the dealer. With the Internet comes the possibility of the customer placing their order online. They can choose their accessories and colour scheme and create a virtual image of how the car will look on-screen before proceeding with their order. Direct electronic communication of the order eradicates possible mis-communication in, and later arguments over, choice of specification. The delivery date can be electronically calculated and arrangements for payment can be finalised. The manufacturer's automated system will communicate electronically with suppliers so that the alloy wheels, or whatever has been selected, can be delivered precisely when needed for fitting to the vehicle. The finished product can be delivered via an intermediary dealer or direct to the customer.

E-MAIL

The spread of electronic communication brings transparency and openness to the management process. The ease with which an e-mail can be forwarded to hundreds or thousands of people changes the nature of communication — one must now assume that not just the recipient will read an e-mail that is sent to him. It is no longer possible to hide behind mountains of paperwork or to 'lose' a letter to which it is currently not desirable to respond. It is common knowledge that the ability to send and forward e-mails causes two main concerns for employers. The first is that time will be wasted — some employees spend several hours each week in sending e-mails to friends. The second is that the employer will incur some liability as a result of the content of such e-

mails. Both these concerns can be met by an 'e-mail policy', which should be clear and comprehensive. It should set out the employer's attitude to e-mail and explain that employees' e-mails will be monitored (if this is desired) to ensure quality of service and appropriateness of communication. The presence of such a policy should have the effect of employees exercising some caution when sending and forwarding e-mails. It should also mean that embarrassing incidents such as the Norton Rose 'blow job' e-mail debacle can be avoided. It should be noted that the monitoring of employee e-mails has data protection and privacy implications that are beyond the scope of this Report.

A related issue is that of pornography. Pornographic images are popular with employees but cause great concern for employers. Not only do they waste employee time, they cause offence and clog vital server space. They could also lead to liability for the employer. To avoid the sort of situation experienced by Orange when it dismissed 30 employees for downloading and distributing pornographic images, a clear statement should be made to employees of the employer's attitude to this activity. To stem the flow closer to its source, the business should consider installing software capable of screening-out unwanted material. E-businesses should remember that an e-mail is a business document in the same way as is a letter or a fax. E-mails must therefore comply with the requirements of the Companies Act 1985, namely that the following information must appear:

- The full name of the company;
- The registered number of the company;
- The address of the registered office; and
- The country of registration of the company.

HUMAN RESOURCES

Given the current climate of short-term working practices and the need to have efficient and productive staff, e-businesses need to focus on staff attraction and retention. E-businesses, generally speaking, need fewer but better staff. This may require a review of working conditions, working practices, pay structures and benefits. The e-business revolution has, significantly, created new jobs that did not exist in the offline economy — examples include website designers, knowledge management officers and e-business managers.

Workers in e-business are increasingly flexible, many working from home or on the road. Rather than being feared, such practices should be embraced by businesses. Not only do they lead to a lowering of overheads (heat, light and a desk in the office are not required for mobile workers) but they can create greater satisfaction and loyalty amongst staff. Appropriate communication infrastructures should be employed so that staff are kept fully informed of events 'back at the office' and are able to communicate with other staff members, as well as customers and suppliers. Where sales teams spend much of their time out of the office, they too should be kept fully informed and accountable.

The lack of office presence of individual employees and the globalisation of businesses mean that the 'drink after work' and the 'training weekend' become increasingly difficult. Many businesses are developing corporate portals for their business to employee (B2E) communications. Such portals can prove invaluable in providing information and training to employees. Larger companies can additionally use the portals as an online job market and to provide maps of buildings and photographs and locations of staff. Some companies motivate employees to log on to the portal on a daily basis by posting the share prices of the company or by displaying a list of employee birthdays for the relevant week. A section of the portal can be used for selling the company's products to the employees.

The Internet can be used for recruiting staff. Potential job applicants can be given up-to-date information on vacancies, be taken on a virtual tour of the business and be invited to apply online. Communications between the HR department and job applicant can be undertaken by e-mail. The expense of using a recruitment agent can be saved.

CUSTOMERS

The Internet gives rise to the possibility of a wider marketplace. Customers may now be located anywhere on the planet but have equal access to the marketing material of the business. Systems should be set up to deal with this new global presence.

Better information on customers is available from the monitoring of their activities whilst visiting the website. Using cookies it is possible to 'personalise' the content of the site for each particular customer. Knowing that a customer lives in Oxford, for example, it is possible to display a banner advertisement for an Oxford-based pizza restaurant on your homepage. Were a different customer to visit the homepage, the advertisement would be of a different kind. Location-specific and targeted advertising have data protection implications and detailed proposals should be submitted to a data protection lawyer before procedures are implemented.

As the technology grows, so do customers' expectations. Customers will expect to be kept informed of the process of their order and will want more information than would have previously been available. Electronic order tracking systems should be available to customers online to provide them with this information. DHL, for example, has a system which enables a customer to see the location of their package at each stage of its journey and to know the precise moment that it is 'signed-for' by the recipient.

INTERNET AND WORLD WIDE WEB TECHNOLOGIES

The microcomputer or Personal Computer (PC) forms the basis of e-Commerce. The Internet would not have the wide reach that it has into peoples' homes in the absence

of the PC. Most office PCs are linked to a computer network and PCs at home are connected to the Internet via modems or dedicated data transfer cables.

LOCAL AREA NETWORKS AND WIDE AREA NETWORKS

PCs on local area networks (LANs) and wide area networks (WANs) are connected to the Internet. LANs and WANs are digital computer networks that connect PCs to enable sharing of information between separate computers. LANs are limited to a geographical area such as an office, whereas WANs connect computers over a wider geographical area. The computers on a LAN or WAN share information in the form of addressed packets of data.

The capacity of a single computer is increased by connecting it to a LAN or WAN, as it can share computer programmes and information. A computer that is linked to a network can download and use computer programmes and share storage and printing facilities, and take advantage of network capabilities for sending and receiving data and information.

CLIENT-SERVER ARCHITECTURE

A computer on a LAN or WAN that provides resources like application programmes or printer connections is known as a 'server'. A server provides other computers known as 'clients' on the LAN or WAN with software and other resources.

This set-up is called client-server computing architecture. A server that is linked to the Internet is called a 'host computer'.

A server computer is capable of running server software, which needs to be compatible with the network operating system. The client and server are independent and perform specialized tasks to process information and run computer applications.

If you use Microsoft Word on your computer, the client, it is probable that a server will have provided the Microsoft Word software for you to use. The client-server architecture is the basis of the Internet too.

THE INTERNET AND PROTOCOLS

The Internet is the sum total connection of LANs, WANs and stand-alone computers around the world. The term 'Internet' is short for 'inter-networking' or an interconnected set of networks.

It is the global network of computers, whether the computer is on a LAN in an office, or a WAN in an industry extranet, or an independent computer in a private home. An Internet-connected computer is known as an Internet host computer. The Internet is also known as the 'Net', the 'Information superhighway' or 'Cyberspace'.

The set of rules for moving information over the Internet is called a 'protocol'. The transmission control protocol (TCP) is used for sending large amounts of information between host computers on the Internet. Vinton Cerf wrote the basic ideas for TCP on

the back of an envelope over lunch, and in 1983 the core protocols of the Internet transmission control protocol/Internet protocol (TCP/IP) became the standard protocol for transmitting information over the Internet.

Intranet

An intranet consists of networked internal connection of computers owned by an organization and makes use of Web technology. An intranet may be mounted on a LAN or a WAN. An intranet makes use of Internet technology such as TCP/IP, HTML, Java and HTTP to make it interoperable and to provide it with Web capability. The basic elements of an intranet are a computer network, a computer designated as a server containing server software and the Internet protocols TCP/IP.

An intranet offers several benefits to an organization: improved sharing and communication of information, open standards and cross-platform collaboration. For example, a car manufacturing company may connect its information systems on purchasing with its accounting information systems to form an intranet for sharing and processing information between two departments in the company.

An intranet is a company's own internal information transfer system that offers: e-mail, communication among different computers, connection to remote offices, a Web browser interface and special-interest news groups. For e-Commerce, an intranet is used to provide a corporate image and unified 'experience' for a customer that combines product information, ordering and customer care.

Extranet

An extranet is used for business-to-business (B2B) e-Commerce. It is the networked connection of computers of two or more companies. It is a private communication system to support trade and is used for communication and transactions between business partners, suppliers or special customers. For example, a car manufacturing company may connect its computers with its various suppliers of components for the cars it makes. Another example is the airline industry's OneWorld network, which enables customers to transfer seamlessly between airlines to reach their destinations.

Design Principles

Intranets and extranets share the same design principles formulated for the Internet. The Internet design principles are:

Interoperability: There are varying operating systems that control computer functions on the many different computers on the Internet. Normally a given operating system cannot communicate with a different one. A standard to enable varying operating systems to communicate with each other was introduced by the US Department of Defence, which originally financed the Internet. Interoperability means that independent implementations of Internet protocols can work seamlessly.

Consequently, operating system developers such as Microsoft or Apple incorporate software into their operating systems that enables them to operate with other computers over the Internet. Internet-compatible systems use the TCP/IP common protocol for communication. Interoperability for e-Commerce means that companies and customers do not have to purchase and upgrade software from the same vendors. Their computers will be able to communicate over the Internet because of the standard protocols.

Layering: The Internet can be regarded as a series of layers of software. The structure of the Internet is layered. It is a five-layer system consisting of: interconnect level (National Access Points, NAPs), national backbone providers, regional Internet providers, local Internet service providers, and the business and consumer market.

The bottom layers of software, interconnect and network, are concerned with operating the computer hardware. The higher layers, business and consumer market, come closer to the needs of the person using the Internet.

The most relevant layer is the layer that processes the information required by the Internet user; this is called the 'application' layer. These different hardware and application layers need to communicate with each other and they do so by using well defined interfaces. The Internet layering standard has resulted in increased reliability of Internet software and it is invisible to the Internet user.

Simplicity: The layering has resulted in simplicity of software design. Each layer is concerned only with its own functionality, making its design simpler. For example, the layer concerned with physical devices like mouse operations are hidden from the higher layers that process information. This kind of simplicity has contributed to increased software reliability too.

Uniform Naming and Addressing: Each Internet host computer has an Internet Protocol (IP) address that uniquely identifies it on the Internet from the millions of other computers also on the Internet. The IP address is expressed in a uniform format or 'dotted quad'. An example is: '17.10.2.3'. As the dotted quad is not meaningful to humans the domain name system (DNS) is used to provide a symbolic name for the dotted quad. An example is: 'sol.brunel.ac.uk'. The DNS is capable of translating the symbolic name into the required IP address to enable Internet operations.

End-to-end Protocols: The Internet does not process information. It enables the transfer of 'packets' of information between computers. This is called end-to-end protocol. The Internet enables the transfer of information or 'content' of the packet. The transmitting or receiving computers, known as the 'end' system, do the actual processing of information.

The combined use of the Internet, intranets and extranets by companies has resulted in the term 'virtual organization' or 'networked organization'. The virtual organization is to be contrasted with the physical organization; the latter is restricted by geographical, physical space. Companies can combine computer-networking technology with information technology (IT) and information systems (IS) to develop a network of computers that can capture, process and share information and knowledge

in virtual or cyberspace, where time and space are defined differently from physical time and space. Amazon.com may be classed as a virtual organization. Virtual organizations tend to be flatter because access to information and knowledge is potentially open to all employees who have access to a PC.

THE WORLD WIDE WEB

The World Wide Web, also known as the 'Web' or 'WWW', consists of pages of information depicted as text, graphics, sound, or video clips. A Web page may contain Java applets - Java programmes that are downloaded from the server and run on the local computer.

The Web consists of over a million Web servers, and an untold number of Web browsers. What distinguishes the Web from other computer media is its ability to link Web pages dynamically; this is known as hypertext links or simply links. The Web combines computer network technology with hypertext to provide a 'global information system'.

One Web page can be linked to another by a hypertext link and the user merely has to click on the link to display related information. A link appears as highlighted on the Web page. It is this hypertext linking capability of the Web that makes it a powerful source of information. The Web is a client-server architecture system, and a Web user can access the Web servers with a Web browser such as Netscape or Microsoft Explorer.

HYPERTEXT

A document on the Web is composed using hypertext technology. Hypertext is an electronic document system that can be read non-sequentially and interactively. The reader does not have to read it like a book, she/he can click on hyperlinks of related information. The Apollo space programme to record its documentation used a hypertext system. e-Commerce Web sites are built using hypertext documents, and other hypermedia.

HYPERTEXT TRANSFER PROTOCOL

The Web differs from the Internet by the protocol it uses to transfer information between computers. The Web uses the hypertext transfer protocol (HTTP) to connect and transfer hypertext documents stored on Web servers, also known as Web sites. All the computers on the Internet that use the HTTP protocol compose the Web.

THE WEB BROWSER

A Web browser is a graphical interface for searching, accessing and viewing hypertext and multimedia files on the Web. The first browser, called Mosaic, was invented by the

US company Netscape Communications, founded in 1994 by Marc Andreessen. He set himself the problem of transferring sound and pictures over the Web. His solution was the first graphical Web browser.

It allowed basic functions such as retrieving and displaying hypertext files. The growth of the Internet and the Web is largely because of the invention of the graphical browser which makes it easier for people to search and view information. Two popular browsers are Netscape Communicator and Internet Explorer. They are more sophisticated than the original Mosaic browser and enable e-mail messaging, HTML authoring, and extensions such as scripting, plug-in and multimedia.

INTERNET SEARCH ENGINES

The Internet is unarguably the most voluminous information store in the world. Finding information on the Internet requires complex computer algorithms. These algorithms form the backbone of Internet search engines designed to help people find the information they want. Organizations need search engines to find information about customers or business competitors or partners. Search engines are computer programmes that locate specific Web pages, files or multimedia items stored on the Internet. Search engines such as Google or Autonomy use different algorithms or techniques to search for Web pages.

UNIFORM RESOURCE LOCATOR

The uniform resource locator (URL) is the address of a particular Web site on the Internet. It consists of domain names and a pathway that locates a particular host computer that is connected to the Internet. An example URL is www.yahoo.com or www.dell.com. A URL consists of the protocol that is to be used to make the Internet connection (the WWW in the examples), the name of the host computer ('Yahoo' or 'Dell') and the domain (.com). In the case of commercial companies, the name of the host computer connected to the Internet is usually the registered name of the company.

HYPERTEXT MARK-UP LANGUAGE

Hypertext mark-up language (HTML) is used to author information on a Web page. HTML documents are text files that are interpreted by a Web browser. The browser reads the HTML file and interprets the HTML instructions. The browser has to do this interpretation each time a hypertext file is loaded. A Web page can be divided into 'frames' or sub-pages.

Each frame is used to display separate but relevant information designed by the Web author. Multimedia can be facilitated by extensions to HTML or by 'plug-ins' or add-on programmes that make the Web client more versatile to deal with additional media types. An example of extension is Secure HTTP that enables sophisticated encryption and decryption algorithms for sensitive e-Commerce data like payments for transactions.

It takes time to learn to write HTML code. Authoring tools are available for automatically converting documents in Microsoft Word to HTML files, ready to load on to a Web server. Hot Dog Pro, Adobe PageMill and Microsoft FrontPage are examples. A Web site consists of one or more Web pages linked together by hyperlinks. The contents of a Web site are known as 'content'. HTML is not capable of processing or interacting with Internet users. To process data captured through a Web form JavaScript is required.

MULTIMEDIA

Multimedia is a tool capable of providing the transfer of sound and images like pictures over the Internet. Multimedia applications require lots of memory and bandwidth - the physical capacity of data cables to carry and transfer data. For example, one minute of music requires 5 MB of data. The transfer of sound and images in real time requires special software which is added to TCP/IP and HTTP protocols. On the client end, a player is required to interpret the data and render it into sound or images.

CHANNEL ALIGNMENT, LOGISTICS, AND SUPPLY CHAINS

The goals of supply chain management have always included cost minimization, increased level of service, improved communication, and increased flexibility in delivery and response time. The growth of the Internet has presented companies with many opportunities to further improve services and reduce costs. The key for success in managing a supply chain is fast, accurate information from a wide range of operating areas including transportation, inventory, purchasing, customer service, production scheduling, order processing, and vendor operations. The ability to quickly react to market changes and to adjust inventory, production, and transportation systems is necessary for cost reduction and improved utilization of assets. The Internet has, and will continue, to provide managers with this type of information and enable them to improve the profitability of the supply chains.

CHANNEL ALIGNMENT

Focus on the economics of a channel is important, but not as crucial as aligning the channel with how the customer wants to do business. Business buyers will order less if you do not meet them where they want to do business. Several steps are required to adopt this approach. First, identifying demonstrated business buyer buying behaviour by tracking prior channel choices. This is evidence of what exists and does not reflect what is ultimately possible. To stay on the cutting edge, the underlying criteria which buyers use, to make channels these choices must be derived and constantly monitored (*e.g.*, expert advice and training, customization, delivery options, order ease/speed,

lower prices, 24/7 support, etc.). Key is the need to understand how these criteria are used in relation to channel alternatives selected. This approach can help push the envelope out from what has already been done to some more innovative and attractive options for business buyers. Finally, it must be recognized that a diversity of buying situations are likely faced by any business marketer even for a single customer. If this is not the case, then a single, well-maintained, efficient channel will do. More likely, though, is that multiple channels may be required to satisfy the influx channel selection habits of served and prospective customer bases.

CHANNEL ECONOMICS

Another of the pressing challenges facing e-businesses today is how to fulfil customer orders and still maintain reasonable profit margins. What has been well known for brick and mortar institutions for years is now being found out for e-companies. Traditional companies are simply better at managing and operating supply chains—years of investment and experience. Doing it well today requires a lot more effort than it did in the past. It requires accurate sales forecasting, increased integration with shipping hubs, real-time integration with back-end systems and call centres, real-time tracking and scheduling, invoice reconciliation, and performance management. Working with exchanges and outsourcing some of the requirements will allow some companies to deal better with these changes.

A popular type of exchange that has seen rise lately has been the logistics exchange, also known as LXs. With businesses extending themselves globally, they encounter a myriad of new logistics challenges, which they are not always capable of handling. LXs, both private and independent, were set up to handle some of these needs.

LXs exist for all modes of transportation, nationally and global, with different business models and currently there are essentially four different types: freight matching services (connect buyers and sellers of transportation services to each other in the form of bulletin board), auctions, portals (manage every transaction that occurs within the portal and offer services for both buyer and seller), and marketplaces (similar to portals, however with additional functions and access to an open marketplace). These LXs vary in the services provided, from the simple sites that connect your need to a potential supplier, all the way to sophisticated transportation management.

As more and more companies move to the Internet and conduct businesses in e-marketplace environments, companies know less and less about their trading partners. Trust issues will become important since there will not be time to spend weeks checking references in the traditional way, especially when these partners are halfway across the world. One of the results has been that there are a lot less transactions occurring between new partners. Rather, existing partnerships have been moved online and automated. A possible emerging solution is the so-called “trust infrastructure”—a set of technologies and e-business practices that will allow buyers to evaluate

suppliers, confirm their identity, and make trusted transactions. One example of a site that would facilitate such needs is Asian-trader.com. The company's site offers price lists of several hundred manufacturers and allows the users to then place orders with Asian-trader thus avoiding language, shipping, and customs barriers.

PRICING, PARRYING, AND PAYING

The common expectation that BMM exchanges were to turn industrial procurement into one big transparent electronic marketplace, where buyers and sellers turn all goods into a commodity that is comparable only on price, has not turned out entirely as anticipated. Analysts think that this view is overly simplistic and that it does not fully take into account the character of real-world business relationships, the importance of customer relationships, and the impact of brand equity. It may also be based on a basic misunderstanding as to how industrial buying actually works.

PRESUMPTUOUS

First is the erroneous assumption that companies would welcome dozens or even hundreds of suppliers promising to be the lowest bidder for a product. To an extent this is naturally true, but only for specific cases such as spot buying of commodities. In most cases, companies were actually going in the other direction—establishing deep relationships with preferred suppliers with a “total cost” approach, where price is only one of several criteria such as quality, order cycle time, augmented services, geographic proximity or desirability, etc.

Another assumption was that the exchanges would be able to control an industry, something which may indeed happen in a highly fragmented industry. That has not happened in most cases because the companies with the purchasing power refused to relinquish their control. What many manufacturers really are doing is in fact no more than automating and computerizing many of their existing processes while improving their interaction with their suppliers/clients, and ironically the only winners so far are the companies who created the software for the market making.

UNEXPECTED INSIGHTS

Nevertheless, it seems that through this process, the networks may have in fact given sellers unprecedented direction into identifying the needs and preferences of buyers, thus a better understanding of how to fulfil them. Also, fixed pricing is thought to completely change or be eliminated by the fact that price lists and catalogues will become customer-specific. Moreover, the use of purchasing agents, reverse auctions, side-by-side price comparisons, and new agent shoppers will probably mean a long-term change for companies and marketers that rely on brands.

Some analysts believe that exchanges will ultimately carry less activity than originally expected for another reason. These ultra efficient exchanges could turn the manufacturing industry into a low-margin commodity business by way of giving the buyers and other competitors great price and product visibility.

COMPLETE COMMODITIZATION UNLIKELY

Nevertheless, besides pricing and differentiation issues, companies should realize that there are a large number of products that simply cannot be commoditized at all. Some of these products are simply too unique or engineering complicated to be handled by these sites, unless under a simple rebuy mechanism perhaps. Struck with the possibility that they would have to compete mainly on price and product specifications, some manufacturing companies are trying to devise strategies that will help them deal with the imminent threat of eroding profits that may be the result of these exchanges.

Some of these efforts are concentrated on the banding together of some independent Internet trading hubs to fight back against the consortiums of the biggest buyers and sellers in the respective industries. To some, it seems that partnership may be the only way for a company to survive the onslaught of competitive pressures it faces on all fronts today. While most companies take control of procurement in particular, there may be many other niche marketplaces in areas such as project management, industry workflow, and design collaboration.

An interesting pricing phenomenon is the rise of the surplus inventory web site. It is generally attributed to the fact that traditional mechanisms for getting rid of excess or surplus inventory have usually resulted in a low, or negative, return on investment.

This market is estimated to be able to generate some 350 billion dollars, and use of these types of sites will have an impact on interday and intraday price fluctuations. These sites allow for different dynamic pricing mechanisms, which appeal to the users because of the following reasons: increase in revenue on inventory because they reach prospects more efficiently; decrease cost by eliminating paper trails and lengthy communications; elimination of costly middlemen; test pricing to get market price; increase inventory turns by removing excess inventory; create new intermediaries for bringing buyers and sellers together; and increase revenue by getting the highest possible price for scarce items.

Finally, monitoring intraday and interday price fluctuations may give great insight into the value-added aspect of pricing through electronic markets the more visible hand of this marketplace.

REGULATION IN MODERATION

Some see it as perhaps the greatest paradox and cruelest of ironies of the Internet economy, where the “great leveller” has the potential to disenfranchise the most. Antitrust experts have three main concerns: the exclusion of some companies, thus

limiting competition; requiring companies to use one exchange exclusively, thus creating a dominant one that will have too much power; and perhaps the greatest concern is that buyers or sellers would collude on pricing and set prices in unison. It is believed that the increasingly rapid exchange of information may mean that companies would be able to signal one another without having the need to coordinate the prices formally, and without having the government pick up on that activity. Nevertheless, privacy and security issues are also of great importance, especially for the small companies.

INDUSTRY

Some of these concerns exactly have been expressed with the creation of the consortium Covisint, an online trading exchange created by General Motors, the Ford Motor, and Daimler Chrysler, and later joined by Renault and Nissan. The name itself is supposed to be a reflection of the service—"Co" represents connectivity, collaboration, and communication. "Vis" represents vision and visibility of the Internet, and "Int" represents integration and international scope. So far, the consortium is estimated at already having been able to aggregate more than 300 billion dollars in purchasing volume, and has raised concerns of oligipsony, where many sellers encounter few buyers with too much power, also termed monopsony—the power of an oligopoly to impose a form of monopoly. Due to Covisint, the five companies are expected to account together (versus some 50,000 suppliers) for 44% of the cost of goods sold and 91% of the net profits. A similar example is the Exostar aerospace consortium of Boeing, Lockheed Martin, BAE, and Raytheon which accounts for 71% of cost of goods sold and 62% of net profits. It seems that for many suppliers, at least those that are not in a niche market, to successfully compete in such an exchange will require either merging or dropping out.

FEDERAL GOVERNMENT

So far, it seems that the Federal Trade Commission (FTC) and the Department of Justice (DOJ) have not found any reason to be overly suspicious. Covisint managed to pass their scrutiny, while the exchanges in the aerospace, automotive, computer, chemical, retail, steel, and other industries, though reviewed, did not seem to draw as much fire. Despite the investigation into Covisint, it seems that the FTC and the DOJ are still far from setting clear, elaborate guidelines for industry. And the FTC indicated it will not regulate the exchanges until the technology that they encompass, and the degree, which allows them to violate antitrust issues, shall be determined.

Companies though, are already looking at ways to protect themselves by hiring antitrust lawyers, setting up clear guidelines whose compliance is routinely audited by third parties, and choosing third party personnel to fill executive positions. In addition, looking for processes and structures that limit accessibility of product and price information to competitors, and making sure the exchange is open to all legitimate

buyers and sellers. Another approach that seems to have worked for Covisint was attempting to distance itself as much as possible from the original companies as a show of independence and neutrality.

Encouragingly, the Department of Commerce (DOC) is to conduct its first survey of business-to-business e-commerce. As part of the plan, the DOC is to survey tens of thousands of companies, see how much they buy and sell over the Internet and come out with a report that will serve as the benchmark for BMM e-commerce. On a final note regarding this important issue, should be given to the fact that we are in a period of transition, where current laws and practices do not always present the solutions to problems that emerge.

PROMOTIONAL MIXTURES

With competition in e-markets growing rapidly, providers of the marketplaces need volume and quickly. To attract the buyers and sellers, these companies need to spend a great deal of money both online and offline. Interestingly enough, most of the money gets spent on advertising in only 25 sites, thus following the trend set beforehand with the B-to-C companies. Forrester Research estimates BMM Internet advertising will grow to 2.5 billion in 2002.

Online promotional campaign management continues to gather importance and to evolve rapidly. Some ways may help companies to automate their marketing function. Among them: automating their data warehouses; making the buy-or-build decision; preparing to test; measuring the whole media mix; focusing on ROI; keeping an eye on the competition; and getting the right data rapidly into the right hands. It is in this context, that Customer Relationship Management (CRM) has become a quickly growing trend. With the marketplace being as competitive as it is companies must get as close to their customer as possible in order to know their needs. Touch-point management is a term often used to describe the approach. CRM means giving your customer many points of contact and though it has always been part of the BMM transaction, its importance and complexity have recently grown. So much so that companies spend millions of dollars on consulting, relevant software, and investment in this field.

Overall, there is a need to discover how traditional BMM promotional forms such as trade shows, personal selling, and direct mail interact with Websites, e-mail, and even fax technology in moving a business buyer to the conclusive act of selecting one business brand over another. A shift is underway that will recognize that integrated marketing communications elements (trade shows, personal selling, advertising, sales promotion, and publicity) and integrated direct marketing elements (face-to-face personal selling, telemarketing, catalogues, direct mail, and online marketing) must be merged into one summative unit for assessing the impact of corporate communications for enhanced buying and selling as well as sustaining contact with publics relevant to the firm.

Linking the various media to effectiveness in achieving traditional communication is a likely first step. Each medium has its own unique attributes that make it best suited for a particular communications task.

Advertising, fax, and e-mail have in common relatively low cost of exposure per thousand, with only the latter providing the opportunity for convenient immediate response on the part of target market member. They are best suited for generating awareness and providing knowledge and, unlike publicity, they are under the marketer's control to a much greater degree. Easy linkage to a Website gives e-mail a leg up on all three as well as direct mail and catalogues that require greater buyer effort to respond.

Direct mail and catalogues are more costly per thousand reached although they likely provide information about a greater variety of items in more depth. In addition, they may be saved for easy reference and have better visuals.

Common to all six is that the marketer "initiates" the contact with the target market and the buyer has to engage in a higher threshold activity to respond (*i.e.*, using an 800 number, call a sales rep, ordering online). The remaining media, except sales promotions, generally require the buyer initiate contact with the marketer.

Online in many ways has the best of all worlds. It likely costs the least per thousand and is available 24/7. However, it is almost entirely buyer initiated. There are no reasonable limits on variety and quantity of information as long as the Website is easy to navigate for all steps of the buying process. As well, it is perhaps the lowest cost tool that can bring a buyer through all stages.

Trade shows do what other mediums cannot by allowing face-to-face contact and product inspection in a relatively safe environment for the buyer. The closest approach to missionary selling, and attended in teams, companies know that they can gather product and company information without the usual pressure of buying. There is safety in numbers.

Promotions or incentives are often used in tandem with any other medium to induce trial leading to preference and intention to repeat purchase. Chemical supply companies for dry cleaners have used this approach for decades. If a new charge system is developed for use in cleaning fluid, plant operators are given enough of the new material to try it in their equipment to determine if new product benefits such as greater brightness, less need to prespot stains, more pounds of clothing per gallon cleaned, etc. will materialize in their environment. Chemicals used for stain removal are sold this way as well. Through positive low cost experience, the dry cleaner has the opportunity to develop preference and intention to buy. The last two media, telemarketing (outbound) and personal selling cost and therefore are often relied upon for the final stages of the process of developing buyer conviction and ultimately purchase.

SECURITY ISSUES

Some of the enthusiasm that companies have experienced about the new degrees of cooperation that can be achieved through the exchanges has been withdrawn by the

realization that they must be prepared to reveal information about themselves, their products and business processes, sometimes even to competitors. Information of a proprietary nature being passed between companies and potential competitors would require exceptional trust, possible safeguards, as well as monitoring the access of the information. Some companies have taken information sharing to an extreme by creating a sort of demilitarized zone, whereby the posted information to be shared cannot be used for the practices of either of the companies involved.

Yet another factor to be taken into account is the fact that companies may pass information due to different security levels, a mismatch of integration between the two. Good security starts within the company and with a well thought-out security policy, and should also be the means to secure your data and communication with your partner. Two basic rules of thumb: good fences make good neighbours and that your security is only as strong as the weakest link.

AREAS FOR FURTHER RESEARCH

There are several areas of interest requiring immediate investigation. So much so that to call this section by its traditional name “future research” gives the wrong connotation regarding the urgency for Internet integration into business marketing tactics. First is the need to study design and maintenance of Website(s) while focusing on the integration of the Internet into the marketing mix. All business marketing mix areas are affected by the Internet, hence site purpose and features, as well as their number and linkages, must be created and sustained so as to make for maximum customer value and satisfaction. Next, and in a sense in tandem with the aforementioned, is the need for an elucidation on the use of the Internet for generating competitive intelligence and its use for collecting traditional forms of market research data. Issues needing examination include: enhancement of in-house Web-based information intelligence gathering and dissemination; monitoring “survey” response rates, the impact on respondent perceived anonymity, and willingness to participate and disclose their views; data quality versus quantity; and MIS as a continuous process with periodic discrete “reporting events.” Finally, implicit throughout our discussion is focus on larger business entities. Only anecdotal reference is made to the Internet’s impact of allowing smaller firms to compete more effectively with larger firms given its ability to offset economies of scope and scale. However, a similar analysis should be conducted with an eye towards smaller firms, enumerating in detail the impact across all mix elements that any firm likely faces.

Supply Chain and Logistics Management in Retail Marketing

INTRODUCTION

The term logistics is often misinterpreted to mean transportation. In fact, the scope of logistics goes well beyond transportation. Logistics forms the system that ensures the delivery of the product in the entire supply pipeline. This includes transportation, packaging, storage and handling methods, and information flow. The impact of logistics in the ability of a company to satisfy its customers cannot be overstated. All other efforts at modernization within a company would not bear fruit until the logistics system is carefully designed to facilitate the smooth and efficient flow of goods in the system. The topic of logistics is relatively new in India.

There have been some companies that have done work in this area, but a large number of companies are only now beginning to realize the benefits of designing and managing the entire supply chain. With India joining the global marketplace, the role of logistics assumes greater importance. The industrial policies in India have prompted manufacturers to build plants in remote, backward areas due to inexpensive land and tax benefits. This poses some serious logistical problems. Apart from a poor road and transportation network, the existing communications system in India leaves a lot to be desired by any international standard. It is in this context that logistics has to be considered in India.

THE VALUE OF LOGISTICS

Material handling and storage are typically labelled as “non-value adding” activities. While one can appreciate the motivation behind such labelling as one directed towards waste reduction, it can lead to is an erroneous assumption that all material handling and storage can be avoided. While manufacturing processes provide “form utility”, logistics related activities provide “time and place” utility to a product. The challenge is to provide the time and place utility at a competitive cost. If a company can achieve this goal, it will gain a significant competitive advantage in the marketplace.

PULL VS. PUSH SYSTEMS

There are two basic approaches of bringing the product to its final destination, *i.e.*, the customer. In a Push system, products are pushed from the manufacturing plants to distribution points based on a sales forecast. The second approach is the Pull system which requires that the product be pulled from the plants based on actual demand.

In a Push system, since all the product is deployed based on the sales forecast for each region, an inaccurate sales forecast incurs several severe penalties which include:

- Increased safety stock
- Larger Distribution Centres/Godowns
- Higher stock transfer rates.

The pre-order deployment of product increases safety stock. Since there is greater uncertainty associated with forecasts, which are often little better than educated guesses, the system must provide for variations in the demand in a particular region serviced by the particular godown. In addition the system must provide for errors in the overall forecast for the country as a whole. These concerns lead to the carrying of larger safety stocks, which necessitate larger godowns.

The irony in the concept of safety stocks is that although sufficient stocks may exist in the system, the product mix demanded in a particular region may not exist in the regional godown. This necessitates inter-godown transfer of goods. The result is an increase in the transportation costs system-wide, in addition to handling and shipping costs, information costs, product loss and damage, and poor customer service. The more points of distribution in the system, the greater the penalties incurred for unpredictable order fluctuations. The goal of any logistics system is to maintain or improve customer service. In the Push mode of operation, the penalties of higher safety stock, larger godowns, and inter-godown transfer are not the only penalties. Stock rotation becomes more difficult to maintain. Handling of all products at each godown involves unloading, staging, storing, picking, staging and loading for shipment. All these activities involve an element of cost. In addition, there is a potential for product damage each time a product is handled.

There are some positive aspects of a Push system as well. These are:

- Small plant warehouses
- Potential for higher customer service
- Lower transportation costs.

Since the majority of the product is stored at the godowns, the plant needs to maintain a low inventory of finished goods. This allows the plant to utilize its space for production and eliminate the need for a full warehouse staff. If the forecast is accurate, the Push system provides the potential for higher customer service by having the product ready for delivery directly to the customer/retailer. Finally, by having the products deployed in the godowns, the plants have the capability of shipping full truckloads and thereby reducing the system-wide transportation costs.

A Push system works best when sales are consistent, the product variety is small, and there are a few regional distribution points.

REVOLUTIONIZING SUPPLY CHAIN MANAGEMENT THROUGH HOLISTIC GOVERNANCE STRUCTURES

Most organizations exist within a complex labyrinth of vendors, suppliers, customers, distributors, and business partners. Understanding and leveraging these supply chain relationships are key success factors in navigating an increasingly interconnected world. Many companies have tried to leverage supply chain relationships via industry consortiums and emerging Internet technology, but these solutions have collectively fallen short of their goal because most supply chain relationships are ill defined and lack a coordinating governance structure.

Deriving exponential greater value from supply chain relationships requires developing a common vision in which companies concurrently cooperate and compete. To meet this goal, companies should formalize a supply chain alliance in which cumulative benefits exceed what an individual company could achieve on its own. This requires exploring supply chain governance structures that define shared purposes and principles for all participants. I will discuss basic supply chain challenges and outline holistic governance structures needed to address those challenges.

REVIEWING THE SUPPLY CHAIN CHALLENGE

Suppliers, business partners, and customers come in many varieties. Many times, institutions underestimate the importance of suppliers to the viability of their enterprise. Understanding who composes your supply chain is step one in understanding how to leverage it.

SUPPLY CHAIN CATEGORIES

1. Raw materials: Include steel, lumber, fuel, and other materials needed to build a product or maintain a work environment.
2. Components: Include preprocessed items needed to produce products that are then sold to your customers. These can range from springs to computer chips.

3. **Products:** Differ from components in that they are procured for use by a company or consumer and not resold.
4. **Services:** Include maintenance, banking, insurance, health care, legal, association, and many other service categories needed to run a business.
5. **Infrastructure:** Includes emergency and other government services, electricity, water, natural gas, and communications capabilities.
6. **Data or information:** Includes any data or information, obtained in paper, electronic data interchange (EDI), or other format, from a third party.
7. **Distributors:** Includes agents, resellers, franchises, wholesalers, retailers, or other distributors of your products or services.
8. **Customers:** Include buyers or users of your products and services. Some institutions call their customers subscribers, constituents, clients, or patients.

Most enterprises rely on raw materials, products, services, infrastructure offerings, and data during a given business cycle. Companies that manufacture or assemble products also rely on component parts. The first six categories in Table 1 present opportunities up the supply chain. Relationships with distributors and customers present opportunities down the supply chain.

Many companies have little knowledge of the entities leading into and out of their supply chain funnel. This not only limits your ability to assess how a given industry may evolve, but it limits your understanding of when this evolution will occur. These complexities are magnified when thousands of entities lie up and down a supply chain. One company may supply a component, which is then sold to a third party, embedded in another product, and ultimately sold back to the originating company. Ending a relationship with a company could cut off your own supplier! Companies are concurrently cooperating and competing as suppliers, business partners, competitors, and customers — all in the same supply chain.

When a link in the supply chain fails unpredictably, numerous entities can suffer. Major damage can result when one company's problem shuts down an entire portion of a supply chain that lacks an industry-wide contingency plan. When two companies in Boeing's supply chain failed to deliver component parts in 1997, it cost Boeing \$1.6 billion. The costs to other companies within the supply chain were never released. Obviously, effective supply chain management, including the creation of industry-wide contingency plans, is not a luxury.

HISTORIC ATTEMPTS AT SUPPLY CHAIN MANAGEMENT

The Year 2000 problem exposed the importance of supply chain management, but it also demonstrated that most industries lacked the ability to track compliance across those supply chains. Supply chain management strategies have lacked the cohesion needed to address minor and major industry challenges, and attempting to automate

these defective strategies will only exacerbate the problem. Historically, industries wishing to track supplier-customer relationships tried to utilize industry associations.

For instance, the Automotive Industry Action Group (AIAG) tried to address Year 2000 supply chain issues for its 1,600+ members by informing members of the problem, providing project guidelines, prodding members into compliance, and urging members to prod their suppliers into compliance.

But because the automotive industry relies on more than 100,000 companies, this effort fell short of what was needed. AIAG lacked the broad-based access and clout needed to deal with this challenge. Ultimately, supply chain readiness efforts fell to the large corporations that dominate the supply chain.

The Securities Industry Association (SIA) coordinated a cross-industry Year 2000 test, but it focused on clearing selected transactions through a short list of centralized entities. Global financial markets and international clearing houses did not manage to coordinate Year 2000 data interchange testing at the same level. Other industry supply chains were less effectively coordinated. The food industry, for example, has numerous associations that did little to ensure cross-industry supply chain compliance for the Year 2000.

The bottom line is that each company had to independently assess Year 2000 readiness across its supply chain. Almost identical research was replicated thousands of times by countless companies across various suppliers.

Most of these efforts never extended beyond the first tier of suppliers and customers. In other words, supply chain readiness was largely unknown going into the Year 2000 because no model existed that defined which companies supplied or bought products and services from other companies.

Another approach to supply chain management is coercion by large companies that dominate the chain. This has worked in limited fashion in the technology sector — at least for the companies with the clout to impose their view on those below. However, few companies have the market penetration to order a supply chain to bow down to their whims. And other companies in the supply chain derive little value from a top-down, command-and-control structure imposed upon them.

As the need for supply chain management grows, vendors are rushing to automated solutions. Early automation came from ERP vendors using traditional IT solutions. But when these systems failed, distribution capacity collapsed. In evidence of this, a series of distribution system failures at Hershey, International Multifoods, and Royal Doulton caused profits to drop and stocks to tumble. These companies lost control of their distribution capability because they lacked a robust supply chain governance structure that could withstand a computer system failure.

Now companies are pursuing Internet-based solutions that claim to manage supply chains through e-business environments, but these initiatives rely on the same old paradigm in which a company views itself as the centre of the supply chain. Within this model, decisions are made with a skewed view of supply chain relationships. What is

needed instead is a holistic view of a supply chain that will enable the creation of holistic, e-business solutions. Developing this view of supply chains requires governance structures that supersede the myopic view of any one company's trying to assess a seemingly endless number of relationships.

SUPPLY CHAIN GOVERNANCE REQUIREMENTS

Supply chain participants should be able to work with other supply chain participants to benefit themselves and their industry sector. This is difficult because supply chains are organic. They are constantly evolving into new and complex structures that cannot be pinned down or defined hierarchically. This requires a holistic view of the supply chain in which the participants can evolve organically around a common governance structure.

For example, a holistic view of a supply chain would suggest patterns where companies could pursue mergers, acquisitions, partnerships, geographic restructuring, direct-to-market options, and other unforeseen opportunities.

And a holistic view would never assume that a single entity dominates a supply chain — a transitory situation that invariably changes over time. Sample principles that can be used to create a holistic supply chain governance structure.

SAMPLE SUPPLY CHAIN GOVERNANCE PRINCIPLES

1. Establish and share a holistic view of the supply chain that allows any member company to derive value from that view.
2. Create an incentive-based supply chain programme in which members work cooperatively to create a complete picture of their own supply chain as input to the whole.
3. Foster an environment where supply chain participants work harmoniously for the betterment of the industry.
4. Ensure that cooperation on certain matters does not detract from free market competition.
5. Provide access to an evolving view of the supply chain that allows individual companies to streamline acquisition, processing, marketing, and distribution costs.
6. Shorten time-to-market for products through improved tracking of internal and external supplier, distribution, and customer processes.
7. Ensure participants in this governance structure share equal rights and that domination by a single entity or small group of large entities is impossible.
8. Work with the collective supply chain to set priorities for industry initiatives, including development of cross-industry contingency plans.
9. Work with supply chain and non-supply chain organizations for the betterment of the overall environment.
10. Ensure that competitive and proprietary information is secured as a component of this approach.

These principles can be distilled down to a single goal: companies should work together where they can collectively benefit but still maintain an environment where they can compete in a free market scenario. This requires the creation of a governance structure — a legal entity — owned and run by all members of the supply chain. This approach allows companies to build a supply chain knowledge base that benefits the entire industry. Members would have an incentive to join the governance structure because they could access and profit from information about the entire supply chain unavailable elsewhere. Member companies would be able to quickly pinpoint supply chain dependencies up and down the chain — something that has eluded most companies for decades.

Any group of members could create substructures, as needed, as long as they were in accordance with the purpose and principles of the governance structure. Initiatives enabled under this governance structure include the development of a centralized, supply chain relationship model and the ability to address industry challenges based on the relationships defined in this model. These initiatives might include foreign market expansion, labour issues, common data sharing, conservation measures, human rights, raw material management, regulatory issues, waste treatment, and development of environmental positions.

CREATING A SUPPLY CHAIN GOVERNANCE STRUCTURE

In order to meet the aforementioned requirements, a supply chain governance structure must have:

1. Accepts all product and service suppliers, customers, and distributors as members
2. Exists to further the goals of its membership
3. Uses incentives, as opposed to coercion, to build participation
4. Concurrently competes and cooperates
5. Is empowered at the periphery and unified at the core
6. Self-organizes and self-governs
7. Constructively channels conflict
8. Seeks opportunities in challenges
9. Fosters diversity and accommodates change
10. Learns, adapts, and innovates in ever-expanding cycles.

These principles are found most commonly in a chaordic® governance structure. Chaordic describes the state of an organization that embraces and leverages chaos and order. The term chaordic comes from a book by Dee Hock, founder and CEO emeritus of Visa International. Visa is a prototype chaordic organization, being member owned and operated. Under a chaordic alliance, a legal entity is formed that binds relevant and affected parties under a civil contract, which, for the purposes of this article, includes suppliers and customers in a given supply chain.

A chaordic governance structure enables supply chain initiatives because it fosters collaboration in the face of disparate or competing agendas. This is accomplished by defining a shared purpose, guiding principles, infrastructure, and a constitution governing the behaviour of participating entities. Suppliers and customers linked to the supply chain join the alliance by committing to the purpose and principles defined in the constitution. Once in, they cannot be removed, unless they do not abide by the constitution. They can leave whenever they wish.

Creating this governance structure requires leading suppliers and customers to assess the value of such an alliance. They would then assemble a representative body to draft a purpose, principles, governance structure, and constitution. Great attention should be paid to institutionalizing incentive-based participation. A small number of large entities must not dominate these proceedings, because the holistic value of chaordic supply chain governance would be lost to the controlling entities.

While supply chains across various industries share common goals, unique market sectors also have unique requirements. 10 principles that a supply chain governance structure might embrace.

The purpose that ties any given set of principles together must clearly state what the governance structure sets out to accomplish. For example, Visa International created the following statement of purpose: “To create the world’s premier system for the exchange of monetary value.” Let’s consider some statements of purpose for other supply chain governance structures. Table below lists sample statements of purpose that industries on the left side of the table might wish to pursue.

Table. Sample industry statements of purpose

<i>Industry</i>	<i>Possible Statement of Purpose</i>
Automotive	Streamline delivery of components and products to enable a successful and environmentally sound automotive industry for consumers worldwide.
Food Processing	Ensure the safe and effective processing and distribution of food products while focusing on the short- and long-term safety and health of consumers.
Open Computing	Maintain an equitable vehicle for developers of open computing technology to share information and strategies to the ultimate benefit of consumers and overall market growth.
Agriculture	Work collectively to ensure that current and planned agricultural efforts adhere to the highest standards of ethics and purity in the face of shifting technological, government, and consumer demands.
Internet	Maintain an open, equitable, and secure exchange of information that adheres to principles of decency, international law, and democracy.

Each purpose must be supported by a set of governing principles. When drafting a purpose and set of principles, it is important to focus on:

- The greater good of the entire industry and the supply chain
- Value to the ultimate customers of the supply chain

- How the supply chain alliance will affect other industries interfacing with the supply chain
- How the alliance will impact the overall environment and human rights worldwide.

If there is a situation in a given industry that suggests the need to create a legal, member-owned, and member-run supply chain governance structure, remember that this goal involves considerable pain and effort to achieve.

Mobilizing a core group to agree to this approach could take months. Drafting a purpose, principles, governance structure, and constitution can take years. But if the benefits, as outlined by the purpose and principles, are clear to supply chain participants, it is worth the effort.

E-BUSINESS AND SUPPLY CHAIN GOVERNANCE

As supply chain governance structures mobilize, the desire to deploy strategies to benefit the supply chain will intensify. Members will each want to contribute lists of suppliers and customers to a unified supply chain model. As more members join, the model grows more complete. Soon the model paints a picture of cross-industry relationships that never existed before. Handling proprietary customer information and security will be a challenge, but such issues should be incorporated into the governance structure's core principles.

As this process evolves, the governance structure is able to leverage supply chain relationships beyond what any single entity in that supply chain could accomplish alone. Instead of each company in a supply chain trying to craft myopic e-business solutions, the governance structure could unify a single e-business solution that suppliers and customers could collectively deploy and leverage. With this in mind, a unified set of e-business objectives might include the following initiatives:

1. Establish preeminent e-commerce Web sites for end-customers.
2. Allow manufacturers to track retail demand down the supply chain.
3. Enable retailers to predict supply chain product availability up the supply chain.
4. Streamline distribution strategies based on a comprehensive model of the supply chain.
5. Allow companies to collectively refine just-in-time inventory management, material usage, production, and related processes.
6. Standardize electronic data interchange, including contracts, for the industry and any entities interfacing with the industry.
7. Ensure that invalid data is checked and corrected at key checkpoints in the supply chain.
8. Define common solutions for linking internal ERP systems with e-business environments.

9. Create shared sales or service strategies that can be marketed through e-commerce sites.
10. Interface with e-business strategies from related industry sectors.

This list is just a sampling of e-business initiatives that could be collectively deployed by a supply chain unified under a common governance structure. Individual companies could pursue these options independently, but holistic governance offers exponential greater value to an entire supply chain.

LONG-TERM SUPPLY CHAIN MANAGEMENT PROJECTIONS

The borders between internal and external business processes will eventually blur. Companies leveraging sophisticated supply chain governance structures, fuelled by e-business solutions, will insource and outsource processes based on holistic views of supplier-customer models. On a grand scale, linking multiple supply chain governance structures under a larger chaordic alliance would reflect the ultimate evolution in supply-chain-to-supply-chain management. Imagine leveraging customer-supplier relationships across multiple industries to the ultimate benefit of private and public sectors across global markets. Creating a holistic governance structure for your supply chain is the first step in this journey.

GLOBALISATION AND ITS EFFECTS ON SCM

Business today is in a global environment. This environment forces companies, regardless of location or primary market base, to consider the rest of the world in their competitive strategy analysis. Firms cannot isolate themselves from or ignore external factors such as economic trends, competitive situations or technology innovation in other countries, if some of their competitors are competing or are located in those countries.

Companies are going truly global with Supply-chain Management (SCM). A company can develop a product in the United States, manufacture in India and sell in Europe. Companies have changed the ways in which they manage their operations and logistics activities. Changes in trade, the spread and modernization of transport infrastructures and the intensification of competition have elevated the importance of flow management to new levels.

DRIVING FORCES OF GLOBALIZATION

The last two decades have seen the evolution of the global manufacturing environment. Majority of the manufacturers have global presence through exports, strategic alliances, joint ventures or as a part of a committed strategy to sell and produce in foreign markets.

GLOBAL MARKET FORCES

There is tremendous growth potential in the foreign developing markets which has resulted in intensified foreign competition in local markets which forces the small - and medium-sized companies to upgrade their operations and even consider expanding internationally. There has also been growth in foreign demand which necessitates the development of a global network of manufacturing bases and markets. When the markets are global, the production-planning task of the manager becomes difficult on one hand and allows more efficient utilization of resources on the other. Few industries remain today in which the international product life-cycle theory still applies. Product markets, particularly in technologically intensive industries, are changing rapidly. Product -cycles are shrinking as customers demand new products faster. In addition, the advances in communication and transportation technology give customers around the world immediate access to the latest available products and technologies.

Thus, manufacturers hoping to capture global demand must introduce their new products simultaneously to all major markets. Furthermore, the integration of product design and the development of related manufacturing processes have become the key success factors in many high-technology industries, where fast product introduction and extensive customization determine market success. As a result, companies must maintain production facilities, pilot production plants, engineering resources and even Research and Development (R & D) facilities all over the world. Apple Computer, for example, has built a global manufacturing and engineering infrastructure with facilities in California, Ireland and Singapore. This network allows Apple to introduce new products simultaneously in the American, European and Asian markets. Companies use the state-of-the-art markets as learning grounds for product development and effective production management, and then transfer this knowledge to their other production facilities worldwide. This rationale explains why Mercedes-Benz decided recently to locate a huge manufacturing plant in Vance, Alabama. The company recognizes that the United States is the state-of-the-art market for sport utility vehicles. It plans to produce those vehicles at the Vance plant and introduce them worldwide by 1997.

TECHNOLOGICAL FORCES

A peculiar trend which was prevalent in the last decade, besides globalization, was a limited number of producers which emerged due to diversity among products and uniformity across national markets. Product diversity has increased as products have grown more complex and differentiated and product life cycles have shortened. The share of the US market for high-technology goods supplied by imports from foreign-based companies rose from a negligible 5 per cent to more than 20 percent with the last decade. Moreover, the sources of such imports expanded beyond Europe to include Japan and the newly industrialized countries of Hong Kong, Singapore, South Korea

and Taiwan. There has been diffusion of technological knowledge and global low-cost manufacturing locations have emerged. In response to this diffusion of technological capability, multinational firms need to improve their ability to tap multiple sources of technology located in various countries. They also must be able to absorb quickly, and commercialize effectively, new technologies that, in many cases, were invented outside the firm thus overcoming the destructive and pervasive 'not-invented-here' attitude and resulting inertia. There has been technology sharing and interfirm collaborations. The well-known joint ventures in the auto industry between US and Japanese firms (GM-Toyota, Chrysler-Mitsubishi, Ford-Mazda) followed a similar pattern. US firms needed to obtain first-hand knowledge of Japanese production methods and accelerated product development cycles, while the Japanese producers were seeking ways to overcome US trade barriers and gain access to the vast American auto market. As competitive priorities in global products markets shift more towards product customization and fast new product development, firms are realizing the importance of co-location of manufacturing and product design facilities abroad. In certain product categories, such as Application Specific Integrated Circuits (ASICs), this was the main motivation for establishing design centres in foreign countries. Other industries such as pharmaceuticals and consumer electronics also have taken this approach.

GLOBAL COST FORCES

New competitive priorities in manufacturing industries, that is product and process conformance quality, delivery reliability and speed, customization and responsiveness to customers, have forced companies to reprioritize the cost factors that drive their global operations strategies. The Total Quality Management (TQM) revolution brought with it a focus on total quality costs, rather than just direct labour costs. Companies realized that early activities such as product design and worker training substantially impact production costs. They began to emphasize prevention rather than inspection. In addition, they quantified the costs of poor design, low input quality and poor workmanship by calculating internal and external failure costs. All these realizations placed access to skilled workers and quality suppliers high on the priority list for firms competing on quality. Similarly, Just-in-time (JIT) manufacturing methods, which companies widely adopted for the management of mass production systems, emphasized the importance of frequent deliveries by nearby suppliers. A number of high-technology industries have experienced dramatic growth in the capital intensity of production facilities. A state-of-the-art semiconductor factory, for instance, costs close to half a billion dollars. When R & D costs are included, the cost of production facilities for a new generation of electronic products can easily exceed \$ 1 billion. Similarly, huge numbers apply for the development and production of new drugs in the pharmaceutical industry. Such high costs drive firms to adopt an economies-of-scale strategy that concentrates production in a single location, typically in a country that has the required

labour and supplier infrastructures. They then achieve high-capacity utilization of the capital-intensive facility by aggressively pursuing the global market. Besides this the host government subsidies also become an important consideration.

POLITICAL AND MACROECONOMIC FORCES

Getting hit with unexpected or unreasonable currency devaluations in the foreign countries in which they operate is a nightmare for global operations managers. Managing exposure to changes in nominal and real exchange rates is a task which the global operations manager must master. If the economics are favourable, the firm may even go so far as to establish a supplier in a foreign country where one does not yet exist. For example, if the local currency is chronically undervalued, it is to the firm's advantage to shift most of its sourcing to local vendors. In any case, the firm may still want to source a limited amount of its inputs from less favourable suppliers in other countries if it feels that maintaining an ongoing relationship may help in the future when strategies need to be reversed. Becton Dickinson has built a global manufacturing network for its disposable syringe business, with production facilities in the United States, Ireland, Mexico and Brazil.

When the Mexican peso was devalued, the company quickly shifted its production to the Mexican plant, thereby gaining a cost advantage over its competitors' US factories. The emergence of trading blocks in Europe (Europe 1992), North America (NAFTA), and the Pacific Rim has serious implications for the way firms A? structure or rationalize their global manufacturing/sourcing networks. These trends are clearly apparent in many industries. For instance, before 1992, 3M's European plants turned out different versions of the same product for the various European countries. Today, 3M manufacturing plants produce goods for all of Europe and, in the process, realize significant cost savings. Similarly, Philips, Thomson, Electrolux and Ford are in the process of creating pan-European networks of factories (producing both components and finished goods). The trade protection mechanisms which exist in the form of tariff and non-tariff barriers effect the global operation strategy; but these are readily losing importance in the new borderless trade regime.

RETAILING IN COMPETITIVE ENVIRONMENT

Our analysis concludes firstly that competition problems are likely to be particularly prevalent in retailing, and secondly that there are significant differences between retailing and other areas of the economy, but that these differences are a matter of degree rather than exclusive to retailing. This means that, while the emphasis of competition policy will often be different in retailing, there *should* be no need for any major change in general competition policy methodologies.

However, in the process of analysing competition policy in retailing, we have discovered certain difficulties in applying some of the existing general methodologies. This does not mean that retailing requires separate methodologies, but rather that the current general methodologies need to be amended. This leads us to recommend that:

- Competition enquiries should always start with an *identification of the relevant competition issues* and then proceed with a preliminary analysis of demand and supply factors, particularly of the general characteristics of consumers and the dimensions of retailer competition;
- *Market definition* should ensure that the identification of the relevant market or markets serve to clarify the competition issue and that the retailer market is separately identified from the product market; and
- Existing approaches to *vertical restraints* need to be refined with respect to the increased role of retailers in their establishment and enforcement.

THE NATURE OF RETAILING

Retailing is the final link between the production of a good and the end-consumer. The economic characteristics of the end-consumer are thus crucial to the economics of retailing. Typically, the end-consumer can be characterised as being:

1. *Small* (in the sense that the size of a given purchase forms a small part both of the consumer's total expenditure and of the retailer's total sales);
2. *Immobile* (in the sense that they are often not able or willing to travel long distances in order to purchase the appropriate product); and
3. *Uninformed* (in the sense that they often do not know which products are available where, and what prices; and they may not be able to observe product quality in advance of purchase). The combination of these characteristics explains essentially why retailers exist.

The fact that consumers' purchases tend to be *small* means that retailers play an important role in preventing the exploitation of consumers by manufacturers: no single consumer would have any bargaining power against powerful manufacturers, whereas retailers can bargain strongly on their behalf.

The fact that consumers are *immobile* means that retailers play an important function in bringing the products to the consumers, to prevent the consumer from having to go to the product.

The fact that consumers are *uninformed* about the availability of products, their prices, and their qualities means that retailers carry out an important service in terms of providing information and quality assurances.

These characteristics also explain many aspects of retailing. For example, consumers' immobility means that geographical location and one-stop shopping will be important, while their lack of information means that price visibility, product range, and reputation/quality assurances will be important.

These aspects of retailing feed straight into a discussion of the nature of competition in retailing. We identify four dimensions of horizontal competition between retailers:

1. Pricing;
2. Geographical location;
3. Product selection; and
4. Retailer service.

These various dimensions in retailing have important consequences for competition policy, and it is crucial, when considering competition in retailing, to examine all of these dimensions.

A Framework for Assessing Competition in Retailing: Competition problems are likely to be particularly prevalent in retailing, since markets will only tend to work imperfectly when there are asymmetries in bargaining power (consumers are small), high transactions costs (consumers are immobile), information asymmetries (consumers are uninformed), or vertical restraints.

Indeed, there have been numerous cases where the OFT and MMC had to look at competition in retailing. The issues typically covered include:

1. *Pricing issues* (including excessive pricing, discriminatory pricing, predatory pricing, and loss leading);
2. *Merger issues* (for which market definition is crucial); and
3. *Vertical issues* (including vertical restraints, differential discounting, and own-brand competition). In order to analyse these various issues, we suggest the following framework:
 - a. Identification of competition issues
 - b. Preliminary analysis of retailing:
 - The characteristics of consumers; and
 - The dimensions of competition between retailers.
 - c. Market definition.
4. Barriers to entry
5. Competition assessment:
 - Market structure and merger issues;
 - Pricing issues; and
 - Vertical issues.

IDENTIFICATION OF COMPETITION ISSUES

We suggest that at the outset of every enquiry the competition issues are clearly identified. Each step in the assessment of competition in retailing has to be considered in the context of the specific competition issue, and cannot be dealt with in isolation. Some competition issues (such as those resulting from a merger between retailers) are easily identified, whereas a more general complaint of excessive prices for a class of products may require an in-depth investigation of both the retailing and the wholesale/manufacturing element of the supply chain.

SUPPLY CHAIN MANAGEMENT TECHNIQUES

VENDOR MANAGED INVENTORY (VMI)

Vendor Managed Inventory (VMI) is not a new idea – this supply chain management technique was established in the mid 1980's. The basic idea of VMI is that the supplier manages the inventory on behalf of the customer including replenishment. As the VMI should be beneficial for both parties, some limitations need to be defined:

- The business relationship between the supplier and the customer has to be established, strong and collaboration oriented, like partnership type for example.
- Deep trust and extensive sharing of information is required.
- The material flow should be ongoing (steady, not erratic) and preferably have some historical statistics (realized sales, usage and inventory figures) available.
- Effective management of VMI increases, if the items or item groups managed are few and substantial of volume. However, with the use of modern information and communication technology (ICT), smaller and numerous items can be managed as well.
- The VMI setup has to be defined jointly in details. The details include: products (or product groups) included, inventory levels with tolerances, demand (or consumption) levels, demand information sharing rules, transportation routes (modes, lead times, costs, etc.), warehousing details and exception handling.

In the pure VMI model the customer does not place purchase orders to the seller. The main tool used to operate the VMI is demand estimate or forecast. The customer is responsible for giving the estimate for a period of time and 'use' the goods according to the estimate within agreed tolerances. The customer is invoiced according to the real usage.

When a substantial amount of the supplier's volumes are included in VMI model, it brings benefits for more stable and predictable resource (production, warehousing, transportation, etc.) planning. It also decreases the Forrester effect; as the safety stock is built purely on known demand, not on assumptions. Especially in industries, which are very capital intensive, the effective and stable load of the machinery creates significant opportunities for increasing profitability. The VMI has potential for increasing customer satisfaction, as it enables the supplier to act beforehand for the high seasons, and thus ensure the delivery reliability. According to AMR Research, there are significant benefits to be achieved with VMI. When done correctly, some companies have extracted big benefits, such as 50% reduction in lead time, 20% to 70% reduction in inventories or in-stock improvements of 1% to 12%.

CONTINUOUS REPLENISHMENT

Another supply chain management method, Continuous Replenishment (CR), emerged in early 1990's. It moves one step further than VMI, so that the visibility to the customer's sales is included. The point of sales (POS) information is used in forecasting, and the forecasting is not purely based on inventory levels. Even though the CR method extends the VMI to cover the inter-company planning; the sales pattern creation is still the weakness of CR. The 2003 GMA Logistics Study indicates that in the Grocery, Food and CPG sector in the U.S. more than 83 percent of the participants of the survey stated that they have implemented either VMI or Continuous Replenishment (CR) or both. The typical benefits of VMI and/or CR include reduction in inventory, increase in inventory turns, increase in sales and reduction in retail out-of-stocks Berger, R.

FORECASTING

There are some thumb rules for demand forecasting: first, forecasts are always inaccurate. There is no process that will repeatedly match forecast to actual. Second, quantifying the error and using this to adapt the forecasts is essential to the process. Third, forecasts that are made at higher level (*e.g.*, product families instead of SKUs) will always be more accurate than at the lower levels. Creating forecasts at the lower levels and then grouping them accordingly for planning purposes is vital to a healthy process. Fourth, the planning horizon should be kept relatively short, since errors tend to be significantly fewer in the short term. And, finally, it must be remembered that forecasts are only the starting point for the planning process – forecasts help provide a basis for further refinements and the selection of a most likely scenario for the future.

Forecasting and demand estimation in companies usually result to multiple views and forecasts. Integrating the planning processes should lead into one commonly accepted forecast. Best-practice companies have implemented an integrated process, where second-guessing is eliminated. The forecast is made across all the functions resulting to an enterprise-wide forecast. Still today, in most companies each department makes its own forecasts based on multiple information sources. This usually results in excess inventory in process, non-optimal utilization of resources and thus decreases financial results of the company.

In most cases the forecasting and demand estimation is based on historical order or delivery information, which might not reflect the actual demand. However, actual consumer demand may be very different from the order stream. Each member of the supply chain observes the demand patterns of its customers and in turn produces a set of demands on its suppliers. But the decisions made in forecasting, setting inventory targets, lot sizing and purchasing act to transform (or distort) the demand picture.

The further a company is 'upstream' in the supply chain (that is, the further it is from the consumer), the more distorted is the order stream relative to consumer demand. This phenomenon is also known as the Forrester effect or 'bull-whip effect'. It is

important to see the meaning of the bull-whip effect both in downstream and upstream of the value chain; the variability caused by the gap (or unbalance) between companies' speculation and postponement of business activities.

This leads to a demand curve with steeper and steeper peaks and downs and with less and less reliability the further up the party is in the value chain. In the upper stream of the value chain the parties are forced to take extreme actions to survive the peaks only to find out that the demand was exaggerated. The total cost of the value chain is increased heavily and the reliability and timelines of the deliveries has suffered.

The cause of the steep demand curve and the fluctuations is not necessarily related to seasonality or economic trend variations. According to Lee *et al.* there are four main causes for the bullwhip effect; demand forecast updating, order batching, price fluctuations and rationing and shortage gaming. The lack of trust for the supplier as well as for the company's internal planning creates these disturbances. Also the fragmented organizations in companies have led to atomistic considerations, *i.e.*, sub-optimization of business activities, which cause the bullwhip effect to occur internally in the company. The multiplied effect of the intra-organizational and cross-enterprise sub-optimization and non-collaborative, non-synchronized, individual processes leads to the bullwhip curve.

The traditional bull-whip definition starts from the basis that each company speculates more in their incoming goods' inventory than in their outgoing goods' inventory, describes a reverse bull-whip effect, where the starting point is the opposite; the company speculates more in the outgoing inventory than in the incoming inventory. If there is a balance between the company's inventory management in incoming and outgoing side, there is no bull-whip effect within that company. In other words that means that the internal forecasting process is operating well, and the company has a common plan or forecast at both ends.

The bullwhip effect can be (at least partially) eliminated through information sharing with suppliers and customers – including intra-company suppliers and customers. By sharing information, a common understanding of the real demand can be achieved. Special attention should be paid to finding the items of information causing the overreactions. The final aim is to have centralized demand information– one forecast. The four material flow principles, which can be used to reduce the bullwhip effect, are control system, time compression, information transparency and echelon elimination.

COLLABORATION

McCarthy *et al.* define a collaborative supply chain as a long-term relationship among organizations actively working together as one towards common objectives. Another definition of collaboration is that it is 'Eliminating the honest mistakes' Bermudez, J. This refers to that business relations will remain arms length transactions, but the collaboration should target to more visibility and more structured ways of transferring information. The ultimate target of collaboration according to Bermudez should be

replacing the inventory with information. The criteria for selecting the collaboration partners are reliability, loyalty and innovation.

In general, managers of companies are far better at the practice of competition than they are at the art of cooperation. Many long-standing barriers exist to thwart successful implementation of collaborative relationships. One major barrier exists incentive systems. While many companies are looking to enhance overall supply chain performance, most company incentive systems still are focused on enterprise or even functional performance. Issues that earlier were regarded and handled internally, have become more problematic due to the growing fragmentation of providers, accompanied with problems of co-ordination. While outsourcing certain functions, companies also outsource the collaboration from company internal to cross-enterprise level. The fragmentation has also resulted in competitive and individualistic rather than collaborative culture and affected in undermining staff morale and created a climate of mistrust Miller, C. *et al.*

CONSUMER PACKED GOODS INDUSTRY AS THE CASE

The end products of the consumer goods industry have a large variation; they include products like fresh, dry or frozen food, detergents, cosmetics, pharmaceutical products, etc. The common denominator for all of them is that the consumer creates the principal demand impulses, which reflect the whole value chain.

Collaborative planning has been researched and also implemented in the consumer goods industry, but mainly in the retail area (*e.g.*, Wal-Mart), close to the consumers. The manufacturers, also called as brand owners, are also involved in numerous initiatives, but the suppliers of the manufacturers are still very much an “un-touched” area. The further one goes upstream, the less information of such initiatives can be found. This was also the main reason for this research project, and thus affected heavily on the case study selection process.

CONSUMER PACKED GOODS VALUE CHAIN

The value chain of the CPG sector can be described in many ways – there is no one right way to do that. Also it must be remembered that each party in the value chain is belonging to several other value chains, and actually the term chain should be understood as a part of a value network. The figure below describes roughly the parties involved in the CPG value chain.

The nature, size and power of the parties in the chain vary greatly. The consumer is by far the smallest party, but is the main source for demand information. The major retailers have significant power in controlling the chain (some talk about chain captaincy). The manufacturers can also be global corporations, but the converters are often very

small private owned companies. The supplier end of the chain often consists of capital intensive industries such as chemical or paper/board producers.

When looking from the upper stream end, the product changes significantly when it goes down the value chain. Therefore the logistics issues even in the transactional planning cannot be compared to distribution logistics when seen throughout the whole chain.

DESCRIPTION OF THE CASE STUDY

The case study involves three parties; three individual companies, representing a brand owner (manufacturer), a 1st tier supplier and 2nd tier supplier. The purpose of this selection is to evaluate the potential benefits of an operating collaborative forecasting business model in a three-entity demand chain. The target is to build a lean and transparent business model, where the key are is demand and supply information management and efficient transfer of it.

The evaluation of key improvement processes concentrates on planning and forecasting, and the transfer from traditional order-to-delivery processes. In the traditional process, the purchase order is the key impulse for the supplier, whereas in the new model the key input is the rolling forecast.

The challenges of the implementation come from change management, forecasting capabilities, openness and trust. Utilization of modern ICT technology also creates some challenges, but can be regarded as less challenging as the ones mentioned previously. As stated earlier, there are numerous examples of business models where collaborative forecasting is implemented between two parties. Therefore a selected starting point for this case study is that the collaborative forecasting business model exists already between two parties, and the model is extended one step further. In this particular case, the extension is made towards up-stream, *i.e.*, to the 2nd tier supplier.

As the forecast is the main impulse for the supply chain, the accuracy and relevance of it has significant importance. In a two-entity chain the forecast of the customer affects the supplier. In this case study, where the 2nd tier supplier is included in the same model, the initial forecast from the brand owner affects another step higher in the up-stream. Also the 1st tier supplier's planning process, where the brand owner's forecast is processed into a raw material forecast to the 2nd tier supplier, plays a key role.

STARTING POINT

As the case study involves three companies, it means that the collaborative cross-enterprise forecasting includes two steps; between the manufacturer and the 1st tier supplier, and between the 1st tier supplier and the 2nd tier supplier. The two steps have differences in the forecasting process, but also in issues like location, warehousing, transportation and replenishment mode. The case study will address these issues in the as-is analysis.

Also the production in all three parties involved is different; it varies from process industry to manufacturing. This implies also that the capacity planning and production cycles vary significantly, as well as the life cycle of the products of each party. Process industry is capital intensive, and the profitability comes from capacity utilisation efficiency. In manufacturing, where the production cycles are shorter, the working capital tied in the process has higher impact on the profitability. This results to that the key drivers for effective planning in each party are not the same, and cannot be compared as such.

EXPECTED RESULTS AND MEASURING

At the starting point of the case study, the collaborative forecasting model between the brand owner and the 1st tier supplier was already in place. Therefore it can be stated that the key metrics related to the efficiency of the demand chain are in better status than in the upstream part of this demand chain. The brand owner – 1st tier supplier part is used as a best practice when defining the targets for the 2nd tier supplier – 1st tier supplier part. The measuring is done comparing the results of the starting point analysis to a situation where the new business model has been in place for some months.

The new business model is expected to reduce the inventory levels and increase the inventory turnover in the 2nd tier supplier – 1st tier supplier part of the demand chain. Other expected benefits are: less out-of-stock situations, less non-optimal transports, better planning possibilities at the 2nd tier supplier resulting in higher production efficiency and increased customer satisfaction due to improved delivery reliability.

In order to obtain these results, a thorough commitment is expected from each partner. In practice it means *e.g.*, implementing new tools to monitor the supply chain, enhanced and even renewed working models as well as structured and defined ways to exchange information. The commitment involves also trust; the new business model must confirm the parties that they can trust both the model and their partners. Without established trust no collaboration will emerge nor survive, and the threat is that each party will again start sub-optimizing their own part of the chain.

At the time of writing this the case study is in progress. After finalizing the implementation of the business model as well as measurements of the key metrics, the final analysis will be made, and results received. The results will provide empirical evidence on the collaborative forecasting model theory. The results can also be used in similar supply chain initiatives as a guide for practical implementation of collaboration.

MANAGING SUPPLY CHAIN DEMAND VARIABILITY: THE BULLWHIP EFFECT

In a typical supply chain, there is a flow of funds, goods and information. All kinds of interactions are important for the well being of supply chain since there are different

parties and their interests involved. Every stage in the chain tries to increase its pie of profit via its operations with downstream and upstream. Moreover, interests of the parties generally conflict with each other.

For instance manufacturers' objective of making large production batches conflicts with the inventory reduction objective of the warehouses and distribution centres. However, the parties that compose the chain have different advantages against each other. Manufacturer has the advantage of supplying the products demanded and lower levels have the advantage of reaching demand information, retailer having the most precise demand information of the end customer since it faces the customer. Coordination is dependent on the arrangements between the links in the chain.

The problem is whether the owner of the information shares it with the others or not. Precise information is very crucial since all the decisions related with orders and production is tied to it. Accurate information will help to make better forecasts, enable lead-time reductions and give way to higher service levels. So, manufacturers will face lower variances and they will be capacity wise more transparent to retailers. Maybe retailers might be given more initiative such as revenue sharing in order to cooperate for information issue.

This proposal basically deals with the information flow and a problem related with it. *Bullwhip effect* (or alternatively whiplash effect) is the distortion in demand information in a supply chain. There is a common situation that retail orders to manufacturer do not coincide with the actual retail sales.

Moreover, as lead times increase, variability also increases. Retailers order from upstream according to the demand realizations occurred; they see realizations as signals of future demand. If forecast of the future demand is done accurately, life will be easier for all the parties in the chain. Furthermore, retailers want economies of scale in pricing so that they batch their orders rather than frequent orders in small amounts; they prefer fewer orders in larger amounts. Rationing is the situation, where demand of retailers outstrips the supply and manufacturer rations the products according to the ordered amounts to the retailers. Price variations lead to an uncertain market and variability in demand and this is reflected to higher levels of the chain in exaggerated amounts. Since bullwhip effect is an important phenomenon for supply chains, there should be a way to quantify it somehow so that the severity of the problem can be understood. There were previous studies that aim at defining and quantifying the bullwhip effect. This research basically aims to focus on quantifying bullwhip effect in a multi retailer supply chain.

LITERATURE REVIEW

There are several studies on bullwhip effect in literature. Lee *et al.* has a research on bullwhip effect, defining the phenomenon and analyzing the causes of it. There are precautions suggested to hedge against bullwhip problem in this study. In this paper, symptoms of bullwhip effect are listed as excessive inventory, poor product forecasts, insufficient or excessive capacities, poor customer service, uncertain production planning

and high costs. This is a basic study in literature that deals with the preliminaries of bullwhip effect. Moreover, Beer Distribution Game, which was developed at MIT in 1960's, is a successful tool to visualize the bullwhip effect. There are four roles, namely retailer, wholesaler, distributor and manufacturer, which try to satisfy demand in lower levels in the chain. The parties do not know the actual demand figures downstream, but the order amount from one level down. The resulting orders are highly variable specifically in upper levels. Because safety inventory held by one level is supposed to be the actual demand figure for the supplier and all plans are made accordingly. One scenario lets all of the players to see the customer demand. This time, information availability decreases the variability. Consequently, this game illustrates the bullwhip phenomenon very effectively.

Disney and Towill have a study on the effects of vendor-managed inventory on bullwhip effect reduction. Simulation models are presented to show manufacturer's ordering decisions. Important results gathered from this study are:

- With VMI implementation two sources of the bullwhip effect may be completely eliminated, *i.e.*, rationing game or the Houlihan effect, and the order batching effect or the Burbidge effect.
- VMI is also significantly better at responding to rogue changes in demand due to the promotion effect or to price induced variations.
- VMI offers a significant opportunity to reduce the bullwhip effect in real-world supply chains.

Lee *et al.* has a study on value of information sharing in a two level supply chain. The authors are dealing with the benefits of information sharing here. They claim that information sharing could provide significant inventory reduction and cost savings to the manufacturer. Moreover, they show that underlying demand process and the lead times have significant impact on the magnitudes of cost savings and inventory reductions that come along with information sharing. Demand is characterized by AR process in this research. Retailer observes inventory level after demand materializes and places its order; the process is the same with the one that we will be applying in our model with multiple retailers.

Backordering is permitted. At manufacturer side, demand is met one way or another. If manufacturer does not have enough resources on hand, it has to meet the demand from an alternative source. Ordering decisions of the parties are analyzed, manufacturer's decision is analyzed with and without information sharing. As a result it is presented that retailer's cost is not affected by information sharing whereas manufacturer's cost is decreased in case where information is shared. Moreover, it is concluded that if demand is highly correlated over time, highly variable or lead-time is long; manufacturer gains large reductions when information is shared among members of the chain.

Cachon has a paper on supply chain demand variability with scheduled ordering policies. There are one-supplier and N retailers that face stochastic demand. The main result of this study is that the supplier's demand variance will generally decline as the

retailers' order interval is lengthened or as their batch size is increased. Moreover, there are five variables presented that influence the supplier's demand variance. Two of the variables are structural: consumer demand variability and the number of retailers.

Remaining three variables are policy parameters, which are the retailers' batch size; the retailers' order interval length, and the alignment of the retailers' order intervals. Furthermore, Cachon and Fisher have a study on supply chain inventory management and value of shared information. They analyzed the value of sharing information in a model with one supplier, N identical retailers, and stationary stochastic consumer demand.

They compare a traditional information policy that does not use shared information with a full information policy that does exploit shared information. In a numerical study they find that supply chain costs are 2.2% lower on average with the full information policy than with the traditional information policy, and the maximum difference is 12.1%. In full information gathering policy, supplier reaches the inventory position data of the retailers. However, this study concludes that cost reductions via full information are not very significant; there can be more reductions via lead-time and batch size related differences.

The inspiring study behind this proposal is the Chen *et al.*'s "Quantifying the Bullwhip Effect in a Simple Supply Chain: The Impact of Forecasting, Lead Times and Information". This study intends to be an extension to that study. The distinguishing feature of the model to be analyzed is the multi retailer environment set here rather than a two-stage chain with one retailer only. An in depth reference will be given to that study since the proposal is an extension. Chen *et al.* considers a supply chain, a single retailer observes its inventory level and places its order to a single manufacturer. After order is given, retailer fills customer demand for the period. Unmet demands can be backlogged. There is a fixed lead-time, L , between placement of orders and receipt of the orders. They assume order-up-to inventory policy. The forecast method is simple moving average, where p is number of periods to base the forecast on.

There are analyses with the scenarios in which excess inventory is returned and not returned. It is found that the way one treats excess inventory has little impact on the increase in variability for different values of p and ρ , which is the correlation parameter for demand process.

Moreover, when forecasts are dependent on larger p values, increase in variability is negligible. Also, one of the results is that when lead times increase, variability increases. Correlation coefficient related results are such that if the demands are positively correlated, variability is smaller and when the demands are negatively correlated for even values of p , the variability will be smaller whereas the variability will be higher for odd values of p .

Furthermore, in a multistage supply chain, where all demand information is shared by all parties, and the same forecasting technique is used by all, the bullwhip effect will not be completely eliminated.

In addition to those results, it is found that for supply chains with centralized demand information, increase in variability at each stage is an additive function of the lead time and the lead time squared, while for supply chains without centralized information increase in variability is a multiplicative. Thus, centralizing demand information can significantly reduce the bullwhip effect. One of the drawbacks of this paper is the lack of complexities related with real supply chains. Ryan has a study on multi retailer case but there are not any other researches on this issue in the literature. The motivation lies in here.

INTERDEPENDENCE IS A HIGHER VALUE THAN INDEPENDENCE

As rightly said in the above book by Stephen Covey, the value of interdependence is as much important in the business of Retail as it is important in the business of life. Standing on the threshold of a retail revolution and witnessing a fast changing retail landscape, the retail sector is poised for a big leap. Currently retail sector in India accounts for Rs. 55,000 crore (\$12.4 billion) business at current prices in the calendar year 2006, increasing its share to 4.6% of the total Indian Retail Value that stood at Rs. 12,00,000 crore (\$270 billion). With the potential of crossing Rs 2,00,000 crore (\$45 billion) business by the Year 2010, generating employment for some 2.5 million people in various retail operations and over 10 million additional workforce in retail support activities including contract production and processing, supply chain and logistics, retail real estate development and management, etc.; the retail sector is growing at a scorching pace of about 37 percent in 2007 and expected to grow by 42 per cent in 2008. With this enormous growth, the retail sector is also facing challenges on the fronts of escalating real estate cost, scarcity of skilled workforce and structured supply of merchandise.

IMPORTANCE OF SUPPLY CHAIN AND LOGISTICS MANAGEMENT

One of the most important challenge in organized retail in India is faced by poor supply chain and logistics management. The importance can be understood by the fact that the logistics management cost component in India is as high as 7% -10% against the global average of 4% - 5% of the total retail price. Therefore, the margins in the retail sector can be improved by 3% - 5% by just improving the supply chain and logistics management.

In India, with demand for end-to-end logistics solutions far outstripping supply, the logistics market for organised retail is pegged at \$50 million and is growing at 16%. It is expected to reach \$120-\$130 million by 2010. Organised retail on the other hand is growing at 400% and is expected to reach around \$30 billion by 2010. Even supply

chain and logistics firms like Hong Kong based Heng Tai Consumables and ABS Procurement Co and ACM China (the greenhouse specialist) is also eyeing the opportunity for managing the supplies.

The supply chain management is logistics aspect of a value delivery chain. It comprises all of the parties that participate in the retail logistics process: Manufacturers, Wholesalers, Third Party Specialists like Shippers, Order Fulfilment House, etc. and the Retailer. Here, logistics is the total process of planning, implementing and coordinating the physical movement of merchandise from manufacturer to retailer to customer in the most timely, effective and cost efficient manner possible. Logistics regards order processing and fulfilment, transportation, warehousing, customer service and inventory management as interdependent functions in the value delivery chain. It oversees inventory management decisions as items travel through a retail supply chain. If a logistics system works well, the retail firm reduces stock outs, hold down inventories and improve customer service – all at the same time.

Logistics and Supply Chain enables an organized retailer to move or store products more effectively. Efficient logistics management not only prevents needless movement of goods, vehicles transferring products back and forth; but also frees up storage space for more productive use. Retail analysts say on-time order replenishments will become even more critical once the Wal-Mart/Bharti combine begins operations - the American retailer works almost entirely on cross-docking and is likely to demand higher service levels, including potential levies for delays in shipment.

The efficiency and effectiveness of supply chain and logistics management can also be understood by the fact that modern retail stores maintain lower inventories than traditional retail. In India, generally in the traditional kirana stores, three weeks inventories are kept; while in a modern retail store like Hypercity, it's nine days and it's under two weeks for Food Bazaar. Now, it is beneficial for both the manufacturer as well as the retailer. If we go through the following food supply chain in India, we find that a lot can be improved by maintaining the supply chain and logistics.

FOOD SUPPLY CHAIN IN INDIA

In India, about 60 percent of food quality is lost in the supply chain from the farm to the final consumer. Consumers actually end up paying approximately about 35 percent more than what they could be paying if the supply chain was improved, because of wastage as well as multiple margins in the current supply structure. The farmer in India gets around 30 percent of what the consumer pays at the retail store. Compare this with the situation obtaining in the USA, where farmers can receive up to 70 percent of the final retail price and wastage levels are as low as 4 to 6 percent. One can easily understand the benefits that could be generated from emulating those practices and tapping that expertise for the supply chain in India.

As supply chain Management involves procuring the right inputs (raw materials, components and capital equipments); converting them efficiently into finished products

and dispatching them to the final destinations; there is a need to study as to how the company's suppliers obtain their inputs. The supply chain perspective can help the retailers identify superior suppliers and distributors and help them improve productivity, which ultimately brings down the customers costs. At the same time, Market logistics helps planning the infrastructure to meet demand, then implementing and controlling the physical flows of material and final goods from point of origin to points of use, to meet customer requirements at a profit.

Till now most retailers in India have invested majority into the front end, but relatively little on the back end and supply chain. Even in countries like the USA, Germany and England, where organized retail is highly developed; supply chain efficiency is a major concern. The nature of retail sector in India is different from other countries around the world. The organized retail sector in India is highly fragmented and there are huge inefficiencies in the supply chain.

The most important part of retailing business is to find a balance between investing in front-end and back-end operations. The channel dynamics is going to change over next couple of years as the retailers start growing in size and their bargaining power is likely to increase. Probably that would bring some kind of mutual understanding between manufactures and retailers to develop strong supply chain network. In such a scenario, both the existing operators and new operators must put collaborative efforts to phase out inefficiencies in the supply chain network.

Now, let us try to find out what efforts are being taken up by the big retailers in India like Future Group with retail stores like Big Bazaar and Pantaloons, Reliance Retail and Wal-Mart and Bharti to improve the efficiency and effectiveness of supply chain and logistics. We will also try to find out the changed role of Agriculture Produce Marketing Cooperatives and third party sourcing firms.

FUTURE GROUP

Future Group is the country's leading retail business group that caters to the entire Indian consumption space. It operates through six verticals: Future Retail (encompassing all lines of retail business), Future Capital (financial products and services), Future Brands (all brands owned or managed by group companies), Future Space (management of retail real estate), Future Logistics (management of supply chain and distribution) and Future Media (development and management of retail media spaces).

The group's flagship company, Pantaloon Retail (India) Limited operates over 5 million square feet through 450 stores in 40 cities. Some of its leading retail formats include, Pantaloons, Big Bazaar, Central, Food Bazaar, Home Town, EZone, Depot, Health and Beauty Malls and on-line retail format, www.futurebazaar.com.

The group's joint venture partners include Italian insurance major, Generali, French retailer ETAM group, US-based stationary products retailer, Staples and UK-based Lee Cooper and India-based Talwalkar's, Blue Foods and Liberty Shoes. Future Group is working on the vendor network as well as the logistics network. The company has

identified up to 40 anchor vendors, each with turnovers of US\$45 million, to achieve economies of scale. The group is also keen to ensure that its smaller vendors are able to reach turnovers of around US\$1 million and a growth rate of 40% annually, to be able to pass on the benefits of scales. The company is also working towards bringing its 1,200 vendors on-line, like Wal-Mart.

Going further in this direction, the Future Group has also launched Future Logistics initially aimed at handling the supply chain logistics of the group. However, sensing immense opportunity in this area, the company is now looking to offer its services to its 1000-odd vendors, spread across consumer related goods, to reach a targeted turnover of about Rs. 700 crore by 2010. The thrust at present will be on modes of surface transport like roads and rail only. However, at a later stage, sea and air modes might also be considered as per the requirement, said sources.

In India, Future group derives significant economies of scale in managing their supply chain. With more than 170000 products, the company maintains a strong supplier relationship in a partnership mode, avoiding the exploitative supplier – buyer transactional philosophy.

The IT enabled back-end operations and supply chain management increases the reliability and efficiency of the business. As part of the operation, Future Group is also undertaking to reduce its warehousing costs through a consolidation process. In a country like India, where most retail stores are located in the heart of the city—where rents are high and storage space is scarce—supply chain management has even more serious business implications. Future Logistics now handles two-and-a-half million SKUs (or stock keeping units) a day across the Future Group's various retail formats around the country. By 2010, this number is expected to increase to more than 30 million SKUs a day. Even with 98% accuracy, some 600,000 pieces will not be delivered correctly, resulting in an estimated sales loss of more than Rs 4 crore a day.

The biggest driver in consumer logistics is going to be zero defect in managing the supply chain. While infrastructure, technology, automation, processes and people will all play an important role, zero defect can only be achieved through vertical integration across the entire supply chain—from raw material supply, production, wholesale and retail. The different parts of the supply chain will no longer be able to work in silos as they do today.

RELIANCE FRESH

Reliance Retail is also going to open one store for every 3,000 families within a radius of 2 km across all locations by 2011. The company is competing directly with the large number of traditional local provision stores. Reliance Retail is either going to set up new stores in the identified areas or take over existing stores. The company has already done that in Mumbai and other cities.

Of the four million sq ft of retail space to be created under the “Reliance Fresh” brand (for groceries), one million will be through acquisitions. The retailer is also moving

into laundry, personal care and apparel product lines, in which it plans to launch private labels. Reliance is planning to roll out its speciality format stores this year, beginning with consumer durables, for which it has struck sourcing deals with companies in Hong Kong, the Chinese mainland and with Videocon in India.

To strengthen its links with farmers, the company is setting up integrated agri-retail business centres, which include three processing and distribution centres, 51 retail outlets for farmers and 75 rural business hubs, all with an investment of US\$445 million. Many companies, looking at the retail boom in food and grocery, are setting up ventures to help retailers source these goods.

Reliance Logistics Ltd. part of Reliance Industries Ltd., currently handles Reliance Retail's logistics services.

WAL-MART AND BHARTI

The success of Wal-Mart is well known all across the world. One of the major factors behind their success is the right implementation of supply and logistics management. Now the same Supply Chain and Logistics Management take a front seat here and that's why Wal-Mart is coming to India in a joint venture with Bharti Group. Here, Wal-Mart is going to manage the back end operation, while Bharti will manage the front end operations. Wal-Mart has also stated that it would replicate its global supply chain model in India, while taking into account the unique features of the Indian market. They are also going to emphasise on local sourcing of goods. Besides sourcing locally, Wal-Mart, through its international operations is also in a position to source globally. The company is set to roll out its first set of stores by the first quarter of 2008, in cities that have a population of one million. Wal-Mart claims it will take 35% of the Indian retail market by 2015.

It is the sheer importance of the logistics management that Wal-Mart's fully-owned logistics arm Gazeley has already confirmed its India foray and is going to look after the Wal-Mart and Bharti retail venture. They are closely studying various logistics providers like Radhakrishnan Foods, before they finally closes on its India model. Again, Bharti Enterprises is directly negotiating with the rail authorities instead of negotiating with a logistics provider.

WAL-MART AND BHARTI FIELD FRESH

Just like Reliance Fresh, Bharti Group in a joint venture with NM Rothschild is launching Field Fresh to provide premium quality fresh produce to markets worldwide. It has over 5,000 acres of land under cultivation all over the country producing many varieties of fruits and vegetables and is planning to double land under cultivation by the end of 2007.

The company is to supply fresh produce to the Bharti-Wal-Mart venture. To ensure best qualities and varieties, Field Fresh has engaged ACM China, an industry leader in

building greenhouses, to set up state-of-the-art glass-based greenhouses at the Field Fresh Agri Centre of Excellence in the Punjab. Field Fresh is also planning investments to the tune of US\$220 million in the backend, including investments in cold chains and warehouses. Bharti's Field Fresh will enter this segment within the next three months. A number of companies are also venturing into this segment to service the backend needs of retailers.

ROLE OF IT IN SUPPLY CHAIN MANAGEMENT

SCM is essentially an information-driven function. Time and responsiveness play a vital role in SCM and information technology has an important role to play towards this goal. The most critical task in any information-driven function is processing of large chunks of data and maintaining a large database. In the past, these records were maintained manually. Today, any global company cannot survive without thinking of integrating its supply chain. Another milestone has been the recent development of the Internet. All these developments are bringing the world closer and closer.

SCM is a function that calls for extensive coordination with both external and internal environment. IT has a substantial role to play in customer servicing. Cargo management companies like DHL, FedEx, etc. are using this medium extensively for better customer servicing and information flow. If a customer wants to know the status of his shipment, all he needs to do is to log on to the website and enter his airway bill number. Immediately, within seconds he gets the status.

Many companies feel that their current information management systems do not provide adequate support for their supply chain initiatives. Consequently there is an ever-increasing need for fully integrated supply chain information management solutions which incorporate all the functionality of network strategy/supply configuration, demand planning, transportation management and warehouse management.

IT- BASED TOOLS FOR SUPPLY CHAIN MANAGEMENT

A number of IT-based SCM tools are now available to provide intelligent decision support and execution management. They can be transaction processing systems focused on day to day operations; operational planning systems or strategic planning tools used to redesign the supply chain infrastructure. Some of the major developments in IT which are transforming the supply chain today are as follows:-

ELECTRONIC DATA INTERCHANGE (EDI)

Electronic Data Interchange (EDI) is the inter-organizational exchange of business documentation in a structured machine-processable form. It consists of standardized

electronic message formats for common business documents such as request for quotations, purchase orders, invoices and other standard business correspondence documents.

These electronic transactions sets enable the computer in one company to communicate with the computer in another organization without actually producing paper documents. All human efforts required to sort and transport the document are eliminated. A fully integrated EDI solution adds speed and efficiency to business processes enabling the organization to maximize resources, minimize waste and increase customer satisfaction. The key benefits of EDI are:

- EDI helps in reducing transaction costs across the supply chain of an organization through a reduction in labour and material costs, communication costs and administrative costs.
- With no data re-entry, EDI ensures grater accuracy of information while reducing the likelihood of costly errors.
- Reduced inventory and its associated costs is one of the biggest advantages of EDI.
- Critical information is available instantly, enabling quicker response to changes in the marketplace.
- EDI increases business opportunities by making it easier for customer to do business with the company.

ENTERPRISE RESOURCE PLANNING (ERP)

ERP is a philosophy where one tries to integrate the value chain of the organization to reduce the wastage or slack of the entire process of delivering the end product right from the sourcing of raw materials. ERP is a comprehensive planning and control framework that has evolved over a thirty year time. It finds its genesis in materials requirement planning (MRP), manufacturing requirement planning (MRP II), relational Database management systems (RDBMS) and 4th generation computer languages (4GL). It also is influenced by just in time (JIT) and computer integrated manufacturing (CIM) and takes advantage of latest IT developments such as client-server computing and Internet.

The key benefits of ERP are:

- ERP links all the activities in the organization with customer orders and thus the customer becomes the key focus of all departments.
- An ERP system regulates the flow of goods from a number of manufacturing sites to the stocking points. It captures and consolidates related data from the retailer that can be used to change the production schedule quickly.
- ERP as a tool can enhance overall performance by reduction of costs, increased per capita productivity and improved quality of goods and services.

INTERNET COMMERCE

Internet is transforming the entire nature of supply chains by eliminating middlemen, making commerce more democratic and creating a frictionless economy. Internet commerce is changing the manner in which the entire supply chain is managed today.

It is reducing entry barriers for the new entrants and the costs of operations for the existing ones at the same time and also offering the customers a wider choice for selection.

BAR CODING

A bar code is a grouping of parallel bars (usually blocks) of varying widths separated by light spaces (usually white) of varying width. Scanner is used to read the bars and spaces and it uses software to interpret their meaning. In the supply chain, the accurate, rapid identification of products and use of this information in controlling the entire process have been key factors. The following are the benefits of bar code technology in supply chain:

- Speed data entry
- Enhances data accuracy
- Reduces materials handling labour
- Verifies orders at receiving and shipping
- Improves customer service.

COMMUNICATION TECHNOLOGY

Information and Communication Technology (ICT) also significantly enhances the supply chain performance through faster and widespread communication. Applications of radio frequency, satellite communications and image processing technologies have overcome the problems caused by product movement and geographic decentralization. Improved customer service is provided in the form of more timely definition of tasks, quicker transport tracing, and faster transfer of sales and inventory information. The availability of shared information has led to increase efficiency and effectiveness across the supply chain. In today's competitive global environment, the onus is now on supply chain management. Companies whose SCM is competitive will gain in the long run. The advances in information technology have transformed the supply chain elements and enabled organizations to make significant improvements in productivity and competitiveness. The IT-enabled options available today for supply chain facilitate real time information flow through networking and electronic data transfer, faster response time and improved decision making, through timely availability of relevant information.

Theories of Supply Chain Management

Currently there is a gap in the literature available on supply chain management studies: there is no theoretical support for explaining the existence and the boundaries of supply chain management. A few authors such as Halldorsson, *et al.* (2003), Ketchen and Hult (2006) and Lavassani, *et al.* (2009) have tried to provide theoretical foundations for different areas related to supply chain by employing organizational theories. These theories include:

- Resource-Based View (RBV)
- Transaction Cost Analysis (TCA)
- Knowledge-Based View (KBV)
- Strategic Choice Theory (SCT)
- Agency Theory (AT)
- Institutional theory (InT)
- Systems Theory (ST)
- Network Perspective (NP)
- Materials Logistics Management (MLM)
- Just-in-Time (JIT)
- Material Requirements Planning (MRP)
- Theory of Constraints (TOC)
- Total Quality Management (TQM)

- Agile Manufacturing
- Time Based Competition (TBC)
- Quick Response Manufacturing (QRM)
- Customer Relationship Management (CRM).

SUPPLY CHAIN CENTROIDS

In the study of supply chain management, the concept of centroids has become an important economic consideration. A centroid is a place that has a high proportion of a country's population and a high proportion of its manufacturing, generally within 500 mi (805 km). In the U.S., two major supply chain centroids have been defined, one near Dayton, Ohio and a second near Riverside, California.

The centroid near Dayton is particularly important because it is closest to the population centre of the US and Canada. Dayton is within 500 miles of 60% of the population and manufacturing capacity of the U.S., as well as 60 percent of Canada's population. The region includes the Interstate 70/75 interchange, which is one of the busiest in the nation with 154,000 vehicles passing through in a day. Of those, anywhere between 30 percent and 35 percent are trucks hauling goods. In addition, the I-75 corridor is home to the busiest north-south rail route east of the Mississippi.

TAX EFFICIENT SUPPLY CHAIN MANAGEMENT

Tax Efficient Supply Chain Management is a business model which consider the effect of Tax in design and implementation of supply chain management. As the consequence of Globalization, business which is cross-nation should pay different tax rates in different countries. Due to the differences, global players have the opportunity to calculate and optimize supply chain based on tax efficiency legally. It is used as a method of gaining more profit for company which owns global supply chain.

SUPPLY CHAIN SUSTAINABILITY

Supply chain sustainability is a business issue affecting an organisation's supply chain or logistics network in terms of environmental, risk, and waste costs. Sustainability in the supply chain is increasingly seen among high-level executives as essential to delivering long-term profitability and has replaced monetary cost, value, and speed as the dominant topic of discussion among purchasing and supply professionals. Supply chains are critical links that connect an organisation's inputs to its outputs. Traditional challenges have included lowering costs, ensuring just-in-time delivery, and shrinking transportation times to allow better reaction to business challenges. However, the increasing environmental costs of these networks and growing consumer pressure for eco-friendly products has led many organisations to look at supply chain sustainability as a new measure of profitable logistics management. This shift is reflected by an

understanding that sustainable supply chains frequently mean profitable supply chains. Many companies are limited to measuring the sustainability of their own business operations and are unable to extend this evaluation to their suppliers and customers. This makes determining their true environmental costs highly challenging and reduces their ability to remove waste from the supply chains. However much progress has been made in defining supply chain sustainability and benchmarking tools are now available that enable sustainability action plans to be developed and implemented.

One of the key requirements of successful sustainable supply chains is collaboration. The practice of collaboration — such as sharing distribution to reduce waste by ensuring that half-empty vehicles do not get sent out and that deliveries to the same address are on the same truck — is not widespread because many companies fear a loss of commercial control by working with others. Investment in alternative modes of transportation — such as use of canals and airships — can play an important role in helping companies reduce the cost and environmental impact of their deliveries.

THREE TIERS OF SUSTAINABILITY

In 2008, The Future Laboratory produced a ranking system for the different levels of sustainability being achieved by organisations. This was called the Three Tiers of Sustainability:

TIER 1: GETTING THE BASICS RIGHT

This is the base level and is the stage in which the majority of organisations are at. Companies employ simple measures such as switching lights and PCs off when left idle, recycling paper, and using greener forms of travel with the purpose of reducing the day-to-day carbon footprint. Some companies also employ self-service technologies such as centralised procurement and teleconferencing.

TIER 2: LEARNING TO THINK SUSTAINABLY

This is the second level, where companies begin to realise the need to embed sustainability into supply chain operations. Companies tend to achieve this level when they assess their impact across a local range of operations. In terms of the supply chain, this could involve supplier management, product design, manufacturing rationalisation, and distribution optimisation.

TIER 3: THE SCIENCE OF SUSTAINABILITY

The third tier of supply chain sustainability uses auditing and benchmarks to provide a framework for governing sustainable supply chain operations. This gives clarity around the environmental impact of adjustments to supply chain agility, flexibility, and cost in the supply chain network. Moving towards this level means being driven by the current

climate (in which companies recognise cost savings through green operations as being significant) as well as pushing emerging regulations and standards at both an industry and governmental level.

COMPONENTS OF SUPPLY CHAIN MANAGEMENT INTEGRATION

THE MANAGEMENT COMPONENTS OF SCM

The SCM components are the third element of the foursquare circulation framework. The level of integration and management of a business process link is a function of the number and level, ranging from low to high, of components added to the link (Ellram and Cooper, 1990; Houlihan, 1985). Consequently, adding more management components or increasing the level of each component can increase the level of integration of the business process link. The literature on business process re-engineering, buyer-supplier relationships, and SCM suggests various possible components that must receive managerial attention when managing supply relationships. Lambert and Cooper (2000) identified the following components:

- Planning and control
- Work structure
- Organization structure
- Product flow facility structure
- Information flow facility structure
- Management methods
- Power and leadership structure
- Risk and reward structure
- Culture and attitude.

However, a more careful examination of the existing literature leads to a more comprehensive understanding of what should be the key critical supply chain components, the “branches” of the previous identified supply chain business processes, that is, what kind of relationship the components may have that are related to suppliers and customers. Bowersox and Closs states that the emphasis on cooperation represents the synergism leading to the highest level of joint achievement (Bowersox and Closs, 1996). A primary level channel participant is a business that is willing to participate in the inventory ownership responsibility or assume other aspects of financial risk, thus including primary level components (Bowersox and Closs, 1996). A secondary level participant (specialized) is a business that participates in channel relationships by performing essential services for primary participants, including secondary level components, which support primary participants. Third level channel participants and components that support the primary level channel participants and are the fundamental branches of the secondary level components may also be included.

Consequently, Lambert and Cooper's framework of supply chain components does not lead to any conclusion about what are the primary or secondary (specialized) level supply chain components (see Bowersox and Closs, 1996, p. 93). That is, what supply chain components should be viewed as primary or secondary, how should these components be structured in order to have a more comprehensive supply chain structure, and how to examine the supply chain as an integrative one (See above sections 2.1 and 3.1).

Reverse supply chain Reverse logistics is the process of managing the return of goods. Reverse logistics is also referred to as "Aftermarket Customer Services". In other words, any time money is taken from a company's warranty reserve or service logistics budget one can speak of a reverse logistics operation.

SUPPLY CHAIN SYSTEMS AND VALUE

Supply chain systems configure value for those that organise the networks. Value is the additional revenue over and above the costs of building the network. Co-creating value and sharing the benefits appropriately to encourage effective participation is a key challenge for any supply system. Tony Hines defines value as follows: "Ultimately it is the customer who pays the price for service delivered that confirms value and not the producer who simply adds cost until that point".

GLOBAL SUPPLY CHAIN MANAGEMENT

Global supply chains pose challenges regarding both quantity and value:

Supply and Value Chain Trends:

- Globalization
- Increased cross border sourcing
- Collaboration for parts of value chain with low-cost providers
- Shared service centres for logistical and administrative functions
- Increasingly global operations, which require increasingly global coordination and planning to achieve global optimums
- Complex problems involve also midsized companies to an increasing degree.

These trends have many benefits for manufacturers because they make possible larger lot sizes, lower taxes, and better environments (culture, infrastructure, special tax zones, sophisticated OEM) for their products. Meanwhile, on top of the problems recognized in supply chain management, there will be many more challenges when the scope of supply chains is global. This is because with a supply chain of a larger scope, the lead time is much longer. Furthermore, there are more issues involved such as multi-currencies, different policies and different laws.

The consequent problems include:

1. Different currencies and valuations in different countries;
2. Different tax laws (Tax Efficient Supply Chain Management);

3. Different trading protocols;
4. Lack of transparency of cost and profit.

SUPPLY CHAIN NETWORK

Due to the rapid advancement of technology such as pervasive or ubiquitous wireless and internet networks, connective product marking technologies like RFID and emerging standards for the use of these defining specific locations using Global Location Number(s), the basic supply chain is rapidly evolving into what is known as a Supply Chain Network.

ORIGINS OF THE CONCEPT

All organizations have or can purchase the components to build a supply chain network, it is the collection of physical locations, transportation vehicles and supporting systems through which the products and services your firm markets are managed and ultimately delivered.

Physical locations included in a Supply Chain Network can be manufacturing plants, storage warehouses, carrier crossdocks, major distribution centres, ports, intermodal terminals whether owned by your company, your suppliers, your transport carrier, a third-party logistics provider, a retail store or your end customer. Transportation modes that operate within a Supply Chain Network can include the many different types of trucks, trains for boxcar or intermodal unit movement, container ships or cargo planes.

The many systems which can be utilized to manage and improve a Supply Chain Network include Order Management Systems, Warehouse Management System, Transportation Management Systems, Strategic Logistics Modelling, Inventory Management Systems, Replenishment Systems, Supply Chain Visibility, Optimization Tools and more. Emerging technologies and standards such as the RFID and the GS1 Global Standards are now making it possible to automate these Supply Chain Networks in a real time manner making them more efficient than the simple supply chain of the past.

SUPPLY CHAIN SECURITY

Supply chain security refers to efforts to enhance the security of the supply chain: the transport and logistics system for the world's cargo. It combines traditional practices of supply chain management with the security requirements of the system, which are driven by threats such as terrorism, piracy, and theft. Some analysts have raised concerns about supply chain security overreach. Typical supply chain security activities include:

- Credentialing of participants in the supply chain.
- Screening and validating of the contents of cargo being shipped.
- Advance notification of the contents to the destination country.

- Ensuring the security of cargo while in-transit via the use of locks and tamper-proof seals.
- Inspecting cargo on entry.

KEY INITIATIVES

There are a number of supply chain security initiatives in the United States and around the world today. These include:

- The Customs Trade Partnership against Terrorism (C-TPAT), a voluntary compliance programme for companies to improve the security of their corporate supply chains.
- The World Customs Organization (WCO) adopted the Framework of Standards to Secure and Facilitate Global Trade in 2005, which consists of supply chain security standards for Customs administrations including Authorized Economic Operator (AEO) programmes.
- The Container Security Initiative (CSI), a programme led by U.S. Customs and Border Protection in the Department of Homeland Security focused on screening containers at foreign ports.
- The World Customs Organization.
- Efforts for countries around the world to implement and enforce the International Ship and Port Facility Security Code (ISPS Code), an agreement of 148 countries that are members of the International Maritime Organization (IMO).
- Pilot initiatives by companies in the private sector to track and monitor the integrity of cargo containers moving around the world using technologies such as RFID and GPS.
- The International Organization for Standardization have released a series of Standards for the establishment and management of supply chain security. ISO/PAS 28000 - Specification for security management systems for the supply chain, offers public and private enterprise an international high-level management standard that enables organisations to utilise a globally consistent management approach to applying supply chain security initiatives.

LOGISTICS MANAGEMENT

The term Logistics Management or supply chain management is that part of Supply Chain Management that plans, implements, and controls the efficient, effective, forward, and reverse flow and storage of goods, services, and related information between the point of origin and the point of consumption in order to meet customers' requirements.

LOGISTICS MANAGEMENT SOFTWARE

Software is used for logistics automation which helps the supply chain industry in automating the workflow as well as management of the system. There are very few generalized software available in the new market in the said topology.

This is because there is no rule to generalize the system as well as work flow even though the practice is more or less the same. Most of the commercial companies do use one or the other of the custom solutions. But there are various software solutions that are being used within the departments of logistics. There are a few departments in Logistics, namely: Conventional Department, Container Department, Warehouse, Marine Engineering, Heavy Haulage, etc.

Software used in these departments:

- Conventional department: CVT software/CTMS software
- Container Trucking: CTMS software
- Warehouse: WMS/WCS.

Improving Effectiveness of Logistics Management:

1. Logistical Network
2. Information
3. Transportation
4. Sound Inventory Management
5. Warehousing, Materials Handling and Packaging.

CUSTOMER-DRIVEN SUPPLY CHAIN

In a customer driven supply chain, end-user demand drives all activities among trading partners.

Customer driven supply chain distinguishes between actions that provide customers with real value, and actions which just add costs. It facilitates and possibly maximizes collaboration with suppliers. It reduces throughout time, touchpoints and decision points. It increases machine and transport effectiveness. It provides quick and easy access to inventory for sales and service. It reduces excess inventory, stockouts, shrinkage, and order delays, as well as automates inventory data transfers between stores and warehouses. It reduces purchase order errors. It accelerates time to market.

DEMAND CHAIN MANAGEMENT

Demand chain management is the management of upstream and downstream relationships between suppliers and customers to deliver the best value to the customer at the least cost to the demand chain as a whole. The term demand chain management is used to denote the concept commonly referred to as supply chain management, however with special regard to the customer pull. In that sense, demand chain management software tools bridge the gap between the customer relationship management and the supply chain management.

The organization's supply chain processes are managed to deliver best value according to the demand of the customers. A study of the university in Wageningen (the Netherlands) sees DCM as an extension of supply chain management, due to its

incorporation of the market orientation perspective on its concept. While the term “demand-driven supply chain or network” denotes a set of concepts, the term “demand-driven execution” or DDE is used to express the means of achieving those concepts.

DEMAND-DRIVEN SUPPLY NETWORK

A *Demand-driven supply network (DDSN)* is one method of supply chain management which involves building supply chains in response to demand signals. The main force of DDSN is that it is driven by customers demand. In comparison with the traditional supply chain, DDSN uses the pull technique. It gives DDSN market opportunities to share more information and to collaborate with others in the supply chain.

DDSN uses a capability model that consist of four levels. The first level is *Reacting*, the second level is *Anticipating*, the third level is *Collaborating* and the last level is *Orchestrating*. The first two levels focus on the internal supply chain while the last two levels concentrate on external relations throughout the Extended Enterprise.

COMPETITIVE ADVANTAGES

To create sustainable competitive advantages with DDSN, companies have to deal with three conditions: *Alignment* (create shared incentives), *Agility* (respond quickly to short-term change) and *Adaptability* (adjust design of the supply chain).

MISCONCEPTIONS

There are five commonly made misconceptions of demand driven (DDSN):.

1. Companies might think they are demand driven because they have a good forecast of their company.
2. They have implemented lean manufacturing.
3. They have great data on all their customers.
4. They think it is a technology project and the corporate forecast is a demand visibility signal.
5. They have a better view of customers demand.

An important component of DDSN is DDM (“real-time” demand driven manufacturing). DDM gives customers the opportunity to say what they want, where and when.

DEMAND-DRIVEN EXECUTION

Demand chain management is the same as supply chain management, but with emphasis on consumer pull vs. supplier push. The demand chain begins with customers, then funnels through any resellers, distributors, and other business partners who help sell the company’s products and services. The demand chain includes both direct and indirect sales forces. Customers demand is hard to detect because out of stock situations

(OOS) falsify data collected from POS-Terminals. According to studies of Corsten/Gruen (2002, 2008) the OOS-rate is about 8%. For products under sales promotion OOS rates up to 30% exist. Reliable information about demand is necessary for DCM therefore lowering OOS is a main factor for successful DCM.

Corsten and Gruen describe key factors for lowering OOS-rates:

- Data accuracy
- Forecast and order accuracy
- Order quantity
- Replenishment
- Capacity (time supply)
- Capacity (Packout) and Planogram Compliance
- Shelf Replenishment.

Implementation of system supported processes leads to the new technology Extreme Transaction Processing described by Gartner Research. This technology allows to process the huge amount of data (POS, RFID) in real time providing information for store managers, shelve managers and the supply chain.

OPERATIONS MANAGEMENT

Operations management is an area of management concerned with overseeing, designing, and redesigning business operations in the production of goods and/or services. It involves the responsibility of ensuring that business operations are efficient in terms of using as little resources as needed, and effective in terms of meeting customer requirements. It is concerned with managing the process that converts inputs (in the forms of materials, labour, and energy) into outputs (in the form of goods and/or services).

The relationship of operations management to senior management in commercial contexts can be compared to the relationship of line officers to the highest-level senior officers in military science. The highest-level officers shape the strategy and revise it over time, while the line officers make tactical decisions in support of carrying out the strategy. In business as in military affairs, the boundaries between levels are not always distinct; tactical information dynamically informs strategy, and individual people often move between roles over time.

Operations traditionally refers to the production of goods and/or services separately, although the distinction between these two main types of operations is increasingly difficult to make as manufacturers tend to merge product and service offerings. More generally, operations management aims to increase the content of value-added activities in any given process. Fundamentally, these value-adding creative activities should be aligned with market opportunity (through marketing) for optimal enterprise performance.

According to the U.S. Department of Education, operations management is the field concerned with managing and directing the physical and/or technical functions of a firm or organization, particularly those relating to development, production, and

manufacturing. Operations management programmes typically include instruction in principles of general management, manufacturing and production systems, plant management, equipment maintenance management, production control, industrial labour relations and skilled trades supervision, strategic manufacturing policy, systems analysis, productivity analysis and cost control, and materials planning. Management, including operations management, is like engineering in that it blends art with applied science. People skills, creativity, rational analysis, and knowledge of technology are all required for success.

ORIGINS

The origins of operations management can be traced back through cultural changes of the 18th, 19th, and 20th centuries, including the Industrial Revolution, the development of interchangeable manufacture, the Waltham-Lowell system, the American system of manufacturing, Fayolism, scientific management, the development of assembly line practice and mass production, industrial engineering, systems engineering, manufacturing engineering, operations research, the Toyota Production System, lean manufacturing, and Six Sigma. Combined, these ideas allow for the standardization of best practices balanced with room for further innovation through continuous improvement of production processes.

Key features of these production systems are the departure from craft production to a more thorough division of labour and the transfer of knowledge from within the minds of skilled, experienced workers into the systems of equipment, documentation, and semiskilled workers, often with an average of less tenure and less experience. The disciplines of organizational studies, industrial and organizational psychology, programme management, project management, and management information systems all ideally inform optimal operations management, although most smart people who work in the corporate world can empirically observe that the reality often falls far short of the ideal in ways that the market nevertheless rewards, based mostly on the fact that in markets, “good-enough-to-scrape-by” methods tend to defeat “proper” ones on cost.

There is a strong tradition of recruiting operations managers simply by promoting the most effective workers, which does work, although its main systemic flaw is the Peter Principle. One of the reasons why competition doesn’t kill businesses that operate this way is that few operate in any more ideal way. Typically the Peter Principle is so pervasive throughout an industry that similarly afflicted businesses face a field of competitors who are more or less equally hobbled by it (the “same circus, different clowns” problem).

There are scores of people who can be viewed as thought leaders whose life’s work laid the foundations for operations management (only some of which have name recognition among the general population). A very cursory list would include (in approximate chronological order) Adam Smith, Jean-Baptiste Vaquette de Gribeauval, Louis de Tousard, Honoré Blanc, Eli Whitney, John H. Hall, Simeon North, Henri Fayol, Frederick Winslow Taylor, Henry Gantt, Henry Ford, Sakichi Toyoda, Alfred P.

Sloan and Bill Knudsen of GM, Frank and Lillian Gilbreth, Tex Thornton and his Whiz Kids team, and W. Edwards Deming and the developers of the Toyota Production System (Taiichi Ohno, Shigeo Shingo, Eiji Toyoda, Kiichiro Toyoda, and others).

Whereas some influences place primary importance on the equipment and too often viewed people as recalcitrant impediments to systems (*e.g.*, Taylor and Ford), over time the need to view production operations as sociotechnical systems, duly considering both humans and machines, was increasingly appreciated and addressed.

Historically, the body of knowledge stemming from industrial engineering formed the basis of the first MBA programmes, and is central to operations management as used across diverse business sectors, industry, consulting and non-profit organizations.

ORGANIZATIONS

The following organizations support and promote operations management:

- Association for Operations Management (APICS) which supports the *Production and Inventory Management Journal*
- European Operations Management Association (EurOMA) which supports the International Journal of Operations and Production Management
- Production and Operations Management Society (POMS) which supports the journal: *Production and Operations Management*
- Institute for Operations Research and the Management Sciences (INFORMS)
- Institute of Operations Management (UK)
- Association of Technology, Management, and Applied Engineering (ATMAE).

REVERSE LOGISTICS

Reverse logistics stands for all operations related to the reuse of products and materials. It is “the process of planning, implementing, and controlling the efficient, cost effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or proper disposal. More precisely, reverse logistics is the process of moving goods from their typical final destination for the purpose of capturing value, or proper disposal. Remanufacturing and refurbishing activities also may be included in the definition of reverse logistics.” The reverse logistics process includes the management and the sale of surplus as well as returned equipment and machines from the hardware leasing business. Normally, logistics deal with events that bring the product towards the customer. In the case of reverse, the resource goes at least one step back in the supply chain. For instance, goods move from the customer to the distributor or to the manufacturer.

BUSINESS IMPLICATIONS

In today’s marketplace, many retailers treat merchandise returns as individual, disjointed transactions. “The challenge for retailers and vendors is to process returns at

a proficiency level that allows quick, efficient and cost-effective collection and return of merchandise. Customer requirements facilitate demand for a high standard of service that includes accuracy and timeliness.

It's the logistic company's responsibility to shorten the link from return origination to the time of resell." By following returns management best practices, retailers can achieve a returns process that addresses both the operational and customer retention issues associated with merchandise returns. Further, because of the connection between reverse logistics and customer retention, it has become a key component within Service Lifecycle Management (SLM), a business strategy aimed at retaining customers by bundling even more coordination of a company's services data together to achieve greater efficiency in its operations. Reverse logistics is more than just returns management, it is "activities related to returns avoidance, gate keeping, disposal and all other after-market supply chain issues". Returns management – increasingly being recognized as affecting competitive positioning – provides an important link between marketing and logistics. The broad nature of its cross-functional impact suggests that firms would benefit by improving internal integration efforts. In particular, a firm's ability to react to and plan for the influence of external factors on the returns management process is improved by such internal integration. Third-party logistics providers see that up to 7% of an enterprise's gross sales are captured by return costs. Almost all reverse logistics contracts are customized to fit the size and type of company contracting. The 3PL's themselves realize 12% to 15% profits on this business.

"Studies have shown that an average of 4% to 6% of all retail purchases are returned, costing the industry about \$40 billion per year."

RETURN OF UNSOLD GOODS

In certain industries, goods are distributed to downstream members in the supply chain with the understanding that the goods may be returned for credit if they are not sold. Newspapers and magazines serve as examples. This acts as an incentive for downstream members to carry more stock, because the risk of obsolescence is borne by the upstream supply chain members. However, there is also a distinct risk attached to this logistics concept. The downstream member in the supply chain might exploit the situation by ordering more stock than is required and returning large volumes. In this way, the downstream partner is able to offer high level of service without carrying the risks associated with large inventories. The supplier effectively finances the inventory for the downstream member. It is therefore important to analyze customers' account for hidden cost.

SUPPLY CHAIN

An illustration of a company's supply chain; the arrows stand for supplier-relationship management, internal SCM and customer-relationship management (cf. Chen/Paulraj, 2004)

A supply chain is a system of organizations, people, technology, activities, information and resources involved in moving a product or service from supplier to customer. Supply chain activities transform natural resources, raw materials and components into a finished product that is delivered to the end customer. In sophisticated supply chain systems, used products may re-enter the supply chain at any point where residual value is recyclable. Supply chains link value chains.

The Council of Supply Chain Management Professionals (CSCMP) defines Supply Chain Management as follows: “Supply Chain Management encompasses the planning and management of all activities involved in sourcing and procurement, conversion, and all logistics management activities. Importantly, it also includes coordination and collaboration with channel partners, which can be suppliers, intermediaries, third-party service providers, and customers. In essence, supply chain management integrates supply and demand management within and across companies. Supply Chain Management is an integrating function with primary responsibility for linking major business functions and business processes within and across companies into a cohesive and high-performing business model. It includes all of the logistics management activities noted above, as well as manufacturing operations, and it drives coordination of processes and activities with and across marketing, sales, product design, finance and information technology.”

A typical supply chain begins with ecological and biological regulation of natural resources, followed by the human extraction of raw material, and includes several production links (*e.g.*, component construction, assembly, and merging) before moving on to several layers of storage facilities of ever-decreasing size and ever more remote geographical locations, and finally reaching the consumer.

Many of the exchanges encountered in the supply chain will therefore be between different companies that will seek to maximize their revenue within their sphere of interest, but may have little or no knowledge or interest in the remaining players in the supply chain. More recently, the loosely coupled, self-organizing network of businesses that cooperates to provide product and service offerings has been called the *Extended Enterprise*.

SUPPLY CHAIN MODELING

There are a variety of supply chain models, which address both the upstream and downstream sides. However the SCOR model is most common. The SCOR Supply-Chain Operations Reference model, developed by the Supply Chain Council, measures total supply chain performance. It is a process reference model for supply-chain management, spanning from the supplier's supplier to the customer's customer. It includes delivery and order fulfilment performance, production flexibility, warranty and returns processing costs, inventory and asset turns, and other factors in evaluating the overall effective performance of a supply chain.

The Global Supply Chain Forum (GSCF) introduced another Supply Chain Model. This framework is built on eight key business processes that are both cross-functional

and cross-firm in nature. Each process is managed by a cross-functional team, including representatives from logistics, production, purchasing, finance, marketing and research and development. While each process will interface with key customers and suppliers, the customer relationship management and supplier relationship management processes form the critical linkages in the supply chain.

The American Productivity and Quality Centre (APQC) Process Classification Framework (PCF) SM is a high-level, industry-neutral enterprise process model that allows organizations to see their business processes from a cross-industry viewpoint. The PCF was developed by APQC and its member companies as an open standard to facilitate improvement through process management and benchmarking, regardless of industry, size, or geography. The PCF organizes operating and management processes into 12 enterprise level categories, including process groups, and over 1,000 processes and associated activities.

SUPPLY CHAIN MANAGEMENT

In the 1980s, the term Supply Chain Management (SCM) was developed to express the need to integrate the key business processes, from end user through original suppliers. Original suppliers being those that provide products, services and information that add value for customers and other stakeholders. The basic idea behind the SCM is that companies and corporations involve themselves in a supply chain by exchanging information regarding market fluctuations and production capabilities.

If all relevant information is accessible to any relevant company, every company in the supply chain has the possibility to and can seek to help optimizing the entire supply chain rather than sub optimize based on a local interest. This will lead to better planned overall production and distribution which can cut costs and give a more attractive final product leading to better sales and better overall results for the companies involved. Incorporating SCM successfully leads to a new kind of competition on the global market where competition is no longer of the company versus company form but rather takes on a supply chain versus supply chain form.

The primary objective of supply chain management is to fulfil customer demands through the most efficient use of resources, including distribution capacity, inventory and labour. In theory, a supply chain seeks to match demand with supply and do so with the minimal inventory. Various aspects of optimizing the supply chain include liaising with suppliers to eliminate bottlenecks; sourcing strategically to strike a balance between lowest material cost and transportation, implementing JIT (Just In Time) techniques to optimize manufacturing flow; maintaining the right mix and location of factories and warehouses to serve customer markets, and using location/allocation, vehicle routing analysis, dynamic programming and, of course, traditional logistics optimization to maximize the efficiency of the distribution side.

There is often confusion over the terms supply chain and logistics. It is now generally accepted that the term Logistics applies to activities within one company/organization

involving distribution of product whereas the term supply chain also encompasses manufacturing and procurement and therefore has a much broader focus as it involves multiple enterprises, including suppliers, manufacturers and retailers, working together to meet a customer need for a product or service.

Starting in the 1990s several companies chose to outsource the logistics aspect of supply chain management by partnering with a 3PL, Third-party logistics provider. Companies also outsource production to contract manufacturers. There are actually four common Supply Chain Models. Besides the two mentioned above, there are the American Productivity and Quality Centre's (APQC) Process Classification Framework and the Supply Chain Best Practices Framework.

An unusual food supply chain operated by Dabbawalas in Mumbai is noted for being extremely reliable without using any computers or modern technology. It has been verified to be a six sigma supply chain.

GAINING RESPECT FOR THE SUPPLY CHAIN DISCIPLINE

To establish the supply chain as a strategic asset, the leader of the supply chain organization must be accorded status on a par with other members of the executive management team. Many companies remain locked in a traditional paradigm in which sales, marketing, and development are considered the architects of the company's strategic direction, whereas purchasing, manufacturing, and distribution are thought of as tactical executors. While bemoaning high inventories, late shipments, or missed revenue opportunities, few companies see the potential value that effective supply chain management can deliver or view supply chain management as a critical area of expertise. Without the support of the executive team and a clear mandate for the supply chain leader to champion improvement efforts, major business-impacting opportunities are left underexploited.

The development of supply chain management as a core competency and an organizational imperative doesn't just happen-a conscious effort is needed. As a discipline, supply chain management is constantly evolving, and its impact on business performance is growing. Staying on top of best practices in supply chain management is challenging enough; ensuring that the rest of your company does the same is even more difficult.

How do you ensure that an integrated supply chain is viewed as being as critical to business success as product development or marketing and sales? An education and awareness programme can be a key driver. Metrics also can be a powerful tool in communicating the benefits of a highly integrated supply chain. Start with a quantitative gap analysis expressed in terms that any executive can relate to-bottom-line impact. For example, research by PRTM's benchmarking subsidiary, The Performance Measurement Group, LLC, shows that best-in-class consumer products companies enjoy

a whopping 5.5 percent (of revenue) advantage in total supply chain management cost over median performers. For a \$150 million company, that's more than an \$8 million difference in cost savings. Results are the ultimate attention-getter. Following the gap analysis, make a change to your supply chain that delivers a sustainable improvement in costs or service-and use it to get attention. This is what Angel Mendez did at palmOne, Inc. palmOne is one of the world's leading developers of personal digital assistants (PDAs).

When Mendez joined the company in 2001 as senior vice president of global operations for the solutions group, he was given the task of transforming the supply chain into a competitive differentiator-no small task considering that palmOne had just been forced to write off nearly \$300 million of inventory. The first item on his agenda: educating palmOne's senior management on the critical role the supply chain plays in executing the company's strategy. "Before I arrived," says Mendez, "palmOne had eight consecutive quarters where it was difficult to produce enough product to keep up with demand. The mind-set had always been building product, then pushing it to the channel. The supply chain organization was viewed as the means to get the product made. There were really no other expectations." Mendez aimed to raise the visibility of his organization and garner support from the senior management team. The recession that was in full swing when he joined the company actually proved an advantage. "Sometimes it takes a shrinking economy to get the CEO to focus on the supply chain," he said. "Catastrophe can be a catalyst. Why not leverage this fact to get what you need?" He quantified the financial benefits an upgraded supply chain would deliver to the company and its shareholders. Then he developed a clear roadmap for getting there.

Mendez's roadmap included a fundamental overhaul of the supply chain processes and organization. "We had a lot to do," he noted. "The plan we put in place was iterative. It allowed us to prioritize, get some initial wins, [and] then justify the resources it would take to execute the next step of the plan." palmOne had a weak demand-management process, making it difficult to see the level of activity within the sales channel. Decisions to build product were based on inaccurate forecasts rather than on current market activity. Manufacturing and material costs were high relative to those of other consumer electronics companies. Relationships with key suppliers and customers were strained. And the company used four different service providers to repair returned units, resulting in few economies of scale, a high repair cost per unit, and a highly complex returns-management process.

Still, Mendez didn't "own" all the core supply chain processes. Product groups were responsible for demand management. Repair and support were managed on a regional basis within the sales organization. "The company had always focused on the 'supply' part of the supply chain," says Mendez. "They weren't used to thinking of an end-to-end process. So the supply chain organization was mostly focused on procurement, with enough planning capability to determine what needed to be purchased. And not everyone within the organization was as highly skilled as what was needed for the

environment in which we operate." Mendez used the Supply-Chain Operations Reference-model (SCOR), to build the roadmap for his organization and for the group process-improvement efforts. He developed a three phased plan to be executed over two years. Then he began to restructure the supply chain organization, taking on responsibility for demand management, repair, and product support.

He knew that his title of senior vice president made the task easier. "My title helped me cut through a lot of red tape," he says, "but so did the fact that I was able to clearly demonstrate the benefits of an integrated organization." palmOne's new supply chain organization was set up as a global organization with localized capabilities. Mendez's staff included a director responsible for each of the core plan, source, make, deliver, and return processes and a well-qualified set of individuals within each group. The skills within the organization reflected the company's strategic imperatives: a demand-management process that could react quickly to current market conditions, a supply base that could provide palmOne with the best value in the materials and services it procures, and ongoing improvements in cost-effectiveness, quality, and customer service. palmOne's approach is a good example of how to establish the importance of your supply chain organization to ensure that it gets the resources it needs. The best way to gain support and respect for the supply chain as a strategic asset is to prove that it already is one. Simply put, results talk.

FOCUS ON THE SKILLS YOU NEED

Clearly, balancing the competencies you need with the competencies you have is critical to executing your strategy. But what happens when the demand for skills outstrips the supply? Should you change the structure of your organization, change your strategy, or accept the imbalance? Common sense tells us that companies with well-trained, knowledgeable people can operate more efficiently, seize market opportunities more readily, and weather economic downturns more effectively. And despite claims to the contrary, state-of-the-art systems and tools that support supply chain management are no replacement for human beings. Although they provide a degree of decision support that was not possible even a few years ago, these systems demand sophisticated users. In fact, today's new knowledge economy requires a new type of supply chain professional—one who can quickly assimilate volumes of information and use it to make sound decisions. In effect, the bar has been raised, not lowered.

Simply put, behind every world-class supply chain are world-class people. The same "total cost of ownership" principles that apply to your material assets also apply to your human assets. Procter and Gamble (P&G) is a case in point. P&G built an organization that merges manufacturing, engineering, purchasing, and customer service. The company sees these functions as an integrated system whose principal purpose is to move materials from suppliers to customers while adding value along the way. P&G hires highly qualified, flexible people and pays them well, reasoning that it will get more for its money than by hiring lower-cost people with limited skills and higher turnover. In

short, a more expensive workforce provides a lower total cost of ownership. New technologies such as the Internet are transforming traditional business functions. But it's important to remember, as you design your organization, that technologies don't deliver success-people do. To get the most out of new technology, you'll want to find and develop people who understand how to leverage it to improve operations and the performance of such routine activities as purchasing, customer service, and logistics management. Advanced Fibre Communications, Inc. (AFC), understands this well. The Petaluma, California-based telecommunications equipment manufacturer is the leading provider of broadband access solutions for the global telecommunications industry. And, although AFC had enjoyed eight years of steady growth, it still operated like a startup when Jeff Rosen joined the company in 2000 as vice president of operations. "The organization was set up to be effective in a startup mode," explains Rosen, "but the processes in place were increasingly strained. Everything was done manually, and there was no focus on building processes and systems that would allow the company to scale." AFC's focus on tactics and day-to-day execution was wholly consistent with the company's strategy for many years. However, the market was changing, the strategy was changing, and the supply chain had to adapt to new business imperatives-immediate product availability, high service levels, and higher product margins.

Rosen knew that he had to upgrade his organization and the systems used for supply chain planning and execution. He also knew, though, that he had to prove himself before he could get approval to make broad changes to his team. Thus he developed a plan to "get some wins and put something on the table to get attention." His first efforts targeted the procurement group, where he replaced some people with more experienced commodity managers. These new hires were tasked with building cost models for key materials, restructuring the supply base, and negotiating with existing vendors to secure better pricing. Rosen avoided making very senior hires. "Reasonably experienced was fine for what we needed," he explained. As a result of these initial efforts, customer service levels and time margins began to improve. These improvements caught the attention of AFC chairman, president, and CEO John Schofield, as well as the company's board of directors. Over the next two years, Rosen was given more responsibility and more resources to continue upgrading the supply chain organization. He replaced many of the people who were focused on day to- day transactions with people who could design entirely new supply chain processes. "My strategy was simple," says Rosen. "Start by hiring smart people capable of designing highly effective processes, [and] then let them run the process they helped architect." His first key hires were chartered with defining new processes for procurement, asset management, planning, and order management.

Once the basic processes were in place, Rosen brought in several very senior managers tasked with putting in place a mind-set of continuous improvement. He also began a systematic update of the systems that supported AFC's supply chain processes. Like many telecom companies, AFC outsources production to a contract manufacturer and

uses sophisticated information systems to provide forecast data and to closely monitor the status of customer orders. Rosen notes that the outsourcing strategy and the new software tools shaped the skills his organization needed. "When I started, I had a lot of people who were good at day-to-day transactions and putting out fires. Many of them were uncomfortable with the idea of using technology for processes they were used to doing manually. I needed people who were not afraid to use technology to do their job." Over a two-year period, Rosen completely restructured the organization, merging procurement and materials planning and forming an umbrella organization that included plan, source, make, deliver, and return processes. Old positions were eliminated and new ones created. Rosen chose his team carefully. Today, his "cost per head" is somewhat higher than in the past, but there are far fewer people, and each one provides critical skills the organization formerly lacked.

Matching the right people to vital roles is no easy task. Many managers are reluctant to jettison an employee who has served the company well, for example, even if his or her skill set does not meet the needs of the redesigned supply chain organization. Other companies view the prospect of finding, recruiting, hiring, and training the people they really need as too complicated or time-consuming, so they transfer someone who did well in one position but is a poor fit for the slot being filled.

It's important to resist the temptation to force-fit a valued employee into a role for which he or she is not qualified. Even worse than making wrong fits is designing an organization that is too strongly influenced by the skills and interests of existing employees. We worked with a computer peripherals company that was moving to an outsourced production model. The company engaged two contract manufacturers, each responsible for a major product line.

The supply chain vice president created two relationship manager positions, knowing that managing each of these new relationships would be a full-time job. Not surprisingly, several current employees saw the new positions as an excellent opportunity. Although the new jobs called for a more sophisticated set of skills, the vice president found himself lobbied aggressively by internal candidates.

He also was under tremendous pressure to "just get someone in here." He ended up hiring both an internal candidate and an external hire—a seasoned industry expert who had been managing a contract manufacturing relationship for several years. Not surprisingly, the "home grown" relationship manager struggled with his new responsibilities.

After six months, he was replaced by a more qualified individual. The lesson? Your supply chain strategy and associated processes will fail to move your company forward if you don't have the right people. Effective execution demands the right skills and capabilities. Define your organization around the end-to-end supply chain processes, identify the skills you need for each role, and then go shopping—both inside and outside your company. Don't let the "inventory on hand" limit the success of your organization.

NEXT-GENERATION ORGANIZATIONAL DESIGN

The end-to-end supply chain typically needs management skills and roles that did not exist previously. Given the relative newness of some of these positions, your human resources department may not know how to find the people you need or how to screen them effectively without some support from your organization. To ensure that you have the right people in the right positions, put together a description of your supply chain strategy, a description of each key role, a clear definition of available career paths, and a comprehensive hiring plan. This is not to say that every reorganization requires a major workforce turnover. Most supply chain organizations have only a few extremely critical roles-before or after an overhaul. These are the jobs that require specific qualifications, and you shouldn't settle for less. Current employees, with appropriate retraining or coaching, can fill most of the other roles in your new organization.

OWENS CORNING PROFILE: REORGANIZING FOR “A BRIGHT FUTURE”

The “big squeeze” of its core industry segments-cost inflation and price deflation-led Owens Corning to rethink its supply chain organization and processes in order to understand the impact of these elements on its ability to be competitive and customer-facing enterprise wide.

Based in Toledo, Ohio, Owens Corning (OC) has manufacturing facilities in more than 25 countries and 165 distribution centres. It's a \$5 billion market leader in building-materials systems and composites solutions known for its innovative portfolio of products-it invented glass fibre and glass-fibre insulation over 50 years ago. Approximately 80 percent of the company's revenues come from its building-materials portfolio of products and systems that are sold through distributors, contractors, and large national retailers mainly in North America. Composites account for the remainder of revenues and are used in the automotive, telecommunications, electronics, and construction industries in a global market. As big as it is and as well known as its brand is, however, OC has faced increasing competition within key markets because of globalization and industry consolidation in recent years. “We must be able to compete on more than price,” notes Sue Hatfield, the company's director of strategy and integration for supply chain and technology. “Our greatest challenge is providing low-cost flexibility to meet increased customer demands.”

This is a turnabout for the industry. Building products is a process manufacturing industry and is highly asset-intensive, making use of assets and production efficiency the primary success factors for competing in the marketplace. Traditionally, flexibility has not been considered a key factor but increasingly is becoming one. OC is organized into business units, each with its own commercial, sales, and supply chain processes and facilities. Having many business segments within the business units, the company

struggled as supply chain requirements got more demanding. In the past, the company had a substantial level of redundancy in processes and personnel across business units and had inherited multiple legacy software applications that were poorly integrated. These factors created a lack of data integrity/integration at the enterprise level. This was seen as a key contributor to the increasing problems that OC was experiencing, such as inaccurate demand forecasts, poor planning, overall customer dissatisfaction, and high-cost supply chain operations.

THE TECHNOLOGY SOLUTION WASN'T ENOUGH

In the mid-1990s, the company deployed SAP's enterprise resource planning (ERP) product across its global enterprise. At the time, OC believed that a common global technology platform would resolve many of the business issues relating to its product-driven, make-to-stock manufacturing environment. This was important because these process issues affected OC's forecasts, materials management, production scheduling, ability to meet deadlines, and cost containment. According to Hatfield, "Our forecast accuracy was not good, we didn't have the right linkage to production schedules, and we weren't doing the right level of demand-supply balancing. Our planning problems were having downstream effects on our customers, impacting on-time delivery and order fill-rate performance."

The SAP system eliminated 500 legacy systems, achieving one of the company's prevailing goals-to keep it "simple, common, and global." OC then implemented SAP's advanced planning and logistics software. However, the information technology applications only brought so much benefit. As David Johns, senior vice president and chief supply chain and information officer, notes: "The progress made was in the areas of back-office operations, logistics, and materials management. But where we still lacked real integration was with our manufacturing and our sales and operations planning processes."

It became apparent that a more customer-centric, enterprise-level approach using best practices and standards adopted by top-performing companies was needed to meet strategic goals. The biggest goal was "to operate as one company." This would prove to be no small order. A supply chain transformation initiative would become one of five top strategy principles championed by OC's new chief executive officer, Dave Brown. OC wanted to make significant gains in customer service levels and a \$250 million improvement in income from operations (IFO) and working capital over a three-year period. It wanted to create a brighter future for its employees and stockholders.

Lifting supply chain transformation to the strategic level was just the beginning of the three-year plan. As Hatfield explains, "What we needed to do was to get better in each supply chain process (plan, source, make, deliver), change our mind-set from manufacturing efficiency to supply chain flexibility, and integrate our customers and suppliers into our supply chain. We also had to expand our thinking beyond the functional level to the enterprise, integrated supply chain level."

ORGANIZATIONAL CHANGES

A first step was to integrate the functions having an impact on the customer experience-materials management, logistics and warehousing, customer service/call centres, receivables management, customer and product master data integrity-within a single group. Today, 350 employees report to this department, headed by Vice President of Customer Supply Chain Operations Meg Ressler.

"We intentionally put 'customer' in front of our organization's name to make sure that we were really driving the culture of the supply chain to be end-to-end and externally focused," she says. "The other thing we're trying to do is distinguish the process of supply chain from the organization that operates many of the processes." Ressler's group sits inside a broader organization called Supply Chain and Technology Solutions (SCTS), headed by David Johns. The reason for the differentiation? "We've been trying to help our company see that the supply chain is a business process that touches everybody in the company. SCTS focuses on the processes, by operating them, deploying technology that supports them, or encouraging process innovation and enhancements." OC has recognized the value in combining the technology and supply chain process function in one organization. Johns explains the rationale: "We wanted to leverage the resources we have across the enterprise, do things faster, and improve the service we can provide." The company also wanted to be more flexible in its response to customers. Because OC had a "do everything for everybody" mentality a decade ago, company costs were too high, and, as Johns says, "We weren't really able to grow, because we didn't have a consistent, coherent strategy."

"In those earlier days, we did understand that 'simple, common, and global' were key operating principles-and they remain so today-but now we understand that flexibility is the key competitive advantage," Johns says. He contrasts this customer-facing perspective with OC's historical focus on manufacturing efficiencies: "Our philosophy as a 60-yearold manufacturer was long product run times. And that strategy had all sorts of implications. But today, with a focus on flexibility, we're looking at market-driven lead times for product delivery and our ability to quickly respond to big changes in demand."

THE VISION THING

In order to ensure that it could respond more flexibly to customer demand, OC conducted an end-to-end supply chain assessment using PRTM's "stages of process maturity model for supply chain excellence" as a reference point. OC found that it was at early stage 2 of functional excellence in process maturity yet aspired to achieve solid stage 3 status, designated "enterprise integration." The company used the Supply-Chain Operations Reference-model (SCOR) to create a context for understanding the relationships among business strategy, supply chain configuration, supply chain practices, and technology. It

knew that it needed to define the value proposition before it could get buy-in by all stakeholders. "The transformation vision provided a common, overall roadmap," says Hatfield, "and provided an umbrella for enterprise-level improvement projects."

The supply chain organization identified four areas for immediate improvement: fixing the end-to-end planning process within materials management and instituting a sales and operations planning process in every business, leveraging warehousing and transportation at an enterprise rather than an individual business level, improving overall service performance with customers, and building greater manufacturing effectiveness and flexibility. At the start, OC benchmarked its then-current performance levels against similar manufacturing organizations in three areas: customer service and responsiveness, cost performance, and asset performance. It set priorities and targets for customer-facing processes such as on-time delivery to request-a downstream result of greater alignment between demand and supply planning processes. Increasing the effectiveness of its 10-step sales and operations planning (S&OP) process also enabled OC to reduce costs by operating in a more integrated way across global operations. Early results have included lower inventory levels and higher inventory quality, improved on-time delivery and order fill-rate performance, and reduced logistics costs. OC estimates that it will generate more than \$165 million in value within the initial three years of the plan.

FROM PRODUCT TO MARKET FOCUS

Another goal was moving the organizational mind-set and execution capabilities from a commodity-based product focus to a solutions-based market focus, even though, as Meg Ressler confirms, OC has customers of both types: "It's true that our customer demands are evolving," she says, "and we must have the capability to deliver on all those demands, which means you have to have a flexible platform." To this end, account teams, transparent to the customer regardless of their request for service, were organized around specific high-value/high-volume partners. The intent is to align the right set of people, skills, and processes around customers' requirements. This might mean self-service for some customers, supported by portal technology or electronic data interchange (EDI) transactions for order management and fulfillment. It might mean a dedicated customer account team for other customers. As Ressler notes, even though OC is exploring use of the Internet for ordering and other customer-facing activities, the company also realizes that many of its major distributors and retailers have an emotional comfort level with their customer service representatives from OC.

The challenges are manifold. Customers want more customized service and more flexibility. They want tailored solutions that take cost out of their supply chains. "Many customers want us to be their partner- where we leverage the full resources of OC to help their business grow," says Hatfield. In customer alliances, this means that OC needs to become more integrated into its customers' supply chains-to understand what its customers are demanding and to create the processes that will ensure that demand is filled on time every time.

THE MECHANICS OF TRANSFORMATION

A key challenge of putting supply chain issues on everyone's radar screen was to redefine and educate employees on what the integrated supply chain meant for OC down to how each employee's role and responsibility fit into the supply chain. Says Hatfield, "We embarked on a whole educational process. One of the most effective things we did, as simple as it sounds, was to hire a design firm to draw an illustration of all the high-level supply chain processes at OC. When they were finished, it was clear that the supply chain is centred on our customers and really encompasses almost everything that we do. Everyone has a role, and the key to success is integration."

The supply chain organization realized that it would take three components to effect the transformational change it was aiming for: the people component, the process component, and the technology component. The people component would require education about the value of the supply chain to create "fully engaged employees" those who would be trained and certified to execute processes with a high level of accountability for performance. A process architecture would have to be created with the unified goal of providing exceptional customer experiences. And last, the technology would have to be "fit for purpose": the enablement of higher levels of performance through greater visibility of information at all levels of management.

Once viewed as the primary driver of business processes, technology was now seen as an enabler of larger, more strategic goals. Evidence of this shift is apparent in the type of employee hired for the information systems organization. Don Kosanka, OC's chief information officer, explains: "The IS folks that we have are really not the technical folks you find in typical IS organizations. They have a really good understanding of the business processes that we operate, and their value comes in helping figure out where technology can enable each process to become more efficient, precise, or accurate."

We outsource the more routine technical skill sets required to operate the IT infrastructure." Kosanka is an example of the shift in thinking at OC. As he says, "When I first joined the firm 25 years ago, fresh out of college, I thought technology was the important driver. After a few years here, I shifted my view to process. Now I see the equally critical importance of people. Without every employee's buy-in, you can't execute the process or realize the value in the technology. So you might say I've moved to a much more integrated approach."

What's especially different about today's perspective on the role of technology, says Kosanka, is that "we actually try to solve our process challenges without technology, if possible, and use technology when it's required and when it actually enhances the project so [that] it justifies the investment." Kosanka has used IT to take costs out of the supply chain.

"Our total investment in IT over the last seven years has dropped every year, and as a percentage of sales it has dropped even faster," he says. "While we've decreased the amount spent in the infrastructure, we've been maintaining or growing our investment

in applications to help the business units operate their processes better. Basically, we've done a good job of managing our costs to get higher performance for lower costs. We now have just a handful of major systems that we use."

OC achieved these gains primarily by outsourcing many operational aspects of the IT infrastructure, from help-desk and data-centre operations to network management. Now the organization looks at technology very differently from even seven years ago. Ressler refers to the common set of technology and process elements as an enterprise wide toolkit. "Flexibility comes from using that toolkit and recognizing the unique characteristics of a particular market segment or customer. Without having the process and IT foundation in place that we have today, it would be difficult, if not impossible, and certainly much more expensive, to customize an approach."

GREATER COLLABORATION WITH SUPPLIERS

As it builds better partnerships with distributors and retailers, OC has developed a new perspective on its suppliers. "We have a point of view that our suppliers are as important to us as our customers, so just as we want to be our customers' best supplier, we want to be our suppliers' best customer," explains David Johns. "We are trying to make our systems, our processes, etc., as transparent as possible with our suppliers and our customers, by eliminating as many inefficiencies as possible in our dealings and taking the costs out of our shared supply chains."

The company has established programmes and metrics for vendor relationship management and supplier development. Although its sourcing organization is separate from the supply chain solutions organization, the two departments leverage resources in support of common goals. For example, both groups collaborated on creating a portal structure that would facilitate interaction with vendors.

HORIZONTAL INTEGRATION

OC also has developed project review boards to help enforce the alignment and integration of key business processes, such as customer operations, materials management, sourcing, production, research and development, IT, and logistics.

Each has representation from the three functional areas of technology, process, and operations to jointly assess and monitor IFO benefits, working capital savings, return on net assets (RONA), supply chain performance metrics, and overall allocation of people and capital resources for approved initiatives.

A new "stage gated" approach ensures that before resources are assigned to specific projects, each has to pass a stringent test for business value.

This is another one of the building blocks to OC's "one company" mantra these days. Currently in the process of coming out of a financial restructuring process, it is reinventing its processes and internal organizations to give customers the experience of a single company that is poised for the increased competitiveness of the future.

BUILD THE RIGHT COLLABORATIVE MODEL

Collaboration is the cornerstone of effective supply chain management. As companies continue to narrow their strategic focus to a smaller number of core competencies, the skills and talents of outside partners become more critical. This creates a growing reliance on resources that you may not control directly and on strategies that you may have no hand in developing. A recent survey of more than 100 international business leaders found that as companies migrate towards more extended supply chains, collaboration becomes their most strategic activity. Despite its importance, there is little consensus about what collaboration means. If you asked 100 supply chain executives for a definition, you'd likely get 100 different answers. Certainly most would agree that collaboration is important, that technology and relationship building are critical components, and that companies with effective collaboration skills are likely to have a competitive edge. However, few executives would be able to offer a clear, unambiguous definition.

Why is it so hard to define collaboration? Because it can be many things and involve many types of partners. It can refer to a wide range of joint activities, from information sharing among business units to complex, long-term product development and marketing projects. We define collaboration as "the means by which companies within the supply chain work together towards mutual objectives through the sharing of ideas, information, knowledge, risks, and rewards."

Why collaborate? Very simply, an effective collaborative relationship can have major strategic and financial benefits. It can accelerate entry into a new market, increase flexibility, and provide access to expertise not available within your own company. It can deliver cost savings or increased revenues-or a combination of both. Collaboration is a business arrangement that changes the overall dynamics between two or more partners. Drivers of collaboration include the desire to access:

- A technology owned by another company
- A technology that is too capital-intensive for one company to invest in alone
- A competency that is too costly to acquire, develop, or maintain
- A new market effectively closed off by high entry costs or preconditions (trade barriers, legislation, etc.)

Collaboration changes the most fundamental of all economic models- the relationship among cost, volume, and profit (C/V/P). For example, a company that needs specialized, capital-intensive equipment for production of a key component might have a C/V/P model with high fixed costs and low per-unit variable costs. This company needs a high volume of sales to be price-competitive and profitable. If an economic recession cuts into volume, the company could soon be operating at a loss.

Collaborating with a partner that focuses on the production of specialized materials similar to the component might allow this company to offload some of its fixed costs, but with an accompanying increase in variable costs associated with the increase in the level of external sourcing. To make this approach pay off, the company must be willing to share any proprietary technology needed to manufacture the component, and its

collaboration partner must be willing to invest in developing the additional capabilities needed to produce it. Since breakeven volume is lower, the company can compete across a wider range of volumes-albeit at the expense of gross margin at high volume. Ongoing collaboration on product designs and production planning can make the company even more agile while continuing to add volume to the specialized manufacturer's business. Both collaboration partners will benefit economically.

As you can see, collaboration is not an altruistic activity. While it may seem a best practice to provide "seamless integration" and "extended visibility" to your supply chain partners, the fact is that true collaboration is very difficult, and there's no point in doing it unless you can achieve financial or strategic gain. For collaboration to be truly successful, therefore, it must deliver quantifiable economic benefit to all partners. Despite the highly touted benefits, supply chain

collaboration has the dubious distinction of being one of the most sought after but disappointing aspects of supply chain strategy. What's going on? To start with, the promise of effective, efficient collaboration is based on Internet technology and its ability to provide new levels of visibility and information sharing. The Internet bubble of the late 1990s gave rise to hundreds of software products that promised seamless interaction and endless visibility among supply chain partners. Do these tools work? Some do, and some don't. Technology doesn't ultimately determine the success or failure of a collaborative relationship. Nor do the underlying processes that govern the use of technology-at least not on their own. Successful collaboration requires two additional components: sharing information and sharing benefits.

Information is at the heart of any collaborative relationship. To collaborate effectively, all partners must provide timely, accurate, and complete information-whatever is needed to achieve their mutual objectives. And each partner must respect the confidentiality and security requirements of the other. Mutual trust is key to a successful collaboration. Just as important, each partner must commit to a joint sharing of benefits-not necessarily an equal sharing but an equitable sharing. The success or failure of a collaborative relationship depends on clearly identified mutual gain.

COLLABORATION IS A SPECTRUM

Potential collaboration partners in supply chain management can be classified in three broad groups-customers, materials suppliers, and suppliers of services that support supply chain operations, such as manufacturing and logistics. Although each group requires a slightly different management approach, the relationships are established and maintained in similar ways.

Not all collaborations are created equal. Relationships between supply chain partners can have very different characteristics and still be considered collaborative in nature. And the results of collaborative relationships may vary widely from one set of partners to another. The horizontal axis plots the relative number of relationships, whereas the

vertical axis measures the relative depth of collaboration. Within this framework, we define four levels of collaboration: transactional, cooperative, coordinated, and synchronized. Note that the boundaries between the different levels of collaboration are blurred. This is so because collaboration is a continuum, not a set of clearly delineated management practices. Note, too, that the dimensions of the two axes are inherently subjective and are used simply to provide a clear graphic view of the collaboration spectrum. Other models use different criteria, such as level of investment or dependence on technology, to describe the depth and breadth of collaborative relationships. It's possible to create a matrix using any combination of these criteria or even to apply a multidimensional approach.

The point is not to worry about picking the right labels for your collaborative relationships but to examine the various characteristics that differentiate each partnership. First, choose the degree to which each characteristic contributes to the likely success of the collaborative relationship, and then put a plan in place to achieve it. Every customer supplier relationship can involve some level of collaboration.

The fact that you're buying from a specific supplier or selling to a specific customer implies a relationship between your two companies, but it doesn't necessarily mean that you are collaborating. And just as not all relationships are created equal, not all collaborations are created equal. Before setting off to systematically establish collaborative relationships with your supply chain partners, take the time to understand the degrees of collaboration along the spectrum and your company's specific needs. Often, a small number of deeply collaborative relationships is preferable to multiple relationships with a wide range of partners. Later in this chapter we'll discuss how to decide which degree of collaboration to set up with each supply chain partner.

TRANSACTIONAL COLLABORATION

Transactional collaboration aims for the efficient and effective execution of transactions between partners. This isn't to say that transactional relationships between supply chain partners offer no strategic value. However, partners in a transactional relationship rarely focus on reducing supply chain management costs or increasing revenues. The focus is usually on improving the ease at which transactions are conducted—for example, by eliminating the need for constant renegotiation. Transactional collaboration usually applies to customer-supplier relationships in which common or maintenance, repair, and overhaul (MRO) materials are purchased, and the decision to deal with a supplier is based mainly on price. With less strategically important supply chain partners, companies tend to focus on minimizing the effort associated with day-to-day transactions rather than on developing long-term relationships.

Transactional relationships rarely require sophisticated information systems. Indeed, many companies involved in this type of relationship lack the systems and infrastructure needed to provide and respond to information electronically. Because of this, many transactions are manual. An example of a transactional relationship is any time a customer

and a supplier agree to a set price for a specific product over a set period of time or until a certain purchase volume is reached. The buyer gets a fixed price over the life of the agreement in exchange for purchasing a minimum quantity of products; this also helps the seller's production planning. Transactional collaboration is the most basic and by far the most widely used collaboration model.

COOPERATIVE COLLABORATION

Cooperative relationships have a higher level of information sharing. Supply chain partners may provide automatic commitments and confirmations or share information on forecasts, inventory availability, purchase orders, or order and delivery status. Usually, one partner posts information that the other partner reviews and acts on- a one-way communication in which data are sent either manually or electronically ("pushed") from one partner to the other or published in a manner that's accessible by the recipient ("pulled").

In a cooperative collaboration, the type and format of data provided usually are standardized. While more sophisticated technologies are available, electronic data interchange (EDI) is the primary method of communication used today, through either a proprietary EDI network or the Internet. For companies without an EDI capability, Internet-based supplier portals or extranets are an excellent alternative. Most of these tools enable document and content management and include embedded workflows to automate the routing of documents, forms, and certain data and tasks.

COORDINATED COLLABORATION

In a coordinated relationship, supply chain partners work more closely together and rely more on each other's capabilities. As such, a coordinated relationship requires a two-way flow of information between partners and tightly synchronized planning and execution processes. Because the infrastructure and processes needed to support this type of information sharing are more complex than in the cooperative model, coordinated collaboration usually is reserved for more strategically critical supply chain partners. Unlike transactional and cooperative relationships, coordinated collaboration requires a high level of negotiation and compromise. Given the more strategic nature of these partnerships and the high level of data sharing, proprietary systems are needed for exchanging information.

Because of this complexity, a coordinated relationship requires a long-term commitment by both partners and is rarely undertaken lightly. Putting the required processes and tools in place takes time and money; the expectation is that both parties will benefit from the expected efficiencies created as part of the ongoing execution of the relationship. Vendor-managed inventory (VMI) programmes are a commonly used method of coordinated collaboration. In a VMI relationship, the supplier is responsible for making sure that the customer never runs out of materials. While some VMI

programmes are manual-the supplier walks through the customer's site to monitor inventory levels-most programmes in place today are automated. In some cases, the supplier can remotely manage inventory at the customer's site based on forecasts and usage. In other cases, the supplier uses current consumption rates and inventory levels to determine if more inventory is needed. In either case, effective data transmission is the key to successful VMI, a hallmark of coordinated collaboration.

SYNCHRONIZED COLLABORATION

The greatest degree of collaboration on the spectrum occurs at the upper right quadrant of our framework-synchronized collaboration. In this model, the collaborative relationship moves beyond supply chain operations to include other critical business processes. Partners may invest in joint research and development projects, supplier development, and intellectual property (IP) development. The sharing of both physical and intellectual assets may even extend to shared personnel. Synchronized collaborations are often called strategic alliances. In a synchronized relationship, information is developed jointly rather than just transmitted or exchanged. Moreover, synchronized collaboration tends to focus on a strategic vision of the future rather than on near-term planning and tactical execution. Long-term commercial commitment is a hallmark of this type of collaboration.

Development projects that consider supply chain requirements when developing the product strategy are good examples of synchronized collaboration. A company that includes key materials suppliers or manufacturing partners as an integral part of its development team is far more likely to have product designs that are compatible with best-in-class supply chain performance. Unlike other types of collaboration, in which partners are apt to exchange product data, synchronized relationships usually include a shared product data management system.

Cost Analysis and Supply Management

COST STRUCTURE OF THE FIRM

We can look at the business firm from at least two points of view: productivity, inputs and outputs (as we have just done) or outputs and costs. In advanced microeconomics, these two points of view are called “duals.” They are equally valid, but they point up different things. They are also opposites from a certain point of view—the higher the productivity, the lower the costs. By looking at the firm from the point of view of costs, we shift our perspective somewhat and gain a much more direct understanding of supply.

We also look more directly at the difference between the long and short run. In the short run, we have two major categories of costs:

- Fixed costs and
- Variable costs.

In the long run, however, all costs are variable. Thus, we must study costs under two quite different headings. Costs will vary quite differently in the long run and in the short.

FIXED AND VARIABLE COSTS

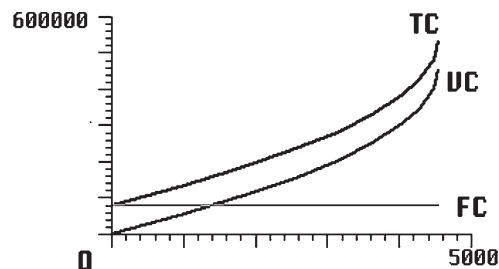
Variable costs are costs that can be varied flexibly as conditions change. In the John Bates Clark model of the firm that we are studying, labour costs are the variable costs.

Fixed costs are the costs of the investment goods used by the firm, on the idea that these reflect a long-term commitment that can be recovered only by wearing them out in the production of goods and services for sale.

The idea here is that labour is a much more flexible resource than capital investment. People can change from one task to another flexibly (whether within the same firm or in a new job at another firm), while machinery tends to be designed for a very specific use. If it isn't used for that purpose, it can't produce anything at all. Thus, capital investment is much more of a commitment than hiring is. In the eighteen-hundreds, when John Bates Clark was writing, this was pretty clearly true. Over the past century, a) education and experience have become more important for labour and have made labour more specialized and b) increasing automatic control has made some machinery more flexible.

So the differences between capital and labour are less than they once were, but all the same, it seems labour is still relatively more flexible than capital. It is this (relative) difference in flexibility that is expressed by the simplified distinction of long and short run.

Of course, productivity and costs are inversely related, so the variable costs will change as the productivity of labour changes. Here is a picture of the fixed costs (FC), variable costs (VC) and the total of both kinds of costs (TC) for the productivity example in the last unit:



Output produced is measured towards the right on the horizontal axis. The cost numbers are on the vertical axis. Notice that the variable and total cost curves are parallel, since the distance between them is a constant number—the fixed cost.

OPPORTUNITY COST

What is the connection between the distinction we have just made—fixed vs variable costs—and opportunity cost, the key concept in some earlier units? In economics, all costs are included—whether or not they correspond to money payments. If we have opportunity costs with no corresponding money payments, they are called implicit costs. The implicit costs (as well as the money costs) are included in the cost analysis we have just given.

There is some correlation between implicit costs and fixed or variable costs, but this correlation will be different in such different kinds of firms as:

A factory owned by an absentee investor: This is the easiest case to understand. All of the labour costs to the absentee investor are money costs, including the manager's salary. If the investor has borrowed some of the money he invested in the factory, then there are some money costs of the capital invested—interest on the loan. However, we must consider the opportunity cost of invested capital as well.

The investor's own money that he has used to buy the factory is money that she could have invested in some other business. The return she could have gotten on another investment is the opportunity cost of her own funds invested in the business. This is an implicit cost and in this case the implicit cost is part of the cost of capital and probably a fixed cost.

A “mom- and-pop” store: A “mom- and-pop” store (family proprietorship or partnership) is a store in which family members are self-employed and supply most of the labour. Typically, “Mom” and “Pop” don't pay themselves a salary—they just take money from the till when they need it, since it is their property anyway. As a result, there are no money costs for their labour. But their labour has an opportunity cost—the salary or wages they could make working similar hours in some other business—and so, in this case, the implicit costs include a large component of variable labour costs.

A large modern corporation: The corporation has relatively few implicit costs, but generally will have some. All labour costs will be expressed in money terms (though benefits and bonuses have to be included), since the shareholders don't supply labour to the corporation as “Mom and Pop” do in a family proprietorship. It will pay interest to bondholders and dividends to shareholders. But the dividends aren't really a cost item—they include profits distributed to the shareholders.

Moreover, the typical corporation will retain some profits and invest them within the business, a “plowback” investment. Conversely, shareholders may take a large part of their payout in appreciation of the stock value—and plowback investment is one reason for the appreciation.

Thus we would say that the corporation has a net equity value, that is, that the corporation “owns” a certain amount of capital that it invests in its own business (very much like the absentee owner in the first example). This capital has an opportunity cost and that opportunity cost is an implicit cost. The stockholders, who own the corporation, ultimately receive (as dividends or appreciation) both the opportunity cost of the equity capital and any profit left over after it is taken out.

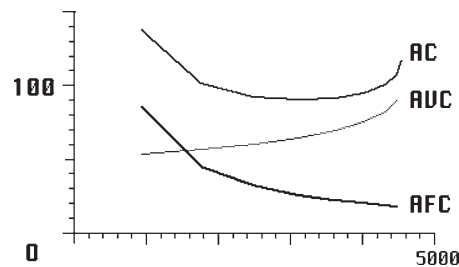
UNIT COST

Here are the average, average variable and average fixed costs for our example firm.

Here are the average cost (AC), average variable cost (AVC) and average fixed cost (AFC) in a diagram. This is a good representative of the way that economists believe firm costs vary in the short run.

Table

Q	AC	AFC	AVC
945	138	85	53
1780	101	45	56
2505	92	32	60
3120	90	26	64
3625	91	22	69
4020	95	20	75
4305	100	19	81
4480	107	18	89
4545	117	18	99



Notice how the average fixed costs decline as the fixed costs are “spread over more units of output.” For large outputs, however, average variable costs rise pretty steeply. The idea is that with a limited capital plant and thus limited productive capacity—in the short run—costs would rise much more than proportionately to output as output goes beyond “capacity.” The average total cost, dominated by fixed costs for small output, declines at first, but as output increases, fixed costs become less important for the total cost and variable costs become more important and so, after reaching a minimum, average total cost begins to rise more and more steeply.

MARGINAL COST

As before, we want to focus particularly on the marginal variation. In this case, of course, it is marginal cost. Marginal cost is defined as

$$MC = \frac{\Delta C}{\Delta Q}$$

As usual, Q stands for (quantity of) output and C for cost, so Q stands for the change in output, while C stands for the change in cost. As usual, marginal cost can be interpreted as the additional cost of producing just one more (“marginal”) unit of output. Let’s have a numerical example of the Marginal Cost definition to help make it clear. In the John Bates Clark style example we have been using, total cost is 280000 for an output of 3120 and it is 330000 for an output of 3625. So we have $\Delta C = 330000 - 280000 = 50000$ and $\Delta Q = 3600 - 3120 = 505$

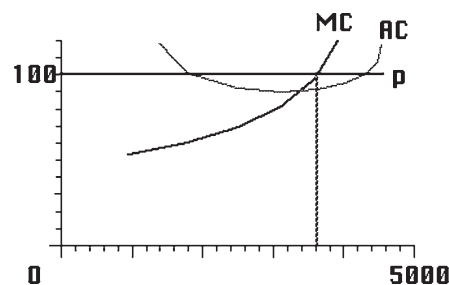
so that $\frac{\Delta C}{\Delta Q} = \frac{50000}{505} = 99.01$ for a marginal cost of \$99.01 for the next unit produced. As usual, this is an approximation and the smaller the change in output we use, the better the approximation is.

Here is the marginal cost for our example firm, along with output and average cost.

Table

Output	Average	Marginal
Cost	Cost	
0	0	9.45
945	137.57	52.91
1780	101.12	59.88
2505	91.82	68.97
3120	89.74	81.30
3625	91.03	99.01
4020	94.53	126.58
4305	99.88	175.44
4480	107.14	285.71
4545	116.61	769.23

Here is a picture of marginal cost for our example firm, together with average cost as output varies.



As before, the output produced is measured by the distance to the right on the horizontal axis. The average and marginal cost are on the vertical axis. Average cost is shown by the curve in yellow and marginal cost in red. Notice how the marginal cost rises to cross average cost at its lowest point.

MAXIMIZATION OF PROFITS

We can now give another rule for the maximization of profits. The new rule is really just the same rule as we saw before, only now we state it in terms of price and costs. It is the equimarginal principle in yet another form.

The question is: “I want to maximize profits. How much output should I sell, at the given price?”

The answer is: increase output until

$$p = MC$$

The point is illustrated by the following table, which extends the marginal cost table in an earlier page to show the price and the profits for the example firm.

Notice how profits are greatest (at 32516.25) when the marginal cost is almost exactly equal to the price of \$100. This occurs at an output of 3625, with marginal cost at 99.01. The profit-maximizing output would be very slightly more than 3625.

MARGINAL COST AND FIRM SUPPLY

We have discovered the principle of supply for the individual firm.

Remember: what is supply? It is the relation between the price and the quantity that people want to sell. For an individual firm, that is: the relation between the price and the quantity the firm wants to sell. So we ask: at a given price, how much will a (profit-maximizing) firm want to sell? The answer: enough so that the price is equal to marginal cost. In other words, the marginal cost curve is the supply curve for the individual firm.

MARGINAL COST

In economics and finance, marginal cost is the change in total cost that arises when the quantity produced changes by one unit.

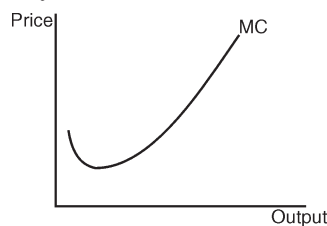


Fig. A typical Marginal Cost Curve

Mathematically, the marginal cost (MC) function is expressed as the derivative of the total cost (TC) function with respect to quantity (Q). Note that the marginal cost may change with volume, and so at each level of production, the marginal cost is the cost of the next unit produced.

$$MC = \frac{dTC}{dQ}$$

In general terms, marginal cost at each level of production includes any additional costs required to produce the next unit. If producing additional vehicles requires, for example, building a new factory, the marginal cost of those *extra* vehicles includes the cost of the new factory. In practice, the analysis is segregated into short and long-run cases, and over the longest run, all costs are marginal. At each level of production and time period being considered, marginal costs include all costs which vary with the level of production, and other costs are considered fixed costs.

It is a general principle of economics that a (rational) producer should always produce (and sell) the last unit if the marginal cost is less than the market price. As the market

price will be dictated by supply and demand, it leads to the conclusion that marginal cost equals marginal revenue. These general principles are subject to a number of other factors and exceptions, but marginal cost and marginal cost pricing play a central role in economic definitions of efficiency.

Marginal cost pricing is the principle that the market will, over time, cause goods to be sold at their marginal cost of production. Whether goods are in fact sold at their marginal cost will depend on competition and other factors, as well as the time frame considered. In the most general criticism of the theory of marginal cost pricing, economists note that monopoly power may allow a producer to maintain prices above the marginal cost; more specifically, if a good has low elasticity of demand (consumers are insensitive to changes in price) and supply of the product is limited (or can be limited), prices may be considerably higher than marginal cost. Since this description applies to most products with established brands, marginal pricing may be relatively rare; an example would be in markets for commodities.

A number of other factors can affect marginal cost and its applicability to real world problems. Some of these may be considered market failures. These may include information asymmetries, the presence of negative or positive externalities, transaction costs, price discrimination and others.

Cost Functions and Relationship to Average Cost

In the most simple case, the total cost function and its derivative are expressed as follows, where $f(Q)$ is a cost function relating cost to production volume and FC represents fixed costs:

$$TC = FC + f(Q)$$

$$MC = \frac{dTC}{dQ} = \frac{d(FC + f(Q))}{dQ} = \frac{df(Q)}{dQ}$$

Since (by definition) the fixed costs do not vary with production volume, the marginal cost *is not related to* fixed costs; the term drops out of the differentiated equation. This can be compared with average cost or AC , which is the total cost divided by the number of units produced and *does* include fixed costs.

$$AC = \frac{FC + f(Q)}{Q}$$

This result has important implications, since it indicates that the producer should choose to produce (sell) even if the market price is less than average cost, because the marginal revenue (the income received from selling the marginal unit) is greater than or equal to marginal cost. For example, an electric powerplant should produce and sell electricity when it is paid \$20 per megawatt if its marginal cost is \$15, despite the fact that its average cost (including the cost of building the plant, salaries, etc.) is \$25 per megawatt. Although the plant is “losing money”, it has lost \$5 *less* than if it had not sold the last marginal unit.

TRANSACTION COST

In economics and related disciplines, a transaction cost is a cost incurred in making an economic exchange. For example, most people, when buying or selling a stock, must pay a commission to their broker; that commission is a transaction cost of doing the stock deal. Or consider buying a banana from a store; to purchase the banana, your costs will be not only the price of the banana itself, but also the energy and effort it requires to find out which of the various banana products you prefer, where to get them and at what price, the cost of travelling from your house to the store and back, the time waiting in line, and the effort of the paying itself; the costs above and beyond the cost of the banana are the transaction costs. When rationally evaluating a potential transaction, it is important to consider transaction costs that might prove significant. A number of kinds of transaction cost have come to be known by particular names.

- *Search and information costs* are costs such as those incurred in determining that the required good is available on the market, who has the lowest price, etc.
- *Bargaining costs* are the costs required to come to an acceptable agreement with the other party to the transaction, drawing up an appropriate contract and so on. In game theory this is analyzed for instance in the game of chicken.
- *Policing and enforcement costs* are the costs of making sure the other party sticks to the terms of the contract, and taking appropriate action (often through the legal system) if this turns out not to be the case.

The term “transaction cost” is frequently thought to have been coined by Ronald Coase, who used it to develop a theoretical framework for predicting when certain economic tasks would be performed by firms, and when they would be performed on the market. However, the term is actually absent from his early work up to the 1970s. While he did not coin the specific term, Coase indeed discussed “costs of using the price mechanism” in his 1937 paper *The Nature of the Firm*, where he first discusses the concept of transaction costs. The term “Transaction Costs” itself can instead be traced back to the monetary economics literature of the 1950s, and does not appear to have been consciously ‘coined’ by any particular individual.

Arguably, transaction cost reasoning became most widely known through Oliver E. Williamson’s *Transaction Cost Economics*. Today, transaction cost economics is used to explain a number of different behaviours. Often this involves considering as “transactions” not only the obvious cases of buying and selling, but also day-to-day emotional interactions, informal gift exchanges, etc.

The determinants of transaction costs are according to Williamson frequency, specificity, uncertainty, limited rationality, and opportunistic behaviour.

At least two definitions of the phrase “transaction cost” are commonly used in literature. Transaction costs have been broadly defined by Steven N. S. Cheung as any costs that are not conceivable in a “Robinson Crusoe economy” — in other words, any costs that arise due to the existence of institutions. To Cheung, “transaction costs”, if the term is not so

popular in economics literatures, should be called “institutional costs”. But many economists seem to restrict the definition to exclude costs internal to an organization. The latter definition parallels Coase’s early analysis of “costs of the price mechanism” and the origins of the term as a market trading fee.

Starting with the broad definition, many economists then ask what kind of institutions (firms, markets, franchises, etc.) minimize the transaction costs of producing and distributing a particular good or service. Often these relationships are categorized by the kind of contract involved. This approach sometimes goes under the rubric of New Institutional Economics.

IT’s Relationship to Transaction Costs

Implementing a new information technology is generally seen as a means for reducing the transaction costs of an organization. However, in practice, implementing new IT often results in higher transaction costs. This is because the amount of information that needs to be processed by the organisation increases.

This can result in information overload. Antonio Cordella and Kai A. Simon call the cost of processing this information coordination cost. If these costs exceed the benefits of IT, then the implementation becomes something negative and expensive. To reduce coordination costs, organizations can do one of two things:

- Improve information processing capabilities. This can be done either through implementing new information systems or creating lateral relations.
- Use IT to reduce the need for coordination through increased slack resources (which reduces the need for extreme precision) or increased reliance on self-contained tasks which provides more of the information to a single point of contact rather than requiring communications and coordination among multiple units. The decreased amount of information to process means lower coordination costs and lower transaction costs.

Technologies like enterprise resource planning (ERP) can provide technical support for these strategies.

Relationship to Transaction Costs

Firms, or more generally, organizations, develop and become larger over time, using more and more computers to work. This growth in the number of computers leads to a growth of software use (operating systems and their applications, for example) and, as a result, to the growth in the number of software use/access licenses to be purchased and managed. For the owners of software intellectual property rights, this process leads to a greater supervision of users to regulate/enforce lawful access to software.

The situation occurs when all of the software used by an organization is proprietary. This results in some costs — transaction costs — that are not usually taken into account by administrators and managers. The use of FLOSS - Free/Libre/Open Source Software

leads to a reduction in transaction costs in terms of computation costs and in terms of the number of managed contracts, which can be numerically reduced by half.

SUNK COST

In economics and in business decision-making, sunk costs are costs that have already been incurred and which cannot be recovered to any significant degree. Sunk costs are sometimes contrasted with variable costs, which are the costs that will change due to the proposed course of action. In microeconomic theory, only variable costs are relevant to a decision. Economics proposes that a rational actor does not let sunk costs influence one's decisions, because doing so would not be assessing a decision exclusively on its own. It is important to note that the decision-maker may make rational decisions according to their own incentives; these incentives may dictate different decisions than would be dictated by efficiency or profitability, and this is considered an incentive problem and distinct from a sunk cost problem.

For example, when one pre-orders a non-refundable movie ticket, the price of the ticket becomes a sunk cost. Even if the ticket-buyer decides that he would rather not go to the movie, there is no way to get back the money he originally paid.

The sunk cost is distinct from the economic loss. For example, when a car is purchased, it can subsequently be resold; however, it will certainly not be resold for the original purchase price. The economic loss is the difference (including transaction costs). The sum originally paid should not affect any rational future decision-making about the car, regardless of the resale value: if the owner can derive more value from selling the car than not selling it, it should be sold, regardless of the price paid.

In this sense, the sunk cost is not a precise quantity, but an economic term for a sum paid, in the past, which should no longer be relevant; it may be used inconsistently in quantitative terms as the original cost or the expected economic loss. It may also be used as shorthand for an error in analysis due to the sunk cost fallacy, non-rational decision-making or, most simply, as irrelevant data.

Economists argue that sunk costs are not taken into account when making rational decisions. In the case of the movie ticket, there are two possible end results. The ticket-buyer will have:

- Paid the price of the ticket and suffered watching a movie that he does not want to see, or;
- Paid the price of the ticket and used the time to do something more fun.

In either case, the ticket-buyer has "paid the price of the ticket" so *that part* of the decision should cancel itself out. If the ticket-buyer regrets buying the ticket, the current decision should be based on whether he wants to see the movie at all, regardless of the price, just as if he were to go to a free movie. The economist will suggest that since the latter option only involves suffering in one way (spent money), while the former involves suffering in two (spent money plus wasted time), the latter is obviously preferable.

Sunk costs may cause cost overrun. In business, an example of sunk costs may be investment into a factory or research that now has a lower value or no value whatsoever. For example, \$20 million has been spent on building a powerplant; the value at present is zero because it is incomplete (and no sale or recovery is feasible). The plant can be completed for an additional \$10 million, or abandoned and a different facility built for \$5 million. It should be obvious that abandonment and construction of the alternative facility is the more rational decision, even though it represents a total loss on the original expenditure - the original sum invested is a sunk cost.

If decision-makers are (economically) irrational, or have the wrong incentives, the completion of the project may be chosen.

For example, politicians or managers may have more incentive to avoid the appearance of a total loss. In practice, there is considerable ambiguity and uncertainty in such cases, and decisions may in retrospect appear irrational that were, at the time, reasonable to the economic actors involved and in the context of their own incentives.

Behavioural economics recognizes that sunk costs often affect economic decisions due to loss aversion: the price paid becomes a benchmark for the value, whereas the price paid should be irrelevant.

This is considered non-rational behaviour (as rationality is defined by classical economics). Economic experiments have shown that the sunk cost fallacy and loss aversion are common, and hence economic rationality — as assumed by much of economics—is limited. This has enormous implications for finance, economics, and securities markets in particular. Daniel Kahneman won the Nobel Prize in Economics in part for his extensive work in this area with his late partner, Amos Tversky.

Features Characterizing the Sunk Cost Heuristic

Two specific features characterizing the Sunk Cost heuristic worth mentioning are:

- An overly optimistic probability bias, whereby after an investment the evaluation of one's investment reaping dividends is increased.
- The requisite of personal responsibility. Sunk Cost appears to operate chiefly in those who feel personal responsibility for the investments that are to be viewed as sunk.

Overly Optimistic Probability Bias

Knox and Inkster (1968), in what is perhaps the classic sunk cost experiment, approached 141 horse bettors. 72 people who had just finished placing a \$2.00 bet within the past thirty seconds, and 69 people who were about to place a \$2.00 bet in the next thirty seconds. Their hypothesis was that people who had just committed themselves to a course of action (betting \$2.00) would reduce post-decisional dissonance by believing more strongly than ever that they had picked a winner. Knox and Inkster asked the bettors to rate their horse's chances of winning on a 7 point scale.

What they found was that people who were about to place a bet rated the chance that their horse would win at an average of 3.48 which corresponded to a “fair chance of winning”, whereas people who had just finished betting gave an average rating of 4.81 which corresponded to a “good chance of winning”. Their hypothesis was confirmed - after making a \$2.00 commitment, people became more confident their bet would pay off. Knox and Inkster performed an ancillary test on the Patrons of the horses themselves and managed (after normalization) to repeat their finding almost identically.

Requisite of Personal Responsibility

In a study of 96 business students Staw and Fox (1977) gave the subjects a choice between making an R&D investment in either an underperforming company department, or in other sections of the hypothetical company. Staw and Fox divided the participants into two groups; a low responsibility condition and a high responsibility condition. In the high responsibility condition the participants were told that they as manager had made an earlier, disappointing R&D investment.

In the low responsibility condition, subjects were told that a former manager had made a previous R&D investment in the underperforming division and were given the same profit data as the other group. In both cases subjects were then asked to make a new \$20 million investment. There was a significant interaction between assumed responsibility and average investment, with the high responsibility condition averaging \$12.97 million and the low condition averaging \$9.43 million.

Loss Aversion and the Sunk Cost Fallacy

Many people have strong misgivings about “wasting” resources. This is called “loss aversion”. Many people, for example, would feel obligated to go to the movie despite not really wanting to, because doing otherwise would be wasting the ticket price; they feel they passed the point of no return. This is sometimes called the *sunk cost fallacy*. Economists would label this behaviour “irrational”: It is inefficient because it misallocates resources by depending on information that is irrelevant to the decision being made. This line of thinking, in turn, may reflect a non-standard measure of utility, which is ultimately subjective and unique to the consumer. When a ticket-buyer purchases a ticket in advance to a bad movie, he has still made a semi-public commitment to watching it.

The ticket-buyer may “save face” by sticking it out, a satisfaction he cannot draw if he leaves. To leave early is to make his lapse of judgement manifest to strangers, an appearance he may rationally choose to avoid. He may, in fact, find some amusement in how bad the movie turned out to be, and take pride that he recognised it to be bad. Or he may feel qualified to criticize the movie in front of his peers. Either way, this mitigates the decision to view the movie, not the decision to purchase the ticket. The idea of sunk costs is often employed when analyzing business decisions.

A common example of a sunk cost for a business is the promotion of a brand name. This type of marketing incurs costs that cannot normally be recovered. It is not typically possible to later “demote” one’s brand names in exchange for cash (except perhaps as an exit strategy from a market).

Decisions about future investments, sales or more advertising should be made based on future possibilities, not biased by the recent large investment in the advertising that the company made last year (or even last week). The sunk cost fallacy is also sometimes known as the “Concorde Effect”, referring to the fact that the British and French government continued to fund the joint development of Concorde even after it became apparent that there was no longer an economic case for the aircraft. The project was regarded privately by the British government as a “commercial disaster” which should never have been started, and was almost cancelled, but political and legal issues ultimately made it impossible for either government to pull out.

By deliberately using the tactic of incurring sunk costs beyond the point of no return, economic actors may get ventures going that otherwise would not have. In his autobiography, film director Elia Kazan explains how he repeatedly used the tactic of sunk costs to get his films started: “My tactic was one familiar to directors ... to get the work rolling, involve actors contractually, build sets, collect props and costumes, expose negative, and so get the studio in deep.

Once money in some significant amount had been spent, it would be difficult for to do anything except scream and holler. If he suspended a film that had been shooting for a few weeks, he’d be in for an irretrievable loss, not only of money but of ‘face.’ The thing to do was get the film going”. The same tactic has been used for the construction of bridges, tunnels, and other infrastructure. Very few partially built bridges exist, because once construction has started sunk costs are too high to revert the decision and stop again. The sunk cost tactic often causes cost overrun.

SHORT AND LONG RUN COSTS

A textbook distinction is made between short-run and long-run marginal cost. The former takes as unchanged, for example, the capital equipment and overhead of the producer, any change in its production involving only changes in the inputs of labour, materials and energy. The latter allows all inputs, including capital items (plant, equipment, buildings) to vary. A long-run cost function describes the cost of production as a function of output assuming that all inputs are obtained at current prices, that current technology is employed, and everything is being built new from scratch. In view of the durability of many capital items this textbook concept is less useful than one which allows for some scrapping of existing capital items or the acquisition of new capital items to be used with the existing stock of capital items acquired in the past.

Long-run marginal cost then means the additional cost or the cost saving per unit of additional or reduced production, including the expenditure on additional capital goods or

any saving from disposing of existing capital goods. Note that marginal cost upwards and marginal cost downwards may differ, in contrast with marginal cost according to the less useful textbook concept. Economies of scale are said to exist when marginal cost according to the textbook concept falls as a function of output and is less than the average cost per unit. This means that the average cost of production from a larger new built-from-scratch installation falls below that from a smaller new built-from-scratch installation. Under the more useful concept, with an existing capital stock, it is necessary to distinguish those costs which vary with output from accounting costs which will also include the interest and depreciation on that existing capital stock, which may be of a different type from what can currently be acquired in past years at past prices. The concept of economies of scale then does not apply.

The long run average cost (LRAC or LAC) curve illustrates - for a given quantity of production - the average cost per unit which a firm faces in the long run (*i.e.*, when no factors of production are fixed). LRAC curve is derived from a series of short run average cost curves. It is also called the 'envelope curve' since it envelops all the short run average cost curve. In perfect competition, the LRAC curve is flat, at the point of equilibrium- there are constant returns to scale. Typical LRACs are U-shaped, which means that up to a certain optimum point, there are economies of scale, and as production increases beyond this, there are diseconomies of scale. LRAC are generally flatter than short run average cost curve. In some industries, the LRAC is L-shaped, and economies of scale increase indefinitely. This means that the largest firm tends to have a cost advantage, and the industry tends naturally to become a monopoly, and hence is called a natural monopoly. Natural monopolies tend to exist in industries with high capital costs in relation to variable costs, such as water supply and electricity supply.

Externalities

Externalities are costs (or benefits) that are not borne by the parties to the economic transaction. A producer may, for example, pollute the environment, and others may bear those costs. A consumer may consume a good which produces benefits for society, such as education; because the individual does not receive all of the benefits, he may consume less than efficiency would suggest. Alternatively, an individual may be a smoker or alcoholic and impose costs on others. In these cases, production or consumption of the good in question may differ from the optimum level.

Negative Externalities of Production

Much of the time, private and social costs do not diverge from one another, but at times social costs may be either greater or less than private costs.

When marginal social costs of production are greater than that of the private cost function, we see the occurrence of a negative externality of production. Productive processes that result in pollution are a textbook example of production that creates negative externalities.

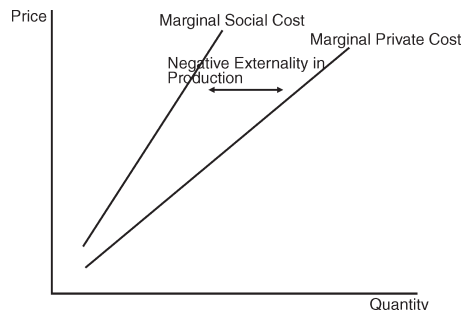


Fig. Negative Externalities of Production

Such externalities are a result of firms externalising their costs onto a third party in order to reduce their own total cost. As a result of externalising such costs we see that members of society will be negatively affected by such behaviour of the firm. In this case, we see that an increased cost of production on society creates a social cost curve that depicts a greater cost than the private cost curve. In an equilibrium state we see that markets creating negative externalities of production will overproduce that good. As a result, the socially optimal production level would be lower than that observed.

Positive Externalities of Production

When marginal social costs of production are less than that of the private cost function, we see the occurrence of a positive externality of production.

Production of public goods are a textbook example of production that create positive externalities. An example of such a public good, which creates a divergence in social and private costs, includes the production of education. It is often seen that education is a positive for any whole society, as well as a positive for those directly involved in the market.

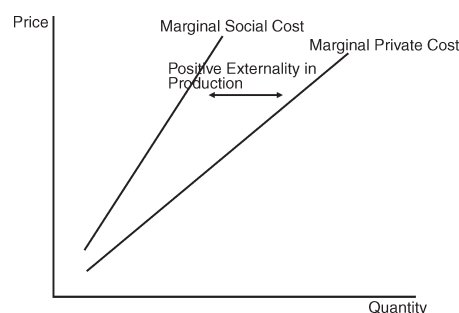


Fig. Positive Externalities of Production

Examining the relevant diagram we see that such production creates a social cost curve that is less than that of the private curve. In an equilibrium state we see that markets creating positive externalities of production will under produce that good.

As a result, the socially optimal production level would be greater than that observed.

Social Costs

Of great importance in the theory of marginal cost is the distinction between the marginal *private* and *social* costs. The marginal private cost shows the cost associated to the firm in question. It is the marginal private cost that is used by business decision makers in their profit maximization goals, and by individuals in their purchasing and consumption choices. Marginal social cost is similar to private cost in that it includes the cost functions of private enterprise but *also* that of society as a whole, including parties that have no direct association with the private costs of production. It incorporates all negative and positive externalities, of both production and consumption. Hence, when deciding whether or how much to buy, buyers take account of the cost to society of their actions *if* private and social marginal cost coincide. The equality of price with social marginal cost, by aligning the interest of the buyer with the interest of the community as a whole is a necessary condition for economically efficient resource allocation.

COST-PLUS PRICING

Cost-plus pricing is a pricing method commonly used by firms. It is used primarily because it is easy to calculate and requires little information. There are several varieties, but the common thread in all of them is that you first calculate the cost of the product, then include an additional amount to represent profit. Cost-plus pricing is often used on government contracts, and has been criticized as promoting wasteful expenditures.

Calculating Price Using the Cost-plus Method

There are several ways of determining cost, and the profit can be added as either a percentage markup or an absolute amount. One example is:

$$P = (\text{AVC} + \text{FC}\%) * (1 + \text{MK}\%)$$

where:

P = price

AVC = average variable cost

FC% = percentage allocation of fixed costs

MK% = percentage markup

For example: If variable costs are 30 yen, the allocation to cover fixed costs is 10 yen, and you feel you need a 50% markup then you would charge a price of 60 yen:

$$P = (30 + 10) \cdot (1 + 0.50)$$

$$P = 40 \cdot 1.5$$

$$P = 60$$

An alternative way of doing a similar calculation is:

$$P = (\text{AVC} + \text{FC}\%) / (1 - \text{MK}\%)$$

To make things simpler, some firms, particularly retailers, ignore fixed costs and just use the purchase price paid to their suppliers as the cost term. They indirectly incorporate

the fixed cost allocation into the markup percentage. To simplify things even further, sometimes a fixed amount is applied rather than a percentage. This fixed amount is usually determined by head-office to make it easy for franchisees and store managers. This is sometimes referred to as turnkey pricing. Another variant of cost plus pricing is activity based pricing. This involves being more careful in determining costs. Instead of using arbitrary expense categories when allocating overhead, every activity is linked to the resources it uses. Cost will need to be recalculated and the percentage markup will likely need to be adjusted as the product goes through its life cycle. This is sometimes referred to as product life cycle pricing, although it is seldom done deliberately or in a planned and organized manner. Price skimming and penetration pricing are also types of product life cycle pricing but they are demand based pricing methods rather cost based.

Advantages of Cost-plus Pricing

- Easy to calculate
- Minimal information requirements
- Easy to administer
- Tends to stabilize markets - insulated from demand Variations and competitive factors
- Insures seller against unpredictable, or unexpected later costs
- Ethical advantages

Disadvantages of Cost-plus Pricing

- Tends to ignore the role of consumers
- Tends to ignore the role of competitors
- Use of historical accounting costs rather than replacement value
- Use of “normal” or “standard” output level to allocate fixed costs
- Inclusion of sunk costs rather than just using incremental costs
- Ignores opportunity costs
- Contractors may not focus on performance because the cost is always covered by the client

Cost-plus Pricing with Elasticity Considerations

One of the most common pricing methods used by firms is cost-plus pricing. In spite of its ubiquity, economists rightly point out that it has serious methodological flaws. It takes no account of demand. There is no way of determining if potential customers will purchase the product at the calculated price. To compensate for this, some economists have tried to apply the principles of price elasticity to cost-plus pricing.

We know that:

$$MR = P + ((dP / dQ) * Q)$$

where:

MR = marginal revenue

P = price

(dP / dQ) = the derivative of price with respect to quantity

Q = quantity

Since we know that a profit maximizer, sets quantity at the point that marginal revenue is equal to marginal cost ($MR = MC$), the formula can be written as:

$$MC = P + ((dP / dQ) * Q)$$

Dividing by P and rearranging yields:

$$MC/P = 1 + ((dP / dQ) * (Q / P))$$

And since (P / MC) is a form of markup, we can calculate the appropriate markup for any given market elasticity by:

$$(P/MC) = (1 / (1 - (1/E)))$$

where:

(P/MC) = markup on marginal costs

E = price elasticity of demand

In the extreme case where elasticity is infinite:

$$(P/MC) = (1 / (1 - (1/9999999999999999)))$$

$$(P/MC) = (1 / 1)$$

Price is equal to marginal cost. There is no markup. At the other extreme, where elasticity is equal to unity:

$$(P/MC) = (1 / (1 - (1/1)))$$

$$(P/MC) = (1 / 0)$$

The markup is infinite. Most business people do not do marginal cost calculations, but one can arrive at the same conclusion using average variable costs (AVC):

$$(P/AVC) = (1 / (1 - (1/E)))$$

Technically, AVC is a valid substitute for MC only in situations of constant returns to scale ($LVC = LAC = LMC$).

When business people choose the markup that they apply to costs when doing cost-plus pricing, they should be, and often are, considering the price elasticity of demand, whether consciously or not.

DIRECT AND INDIRECT COST

Identification with the sponsored work rather than the nature of the goods and services involved is the determining factor in distinguishing direct from indirect cost of sponsored agreements. Direct Costs are those costs that can be identified specifically with a particular sponsored project, an instructional activity, or any other institutional activity, or that can be directly assigned to such activities relatively easily with a high degree of accuracy. Costs incurred for the same purpose in like circumstances must be treated consistently as either direct or indirect costs. Where an institution treats a particular type of cost as a direct cost of sponsored agreements, all costs incurred for the same purpose in like circumstances shall be treated as direct costs of all activities of the institution.

“*Indirect cost*” is defined in the same document as “expenditure on labour, materials or services which cannot be economically identified with a specific saleable cost unit”. These costs are incurred in simply providing a university infrastructure that can support research, such as departmental staff, general equipment, space, etc. The costs are typically charged to a central code and, being invisible to the individual research budget holder, are often mistakenly regarded as notional. Unfortunately, although the technical definitions of direct and indirect are fairly simple, sponsors tend to have a different approach and thus have a different meaning for direct and indirect costs.

LONG RUN PRODUCTION FUNCTIONS AND COST FUNCTIONS

$$Q = f(K, L).$$

Quantity is a function of the inputs used to produce it: in this example capital and Labour. Quantity is measured as a rate of production (flow) as are capital and Labour, *e.g.*, the amount of cars washed per day is a function of the amount of Labour and capital used each day.

- The production function specifies a technically efficient use of Labour and capital necessary to produce output, *i.e.*, no resources are “wasted.”
- The cost function specifies an economically efficient use of resources, *i.e.*, the firm chooses the least cost combination of inputs, to produce a given output.
- This yields the long run cost function: total costs (C) = $g(Q)$, which depends on the prices of inputs. This function can be a pretty good proxy for the opportunity cost of delivering Q , at least where we measure costs in units of present value: *i.e.*, the change in present value or owner’s equity caused by some specified action (and where for purposes of measurement the attendant increase in wealth is excluded from the computation of equity).

The following long-run functional relationships traditionally obtain in the single product case: Cost varies as a function of total production volume (V), the rate of output (x), the date of first delivery (T), and the date of completion of the full production run (m), where $x(t)$ denotes the rate of output at moment t . Moreover, as the total quantity of units produced increases, the cost of future output tends to decline because production-knowledge increases as a result of production experience (this proposition is known as the ‘learning’ or ‘progress’ curve”). That is: $dC / dT \mid x = x_0, V = V_0 < 0$. This relationship probably holds for most products produced in large batches using traditional mass-production methods. Moreover it is usually assumed that:

- $dC / dx(t) \mid T = T_0, V = V_0 > 0$
- $d^2C / dx(t)^2 \mid T = T_0, V = V_0 > 0$
- $dC / dV \mid x = x_0, T = T_0 > 0$
- $d^2C / dV^2 \mid x = x_0, T = T_0 < 0$

Many of these functional relationships have been attenuated by the rise of computer assisted design and manufacturing technology and modern information.

Long Run Cost Minimization: Marginal Analysis

Here we will assume that the firm can choose any level of capital and Labour to produce output, Q . More capital leads to more output; less capital to less output. More Labour leads to more output; less Labour to less output. This is known as a variable proportions production technology because Labour can substitute for capital, and vice-versa, in production.

- The marginal product of Labour is the additional output from one extra unit of Labour.
- The marginal product of capital is the additional output from one more unit of capital.

The cost minimization rule for producing a given quantity: choose Labour and capital such that $(MP \text{ of Labour})/(\text{Price of Labour}) = (MP \text{ of capital})/(\text{price of capital})$.

Proof: dividing the marginal products of each input by the price of the input tells you how much output you can produce for a dollar. If it costs more to produce output using Labour than it does using capital, then sell Labour and buy capital. This allows you to produce the same amount at lower cost. Only when the costs of production using each input are the same are no further cost savings possible.

- If $(MP \text{ of labour})/(\text{Price of labour})$ is greater than $(MP \text{ of capital})/(\text{price of capital})$, sell capital, buy Labour.
- If $(MP \text{ of labour})/(\text{Price of labour})$ is less than $(MP \text{ of capital})/(\text{price of capital})$, sell Labour, buy capital.

Long Run Cost Minimization: Incremental Analysis

Incremental analysis considers large discrete changes in input mix, whereas marginal analysis considers small continuous changes in input mix.

Example: 1985: John Deere acquisition of Versatile. John Deere had an old fashioned production line for making farm tractors. Very high fixed costs, but low marginal costs. Versatile had a “garage” style production facility with much lower fixed costs, but higher marginal costs.

It is easy to see that if production is less than or equal to 6 units, then Versatile has lower costs of production. If production is larger than or equal to 7 units, John Deere has lower costs of production.

- The long run decision between these two production processes would depend on how many units you thought you would sell. If you anticipated selling 7 or more units, use the John Deere production process, if you thought you would sell 6 or fewer units, use the Versatile production process.

Sensitivity Analysis

Suppose you were uncertain about how many units you anticipated selling. Sensitivity analysis allows you to build in uncertainty to your analysis by determining the costs of various scenarios.

Suppose you thought that your uncertainty was best described by a trinomial random variable:

- With $p_1 = .4$, $Q = 7$
- With $p_2 = .3$, $Q = 10$
- With $p_3 = .3$, $Q = 4$; note that $p_3 = 1 - p_1 - p_2$

What's the best technology to choose?

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
1	Q	Prob(Q)	AC of Deere	AC of Versatile
2	4	0.3	\$18.50	\$16.25
3	7	0.4	\$13.14	\$13.57
4	10	0.3	\$11.00	\$12.50
5	Expected[AC]		\$14.11	\$14.05

Answer: The expected output is 7 ($.4 \cdot 7 + .3 \cdot 4 + .3 \cdot 10 = 7$), the average cost of the expected output is not the same as the expected average cost, because Versatile has a large advantage at small outputs, while Deere has a small advantage at high outputs.

The table below shows how to compute expected average costs, which is the usually right criterion to use for deciding which technology to adopt.

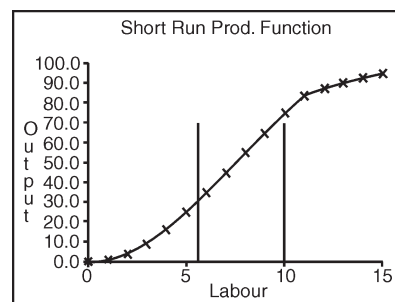
Short run Production Functions

A long run production function relates the output produced to the inputs used, *e.g.*, $Q = f(\text{capital, Labour})$. In the short run, some inputs cannot be varied, so the firm does not have as much flexibility as in the long run. In this case, the short run production function is a function of only the inputs that can be varied. Suppose that capital is fixed in the short run. Then $Y = g(\text{Labour})$

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
1.	Labour Output	Output	Marg. Tot. cost	Tot.Cost Cost	Av. Cost	Av. Var.	Marg.
2.	0	0.0		\$20			
3.	1	1.0	1.00	\$25.00	\$25.00	\$5.00	\$5.00
4.	2	4.0	3.00	\$30.00	\$7.50	\$2.50	\$1.67
5.	3	9.0	5.00	\$35.00	\$3.89	\$1.67	\$1.00
6.	4	16.0	7.00	\$40.00	\$2.50	\$1.25	\$0.71
7.	5	25.0	9.00	\$45.00	\$1.80	\$1.00	\$0.56
8.	6	35.0	10.00	\$50.00	\$1.43	\$0.86	\$0.50
9.	7	45.0	10.00	\$55.00	\$1.22	\$0.78	\$0.50
0.	8	55.0	10.00	\$60.00	\$1.09	\$0.73	\$0.50
11.	9	65.0	10.00	\$65.00	\$1.00	\$0.69	\$0.50
12.	10	75.0	10.00	\$70.00	\$0.93	\$0.67	\$0.50
13.	11	84.0	9.00	\$75.00	\$0.89	\$0.65	\$0.56
14.	12	87.7	3.73	\$80.00	\$0.91	\$0.68	\$1.34
15.	13	90.6	2.86	\$85.00	\$0.94	0.72	\$1.75
16.	14	93.0	2.41	\$90.00	\$0.97	0.75	\$2.07
17.	15	95.1	2.12	\$95.00	\$1.00	\$0.79	\$2.35

The “usual” shape of the short run production function:

- In the short run, output at first increases at an increasing rate with increases in Labour (increasing returns to labour)
- Then output increases at a constant rate with increases in Labour (constant returns to Labour).
- Finally, output increases at a decreasing rate with increases in Labour (diminishing returns to Labour).

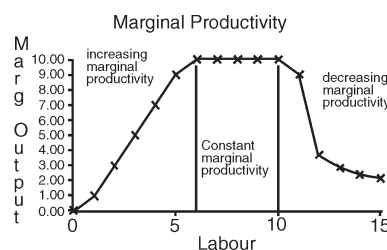


Short run cost functions are larger than long run cost functions because, in the short run, fixed inputs can not be varied.

In the long run, all inputs can be varied, and this greater flexibility allows you to achieve lower costs, *i.e.*, $h(Q)$ is greater than or equal to $g(Q)$.

A Numerical Example

Assume, the following short run production function: Fixed costs = \$20/hour, Labour costs \$5/hour. The marginal productivity of Labour increases, then is constant, and then decreases.



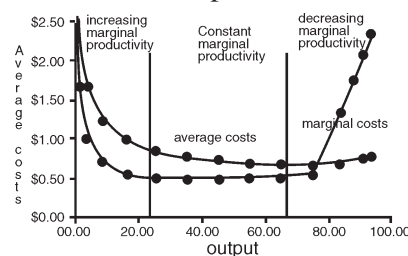
Suppose that the output sells for \$1 per unit and that the firm can sell all it wants at a price of \$1 (infinite elasticity, or a perfectly competitive firm). How much Labour should the firm hire?

Answer: Keep hiring as long as the benefit of hiring another worker is greater than the cost of another worker.

The benefit equals the marginal revenue of the worker (price times the marginal production), Benefit = \$1 (marginal production); the cost is the wage. Cost = \$5.

1.	A Labour	B Output	C Marg. Output	D Benefit of Extra Labour	E Cost of Extra Labour	F Profits
2.	0	0.0				-\$20.00
3.	1	1.0	1.00	\$1.00	\$5.00	-\$24.00
4.	2	4.0	3.00	\$3.00	\$5.00	-\$26.00
5.	3	9.0	5.00	\$5.00	\$5.00	-\$26.00
6.	4	16.0	7.00	\$7.00	\$5.00	-\$24.00
7.	5	25.0	9.00	\$9.00	\$5.00	-\$20.00
8.	6	35.0	10.00	\$10.00	\$5.00	-\$15.00
9.	7	45.0	10.00	\$10.00	\$5.00	-\$10.00
10.	8	55.0	10.00	\$10.00	\$5.00	-\$5.00
11.	9	65.0	10.00	\$10.00	\$5.00	\$0.00
12.	10	75.0	10.00	\$10.00	\$5.00	\$5.00
13.	11	84.0	9.00	\$9.00	\$5.00	\$9.00
14.	12	87.7	3.73	\$73	\$5.00	\$7.73
15.	13	90.6	2.86	\$2.86	\$5.00	\$5.59
16.	14	93.0	2.41	\$2.41	\$5.00	\$3.00
17.	15	95.0	2.12	\$2.12	\$5.00	\$0.12

- It is easy to see that profits are maximized with 11 labourers.
- Can we derive the optimal production decision (how much Labour to produce) from the cost curves instead of the production curves?



Yes: keep producing as long as the benefit of producing another unit (\$1) exceeds the cost of producing another unit. Looking at the graph, you can see that \$1 intersects the marginal cost curve somewhere between 10 and 11 labourers (between 75 and 84 units). To determine whether to produce at 75 or 84, you must look at the spread sheet.

Shut Down Analysis

In the long run, stop producing if economic profits are negative. In the short run, stop producing if revenue is less than total variable cost.

In the short run, you don't have to cover your fixed costs, but you must make enough to cover your variable costs. If not, then you can shut down. You will still have to pay your fixed costs, but at least you can avoid paying your variable costs (which are greater than revenue).

Break-even Analysis

Assumptions: constant price, constant average variable cost. (P-AVC) is sometimes called “contribution margin” because it represents profit per unit sold (ignoring fixed costs).

Set profits equal to zero to solve for how much output would be required to “break even.”

Another way of asking the same question is to ask how much quantity would be required to produce enough profit to cover fixed costs. Revenue-variable costs-fixed Costs = 0 $P*Q - AVC*Q - \text{fixed Costs} = 0$ $(P-AVC)*Q - \text{fixed Costs} = 0$ $Q = \text{fixed Costs} / (P-AVC)$.

THE SUPPLY CURVE

Price usually is a major determinant in the quantity supplied. For a particular good with all other factors held constant, a table can be constructed of price and quantity supplied based on observed data. Such a table is called a supply schedule, as shown in the following example:

Table. Supply Schedule

Price	Quantity Supplied
1	12
2	28
3	42
4	52
5	60

By graphing this data, one obtains the supply curve:

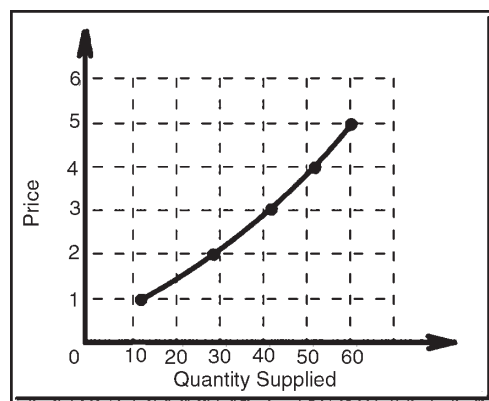


Fig. Supply Curve

As with the demand curve, the convention of the supply curve is to display quantity supplied on the x-axis as the independent variable and price on the y-axis as the dependent variable.

The law of supply states that the higher the price, the larger the quantity supplied, all other things constant. The law of supply is demonstrated by the upward slope of the supply curve. As with the demand curve, the supply curve often is approximated as a straight line to simplify analysis. A straight-line supply function would have the following structure:

$$\text{Quantity} = a + (b \times \text{Price})$$

where a and b are constant for each supply curve.

A change in price results in a change in quantity supplied and represents movement along the supply curve.

SHIFTS IN THE SUPPLY CURVE

While changes in price result in movement along the supply curve, changes in other relevant factors cause a shift in supply, that is, a shift of the supply curve to the left or right.

Such a shift results in a change in quantity supplied for a given price level. If the change causes an increase in the quantity supplied at each price, the supply curve would shift to the right:

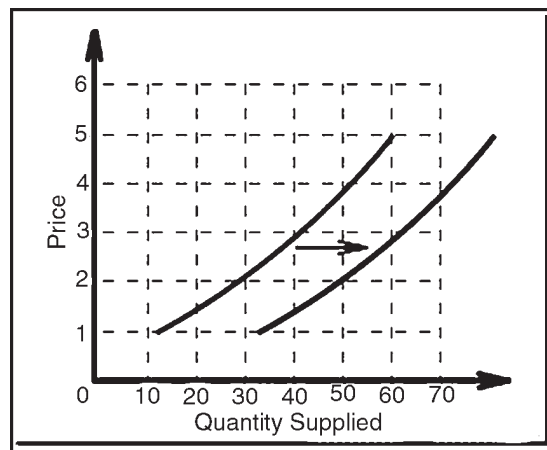


Fig. Supply Curve Shift

There are several factors that may cause a shift in a good's supply curve. Some supply-shifting factors include:

- *Prices of other goods:* The supply of one good may decrease if the price of another good increases, causing producers to reallocate resources to produce larger quantities of the more profitable good.
- *Number of sellers:* More sellers result in more supply, shifting the supply curve to the right.
- *Prices of relevant inputs:* If the cost of resources used to produce a good increases, sellers will be less inclined to supply the same quantity at a given price, and the supply curve will shift to the left.
- *Technology:* Technological advances that increase production efficiency shift the supply curve to the right.

- *Expectations*: If sellers expect prices to increase, they may decrease the quantity currently supplied at a given price in order to be able to supply more when the price increases, resulting in a supply curve shift to the left.

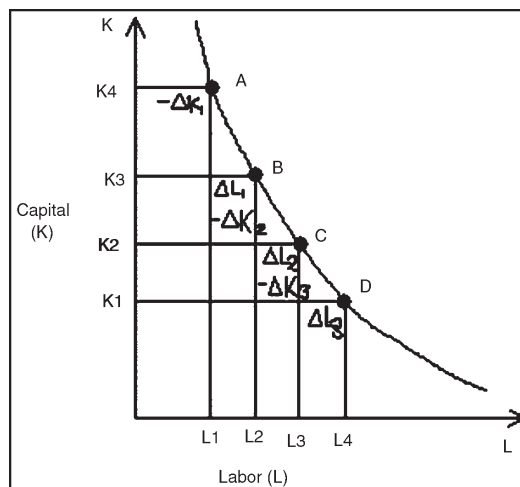
ISOQUANT CURVE

The term “isoquant” has been derived from the Greek word “iso” meaning equal and Latin word “quantus” meaning “quantity”. The “isoquant curve” is, therefore, also known as “Equal product curve” or “Production Intelligence Curve”. An isoquant curve is locus of point representing various combinations of two inputs - capital and labour - yielding the same output. *Isoquant curves are drawn on the basis of the following assumptions*:

- There are only two inputs, viz, labour (L) and capital (K) to produce a commodity X.
- The two inputs - L and K - can substitute each other but at diminishing rate.
- The technology of production is given.

Given these assumptions, it is always possible to produce given quantity of commodity X with various combination of labour and capital. The factor combinations are so formed that the substitution of one factor for the other leaves the output unaffected. The technology is presented through an isoquant curve (IQ1 = 100). The curve IQ1 all along its length represents a fixed quantity, 100 units of product X. This quantity of output can be produced with a number of labour-capital combination.

For example: Points A, B, C and D on the isoquant curve IQ1 shows four different combinations of inputs, K and L, all yielding the same output - 100 units. The movement from A to B indicates decreasing Quantity Of K and increasing number of L. This implies substitution of labour for capital such that all the input combinations yield the same quantity of commodity X i.e., IQ1 = 100.



INDIFFERENCE CURVE ANALYSIS

The aim of indifference curve analysis is to analyse how a rational consumer chooses between two goods. In other words, how the change in the wage rate will affect the choice between leisure time and work time. Indifference analysis combines two concepts; indifference curves and budget lines (constraints)

THE INDIFFERENCE CURVE

An indifference curve is a line that shows all the possible combinations of two goods between which a person is indifferent. In other words, it is a line that shows the consumption of different combinations of two goods that will give the same utility to the person. For instance, in Figure the indifference curve is I1.

A person would receive the same utility from consuming 4 hours of work and 6 hours of leisure, as they would if they consumed 7 hours of work and 3 hours of leisure.

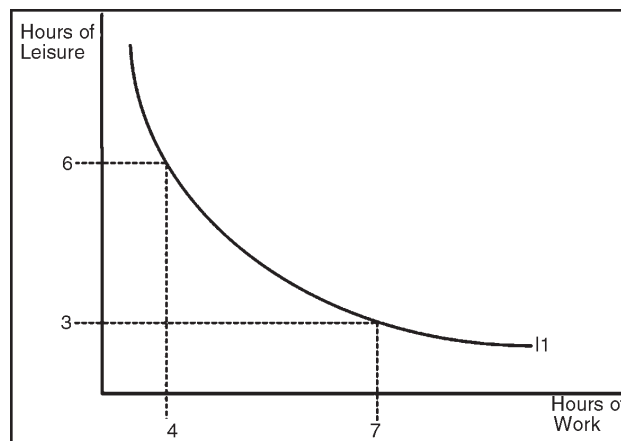


Fig. An Indifference Curve for Work and Leisure

An important point is to remember that the use of an indifference curve does not try to put a physical measure onto how much utility a person receives.

THE SHAPE OF THE INDIFFERENCE CURVE

Figure highlights that the shape of the indifference curve is not a straight line. It is conventional to draw the curve as bowed. This is due to the concept of the diminishing marginal rate of substitution between the two goods. The marginal rate of substitution is the amount of one good that has to be given up if the consumer is to obtain one extra unit of the other good.

The equation is: The marginal rate of substitution (MRS) = change in good X/ change in good Y. Using Figure, the marginal rate of substitution between point A and Point B is;

$$\text{MRS} = -3/3 = -1 = 1$$

Note, the convention is to ignore the sign.

The reason why the marginal rate of substitution diminishes is due to the principle of diminishing marginal utility. Where this principle states that the more units of a good are consumed, then additional units will provide less additional satisfaction. Therefore, as a person consumes more of one good then they will receive diminishing utility for that extra unit, hence, they will be willing to give up less of their leisure to obtain one more unit of work. The relationship between marginal utility and the marginal rate of substitution is often summarised with the following equation;

$$MRS = Mu_x / Mu_y$$

It is possible to draw more than one indifference curve on the same diagram. If this occurs then it is termed an indifference curve map.

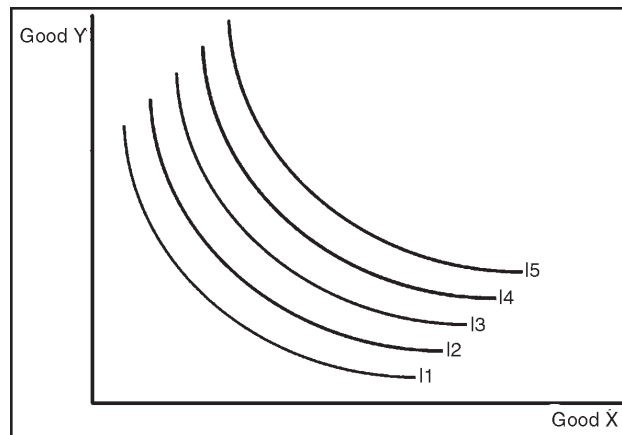


Fig. An Indifference Map

The general rule is that indifference curves further to the right show combinations of the two goods that yield a higher utility, while curves to the left show combinations that yield lower levels of utility.

A BUDGET LINE

The budget line is an important component when analysing consumer behaviour. The budget line illustrates all the possible combinations of two goods that can be purchased at given prices and for a given consumer budget. Remember, that the amount of a good that a person can buy will depend upon their income and the price of the good. This discussion outlines the construction of a budget line and how the change in the determinants will affect the budget line.

With a limited budget the consumer can only consume a limited combination of x and y. A change in consumer income and the budget line. If consumer income increases then the consumer will be able to purchase higher combinations of goods. Hence an increase in consumer income will result in a shift in the budget line. This is illustrated in Figure. Note that the prices of the two goods have remained the same, therefore, the increase in income will result in a parallel shift in the budget line.

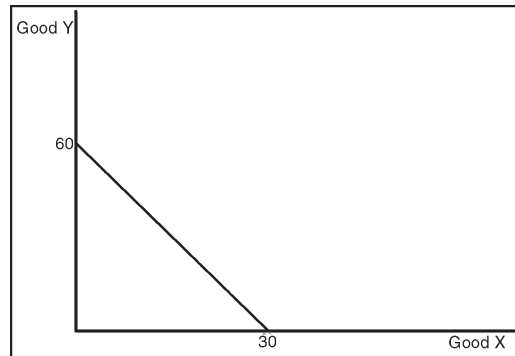


Fig. Constructs a Budget Line for a given Budget of £60, £2 per Unit of x and £1 per Unit of y.

Assume consumer income increased to £90.

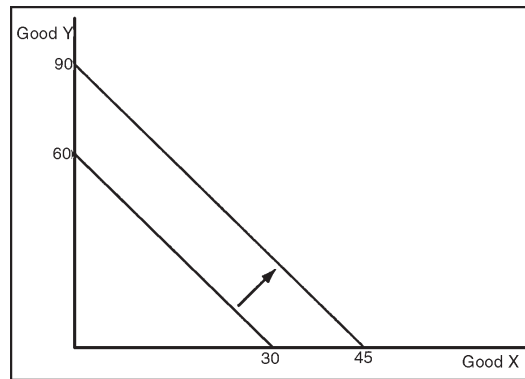


Fig. An Increase in Consumer Income

If consumer income fell then there would be a corresponding parallel shift to the left to represent a fall in the potential combinations of the two goods that can be purchased. A change in the price of a good and the budget line. If income is held constant, and the price of one of the goods changes then the slope of the curve will change. In other words, the curve will pivot. This is illustrated in Figure.

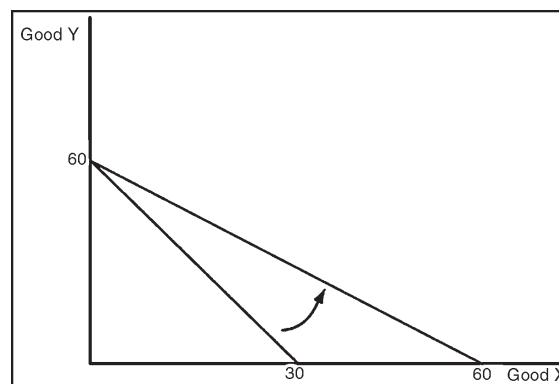


Fig. A Change in Price

The reduction of the price of good x from £2 to £1 means that on a fixed budget of £60, the consumer could purchase a maximum of 60 units, as opposed to 30. Note that the price of good y has remained fixed, hence the maximum point for good y will remain fixed.

INDIFFERENCE ANALYSIS COMBINES TWO CONCEPTS; INDIFFERENCE CURVES AND BUDGET LINES

The first stage is to impose the indifference curve and the budget line to identify the consumption point between two goods that a rational consumer with a given budget would purchase. The optimum consumption point is illustrated on Figure.

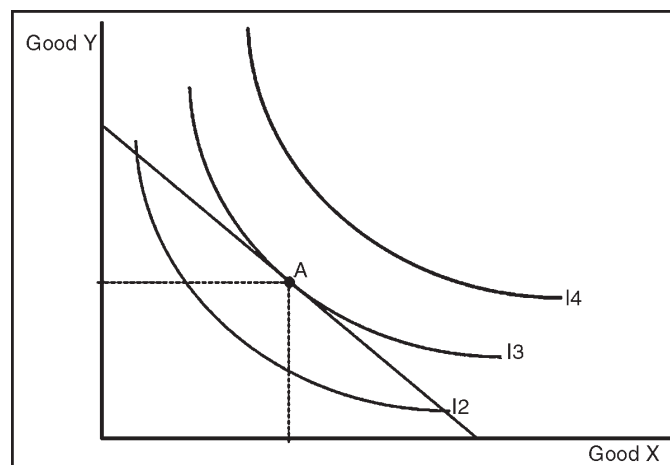


Fig. The Optimum Consumption Point

A rational, maximising consumer would prefer to be on the highest possible indifference curve given their budget constraint.

This point occurs where the indifference curve touches the budget line. In the case of Figure, the optimum consumption point occurs at point A on indifference curve I3.

Indifference analysis can be used to analyse how a consumer would change the combination of two goods for a given change in their income or the price of the good.

The next part looks at the income and substitution effects of a change in price. If we assume that the good is normal, then the increase in price will result in a fall in the quantity demanded. This is for two reasons; the income effect and the substitution effect. These two processes can be visualised using indifference analysis.

Due to the price of good x increasing, the budget line has pivoted from B1 to B2 and the consumption point has moved.

The decrease in the quantity demanded can be divided into two effects;

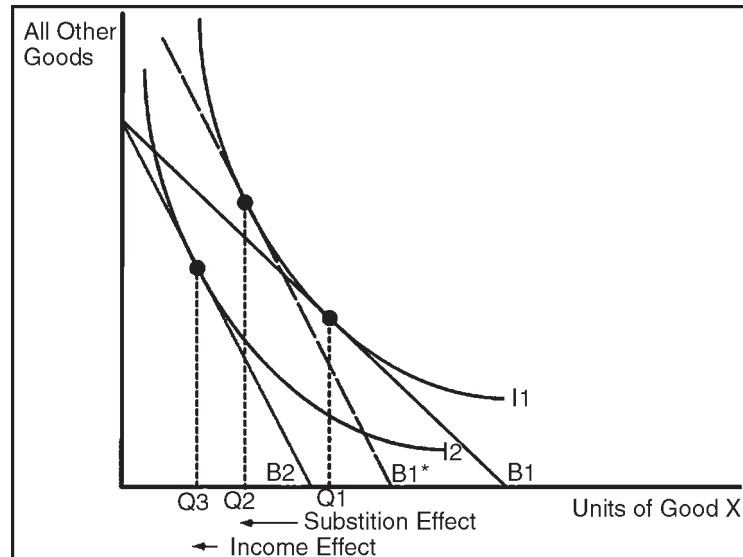


Fig. An Increase in the Price of Good x

THE SUBSTITUTION EFFECT

- The substitution effect is when the consumer switches consumption patterns due to the price change alone but remains on the same indifference curve. To identify the substitution effect a new budget line needs to be constructed. The budget line B1* is added, this budget line needs to be parallel with the budget line B2 and tangential to I1.

Therefore, the movement from Q1 to Q2 is purely due to the substitution effect.

THE INCOME EFFECT

- The income effect highlights how consumption will change due to the consumer having a change in purchasing power as a result of the price change. The higher price means the budget line is B2, hence the optimum consumption point is Q2. This point is on a lower indifference curve.

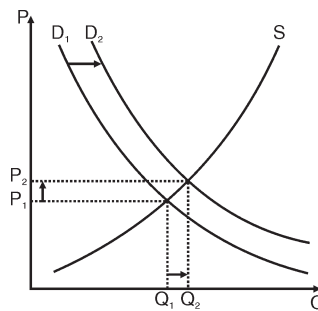
Therefore, in the case of a normal good, the income and substitution effects work to reinforce each other.

SUPPLY

What determines the amount of a good or service that people are willing and ready to sell during some period of time? A discussion of exchange suggested that people sell things because it is a way, indirect but effective, of obtaining other things that they prefer. Sellers intend to make a profit from their sales, and economists assume that they want their profits to be as large as possible. Because profit is the difference

between benefits in the form of revenues and costs, anything that influences revenues or costs can influence the amounts sellers want to sell.

Revenue is found by multiplying the price of the product by the amount sold. A change in price changes revenues and hence profits, so it is a major determinant of the amount sellers will want to sell. Because a higher price leads to higher profit, and a higher profit leads to a larger amount that sellers will want to sell,



one expects that a greater quantity should be supplied when the price is higher. Thus, the relationship between quantity that sellers will sell and price should be direct or positive.

Though the positive relationship is almost always the case, there are a few exceptions. An example is Labour; as wages go up, people may decide to enjoy their higher wages and work less. As a result, there is no law of supply that matches the law of demand.

The theory of supply and demand describes how prices vary as a result of a balance between product availability at each price (supply) and the desires of those with purchasing power at each price (demand). The graph depicts an increase in demand from D_1 to D_2 along with the consequent increase in price and quantity required to reach a new market-clearing equilibrium point on the supply curve. A curve or a schedule showing the total quantity of a good that sellers want to sell at each price. (S).

In microeconomic theory, the partial equilibrium supply and demand economic model originally developed by Antoine Augustin Cournot (published in a book in 1838) and thirty years later broadly publicized by Alfred Marshall attempts to describe, explain, and predict changes in the price and quantity of goods sold in competitive markets. The model is only a first approximation for describing an imperfectly competitive market.

It formalizes the theories used by some economists before Marshall and is one of the most fundamental models of some modern economic schools, widely used as a basic building block in a wide range of more detailed economic models and theories. The theory of supply and demand is important for some economic schools' understanding of a market economy in that it is an explanation of the mechanism by which many resource allocation decisions are made. However, unlike general equilibrium models, supply schedules in this partial equilibrium model are fixed by unexplained forces.

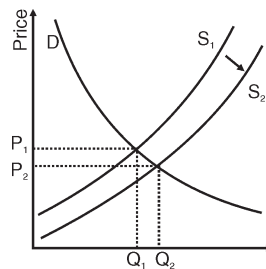
Supply is the amount of output available in the market. *i.e.*, it is the plan expressing quantities of a product that producers are willing to sell at given prices. For example, the potato growers may be willing to sell 1 million lbs of potatoes if the price is \$0.75

per lb and substantially more if the market price is \$0.90 per lb. The main determinants of supply will be the market price of the good and the cost of producing it. The supply curve for a given firm is derived directly from its marginal cost curve where the price is greater than or equal to the minimum cost on the average variable cost curve. Supply curves are traditionally represented as upward-sloping because of the law of diminishing marginal returns. This need not be the case, however, as described below.

Special Cases of a Supply Curve

As described above, the general form of a supply curve is upward sloping. There are cases, however, when supply curves do not slope upwards. A well known example is for the supply curve for Labour: backward bending supply curve of labour. As a person's wage increases, they are willing to supply a greater number of hours working, but when the wage reaches an extremely high amount (say a wage of \$4,000 per hour), the amount of Labour supplied actually decreases. Another example of a non-traditional supply curve is generally the supply curve for utility production companies. Because a large portion of their total costs are in the form of fixed costs, the marginal cost (supply curve) for these firms is often depicted as a constant.

Supply Curve Shifts



An out- or right- shift in supply changes the equilibrium price and quantity

When the suppliers' costs change the supply curve will shift (or fly). For example, assume that someone invents a better way of growing wheat so that the amount of wheat that can be grown for a given cost will increase. Producers will be willing to supply more wheat at every price and this shifts the supply curve S_0 to the right, to S_1 —an *increase in supply*. This causes the equilibrium price to decrease from P_0 to P_1 . The equilibrium quantity increases from Q_0 to Q_1 as the quantity demanded increases at the new lower prices. Notice that in the case of a supply curve shift, the price and the quantity move in opposite directions.

Conversely, if the quantity supplied *decreases*, the opposite happens. If the supply curve starts at S_1 and then shifts to S_0 , the equilibrium price will increase and the quantity will decrease. Notice that this is purely an effect of supply changing. The quantity demanded at each price is the same as before the supply shift (at both Q_0 and Q_1). The reason that the equilibrium quantity and price are different is the *supply* is different.

There are only 4 possible movements to a demand/supply curve diagram. The demand curve can move to the left and right, and the supply curve can also move only to the left or right. If they do not move at all then they will stay in the middle where they already are.

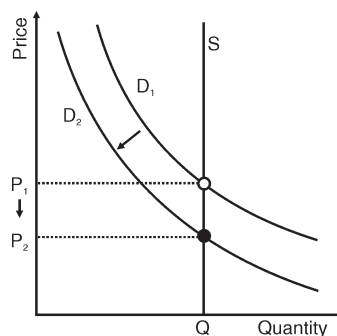
Market Clearance

The market “clears” at the point where all the supply and demand at a given price are balanced. That is, the amount of a commodity available at a given price equals the amount that buyers are willing to purchase at that price. It is assumed that there is a process that will result in the market reaching this point, but exactly what the process is in a real situation is an ongoing subject of research. Markets which do not clear will react in some way, either by a change in price, or in the amount produced, or in the amount demanded.

Graphically the situation can be represented by two curves: one showing the price-quantity combinations buyers will pay for, or the demand curve; and one showing the combinations sellers will sell for, or the supply curve. The market clears where the two are in equilibrium, that is, where the curves intersect. In a general equilibrium model, all markets in all goods clear simultaneously and the “price” can be described entirely in terms of tradeoffs with other goods. For a century most economists believed in Say’s Law, which states that markets, as a whole, would always clear and thus be in balance.

For traditional economics, the market clearing price contains no maximization basis. As a result, any disequilibrium (excess demand or excess supply) is just a matter of graphical exercises. Some economists regarded that a demand curve could be represented by a diminishing marginal use value curve (the schedule of a consumer’s maximum willing to pay with different quantity endowments). By this framework, people will buy when the market price is low enough, and they will sell when the price is high enough. In market clearing condition, the market price implies that the marginal use value of all participants are equalized. In other words, mutual gain of exchange is exhausted. Here come the basis of maximization for the concept of market equilibrium.

Vertical Supply Curve



When demand D_1 is in effect, the price will be P_1 . When D_2 is occurring, the price will be P_2 . Notice that at both values the quantity is Q . Since the supply is fixed, any shifts in demand will only affect price.

It is sometimes the case that the supply curve is vertical: that is the quantity supplied is fixed, no matter what the market price. For example, the amount of land in the world can be considered fixed. In this case, no matter how much someone would be willing to pay for a piece of land, the extra cannot be created. Also, even if no one wanted all the land, it still would exist. If land is considered in this way, then it warrants a vertical supply curve, giving it zero elasticity (*i.e.*, no matter how large the change in price, the quantity supplied will not change). On the other hand, the supply of useful land can be increased in response to demand — by irrigation. And land that otherwise would be below sea level can be kept dry by a system of dikes, which might also be regarded as a response to demand. So even in this case, the vertical line is a bit of a simplification.

In the short run near vertical supply curves are more common. For example, if the Super Bowl is next week, increasing the number of seats in the stadium is almost impossible. The supply of tickets for the game can be considered vertical in this case. If the organizers of this event underestimated demand, then it may very well be the case that the price that they set is below the equilibrium price. In this case there will likely be people who paid the lower price who only value the ticket at that price, and people who could not get tickets, even though they would be willing to pay more. If some of the people who value the tickets less sell them to people who are willing to pay more (*i.e.*, scalp the tickets), then the effective price will rise to the equilibrium price.

Other Market Forms

In a situation in which there are many buyers but a single monopoly supplier that can adjust the supply or price of a good at will, the monopolist will adjust the price so that his profit is maximized given the amount that is demanded at that price. This price will be higher than in a competitive market. A similar analysis using supply and demand can be applied when a good has a single buyer, a monopsony, but many sellers.

Where there are both few buyers or few sellers, the theory of supply and demand cannot be applied because both decisions of the buyers and sellers are interdependent—changes in supply can affect demand and vice versa. Game theory can be used to analyze this kind of situation.

The supply curve does not have to be linear. However, if the supply is from a profit-maximizing firm, it can be proven that supply curves are not downward sloping (*i.e.*, if the price increases, the quantity supplied will not decrease). Supply curves from profit-maximizing firms can be vertical, horizontal or upward sloping. While it is possible for industry supply curves to be downward sloping, supply curves for individual firms are never downward sloping.

Standard microeconomic assumptions cannot be used to prove that the demand curve is downward sloping. However, despite years of searching, no generally agreed upon

example of a good that has an upward-sloping demand curve has been found (also known as a Giffen good). Non-economists sometimes think that certain goods would have such a curve.

For example, some people will buy a luxury car because it is expensive. In this case the good demanded is actually prestige, and not a car, so when the price of the luxury car decreases, it is actually changing the amount of prestige so the demand is not decreasing since it is a different good. Even with downward-sloping demand curves, it is possible that an increase in income may lead to a decrease in demand for a particular good, probably due to the existence of more attractive alternatives which become affordable: a good with this property is known as an inferior good.

An example: Supply and demand in a 6-person economy

Supply and demand can be thought of in terms of individual people interacting at a market. Suppose the following six people participate in this simplified economy:

- Alice is willing to pay \$10 for a sack of potatoes.
- Bob is willing to pay \$20 for a sack of potatoes.
- Cathy is willing to pay \$30 for a sack of potatoes.
- Dan is willing to sell a sack of potatoes for \$5.
- Emily is willing to sell a sack of potatoes for \$15.
- Fred is willing to sell a sack of potatoes for \$25.

There are many possible trades that would be mutually agreeable to both people, but not all of them will happen. For example, Cathy and Fred would be interested in trading with each other for any price between \$25 and \$30. If the price is above \$30, Cathy is not interested, since the price is too high. If the price is below \$25, Fred is not interested, since the price is too low. However, at the market Cathy will discover that there are other sellers willing to sell at well below \$25, so she will not trade with Fred at all. In an efficient market, each seller will get as high a price as possible, and each buyer will get as low a price as possible. Imagine that Cathy and Fred are bartering over the price. Fred offers \$25 for a sack of potatoes. Before Cathy can agree, Emily offers a sack of potatoes for \$24. Fred is not willing to sell at \$24, so he drops out. At this point, Dan offers to sell for \$12. Emily won't sell for that amount so it looks like the deal might go through. At this point Bob steps in and offers \$14. Now we have two people willing to pay \$14 for a sack of potatoes, but only one person willing to sell for \$14. Cathy notices this and doesn't want to lose a good deal, so she offers Dan \$16 for his potatoes.

Now Emily also offers to sell for \$16, so there are two buyers and two sellers at that price (note that they could have settled on any price between \$15 and \$20), and the bartering can stop. But what about Fred and Alice? Well, Fred and Alice are not willing to trade with each other, since Alice is only willing to pay \$10 and Fred will not sell for any amount under \$25. Alice can't outbid Cathy or Bob to purchase from Dan, so Alice will not be able to get a trade with them. Fred can't underbid Dan or Emily, so he will not be able to get a trade with Cathy. In other words, a stable equilibrium has been reached.

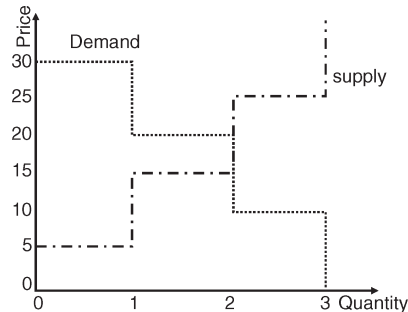


Fig. Discrete Supply and Demand Curves

A supply and demand graph could also be drawn from this. The demand would be:

- 1 person is willing to pay \$30 (Cathy).
- 2 people are willing to pay \$20 (Cathy and Bob).
- 3 people are willing to pay \$10 (Cathy, Bob, and Alice).

The supply would be:

- 1 person is willing to sell for \$5 (Dan).
- 2 people are willing to sell for \$15 (Dan and Emily).
- 3 people are willing to sell for \$25 (Dan, Emily, and Fred).

Supply and demand match when the quantity traded is two sacks and the price is between \$15 and \$20. Whether Dan sells to Cathy, and Emily to Bob, or the other way round, and what precisely is the price agreed cannot be determined. This is the only limitation of this simple model. When considering the full assumptions of perfect competition the price would be fully determined, since there would be enough participants to determine the price. For example, if the “last trade” was between someone willing to sell at \$15.50 and someone willing to pay \$15.51, then the price could be determined to the penny. As more participants enter, the more likely there will be a close bracketing of the equilibrium price.

It is important to note that this example violates the assumption of perfect competition in that there are a limited number of market participants. However, this simplification shows how the equilibrium price and quantity can be determined in an easily understood situation. The results are similar when unlimited market participants and the other assumptions of perfect competition are considered.

History of Supply and Demand

Attempts to determine how supply and demand interact began with Adam Smith’s *The Wealth of Nations*, first published in 1776. In this book, he mostly assumed that the supply price was fixed but that the demand would increase or decrease as the price decreased or increased. David Ricardo in 1817 published the book *Principles of Political Economy and Taxation*, in which the first idea of an economic model was proposed. In this, he more rigorously laid down the idea of the assumptions that were used to build his ideas of supply and demand.

During the late 19th century the marginalist school of thought emerged. This field mainly was started by Stanley Jevons, Carl Menger, and Léon Walras. The key idea was that the price was set by the most expensive price, that is, the price at the margin. This was a substantial change from Adam Smith's thoughts on determining the supply price.

Finally, most of the basics of the modern school theory of supply and demand were finalized by Alfred Marshall and Léon Walras, when they combined the ideas about supply and the ideas about demand and began looking at the equilibrium point where the two curves crossed. They also began looking at the effect of markets on each other. Since the late 19th century, the theory of supply and demand has mainly been unchanged. Most of the work has been in examining the exceptions to the model (like oligarchy, transaction costs, non-rationality).

Criticism of Marshall's Theory of Supply and Demand

Marshall's theory of supply and demand runs counter to the ideas of economists from Adam Smith and David Ricardo through the creation of the marginalist school of thought. Although Marshall's theories are dominant in universities today, other economists have disagreed with it. One theory counter to Marshall is that price is already known in a commodity before it reaches the market, negating his idea that some abstract market is conveying price information. The only thing the market communicates is whether or not an object is exchangeable or not (in which case it would change from an object to a commodity). This would mean that the producer creates the goods without already having customers — blindly producing, hoping that someone will buy them ("buy" meaning exchange money for the commodities). Modern producers often have market studies prepared well in advance of production decisions; however, misallocation of factors of production can still occur.

Keynesian economics also runs counter to the theory of supply and demand. In Keynesian theory, prices can become "sticky" or resistant to change, especially in the case of price decreases. This leads to a market failure. Modern supporters of Keynes, such as Paul Krugman, have noted this in recent history, such as when the Boston housing market dried up in the early 1990s, with neither buyers nor sellers willing to exchange at the price equilibrium.

Gregory Mankiw's work on the irrationality of actors in the markets also undermines Marshall's simplistic view of the forces involved in supply and demand.

SHUTTING DOWN AND BANKRUPTCY

As long as the firm produces something, it will maximize its profits by producing "on the marginal cost curve." But it might produce nothing at all. When will the firm shut down? The answer goes a bit against common sense. The firm will shut down if it cannot cover its variable costs. So long as it can cover the variable costs, it will continue to produce.

This is an application of the opportunity cost principle. Just because fixed costs are fixed, they are not opportunity costs in the short run—so they are not relevant to the decision to shut down. Even if the company shuts down, it must pay the fixed costs anyway. But the variable costs are avoidable—they are opportunity costs! So the firm will shut down if it cannot meet the variable (short run opportunity) costs. But as long as it can pay the variable costs and still have something to apply towards the fixed costs, it is better off continuing to produce.

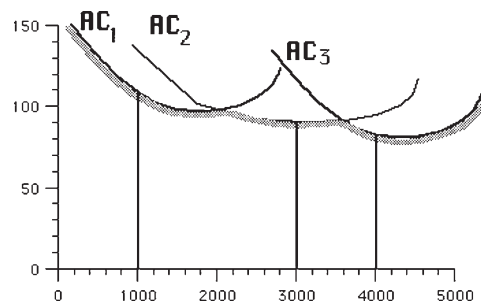
It is important not to confuse shut-down with bankruptcy. They are two different things. If a company cannot pay its interest and debt payments (usually fixed costs), then it is bankrupt. But that doesn't mean it will shut down. Bankrupt firms are often reorganized under new ownership and continue to produce—just because they can cover their variable costs and so the new owners do better to continue producing than to shut down.

LONG RUN COST

Thus far we have not considered the long run in cost theory. We will now think a bit about the long run, using the concept of average cost. We have defined “the long run” as “a period long enough so that all inputs are variable.” This includes, in particular, capital, plant, equipment and other investments that represent long-term commitments. Thus, here is another way to think of “the long run:” it is the perspective of investment planning. So let's approach it this way: Suppose you were planning to build a new plant—perhaps to set up a whole new company—and you know about how much output you will be producing. Then you want to build your plant so as to produce that amount at the lowest possible average cost. To make it a little simpler we will suppose that you have to pick just one of three plant sizes: small, medium and large. Here's the way they look in a picture:

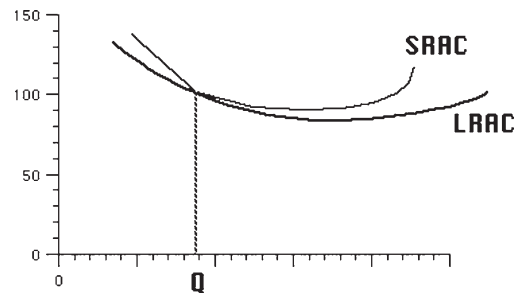
ENVELOPE CURVE

Here are the average cost curves for the small (AC_1), medium (AC_2) and large (AC_3) plant sizes:



If you produce 1000 units, the small plant size gives the lowest cost. If you produce 3000 units, the medium plant size gives the lowest cost. If you produce 4000 units, the large plant size gives you the lowest cost. Therefore, the long run average cost (LRAC)—

the lowest average cost for each output range—is described by the “lower envelope curve,” shown by the thick, shaded curve that follows the lowest of the three short run curves in each range.

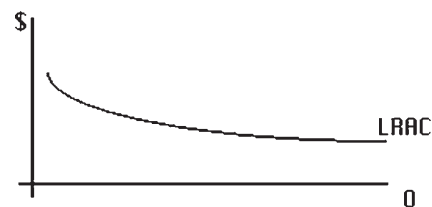


LONG RUN AVERAGE COST IN GENERAL

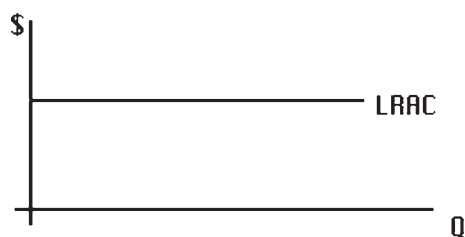
More realistically, an investment planner will have to choose between many different plant sizes or firm scales of operation and so the long run average cost curve will be smooth, something like this: As shown, each point on the LRAC corresponds to a point on the SRAC for the plant size or scale of operation that gives the lowest average cost for that scale of operation.

RETURNS TO SCALE

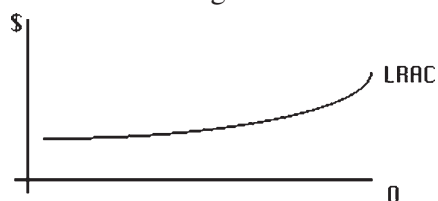
cases: Increasing returns to scale = decreasing cost



Constant returns to scale = constant costs



Decreasing returns to scale = increasing costs



In our pictures of long run average cost, we see that the cost per unit changes as the scale of operation or output size changes. Here is some terminology to describe the changes: increasing returns to scale = decreasing cost average cost decreases as output increases in the long run constant returns to scale = constant costs average cost is unchanged as output varies in the long run decreasing returns to scale = increasing costs average cost increases as output increases in the long run Here are pictures of the average cost curves for the three.

INCREASING RETURNS TO SCALE

Economists usually explain “increasing returns to scale” by indivisibility. That is, some methods of production can only work on a large scale—either because they require large-scale machinery, or because they require a great deal of division of labour. Since these large-scale methods cannot be divided up to produce small amounts of output, it is necessary to use less productive methods to produce the smaller amounts. Thus, costs increase less than in proportion to output—and average costs decline as output increases.

Increasing Returns to Scale is also known as “economies of scale” and as “decreasing costs.” All three phrases mean exactly the same.

CONSTANT RETURNS TO SCALE

We would expect to observe constant returns where the typical firm (or industry) consists of a large number of units doing pretty much the same thing, so that output can be expanded or contracted by increasing or decreasing the number of units. In the days before computer controls, machinery was a good example. Essentially, one machinist used one machine tool to do a series of operations to produce one item of a specific kind—and to double the output you had to double the number of machinists and machine tools.

Constant Returns to Scale is also known as “constant costs.” Both phrases mean exactly the same.

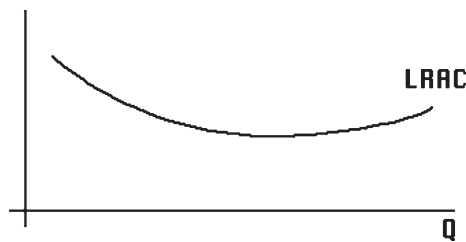
DECREASING RETURNS TO SCALE

Decreasing returns to scale are associated with problems of management of large, multi-unit firms. Again with think of a firm in which production takes place by a large number of units doing pretty much the same thing—but the different units need to be coordinated by a central management. The management faces a trade-off. If they don’t spend much on management, the coordination will be poor, leading to waste of resources and higher cost. If they do spend a lot on management, that will raise costs in itself. The idea is that the bigger the output is, the more units there are and the worse this trade-off becomes—so the costs rise either way.

Decreasing Returns to Scale is also known as “diseconomies of scale” and as “increasing costs.” All three phrases mean exactly the same.

THE LRAC CURVE

In our examples, the LRAC is (more or less roughly) u-shaped, like this:



The idea is that:

- For small outputs, indivisibilities predominate and so long run average cost declines with increasing output
- For intermediate outputs, operations can be expanded roughly proportionately, while tendencies to increasing and decreasing costs—if any—offset one another.
- For large outputs, the problems of management predominate and so long run average cost increases with increasing output.

That’s reasonable—but we should recall that it is pretty much a guess and may or may not apply in a particular case.

TYPES OF COST ANALYSIS

There is a variety of approaches to cost analysis, the suitability of any of which depends upon the purpose of an assessment and the availability of data and other resources. It is rarely possible or necessary to identify and quantify all costs and all benefits and the units used to quantify these may differ.

Main types of cost analysis include the following:

- *Cost-of-illness analysis:* a determination of the economic impact of an illness or condition *e.g.*, of smoking, arthritis or bedsores, including associated treatment costs.
- *Cost-minimization analysis:* a determination of the least costly among alternative interventions that are assumed to produce equivalent outcomes.
- *Cost-effectiveness analysis:* a comparison of costs in monetary units with outcomes in quantitative non-monetary units, *e.g.*, reduced mortality or morbidity.
- *Cost-utility analysis:* a form of cost-effectiveness analysis that compares costs in monetary units with outcomes in terms of their utility, usually to the patient, measured, *e.g.*, in QALYs.

- *Cost-consequence analysis*: a form of cost-effectiveness analysis that presents costs and outcomes in discrete categories, without aggregating or weighting them.
- *Cost-benefit analysis*: compares costs and benefits, both of which are quantified in common monetary units.

Contrasts the valuation of costs and outcomes among these alternative economic analyses.

Table. Different Types of Economic Analysis

	Valuation of Costs		Valuation of Outcomes
Cost of Illness	\$	vs.	None
Cost Minimization	\$	vs.	Assume same
Cost Effectiveness	\$	÷	Natural units
Cost Utility	\$	÷	Utiles
Cost Benefit	\$	÷ or -	\$

Cost-minimization analysis, CEA and CUA necessarily involve comparisons of alternative interventions. A technology cannot be simply cost effective, though it may be cost effective compared to something else. Although CBA typically involves comparisons of alternative technologies, this is not necessary.

Because it measures costs and outcomes in monetary terms, CBA enables comparison of disparate technologies, *e.g.*, coronary artery bypass graft surgery and screening for breast cancer. A drawback of CBA is the difficulty of assigning monetary values to all pertinent outcomes, including changes in the length or quality of human life. CEA avoids this limitation by using more direct or natural units of outcomes such as lives saved or strokes averted. As such, CEA can only compare technologies whose outcomes are measured in the same units. In CUA, estimates of utility are assigned to health outcomes, enabling comparisons of disparate technologies.

Two basic approaches for cost-benefit analysis are ratio approach and the net benefit approach. The ratio approach indicates the amount of benefits that can be realised per unit expenditure on a technology vs. a comparator. In the ratio approach, a technology is cost beneficial vs. a comparator if the ratio of the change in costs to the change in benefits is less than one. The net benefits approach indicates the absolute amount of money saved or lost due to a use of a technology vs. a comparator. In the net benefits formulation, a technology is cost-beneficial vs. a comparator if the net change in benefits exceeds the net change in costs. The choice between a net benefits approach or a benefit/cost approach for a CBA can affect findings.

The approach selected may depend upon such factors as whether costs must be limited to a certain level, whether the intent is to maximize the absolute level of benefits, whether the intent is to minimize the cost/benefit ratio regardless of the absolute level of costs, etc. Indeed, under certain circumstances these two basic approaches may yield different preferences among alternative technologies.

Basic formulas for determining CEA, CUA, and CBA.

Basic Formulas for CEA, CUA, and CBA

Int: Intervention; *Comp*: Comparator

Cost-Effectiveness Ratio:

$$\text{CE Ratio} = \frac{\$Cost_{Int} - \$Cost_{Comp}}{\$Effect_{Int} - \$Effect_{Comp}}$$

For Example: “\$45,000 per life-year saved” or “\$10,000 per lung cancer averted”

Cost-Utility Ratio:

$$\text{CU Ratio} = \frac{\$Cost_{Int} - \$Cost_{Comp}}{\$Utile_{Int} - \$Utile_{Comp}}$$

Utiles, units of utility or preference, are often measured in QALYs. So, for example:

“\$45,000 per life-year saved” or “\$10,000 per lung cancer case averted”

Cost-Benefit, Ratio Approach:

$$\text{CE Ratio} = \frac{\$Cost_{Int} - \$Cost_{Comp}}{\$Benefit_{Int} - \$Benefit_{Comp}}$$

For example: “Cost-benefit ratio of 1.5”

Cost-Benefit, Net Benefit Approach:

$$\text{CB Net} = (\$Cost_{Int} - \$Cost_{Comp}) - (\$Benefit_{Int} - \$Benefit_{Comp})$$

For example: “Net cost of \$5,000.”

QUADRANTS OF COST-EFFECTIVENESS

A basic approach to portraying a cost-effectiveness comparison of a new intervention to a standard of care is to consider the cost and effectiveness of a new intervention in the space of four fields as shown in Box 20, starting with the upper figure. The level of costs and the level of effectiveness for the standard of care are indicated by the “X” in the middle of the figure. A new intervention may have higher or lower costs, and higher or lower effectiveness, such that its plot may fall into one of the four quadrants surrounding the costs and effectiveness of the standard of care. If it is known that the plot of the new intervention falls into either of two of the quadrants, *i.e.*, where the new intervention has higher costs and lower effectiveness or it has lower costs and higher effectiveness then no further analysis may be required.

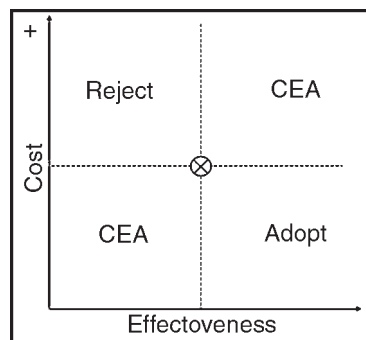
If it is known that the plot of the new intervention falls into either of the other two quadrants, *i.e.*, where the new intervention has higher costs and higher effectiveness, or it has lower costs and lower effectiveness, then further analysis weighing the marginal costs and effectiveness of the new intervention compared to the standard of care may be required. Within either of the two quadrants that entail weighing tradeoffs

of costs and effectiveness, it may be apparent that the marginal trade-off of costs and outcomes is so high or low as to suggest rejection or adoption. This arises when the new intervention yields only very low marginal gain in effectiveness at a very high marginal cost or yields very high marginal improvements in effectiveness at a very low marginal cost.

KEY ATTRIBUTES OF COST ANALYSES

The approaches to accounting for costs and outcomes in cost analyses can vary in a number of important respects. These should be carefully considered by assessors, as well as the policymakers who intend to make use of assessment findings. Given the different ways in which costs and outcomes may be determined, all studies should make clear their methodology in these respects. **Comparator.** Any cost analysis of one intervention versus another must be specific about the comparator. This may be standard of care, minimum practice, or no intervention. Some analyses that declare the superiority of a new intervention may have used a comparator that is no longer in practice or is considered sub-standard care or that is not appropriate for the patient population of interest.

Perspective. The perspective of a cost analysis refers to the standpoint at which costs and outcomes are realised. For instance, the perspective of an analysis may be that of society overall, a third-party payer, a physician, a hospital, or a patient. Clearly, costs and outcomes are not realised in the same way from each of these perspectives. Many analysts favour using the broad perspective of society and identifying all costs and all outcomes accordingly. However, “society” as such may not be the decisionmaker, and what is cost effective from that perspective may not be what is cost effective from the standpoint of a ministry of health, third-party payer, hospital manager, patient, or other decisionmaker. It is possible that this perspective may resemble that of a national or regional government, if indeed that government experiences all of the costs and outcomes that are included in a societal perspective.



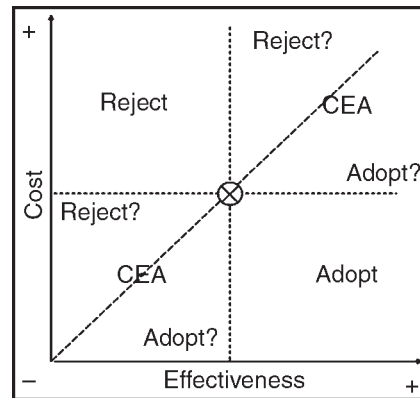


Fig. Quadrants of Cost-Effectiveness

Direct Costs. Depending upon the perspective taken, cost analyses should identify two types of direct costs. Direct costs represent the value of all goods, services, and other resources consumed in providing health care or dealing with side effects or other current and future consequences of health care. Two types of direct costs are direct health care costs and direct non-health care costs.

Direct health care costs include costs of physician services, hospital services, drugs, etc. involved in delivery of health care. Direct non-health care costs are incurred in connection with health care, such as for care provided by family members and transportation to and from the site of care. In quantifying direct health care costs, many analyses use readily available hospital or physician charges rather than true costs, whose determination may require special analyses of resource consumption.

However, charges tend to reflect provider cost shifting and other factors that decrease the validity of using charges to represent the true costs of providing care.

Indirect Costs. Analyses should account for indirect costs, sometimes known as “productivity losses”. These include the costs of lost work due to absenteeism or early retirement, impaired productivity at work, and lost or impaired leisure activity. Indirect costs also include the costs of premature mortality. Intangible costs of pain, suffering, and grief are real, yet very difficult to measure and are often omitted from cost analyses.

Time Horizon. Interpretation of cost analyses must consider that the time horizon of a study is likely to affect the findings regarding the relative magnitudes of costs and outcomes of a health care intervention. Costs and outcomes usually do not accrue in steady streams over time. Comparisons of costs and outcomes after one year may yield much different findings than comparisons made after 5, 10, or 25 years.

The meaningful time horizons for assessing the cost horizons of each of emergency appendectomies, cholesterol-lowering in high-risk adults, and smoking cessation in teenagers are likely to be quite different.

For example, an analysis conducted for the Medicare programme in the US to determine cost and time tradeoffs of hemodialysis and kidney transplantation showed that the annualized expenditure by the Medicare End-Stage Renal Disease Programme

for a dialysis patient was \$32,000. Although patients with functioning transplanted kidneys required a first-year expenditure of \$56,000, they cost Medicare only an average of \$6,400 in succeeding years.

On average, estimated cumulative dialysis and transplantation costs reach a break-even point in about three years, after which transplantation provides a net financial gain compared to dialysis.

Time horizons should be long enough to capture streams of health and economic outcomes. These could encompass a disease episode, patient life, or even multiple generations of life.

Quantitative modelling approaches may be needed to estimate costs and outcomes that are beyond those of available data. Of course, the higher the discount rate used in an analysis, the less important are future outcomes and costs.

Average Costs vs. Marginal Costs. Assessments should make clear whether average costs or marginal costs are being used in the analysis.

Whereas average cost analysis considers the total costs and outcomes of an intervention, marginal cost analysis considers how outcomes change with changes in costs, which may provide more information about how to use resources efficiently. Marginal cost analysis may reveal that, beyond a certain level of spending, the additional benefits are no longer worth the additional costs.

For example, as shown in Box 21, the average cost per desired outcome of an iterative screening test may appear to be quite acceptable whereas marginal cost analysis demonstrates that the cost of adding the last test to detect another case of cancer would be astronomical.

AVERAGE COST ANALYSIS VS. MARGINAL COST ANALYSIS

The importance of determining marginal costs is apparent in the analysis by Neuhauser and Lewicki of a proposed protocol of sequential stool guaiac testing for colon cancer.

Here, average cost figures obscure a steep rise in marginal costs of testing because the high detection rate from the initial tests is averaged over subsequent tests that contribute little to the detection rate.

This type of analysis helps to demonstrate how it is possible to spend steeply increasing health care resources for diminishing returns in health benefits.

Table. Cancer Screening and Detection Costs with Sequential Guaiac Tests

No. of cancers detected	Additional cancers detected	Total cost (\$) of diagnosis	Additional (\$) cost of diagnosis	Average cost (\$) per cancer detected	Marginal cost (\$) per cancer detected
65.9469	65.9469	77,511	77,511	1,175	1,175
71.4424	5.4956	107,690	30,179	1,507	5,492
71.9004	0.4580	130,199	22,509	1,810	49,150
71.9385	0.0382	148,116	17,917	2,059	469,534
71.9417	0.0032	163,141	15,024	2,268	4,724,695
71.9420	0.0003	176,331	13,190	2,451	47,107,214

This analysis assumed that there were 72 true cancer cases per 10,000 population. The testing protocol provided six stool guaiac tests per person to detect colon cancer. If any one of the six tests was positive, a barium-enema test was performed, which was assumed to yield no false positive and no false-negative results.

Other assumptions: the true-positive cancer detection rate of any single guaiac test was 91.667%; the false-positive rate of any single guaiac test was 36.508%; the cost of the first stool guaiac test was \$4 and each subsequent guaiac test was \$1; the cost of a barium-enema was \$100. The marginal cost per case detected depends on the population screened and the sensitivity of the test used.

Discounting. Cost analyses should account for the effect of the passage of time on the value of costs and outcomes. Costs and outcomes that occur in the future usually have less present value than costs and outcomes realised today. Discounting reflects the time preference for benefits earlier rather than later; it also reflects the opportunity costs of capital, *i.e.*, whatever returns on investment that could have been gained if resources had been invested elsewhere. Thus, costs and outcomes should be discounted relative to their present value. Discounting allows comparisons involving costs and benefits that flow differently over time. It is less relevant for “pay as you go” benefits, such as if all costs and benefits are realised together within one year. It is more relevant in instances where these do not occur in parallel, such as when most costs are realised early and most benefits are realised in later years. Discount rates used in cost analyses are typically based on interest rates of government bonds or the market interest rates for the cost of capital whose maturity is about the same as the duration of the effective time horizon of the health care intervention of programme being evaluated. Box 22 shows the basic formula for calculating present values for a given discount rate, as well as how the present value of a cost or benefit that is discounted at selected rates is affected over time.

Cost analyses should also correct for the effects of inflation, such as when costs or cost-effectiveness for one year are compared to another year.

Sensitivity Analysis. Any estimate of costs, outcomes, and other variables used in a cost analysis is subject to some uncertainty.

Therefore, sensitivity analysis should be performed to determine if plausible variations in the estimates of certain variables thought to be subject to significant uncertainty affect the results of the cost analysis. A sensitivity analysis may reveal, for example, that including indirect costs, or assuming the use of generic as opposed to brand name drugs in a medical therapy, or using a plausible higher discount rate in an analysis changes the cost-effectiveness of one intervention compared to another.

COLLECTING COST DATA ALONGSIDE CLINICAL STUDIES

The validity of a cost-related study depends upon the sources of the data for costs and outcomes. Increased attention is being given to collection of cost data in more

rigorous, prospective studies, particularly RCTs. The closer integration of economic and clinical studies raises important methodological issues.

In order to promote more rational diffusion of new technologies, it would be desirable to generate reliable cost and outcomes data during the early part of a technology's lifecycle, such as during RCTs required prior to marketing approval. An RCT would be expected to yield the most reliable data concerning efficacy of an intervention; however, the care given in an RCT and the costs of providing it may be atypical compared to more general settings. For example, RCTs may involve more extensive and frequent laboratory tests and other patient monitoring, and may occur more often in academic medical centres whose costs tend to be higher than in community health care institutions.

Other aspects of trial design, sample size, choice of outcome measures, identification and tabulation of costs, burden on investigators of data collection and related matters affect the usefulness of clinical trial data for meaningful economic studies.

Also, the growth of multinational clinical trials of drugs and other technologies raises challenges of estimating country-specific treatment effects and cost-effectiveness, given differences in epidemiological factors, health care delivery models, resource use, and other factors.

Discount Rate Calculation and Use in Determining Present Value of Future Costs and Benefits.

Discount rate calculation: Compelling the discounted stream of costs (or benefits) over time

$$P = \sum_{n=1}^n \frac{F_n}{(1+r)^n}$$

P = present value

F = future cost (or benefits) at year n

r = annual discount rate

Present value (P) of future cost (F) occurring at year n at selected annual discount rate (r)

Table. Discount Rate

Year	3%	5%	10%
1	0.97	0.95	0.91
5	0.86	0.78	0.62
25	0.48	0.30	0.09
50	0.23	0.09	0.009

For example, the present value of a cost (or benefit) of \$1,000 occurring:

- 1 year in the future, using 10% discount rate, is \$910
- 5 years in the future, using 3% discount rate, is \$860
- 50 yrs in the future, using 5% discount rate, is \$90

In practice, there is wide variation in economic study methodologies. Although some variation is unavoidable, many differences in perspective, accounting for direct and indirect costs, time frames, discounting and other aspects are often arbitrary, result from lack of expertise, and may reflect biases on the part of investigators or sponsors.

This diminishes comparability and transferability of study results as well as credibility of findings. National and international groups have developed and revised voluntary standards for conducting and reporting economic studies of health care technologies.

A recent review of 25 guidelines from North America, Europe, and Australia found a general trend towards harmonization in most methodological aspects, although there were more differences in such dimensions as choice of economic perspective, resources, and costs to be included in analysis.

RELATIONSHIP BETWEEN SHORT RUN AND LONG RUN COST FUNCTIONS

- The SRTC can be tangent to the LRTC at only one point. The SRTC cannot intersect the curve. The SRTC can lie wholly “above” the curve with no tangency point.
- The SRTC curve is tangent to LRTC at long run cost minimizing level of production. At the point of tangency $LRTC = SRTC$. At all other levels of production SRTC will exceed LRTC.
- Average cost functions are the total cost function divided by the level of output. Therefore the LRATC is also tangent to the SRATC at cost minimizing level of output. At the point of tangency $LRATC = SRATC$. At all other levels of production $SRATC > LRATC$.
- The slope of the total cost curves equals marginal cost. Therefore when LRTC is tangent to SRTC, $SRMC = LRMC$.
- At the long run cost minimizing level of output $LRTC = SRTC$; $SRATC = LRATC$ and $SRMC = LRMC$.
- The long run cost minimizing level of output may be different from minimum SRATC.
- If constant returns to scale then (min) $SRATC = LRATC = SRMC = LRMC$.
- If increasing returns to scale (min) SRATC will occur at higher level of production than long run cost minimizing level of production. While $LRTC = SRTC$; $SRATC = LRATC$ and $SRMC = LRMC$. SRMC does not equal LRMC and LRMC does not equal LRAC.

- With decreasing returns (min) SRATC will be tangent to the LRAC curve at a lower level of production lower than long run cost minimizing level of production. While $LRTC = SRTC$; $SRATC = LRATC$ and $SRMC = LRMC$. $SRMC$ does not equal $LRMC$ and $LRMC$ does not equal $LRAC$
- A firm that is experiencing increasing returns to scale and is producing at min SRAC can always reduce average cost by expanding the use of the fixed input.
- $LRATC$ will always equal or be less than $SRATC$.

Bibliography

- Ayrault Dodge Theodore, *Hannibal: A History of the Art of War Among the Carthaginians and Romans Down to the Battle of Pydna, 168 BC.*, Da Capo Press, 1995.
- B.S. Blanchard: *Logistics Engineering and Management*, Pearson Prentice Hall, New York, 2004.
- Baziotopoulos: *An Investigation of Logistics Outsourcing Practices in the Greek Manufacturing Sector*, PhD thesis, 2008.
- Betty A. Kildow: *Supply Chain Management Guide to Business Continuity*, American Management Association, 2011.
- Donald W. Engels: *Alexander the Great and the Logistics of the Macedonian Army*, University of California, 1980.
- Engels, Donald W.: *Alexander the Great and the Logistics of the Macedonian Army*, University of California Press, 1980.
- FAO: *Agro-industrial supply chain management: Concepts and applications*, AGSF Occasional Paper 17 Rome, 2007.
- Ferenc Szidarovszky and Sándor Molnár: *Introduction to Matrix Theory: With Applications to Business and Economics*, World Scientific Publishing, 2002.
- Goldberg, B and Sifonis, JG: *Dynamic planning: the art of managing beyond tomorrow*, Oxford University Press, New York, 1994.
- Haag, S., Cummings, M., McCubbrey, D., Pinsonneault, A., & Donovan, R.: *Management Information Systems For the Information Age (3rd Canadian Ed.)*, McGraw Hill Ryerson, Canada, 2006.

- Halldorsson, A., Kotzab, H., Mikkola, J. H., Skjoett-Larsen, T.: *Complementary theories to supply chain management: Supply Chain Management, Volume 12 Issue 4*, 2007.
- Hines, T.: *Supply chain strategies: Customer driven and customer focused*, Elsevier, Oxford, 2004.
- Hopp, W.: *Supply Chain Science*, Waveland Press, Chicago, 2011.
- Huston, James A.: *The Sinews of War: Army Logistics, 1775–1953*, United States Army, 1966.
- Kouvelis, P.; Chambers, C.; Wang, H.: *Supply Chain Management Research and Production and Operations Management: Review, Trends, and Opportunities*, Production and Operations Management, 2006.
- L. Torre, I.S. Dolinskaya, K.R. Smilowitz: *Disaster relief routing: Integrating research and practice*, Socio-Economic Planning Sciences, 2012.
- Larson, P.D. and Halldorsson, A.: *Logistics versus supply chain management: an international survey*, International Journal of Logistics: Research & Application, 2004.
- Nyhuis P., Wiendahi Hans-Peter: *Fundamentals of Production Logistics*, Springer Berlin Heidelberg, 2009.
- R.G. Poluha: *The Quintessence of Supply Chain Management: What You Really Need to Know to Manage Your Processes in Procurement, Manufacturing, Warehousing, and Logistics (Quintessence Series)*, Springer Heidelberg New York Dordrecht London, 2016.
- Raue, J.S. & Wieland, A.: *The interplay of different types of governance in horizontal cooperations: a view on logistics service providers*, The International Journal of Logistics Management, 2015.
- Ronald H. Ballou, Samir K. Srivastava: *Business Logistics: Supply Chain Management*, Pearson Education, 2007.
- S. H. Ma, Y. Lin: *Supply chain management*, Machinery Industry Press, Beijing, 2005.
- Simchi-Levi D., Kaminsky P., Simchi-levi E.: *Designing and Managing the Supply Chain, third edition*, Mcgraw Hill, 2007.
- Stanton, D.: *Supply Chain Management For Dummies, First Edition*, Wiley New York, 2017.
- Troy T. Kirby: *The Duke of Wellington and the Supply System During the Peninsula War*, CreateSpace Independent Publishing Platform, 2014.
- V. Misra, M.I. Kahn, U.K. Singh: *Supply Chain Management Systems: Architecture, Design and Vision*, North American Business Press, New York 2010.

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